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Tensor Formats in Banach Spaces

Minimal Subspaces, Geometry, and Applications to Open Quantum Systems

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Contents

Preface

This monograph presents a unified and self-contained treatment of the mathematical foundations of low-rank tensor formats in Banach spaces. It synthesises and extends the results of four research papers by the author and his collaborators Wolfgang Hackbusch and Anthony Nouy, published between 2012 and 2023.

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Antonio Falcó

Notation and Conventions

See the file `notation.tex` for the complete list of macros and symbols used throughout.

Part I
Algebraic and Topological Foundations

Chapter 1
Introduction and Motivation

1.1 The curse of dimensionality

Many of the central problems in modern mathematics and its applications involve functions, operators, or data defined on high-dimensional product domains. A d-variate function $f : I_1 \times \cdots \times I_d \to \mathbb{R}$ defined on a product of d intervals, for example, lives in a space whose dimension grows exponentially with d: if each factor I_α is discretised with n grid points, the natural approximation space has dimension n^d. For $d = 100$ and $n = 10$ this is 10^{100}, a number that no computer will ever handle directly. This exponential explosion of complexity was famously termed by Bellman the *curse of dimensionality.*

High-dimensional problems of this kind arise throughout science and engineering:

- *Quantum chemistry:* the electronic Schrödinger equation for an N-electron molecule lives in $\mathbb{R}^{3N}$; realistic molecules have N in the hundreds.
- *Uncertainty quantification:* the solution of a PDE with d uncertain parameters is a function on $\mathbb{R}^d \times \Omega$, where Ω is the spatial domain.
- *Stochastic control and finance:* value functions in optimal control problems typically satisfy PDEs in $\mathbb{R}^d$ for large d.
- *Machine learning:* the approximation of unknown functions from data in $\mathbb{R}^d$ is the fundamental task of supervised learning.

The common mathematical structure underlying all these problems is that the object of interest — a function, a quantum state, a probability density — is an element of a *tensor product space* ${}_a\bigotimes_{\alpha\in D} V_\alpha$, where $D = \{1, \ldots, d\}$ and V_α is a function space on the α-th variable. The curse of dimensionality is the statement that the dimension of this space grows exponentially with $\#(D) = d$.

The strategy for circumventing the curse is to restrict attention to *structured low-rank tensors*: elements of $\mathbf{V}_D$ that can be represented using far fewer parameters than n^d. The systematic study of such representations — their algebraic structure, their topological properties in function spaces, and their geometry as differentiable manifolds — is the subject of this monograph.

1.2 Tensor formats and the role of subspaces

From matrices to tensors

The simplest non-trivial case is $d = 2$: here $\mathbf{V}_D = V_1 \otimes_a V_2$, and a tensor $\mathbf{v} \in \mathbf{V}_D$ corresponds to a linear map (a "matrix") from V_1^* to V_2. The *rank* of $\mathbf{v}$ — the minimal r such that $\mathbf{v} = \sum_{\nu=1}^{r} u_\nu \otimes w_\nu$ — determines the complexity of the representation. In finite dimensions this is the rank of the associated matrix; the Eckart–Young theorem gives the best rank-r approximation in the spectral or Frobenius norm.

The higher-order analogue of rank, however, is not straightforward. For $d \geq 3$ there is no single notion of "tensor rank" with all the good properties of matrix rank. Instead, several notions of rank have proven useful, each corresponding to a different way of decomposing the factors of $\mathbf{V}_D$.

Tucker format

The *Tucker format* Tucker [1966] represents a tensor $\mathbf{v}$ as

$$\mathbf{v} = \sum_{\substack{1 \leq i_\alpha \leq r_\alpha \\ \alpha \in D}} C^{(D)}_{(i_\alpha)_{\alpha \in D}} \bigotimes_{\alpha \in D} u^{(\alpha)}_{i_\alpha},$$

where $u^{(\alpha)}_{i_\alpha} \in V_\alpha$ and $C^{(D)} \in \mathbb{R}^{r_1 \times \cdots \times r_d}$ is the *core tensor*. The storage cost is $\sum_\alpha r_\alpha n_\alpha + \prod_\alpha r_\alpha$, which is linear in d when the ranks r_α are bounded. The Tucker format is equivalent to the *multilinear SVD* (HOSVD) of De Lathauwer, De Moor, and Vandewalle De Lathauwer et al. [2000].

The correct functional-analytic description of the Tucker format is through *minimal subspaces*: the minimal subspace $U^{\min}_\alpha(\mathbf{v}) \subset V_\alpha$ of $\mathbf{v}$ is the smallest subspace such that $\mathbf{v} \in U^{\min}_\alpha(\mathbf{v}) \otimes_a V_{[\alpha]}$ (Chapter 4). The Tucker rank of $\mathbf{v}$ is the tuple $(r_\alpha)_{\alpha \in D}$ where $r_\alpha = \dim U^{\min}_\alpha(\mathbf{v})$.

Hierarchical Tucker and Tensor Train formats

The Tucker format still suffers from the curse of dimensionality through its core tensor: $\prod_\alpha r_\alpha$ grows exponentially with d. The *Hierarchical Tucker* (HT) format Hackbusch and Kühn [2009] and the *Tensor Train* (TT) format Oseledets [2011] resolve this by organising the dimensions into a *hierarchy*, encoded by a rooted tree T_D. Each format restricts the tensor to satisfy rank constraints at every level of the tree, reducing the total storage to $O(dr^3)$ (HT) or $O(dr^2 n)$ (TT), where r is the maximal rank.

Tree-based tensor formats

This monograph studies all these formats within a single unified framework: *tree-based Tucker formats*, introduced in Falcó et al. [2021]. Given a dimension partition tree T_D over D and a rank tuple $\mathfrak{r}$, the set

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) = \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$$

of tensors with fixed tree-based rank $\mathfrak{r}$ is an intersection of Tucker sets at each level of the tree. The Tucker format is the special case $T_D = T_D^{\mathrm{Tucker}}$ (depth 1); HT and TT correspond to binary trees of depth $\lceil \log_2 d \rceil$ and $d-1$ respectively.

1.3 Three fundamental questions

The results of this monograph are organised around three fundamental mathematical questions, each corresponding to one part of the book.

Question 1 (Chapter 4).

Does every tensor $\mathbf{v}$ *in an infinite-dimensional Banach tensor space* $\mathbf{V}_{D,\|\cdot\|_D}$ *have a best approximation from the set* $\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D)$*?*

For convex sets in reflexive Banach spaces, best approximations always exist. The set $\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D)$ is not convex (it is not even a subspace), so the existence of best approximations is non-trivial. The answer, established in Chapters 4 and 7, is *yes*: under a natural condition on the norm (condition (Inj), which holds for all L^p and Sobolev spaces), $\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D)$ is weakly closed in $\mathbf{V}_{D,\|\cdot\|_D}$, and reflexivity gives the best approximation.

Question 2 (Chapters 8 and 9).

Does the set $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ *(or* $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$*) of tensors with fixed rank carry the structure of a differentiable manifold, even in infinite dimensions?*

In finite dimensions, $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is known to be a smooth manifold (a quotient of a matrix group). In infinite dimensions, the standard quotient construction fails because the stabiliser group is infinite-dimensional and the quotient topology may not be well-behaved. The answer, established in Falcó et al. [2019] for Tucker formats and in Falcó et al. [2023] for tree-based formats, is *yes*: an explicit atlas with $\mathcal{C}^\infty$ transition maps can be constructed using the Grassmann–Banach manifold structure of the minimal subspaces.

Question 3 (Chapter 10).

Can the Dirac–Frenkel variational principle — the projection of the vector field onto the tangent space of the manifold — be extended from Hilbert to Banach tensor spaces?

The answer is *yes*, provided the tangent space $Z^{(D)}(\mathbf{v})$ is a *split subspace* of $\mathbf{V}_{D,\|\cdot\|_D}$ (i.e., has a bounded projection onto it). The immersion theorems of Chapters 8 and 9 — the deepest results of the monograph — establish exactly this.

Question 4 (Chapters 11 and 12).

Can the tensor format framework be used to describe and approximate quantum states, and to reduce the Lindblad master equation to a finite-rank manifold?

The answer is *yes*, through a single observation that connects the two theories: for a non-zero vector v in a Hilbert tensor space $H = \bigotimes_j H_j$, the minimal subspace $U_\alpha^{\min}(v)$ of Chapter 4 coincides exactly with the support of the reduced density operator $\rho_\alpha = \mathrm{Tr}_{\alpha^c}(\rho_v)$. The Tucker rank of v is the Schmidt rank of the quantum state; the Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(H)$ is the natural variational class for states with prescribed entanglement structure. The Dirac–Frenkel principle of Chapter 10 then yields a well-posed projected dynamics for both the von Neumann equation (closed systems) and the Lindblad equation (open systems) on the finite-rank manifold $\mathcal{D}_r(H) \subset \mathcal{T}_1^{sa}(H)$.

1.4 The four source papers

This monograph synthesises and expands four research papers written by the author and his collaborators Wolfgang Hackbusch and Anthony Nouy:

[FH2012] *On Minimal Subspaces in Tensor Representations* Falcó and Hackbusch [2012]. This paper introduces minimal subspaces as the fundamental algebraic object in tensor representations, proves their existence and uniqueness in general vector spaces, and establishes the existence of best Tucker approximations in reflexive Banach tensor spaces. It constitutes Part I (Chapters 3–4) of this monograph.

[FHN2019] *On the Dirac–Frenkel Variational Principle on Tensor Banach Spaces* Falcó et al. [2019]. This paper constructs the $\mathcal{C}^\infty$-Banach manifold structure of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ via explicit local charts, proves that the manifold is immersed in $\mathbf{V}_{D,\|\cdot\|_D}$ under condition (Inj), and extends the Dirac–Frenkel variational principle to Banach tensor spaces. It forms the basis of Chapters 8 and 10.

[FHN2021] *Tree-based Tensor Formats* Falcó et al. [2021]. This paper introduces the general framework of dimension partition trees, proves the existence of best tree-based approximations under conditions weaker than Falcó and Hackbusch [2012], and characterises tensors of fixed tree-based rank by means of transfer tensors. It underlies Chapters 5–7.

[FHN2023] *Geometry of Tree-based Tensor Formats in Tensor Banach Spaces* Falcó et al. [2023]. This paper provides the geometric description of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ as an intersection of Tucker manifolds, proves that it is a $\mathcal{C}^\infty$-Banach manifold compatible with the Tucker structure at each level of the tree, and extends the Dirac–Frenkel principle to tree-based formats. It forms the basis of Chapter 9.

[AF2026] *Fixed-Rank Density Operators in Trace Norm: Grassmann–Banach Manifolds and Dirac–Frenkel Projection for Lindblad Dynamics* Acevedo and Falcó [2026]. Written in collaboration with Alberto Acevedo, this paper develops the Grassmann–Banach manifold structure of fixed-rank density operators in the trace-norm topology, establishes the bounded tangent projector, and proves well-posedness of the projected Lindblad dynamics on the finite-rank manifold $\mathcal{D}_r(H)$. The fundamental observation motivating this work — that the minimal subspace $U_\alpha^{\min}(v)$ of a tensor v equals the support of the corresponding partial trace $\rho_\alpha = \mathrm{Tr}_{\alpha^c}(\rho_v)$ — connects the tensor format theory of the earlier papers with quantum mechanics. It forms the basis of Part **??** (Chapters 11 and 12).

1.5 What this monograph adds

Beyond collecting and unifying the results of the four source papers, this monograph contributes the following:

1. **Unified notation.** The four papers were written over a decade and use slightly different notational conventions. This monograph adopts a single coherent system (detailed in the Notation chapter), in which the Tucker format is the special case $T_D = T_D^{\mathrm{Tucker}}$ of the tree-based format, minimal subspaces are always denoted $U_\alpha^{\min}(\mathbf{v})$, and the model Banach space of the Tucker manifold is always $\mathbf{E}_D$.
2. **Expanded proofs.** Several results in the papers are stated with proof sketches or references to earlier work. This monograph provides complete proofs of the central theorems, including the full proof of the local chart system (Lemma 8.2.1), the $\mathcal{C}^\infty$ transition maps (Lemma 8.3.1), the tangent space identification (Proposition 8.4.1), and the immersion theorems (Theorems 8.5.1 and 9.6.1).
3. **Self-contained background.** Chapters 2 and 3 provide the Banach space and algebraic tensor product prerequisites in the form and notation used throughout, so that the monograph is accessible without consulting external references for the background.
4. **Historical context and comparison.** Each chapter closes with historical notes that place the results in context and compare them with alternative approaches in the literature.

5. **Open problems.** Chapter 13 collects concrete open problems arising naturally from the theory, ranging from the global topology of tensor manifolds to learning theory and the extension of the Lindblad reduction to unbounded generators.

1.6 Overview of the main results

We summarise the principal theorems of the monograph, grouped by theme.

Existence of best approximations (Chapters 4, 7).

- *Theorem 4.5.2* [FH2012]: In a reflexive Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$ satisfying (Inj), every $\mathbf{u} \in \mathbf{V}_{D,\|\cdot\|_D}$ has a best Tucker approximation of any prescribed rank.
- *Theorem 7.5.3* [FHN2021]: The same holds for tree-based formats under the weaker local condition (Inj)$[T_D]$.

Manifold structure (Chapters 8, 9).

- *Theorem 8.3.1* [FHN2019]: If the tensor product map is continuous, $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a $\mathcal{C}^\infty$-Banach manifold modelled on $\mathbf{E}_D = \prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \times \mathbb{R}^{\times r_\alpha}$. It is a fibre bundle over $\prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ with fibre $\mathbb{R}_*^{\times r_\alpha}$.
- *Theorem 9.4.1* [FHN2023]: Under (Inj) at each level, $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is a $\mathcal{C}^\infty$-Banach manifold, with an atlas compatible with the Tucker atlases at each level of T_D.

Immersion theorems (Chapters 8, 9).

- *Theorem 8.5.1* [FHN2019]: Under (Inj), the tangent subspace $Z^{(D)}(\mathbf{v})$ is a split subspace of $\mathbf{V}_{D,\|\cdot\|_D}$, and $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is an immersed submanifold of $\mathbf{V}_{D,\|\cdot\|_D}$.
- *Theorem 9.6.1* [FHN2023]: The same holds for $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, with tangent space $\bigcap_k Z^{(D)}([])\mathcal{P}_k(T_D)]\mathbf{v}$.

Dirac–Frenkel variational principle (Chapter 10).

- *Theorem 10.4.1* [FHN2019]: In a reflexive, strictly convex $\mathbf{V}_{D,\|\cdot\|_D}$ satisfying (Inj), the Dirac–Frenkel ODE on $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ has local solutions, and they are characterised by the metric projection onto $Z^{(D)}(\mathbf{v}_r(t))$.
- *Theorem 10.6.1* [FHN2023]: The same holds on $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$.

Quantum states and Lindblad dynamics (Chapters 11–12).

- *Theorem 11.1.1* [AF2026]: For $v \in \bigotimes_j H_j$ with $\|v\| = 1$, the support of the partial trace satisfies $\operatorname{supp}(\mathrm{Tr}_{\alpha^c}(\rho_v)) = U_\alpha^{\min}(v)$ and $\operatorname{rank}(\rho_\alpha) = r_\alpha(v)$.
- *Theorem 11.4.1* [AF2026]: The fixed-rank stratum $\mathcal{D}_r(H)$ is a $\mathcal{C}^\infty$-Banach manifold in the trace-norm topology, with bounded tangent projector $\hat{P}_\rho : \mathcal{T}_1^{sa}(H) \to T_\rho \mathcal{D}_r(H)$.
- *Theorem 12.3.1* [AF2026]: The Dirac–Frenkel projected Lindblad dynamics on $\mathcal{D}_r(H)$ has unique local solutions preserving trace, rank, and positivity.

1.7 Guide to the reader

The monograph is organised into five parts.

Part I: Algebraic and Topological Foundations (Chapters 2–4). Chapter 2 collects the necessary background from Banach space theory and infinite-dimensional differential geometry. Chapter 3 develops the algebraic theory of tensor products. Chapter 4 is the algebraic core of the monograph: it introduces minimal subspaces, proves their fundamental properties, and establishes the existence of best approximations. *Readers familiar with Banach manifolds may skip Chapter 2 and refer back as needed.*

Part II: Tree-Based Tensor Formats (Chapters 5–7). Chapter 5 introduces dimension partition trees and proves the intersection representation $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) = \bigcap_k \mathfrak{M}_{\mathfrak{r}_k}()$. Chapter 6 develops the algebraic theory of tree-based formats, culminating in the transfer tensor representation (Theorem 6.4.1). Chapter 7 treats the topological theory: the T_D-continuity condition, the identification theorem, and the existence of best tree-based approximations. *Readers primarily interested in the Tucker format may focus on Chapters 4 and then jump to Part III.*

Part III: Differential Geometry of Tensor Formats (Chapters 8–9). These two chapters contain the geometrically deepest results of the monograph. Chapter 8 constructs the $\mathcal{C}^\infty$ Tucker manifold and proves its immersion in $\mathbf{V}_{D,\|\cdot\|_D}$. Chapter 9 extends these results to tree-based formats via the Laplacian map and the intersection of Tucker atlases. *The key technical result is the immersion theorem (Theorems 8.5.1 and 9.6.1), which relies on Lemmas 8.5.1 and 8.5.2.*

Part IV: Variational Principles and Applications (Chapters 10–13). Chapter 10 applies the geometric results to the Dirac–Frenkel variational principle, establishing the existence and characterisation of solutions to the reduced ODE on tensor manifolds, and derives the explicit kinematic equations (the MCTDH system) in local chart coordinates. Chapter 13 collects open problems.

Part ??: Quantum States and Open Systems (Chapters 11–12). Chapter 11 develops the connection between tensor formats and quantum mechanics: the minimal subspace $U_\alpha^{\min}(v)$ equals the support of the reduced density operator ρ_α, the Tucker manifold is the natural variational class for states with fixed entanglement structure, and the best-approximation theorem becomes quantum state compression. Chapter 12 applies the Dirac–Frenkel principle to the Lindblad master equation, establishing well-posedness of the projected dynamics on the finite-rank manifold $\mathcal{D}_r(H)$ and deriving the kinematic equations in Stiefel-core coordinates. *Readers without background in quantum mechanics should read Appendix C before Part* **??**. *No physics background is assumed: Appendix C develops all operator-theoretic prerequisites from first principles.*

The five appendices provide quick reference material. Appendix A is an index to the Banach manifold theory of Chapter 2, plus the ODE existence theorem on Banach manifolds (used in Chapter 10) and material on connections and curvature (for Chapter 13). Appendix B is an index to the functional analysis of Chapters 2–4, plus the Hahn–Banach and Picard–Lindelöf theorems in Banach spaces. Appendix C develops the operator-theoretic background for Part **??**: Hilbert–Schmidt operators, trace-class operators, density operators, and their connection to tensor products. Appendix D collects the main algorithms for tensor approximation (HOSVD, ALS, projector-splitting integrator, DMRG). Appendix E is the complete index of notation.

Prerequisites.

The reader is assumed to be familiar with functional analysis at the level of a standard graduate course (Banach spaces, weak convergence, Hilbert spaces) and with differential manifolds at the level of a first course using smooth finite-dimensional manifolds. The infinite-dimensional theory (Banach manifolds, Grassmannians) is developed from scratch in Chapter 2.

Notation.

A complete list of symbols is given in the Notation chapter following this Introduction. All macros used in the text are defined in the file `notation.tex`.

1.8 Historical context

The mathematical study of tensor approximation has roots in several independent traditions.

In *multilinear algebra*, Hitchcock Hitchcock [1927] introduced the notion of tensor rank in 1927. Tucker Tucker [1966] introduced the Tucker format in psychometrics in 1966. The HOSVD was introduced by De Lathauwer, De Moor, and Vandewalle De Lathauwer et al. [2000] in 2000.

In *quantum physics*, the matrix product state (MPS) representation appeared in Vidal's work Vidal [2003] on quantum simulation and was independently proposed as the TT decomposition by Oseledets and Tyrtyshnikov Oseledets [2011]. The DMRG algorithm of White, which is equivalent to optimisation over MPS/TT manifolds, dates from 1992.

In *numerical analysis*, the HT format was introduced by Hackbusch and Kühn Hackbusch and Kühn [2009] with the explicit goal of controlling storage costs for high-dimensional PDEs. The existence of best approximations in infinite-dimensional tensor spaces was first established in the present framework by Falcó and Hackbusch Falcó and Hackbusch [2012].

The *differential-geometric* treatment of tensor manifolds began with Koch and Lubich Koch and Lubich [2007] in the finite-dimensional Hilbert space setting, motivated by the MCTDH method. The extension to infinite-dimensional Banach spaces in Falcó et al. [2019, 2023] required new ideas — notably the explicit chart construction via Grassmann charts and the immersion theorem — that form the core of this monograph.

Chapter 2
Banach Space Prerequisites

This chapter collects the functional-analytic and differential-geometric background required throughout the monograph. The reader familiar with Banach space theory and infinite-dimensional differential geometry may proceed directly to Chapter 3, referring back as needed. All vector spaces are over the field $\mathbb{K}$, which is either $\mathbb{R}$ or $\mathbb{C}$; unless stated otherwise, $\mathbb{K} = \mathbb{R}$.

2.1 Normed spaces and Banach spaces

We begin by fixing notation and recalling the central notions of normed and Banach space theory that will be used throughout.

Basic definitions

Let X be a vector space over $\mathbb{K}$. A *norm* on X is a map $\|\cdot\| : X \to [0, \infty)$ satisfying:

(N1) $\|x\| = 0$ if and only if $x = 0$,
(N2) $\|\lambda x\| = |\lambda| \, \|x\|$ for all $\lambda \in \mathbb{K}$, $x \in X$,
(N3) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$.

The pair $(X, \|\cdot\|)$ is called a *normed space*. A sequence $(x_n)_{n \in \mathbb{N}}$ in X is a *Cauchy sequence* if $\|x_n - x_m\| \to 0$ as $n, m \to \infty$. A normed space in which every Cauchy sequence converges is called a *Banach space*.

Two norms $\|\cdot\|$ and $\|\cdot\|'$ on X are *equivalent* if there exist constants $c, C > 0$ such that $c\|x\| \leq \|x\|' \leq C\|x\|$ for all $x \in X$. Equivalent norms define the same topology, and hence the same Cauchy sequences, so they produce the same Banach space structure.

Example 2.1.1 Let $I \subset \mathbb{R}$ be a measurable set and $1 \le p < \infty$. The Lebesgue space $L^p(I)$, equipped with the norm $\|f\|_{L^p} = \left(\int_I |f(x)|^p \, dx\right)^{1/p}$, is a Banach space. For $p = \infty$ one uses the essential supremum norm. The case $p = 2$ gives a Hilbert space with inner product $\langle f, g \rangle = \int_I f(x)\overline{g(x)} \, dx$.

Example 2.1.2 For a bounded open set $\Omega \subset \mathbb{R}^n$, $1 \le p < \infty$, and $k \in \mathbb{N}_0$, the Sobolev space $W^{k,p}(\Omega)$ consists of all functions $f \in L^p(\Omega)$ whose distributional derivatives $D^\alpha f$ of order $|\alpha| \le k$ belong to $L^p(\Omega)$. Equipped with the norm

$$\|f\|_{W^{k,p}} = \Big(\sum_{|\alpha| \le k} \|D^\alpha f\|_{L^p}^p \Big)^{1/p},$$

it is a Banach space. For $p = 2$ one writes $H^k(\Omega) = W^{k,2}(\Omega)$, which is a Hilbert space. These spaces appear as natural component spaces in high-dimensional tensor problems; see Example 4.5.1 and Section 7.5.

Continuous linear maps

Let X and Y be normed spaces. A linear map $A : X \to Y$ is *continuous* if and only if it is *bounded*, meaning

$$\|A\|_{Y \leftarrow X} \coloneqq \sup_{x \ne 0} \frac{\|Ax\|_Y}{\|x\|_X} = \sup_{\|x\|_X \le 1} \|Ax\|_Y < \infty.$$

The quantity $\|A\|_{Y \leftarrow X}$ is the *operator norm* of A. The space of all continuous linear maps from X to Y, denoted $\mathcal{L}(X, Y)$, is itself a normed space under the operator norm; it is a Banach space whenever Y is complete. We write $\mathcal{L}(X) \coloneqq \mathcal{L}(X, X)$. The set of bounded linear automorphisms (bounded invertible maps with bounded inverse) is

$$\mathrm{GL}(X) \coloneqq \{A \in \mathcal{L}(X) : A \text{ invertible}\}.$$

This is an open subset of $\mathcal{L}(X)$, and hence a Banach manifold modelled on $\mathcal{L}(X)$; see Example 2.4.2.

Dual spaces and reflexivity

The *topological dual* of a normed space $(X, \|\cdot\|)$ is $X^* \coloneqq \mathcal{L}(X, \mathbb{K})$, equipped with the dual norm

$$\|\varphi\|_{X^*} \coloneqq \sup_{\|x\| \le 1} |\varphi(x)|. \tag{2.1}$$

The dual X^* is always a Banach space, regardless of whether X is complete. The *bidual* $X^{**} := (X^*)^*$ contains X as a canonically embedded subspace via the isometric embedding $\iota : X \to X^{**}$, $\iota(x)(\varphi) = \varphi(x)$. When ι is surjective, X is called *reflexive*. All L^p spaces with $1 < p < \infty$ are reflexive; L^1 and L^∞ are not.

Reflexivity plays an important role in the existence of best approximations. Recall that a closed set C in a normed space X is *proximinal* if every point of X has at least one nearest point in C. In a reflexive Banach space, every closed convex set is proximinal (see, e.g., [Hackbusch, 2019, p. 61]). This fact will be used extensively in Chapter 4 to prove the existence of best low-rank tensor approximations.

Convexity and smoothness

Two geometric properties of the norm are of particular importance for the Dirac–Frenkel variational principle in Chapter 10.

Definition 2.1.1 A normed space $(X, \|\cdot\|)$ is *strictly convex* (or *rotund*) if $\|x + y\|/2 < 1$ for all $x, y \in X$ with $\|x\| = \|y\| = 1$ and $x \neq y$.

Definition 2.1.2 A normed space $(X, \|\cdot\|)$ is *uniformly convex* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|x+y\|/2 \leq 1 - \delta$ whenever $\|x\|, \|y\| \leq 1$ and $\|x - y\| \geq \varepsilon$.

Every uniformly convex Banach space is reflexive and strictly convex (Clarkson's theorem). In particular, L^p spaces with $1 < p < \infty$ are uniformly convex. Strict convexity guarantees uniqueness of metric projections onto closed convex sets, a property needed for the well-posedness of the Dirac–Frenkel equation.

Definition 2.1.3 A Banach space X is *smooth* if the limit $\lim_{t\to 0}\big(\|x + ty\| - \|x\|\big)/t$ exists for all $x, y \in X$ with $\|x\| = 1$.

Smoothness is the dual property to strict convexity in the following sense: X is smooth if and only if X^* is strictly convex, and X is strictly convex if and only if X^* is smooth.

2.2 Complemented subspaces and projections

The central geometric notion underlying all manifold structures in this monograph is that of a complemented (or split) subspace. Unlike the Hilbert space setting, complementability is a non-trivial condition in general Banach spaces.

Projections

A bounded linear map $P \in \mathcal{L}(X)$ is a *projection* if $P^2 = P$. In that case $\operatorname{Im} P$ is a closed subspace, $\operatorname{Ker} P$ is closed, and $X = \operatorname{Im} P \oplus \operatorname{Ker} P$ (as a direct sum of closed subspaces). The complementary projection is $\mathrm{id}_X - P$.

Definition 2.2.1 Let X be a Banach space. A closed subspace $U \subset X$ is a *split subspace* (or *complemented subspace*) of X if there exists a closed subspace $W \subset X$ such that $X = U \oplus W$. The pair (U, W) is then called a *pair of complementary subspaces*, and W is a *(topological) complement* of U in X. We write $P_{U \oplus W}$ for the unique continuous projection from X onto U along W.

The following proposition characterises split subspaces in terms of projections and provides three equivalent conditions that will each be used at different points of the text.

Proposition 2.2.1 ([Falcó et al., 2015, Proposition 4.2]) *Let X be a Banach space and $U \subset X$ a closed subspace. The following conditions are equivalent:*

(a) *U is a split subspace of X.*
(b) *There exists $P \in \mathcal{L}(X)$ with $P^2 = P$ and $\operatorname{Im} P = U$.*
(c) *There exists $Q \in \mathcal{L}(X)$ with $Q^2 = Q$ and $\operatorname{Ker} Q = U$.*

In condition (b), the projection P is none other than $P_{U \oplus W}$ for any choice of closed complement W of U; in (c), one has $Q = P_{W \oplus U}$.

Proposition 2.2.2 ([Falcó et al., 2015, Theorem 4.5]) *Let X be a Banach space. Every finite-dimensional subspace of X is a split subspace.*

In a Hilbert space, every closed subspace is complemented (the orthogonal complement provides the split). This is no longer true in general Banach spaces: by a classical result of Lindenstrauss and Tzafriri, a Banach space in which every closed subspace is complemented is necessarily isomorphic to a Hilbert space. The failure of complementability in non-Hilbert spaces is precisely what makes the immersion theorems of Chapter 8 non-trivial.

The nilpotent algebra associated to a splitting

The following result, central to the construction of local charts in Chapters 8 and 9, identifies a particularly well-behaved sub-algebra of $\mathcal{L}(X)$ associated to any splitting $X = U \oplus W$.

Proposition 2.2.3 *Let X be a Banach space and $U, W \in \mathbb{G}(X)$ with $X = U \oplus W$. Define*

$$\mathcal{L}_{(U,W)}(X) \coloneqq \left\{ P_{W \oplus U} \circ S \circ P_{U \oplus W} : S \in \mathcal{L}(X) \right\} \cong \mathcal{L}(U, W).$$

Then:

(i) *The natural map* $\mathcal{L}(U,W) \to \mathcal{L}_{(U,W)}(X)$, $L \mapsto P_{W\oplus U} \circ L \circ P_{U\oplus W}$, *is an isometric isomorphism.*
(ii) $\mathcal{L}_{(U,W)}(X)$ *is a nilpotent sub-algebra of* $\mathcal{L}(X)$*: for every* $L, L' \in \mathcal{L}_{(U,W)}(X)$ *one has* $L \circ L' = 0$.
(iii) *As a consequence, for every* $L \in \mathcal{L}_{(U,W)}(X)$*:*

$$\exp(L) \coloneqq \sum_{n=0}^{\infty} \frac{L^n}{n!} = \mathrm{id}_X + L, \qquad \exp(-L) = \mathrm{id}_X - L = (\mathrm{id}_X + L)^{-1}. \tag{2.2}$$

Proof. For the isometry in (i), note that $\|P_{W\oplus U} \circ L \circ P_{U\oplus W}\|_{X\leftarrow X} = \|L\|_{W\leftarrow U}$ by the definition of the operator norm. For (ii): if $L = P_{W\oplus U} \circ S \circ P_{U\oplus W}$ and $L' = P_{W\oplus U} \circ S' \circ P_{U\oplus W}$, then

$$L \circ L' = P_{W\oplus U} \circ S \circ (P_{U\oplus W} \circ P_{W\oplus U}) \circ S' \circ P_{U\oplus W} = 0,$$

since $P_{U\oplus W} \circ P_{W\oplus U} = 0$. Statement (iii) follows immediately from (ii) and the series definition of the exponential. □

The identity $\exp(L) = \mathrm{id}_X + L$ means that the exponential map on the nilpotent algebra $\mathcal{L}_{(U,W)}(X)$ is a polynomial of degree one, and in particular a global analytic diffeomorphism. This is the key that makes the local charts of the Grassmann–Banach manifold and the Tucker manifold globally well-defined; see Section 2.3.

Common complements and chart compatibility

The following lemma, which underlies the compatibility of overlapping charts in the Grassmann–Banach manifold, characterises when two split subspaces share a common complement.

Lemma 2.2.1 ([Dixmier and Douady, 1963, Lemma 2.1]) *Let* X *be a Banach space and let* $U, U', W \in \mathbb{G}(X)$ *with* $X = U \oplus W = U' \oplus W$. *The following conditions are equivalent:*

(a) U *and* U' *have* W *as a common complement, i.e.,* $X = U \oplus W = U' \oplus W$.
(b) *The restriction* $P_{U\oplus W}\big|_{U'} : U' \to U$ *is a linear isomorphism (with bounded inverse).*

Moreover, in this case $\left(P_{U\oplus W}\big|_{U'}\right)^{-1} = P_{U'\oplus W}\big|_{U}$.

2.3 The Grassmann–Banach manifold

The Grassmannian of a Banach space is the natural parameter space for subspaces, and it serves as the base manifold for the Tucker and tree-based tensor manifolds.

Definition 2.3.1 Let X be a Banach space. The *Grassmann manifold* (or *Grassmannian*) of X is

$$\mathbb{G}(X) \coloneqq \{U \subset X : U \text{ is a split subspace of } X\}.$$

For $r \in \mathbb{N}_0$, the *finite Grassmannian* is

$$\mathbb{G}_r(X) \coloneqq \{U \in \mathbb{G}(X) : \dim U = r\}.$$

Both X and $\{0\}$ belong to $\mathbb{G}(X)$. By Proposition 2.2.2, every r-dimensional subspace of X belongs to $\mathbb{G}(X)$, so $\mathbb{G}_r(X)$ is well defined.

Theorem 2.3.1 (Grassmann–Banach manifold, Douady [1966]) *Let X be a Banach space. The Grassmannian $\mathbb{G}(X)$ is an analytic Banach manifold, with atlas $\{\mathbb{G}(W,X),\, \Psi_{U\oplus W}\}_{U\in\mathbb{G}(X)}$, where for each splitting $X = U \oplus W$:*

(i) $\mathbb{G}(W,X) \coloneqq \{V \in \mathbb{G}(X) : X = V \oplus W\}$ *is the chart domain at U.*
(ii) *The chart map $\Psi_{U\oplus W} : \mathbb{G}(W,X) \to \mathcal{L}(U,W)$ is defined by*

$$\Psi_{U\oplus W}(V) \coloneqq P_{W\oplus U}\big|_V \circ \left(P_{U\oplus W}\big|_V\right)^{-1}, \tag{2.3}$$

with inverse $\Psi^{-1}_{U\oplus W}(L) = G(L) \coloneqq \{(\mathrm{id}_X + L)(u) : u \in U\}$.

The transition maps are analytic. Each chart domain $\mathbb{G}(W,X)$ is a Banach manifold modelled on $\mathcal{L}(U,W)$.

The chart map $\Psi_{U\oplus W}$ sends a subspace $V \in \mathbb{G}(W,X)$ to the linear map $L : U \to W$ whose graph $G(L)$ equals V. The identity $\Psi^{-1}_{U\oplus W}(L) = \exp(L)(U)$, combined with Proposition 2.2.3, shows that the inverse chart is a degree-one polynomial map (hence analytic). Analyticity of the transition maps follows by a direct computation using Lemma 2.2.1.

Remark 2.3.1 The full Grassmannian $\mathbb{G}(X)$ is a Banach manifold but is *not* modelled on a single Banach space: if U and U' are finite-dimensional with $\dim U \neq \dim U'$, then $\mathcal{L}(U,W)$ and $\mathcal{L}(U',W')$ are non-isomorphic. This is not a deficiency but a structural fact: $\mathbb{G}(X)$ decomposes into connected components $\mathbb{G}(X) = \bigsqcup_{r\in\mathbb{N}_0} \mathbb{G}_r(X)$, each of which *is* modelled on a single Banach space.

Proposition 2.3.1 *For each $r \in \mathbb{N}_0$, the finite Grassmannian $\mathbb{G}_r(X)$ is a connected component of $\mathbb{G}(X)$, and it is an analytic Banach manifold modelled on $\mathcal{L}(U,W)$ for any $U \in \mathbb{G}_r(X)$ and complement W (the isomorphism class of $\mathcal{L}(U,W)$ does not depend on the choice of U or W).*

Remark 2.3.2 (Grassmannian of a normed space) The construction extends to normed (non-complete) spaces. Let $(X, \|\cdot\|)$ be a normed space with completion $\bar{X}$. Define $\mathbb{G}_r(X)$ as the set of r-dimensional subspaces of X that are also split in $\bar{X}$. By Lemma 2.5.1, every chart domain $\mathbb{G}(W, \bar{X})$ is contained in $\mathbb{G}_r(X)$, so the atlas of $\mathbb{G}_r(\bar{X})$ restricts to an analytic atlas on $\mathbb{G}_r(X)$. This variant is needed in Section 8.2, where the component spaces V_α are normed but not necessarily complete.

The tangent space to $\mathbb{G}(X)$ at U is identified as follows.

Proposition 2.3.2 *For $U \in \mathbb{G}(X)$ with complement W, the tangent space $T_U\mathbb{G}(X) \cong \mathcal{L}(U, W)$.*

This identification is consistent with the finite-dimensional case: in $\mathbb{R}^n$, the tangent space to the Grassmannian $Gr(r, n)$ at a subspace U is the space of linear maps from U to its orthogonal complement, recovering the classical description.

2.4 Banach manifolds

We recall the definition of a Banach manifold in its general form, since the full Grassmannian $\mathbb{G}(X)$ and the full tensor space $\mathbf{V}_{D,\|\cdot\|_D}$ are manifolds not modelled on a fixed Banach space.

Definition 2.4.1 Let $\mathbb{M}$ be a set. An *atlas of class* $\mathcal{C}^p$ (for $p \geq 0$ or $p = \infty$ or $p = \omega$, the analytic case) on $\mathbb{M}$ is a family of *charts* $\{(M_i, u_i)\}_{i \in A}$ satisfying:

(AT1) $\{M_i\}_{i \in A}$ covers $\mathbb{M}$.
(AT2) For each $i \in A$, the chart $u_i : M_i \to U_i$ is a bijection onto an open set U_i in some Banach space X_i, and for any $i, j \in A$, the set $u_i(M_i \cap M_j)$ is open in X_i.
(AT3) Whenever $M_i \cap M_j \neq \emptyset$, the *transition map* $u_j \circ u_i^{-1} : u_i(M_i \cap M_j) \to u_j(M_i \cap M_j)$ is a diffeomorphism of class $\mathcal{C}^p$.

Two atlases are *compatible* if their union is again an atlas. A *$\mathcal{C}^p$-Banach manifold* is a set $\mathbb{M}$ together with a maximal (complete) atlas of class $\mathcal{C}^p$. If all X_i are isomorphic to a single Banach space X, we say $\mathbb{M}$ is *modelled on* X.

Example 2.4.1 Every Banach space X is a $\mathcal{C}^\infty$-Banach manifold modelled on itself, with the single chart (X, id_X).

Example 2.4.2 If X is a Banach space, then $\mathrm{GL}(X)$ is an open subset of $\mathcal{L}(X)$, and hence an analytic Banach manifold modelled on $\mathcal{L}(X)$. The maps $A \mapsto A^{-1}$ and $(A, B) \mapsto AB$ are analytic (see [Upmeier, 1985, 2.7, 2.8]).

Definition 2.4.2 Let $\mathbb{M}$ be a $\mathcal{C}^p$-Banach manifold with $p \geq 1$ and $m \in \mathbb{M}$. A *tangent vector* at m is an equivalence class of triples (U_i, u_i, v) where (U_i, u_i) is a chart at m and $v \in X_i$, under the equivalence $(U_i, u_i, v) \sim (U_j, u_j, w)$ if $D(u_j \circ u_i^{-1})(u_i(m))v = w$. The *tangent space* $T_m\mathbb{M}$ is the set of all tangent vectors at m, which inherits the structure of a Banach space from any chart.

Definition 2.4.3 A $\mathcal{C}^1$-map $f : \mathbb{M} \to \mathbb{N}$ between Banach manifolds is an *immersion* at m if $Df(m) : T_m\mathbb{M} \to T_{f(m)}\mathbb{N}$ is injective with closed split image. It is an *immersion* if it is an immersion at every point. An *immersed submanifold* of $\mathbb{N}$ is the image of an injective immersion.

The closedness and split condition on the image of $Df(m)$ is automatic in finite dimensions and in Hilbert spaces (where every closed subspace is split), but is a genuine constraint in Banach spaces. This is the key difficulty resolved in Section 8.5.

2.5 Multilinear maps and Fréchet differentiability

Since the tensor product map is multilinear, the following general results will be used repeatedly.

Definition 2.5.1 A map $M : X_1 \times \cdots \times X_d \to Y$ between normed spaces is *multilinear* if it is linear in each argument separately. It is *bounded* (equivalently, *continuous*) if

$$\|M\| := \sup_{\substack{x_j \in X_j \\ \|x_j\|_j \le 1}} \|M(x_1, \ldots, x_d)\|_Y < \infty.$$

The following proposition shows that continuity implies a much stronger regularity.

Proposition 2.5.1 ([Hilgert and Neeb, 2012, Proposition 79]) *Let $X_1, \ldots, X_d$ and Y be normed spaces and let $M : X_1 \times \cdots \times X_d \to Y$ be a continuous multilinear map. Then M is $\mathcal{C}^\infty$-Fréchet differentiable, and $D^k M = 0$ for all $k > d$.*

The vanishing of derivatives beyond order d reflects the polynomial nature of multilinear maps. In the complex case, a stronger conclusion holds.

Proposition 2.5.2 ([Hilgert and Neeb, 2012, Theorem 160]) *Let X, Y be complex Banach spaces and $U \subset X$ open. A map $f : U \to Y$ is analytic if and only if it is $\mathcal{C}^1$-Fréchet differentiable.*

Corollary 2.5.1 *Every continuous multilinear map between complex Banach spaces is analytic.*

In real Banach spaces, $\mathcal{C}^\infty$ and analytic are distinct notions; the tensor product map is $\mathcal{C}^\infty$ but need not be analytic over $\mathbb{R}$. For the real case, $\mathcal{C}^\infty$-regularity is sufficient for the manifold constructions in Chapters 8 and 9.

Remark 2.5.1 Proposition 2.5.1 is the reason why the Tucker and tree-based tensor manifolds are $\mathcal{C}^\infty$-Banach manifolds (and analytic over $\mathbb{C}$): their chart transition maps are compositions of continuous multilinear maps with bounded linear projections, all of which are $\mathcal{C}^\infty$-Fréchet differentiable.

The Grassmannian revisited: Lemma on normed spaces

For later use in Section 8.2, we record the following lemma about Grassmannians of normed (non-complete) spaces.

Lemma 2.5.1 *Let $(X, \|\cdot\|)$ be a normed space with completion $\bar{X}$. Let $U \in \mathbb{G}_r(\bar{X})$ with $U \subset X$, and suppose $\bar{X} = U \oplus W$ for some $W \in \mathbb{G}(\bar{X})$. Then every $V \in \mathbb{G}(W, \bar{X})$ satisfies $V \subset X$.*

Proof. Observe that $X = U \oplus (W \cap X)$, where $W \cap X$ is dense in W. Suppose for contradiction that there exists $V \in \mathbb{G}(W, \bar{X})$ with $V \not\subset X$. Since V is finite-dimensional and $\bar{X} = V \oplus W$, the subspace $V \cap X$ is a proper subspace of V. On the other hand, $X = (V \cap X) \oplus (W \cap X)$ implies $\bar{X} = (V \cap X) \oplus W$, so $V \cap X \in \mathbb{G}(W, \bar{X})$. But $\dim(V \cap X) < \dim V = r$, contradicting the fact that every element of $\mathbb{G}(W, \bar{X})$ has dimension r. □

2.6 Curves in the Grassmannian and retractions

The Tucker and tree-based tensor manifolds live over Grassmannians, and the numerical algorithms of Appendix D — the projector-splitting integrator, the DMRG sweep, and the geometric ALS — require the ability to move along curves in $\mathbb{G}_r(X)$ efficiently. This section develops the necessary tools: explicit curves in $\mathbb{G}_r(L^2)$, the notion of retraction, and the connection between the chart-based retraction and the exponential map.

Curves in the Grassmannian via the chart

Fix $U_0 \in \mathbb{G}_r(X)$ with complement W_0, $X = U_0 \oplus W_0$. By Theorem 2.3.1, a $\mathcal{C}^1$-curve $\gamma : (-\varepsilon, \varepsilon) \to \mathbb{G}_r(X)$ with $\gamma(0) = U_0$ lying in the chart domain $\mathbb{G}(W_0, X)$ is uniquely represented by a $\mathcal{C}^1$-curve $L : (-\varepsilon, \varepsilon) \to \mathcal{L}(U_0, W_0)$ with $L(0) = 0$ and $\gamma(t) = G(L(t)) = (\mathrm{id}_X + L(t))(U_0)$. The velocity at $t = 0$ is $\dot{\gamma}(0) = \dot{L}(0) \in \mathcal{L}(U_0, W_0) = T_{U_0}\mathbb{G}_r(X)$.

Example 2.6.1 (Curves in $\mathbb{G}_1(L^2([0,1]))$) Let $X = L^2([0,1])$, $r = 1$, $U_0 = \mathrm{span}\{\phi_1\}$, where $\phi_k(x) = \sqrt{2}\sin(k\pi x)$ are the Dirichlet eigenfunctions of $-d^2/dx^2$.

Rotation between eigenfunctions.

For fixed $k \geq 2$, the curve

$$\gamma(t) = \mathrm{span}\{\cos(t)\phi_1 + \sin(t)\phi_k\}, \quad t \in (-\pi/2, \pi/2),$$

is $\mathcal{C}^\infty$ in $\mathbb{G}_1(L^2)$. Its chart coordinate is $L(t) = \tan(t)\phi_k$ (in $\mathcal{L}(\mathrm{span}\{\phi_1\}, W_0)$, identifying ϕ_k with the map $\lambda\phi_1 \mapsto \lambda\phi_k$), and $\dot{L}(0) = \phi_k$.

Propagating Gaussian support.

Define $\psi_a(x) = C_a e^{-a(x-1/2)^2}$ (normalised). The curve $t \mapsto \mathrm{span}\{\psi_{a+t}\}$ traces the evolution of the support subspace as the Gaussian width changes. At $t = 0$, the velocity $\dot{L}(0) = \frac{d}{dt}\big|_0 \Psi_{U_0 \oplus W_0}(\mathrm{span}\{\psi_{a+t}\}) \in \mathcal{L}(\mathrm{span}\{\psi_a\}, \{\psi_a\}^\perp)$ encodes the instantaneous rate of change of the dominant mode.

Example 2.6.2 (Curves in $\mathbb{G}_r(L^2)$ for $r = 2$) For $r = 2$ and $U_0 = \mathrm{span}\{\phi_1, \phi_2\}$, the curve

$$\gamma(t) = \mathrm{span}\big\{\cos(t)\phi_1 + \sin(t)\phi_3,\ \cos(t)\phi_2 + \sin(t)\phi_4\big\}$$

has chart matrix $[L(t)]_{ij} = \tan(t)\delta_{ij}$: a uniform rotation of both basis vectors. For the Tucker manifold $\mathfrak{M}_{(2,2)}(L^2 \otimes L^2)$, a pair of such curves (one per factor) gives a curve in the base $\mathbb{G}_2(L^2) \times \mathbb{G}_2(L^2)$ of the Tucker bundle.

Retractions on Banach manifolds

Definition 2.6.1 A *retraction* on a $\mathcal{C}^2$-Banach manifold $\mathbb{M}$ is a $\mathcal{C}^1$-map $R : T\mathbb{M} \to \mathbb{M}$ such that, for every $m \in \mathbb{M}$, the restriction $R_m := R|_{T_m\mathbb{M}}$ satisfies:

(R1) $R_m(0_m) = m$.
(R2) $DR_m(0_m) = \mathrm{id}_{T_m\mathbb{M}}$.

Condition (R2) says the retraction is a first-order approximation to the exponential map. Retractions are the key ingredient for geometric optimisation (gradient descent, Newton's method) and geometric integration on manifolds; they replace the exponential map whenever no Riemannian metric is available.

Example 2.6.3 (Retraction on $\mathbb{G}_r(X)$) For $U \in \mathbb{G}_r(X)$ and $L \in T_U\mathbb{G}_r(X) = \mathcal{L}(U, W)$:

$$R_U(L) := (\mathrm{id}_X + L)(U). \tag{2.4}$$

Conditions (R1)–(R2) are immediate from the chart map (Theorem 2.3.1).

Remark 2.6.1 (Retraction vs. geodesic) In the Hilbert case, the geodesic on $\mathbb{G}_r(H)$ from U with velocity $L \in \mathcal{L}(U, U^\perp)$ is $\gamma(t) = \cos(t|L|)(U) \oplus \sin(t|L|)\mathrm{Im}(L/|L|)$, where $|L| = (L^*L)^{1/2}$. The chart-based retraction (2.4) equals the geodesic to first order in t (agreeing at $t = 0$ and having the same velocity). For $r = 1$, the geodesic is the great circle on $\mathbb{P}(H)$ — the Rabi oscillation in quantum mechanics.

In the non-Hilbert Banach setting, there is no canonical Riemannian metric; the chart-based retraction is the natural substitute and is used in the projector-splitting integrator Lubich et al. [2015].

Example 2.6.4 (Retraction on the Tucker manifold) For $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and $\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v})$, writing $\dot{\mathbf{v}} = d\Phi_{\mathbf{v}}^{-1}[\dot{\mathbf{v}}]$ in chart coordinates $((\delta L_\alpha)_\alpha, \delta C^{(D)})$:

$$R_{\mathbf{v}}(\dot{\mathbf{v}}) = \sum_{(i_\alpha)} (C^{(D)} + \delta C^{(D)})_{(i_\alpha)} \bigotimes_{\alpha} (\mathrm{id} + \delta L_\alpha)(u_{i_\alpha}^{(\alpha)}). \tag{2.5}$$

This satisfies (R1)–(R2) by construction and is the retraction used in the geometric ALS and DMRG algorithms of Appendix D.

Chapter 3
Algebraic Tensor Spaces

This chapter develops the purely algebraic foundations of tensor spaces. We introduce the algebraic tensor product via its universal property, establish the key notions of elementary tensor, matricisation, and multilinear rank, and define the Tucker format and the set $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ of tensors of fixed Tucker rank. No topology is assumed in this chapter; that enters in Chapter 7.

Throughout, $D = \{1, \ldots, d\}$ with $d \geq 2$ is the *dimension index set*, and V_α ($\alpha \in D$) are vector spaces over $\mathbb{K}$. Concerning the foundational construction of tensor products we follow Greub Greub [1981].

3.1 The algebraic tensor product

Universal property and construction

Definition 3.1.1 The *algebraic tensor product* of the spaces V_α, $\alpha \in D$, is a pair $(\mathbf{V}_D, \bigotimes)$ consisting of a vector space $\mathbf{V}_D$ and a multilinear map

$$\bigotimes : \prod_{\alpha \in D} V_\alpha \longrightarrow \mathbf{V}_D, \qquad (v_\alpha)_{\alpha \in D} \mapsto \bigotimes_{\alpha \in D} v_\alpha, \tag{3.1}$$

satisfying the following *universal property*: for every vector space Z and every multilinear map $M : \prod_{\alpha \in D} V_\alpha \to Z$, there exists a unique linear map $\widehat{M} : \mathbf{V}_D \to Z$ such that $M = \widehat{M} \circ \bigotimes$.

We write $\mathbf{V}_D = {}_a\bigotimes_{\alpha \in D} V_\alpha$ and refer to elements of the form $\bigotimes_{\alpha \in D} v_\alpha$ (with $v_\alpha \in V_\alpha$) as *elementary tensors*. Every element of $\mathbf{V}_D$ is a *finite* linear combination of elementary tensors.

The algebraic tensor product exists and is unique up to isomorphism; see Greub [1981] for the standard construction. The prefix 'a' in ${}_a\bigotimes$ emphasises the algebraic (non-topological) nature. The underlying field is $\mathbb{K} = \mathbb{R}$ throughout, but all results hold equally for $\mathbb{K} = \mathbb{C}$.

Remark 3.1.1 The set of elementary tensors is *not* a basis of $\mathbf{V}_D$; it generates $\mathbf{V}_D$ but with many linear dependencies. For example, $\lambda v \otimes w = v \otimes \lambda w$ for all $\lambda \in \mathbb{K}$. If $\{e_i\}$ and $\{f_j\}$ are bases of V_1 and V_2 respectively, then $\{e_i \otimes f_j\}$ is a basis of $V_1 \otimes_a V_2$.

Bilinear identification and the α-unfolding

For each $\alpha \in D$, the algebraic tensor space can be rewritten as a two-factor product by grouping all indices except α:

$$\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}, \qquad \mathbf{V}_\alpha^{[\alpha]} := {}_a\bigotimes_{\beta \in D\setminus\{\alpha\}} V_\beta. \tag{3.2}$$

Concretely, there is a canonical linear isomorphism $\Phi_\alpha : \mathbf{V}_D \xrightarrow{\sim} V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ and throughout this monograph we identify $\mathbf{v} \in \mathbf{V}_D$ with $\Phi_\alpha(\mathbf{v}) \in V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ without further comment. An elementary tensor splits as $\bigotimes_{\beta \in D} v_\beta \leftrightarrow v_\alpha \otimes \bigotimes_{\beta \neq \alpha} v_\beta$.

Tensor products of linear maps

Given linear maps $A_\alpha : V_\alpha \to W_\alpha$ for $\alpha \in D$, their tensor product is the unique linear map

$$\bigotimes_{\alpha \in D} A_\alpha : \mathbf{V}_D \longrightarrow \mathbf{W}_D := {}_a\bigotimes_{\alpha \in D} W_\alpha, \tag{3.3}$$

determined on elementary tensors by $\left(\bigotimes_{\alpha \in D} A_\alpha\right)\left(\bigotimes_{\alpha \in D} v_\alpha\right) = \bigotimes_{\alpha \in D} A_\alpha v_\alpha$. This is the unique linear extension guaranteed by the universal property. We denote the algebraic dual of V by $V' = L(V, \mathbb{K})$ and the space of all linear maps from V to W by $L(V, W)$.

Partitions and iterated tensor products

Definition 3.1.2 Let $\mathcal{P}$ be a partition of D (i.e., a collection of non-empty, pairwise disjoint subsets of D whose union is D). For each $\alpha \in \mathcal{P}$, set $\mathbf{V}_\alpha := {}_a\bigotimes_{j \in \alpha} V_j$ (with the convention $\mathbf{V}_{\{j\}} = V_j$). Then the *iterated tensor product* associated to $\mathcal{P}$ is

$${}_a\bigotimes_{\alpha \in \mathcal{P}} \mathbf{V}_\alpha \cong \mathbf{V}_D, \tag{3.4}$$

where the isomorphism is canonical and consistent with the identifications of elementary tensors.

This associativity-of-tensor-products fact is fundamental: it allows us to regroup the factors of $\mathbf{V}_D$ according to any hierarchical structure, which is precisely what the dimension partition trees of Chapter 5 will exploit.

Example 3.1.1 For $D = \{1, 2, 3, 4\}$ and the partition $\mathcal{P} = \{\{1, 2\}, \{3, 4\}\}$:

$$(V_1 \otimes_a V_2) \otimes_a (V_3 \otimes_a V_4) \cong V_1 \otimes_a V_2 \otimes_a V_3 \otimes_a V_4.$$

The partition $\mathcal{P}' = \{\{1\}, \{2\}, \{3\}, \{4\}\}$ (trivial) recovers the original four-fold product. The non-trivial partition $\mathcal{P}$ is the structure underlying the Tucker format (depth-2 tree), while more refined partitions correspond to hierarchical formats; see Chapter 5.

3.2 Matricisation and multilinear rank

Matricisation in finite dimensions

When the component spaces are finite-dimensional, the tensor $\mathbf{v} \in \mathbf{V}_D$ can be represented as a multi-dimensional array, and the identification (3.2) induces a matrix.

Definition 3.2.1 Let $D = \{1, \dots, d\}$, $\dim V_\alpha = n_\alpha < \infty$, and fix bases of each V_α. Identify $\mathbf{v} \in \mathbf{V}_D$ with its coordinate array $C \in \mathbb{R}^{n_1 \times \cdots \times n_d}$. For each $\alpha \in D$, the *α-matricisation* (or *α-unfolding*) of $\mathbf{v}$ is the matrix

$$\mathcal{M}_\alpha(\mathbf{v}) \in \mathbb{R}^{n_\alpha \times \prod_{\beta \neq \alpha} n_\beta}$$

obtained by reordering the coordinate array C so that the α-index labels the rows and all other indices, in lexicographic order, label the columns. The rank of this matrix, $\mathrm{rank}\mathcal{M}_\alpha(\mathbf{v}) \in \{0, 1, \dots, n_\alpha\}$, is the *$\alpha$-rank* of $\mathbf{v}$.

The notion of α-rank was introduced by Hitchcock Hitchcock [1927] in 1927 under the name "rank on the jth index". In the infinite-dimensional case, the correct replacement of $\mathrm{rank}\mathcal{M}_\alpha(\mathbf{v})$ is the dimension of the minimal subspace $U_\alpha^{\min}(\mathbf{v})$, developed fully in Chapter 4.

Explicit example: three-mode matricisation

Example 3.2.1 (Matricisation for $d = 3$) Let $n_1 = n_2 = n_3 = 2$. Consider $\mathbf{v} \in \mathbb{R}^2 \otimes \mathbb{R}^2 \otimes \mathbb{R}^2$ with coordinate array $C_{111} = C_{122} = C_{212} = C_{221} = 1$, all other entries zero; equivalently,

$$\mathbf{v} = e_1 \otimes e_1 \otimes e_1 + e_1 \otimes e_2 \otimes e_2 + e_2 \otimes e_1 \otimes e_2 + e_2 \otimes e_2 \otimes e_1.$$

The three matricisations, with columns ordered lexicographically by the remaining indices, are:

$$\mathcal{M}_1(\mathbf{v}) = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 4},$$
$$\mathcal{M}_2(\mathbf{v}) = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 4},$$
$$\mathcal{M}_3(\mathbf{v}) = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 4}.$$

All three have rank 2, so the Tucker rank of $\mathbf{v}$ is $(2, 2, 2)$ and $U_\alpha^{\min}(\mathbf{v}) = \mathbb{R}^2 = V_\alpha$ for each α.

CP rank vs. Tucker rank

Example 3.2.2 (CP rank strictly exceeds Tucker rank) Consider the tensor $\mathbf{v}$ of Example 3.2.1. Its Tucker rank is $(2, 2, 2)$, yet its CP rank (the minimum number of elementary summands) equals 4: the representation above has 4 terms, and a 3-term representation can be ruled out by direct computation.

By contrast, the tensor $\mathbf{w} = e_1 \otimes e_1 \otimes e_1 + e_2 \otimes e_2 \otimes e_2 \in \mathbb{R}^2 \otimes \mathbb{R}^2 \otimes \mathbb{R}^2$ has Tucker rank $(2, 2, 2)$ (each matricisation has rank 2) but CP rank 2.

This illustrates a fundamental distinction: the Tucker rank $(r_1, \ldots, r_d)$ describes the subspace structure ($U_\alpha^{\min}(\mathbf{v})$ has dimension r_α), while the CP rank counts minimal summands. One always has $r_\alpha \leq r_{\mathrm{CP}}(\mathbf{v})$, but the inequality is strict in general for $d \geq 3$. Computing Tucker rank is tractable (SVD of the matricisations); computing CP rank is NP-hard (Håstad, 1990). This computational advantage is a key motivation for the Tucker-format framework of this monograph.

Definition 3.2.2 For a rank tuple $\mathfrak{r} = (r_\alpha)_{\alpha \in D} \in \mathbb{N}^d$, define

$$\mathbb{R}_*^{\times_{\alpha\in D} r_\alpha} := \left\{ C \in \mathbb{R}^{\times_{\alpha\in D} r_\alpha} : \mathrm{rank}\mathcal{M}_\alpha(C) = r_\alpha \ \text{ for all } \alpha \in D \right\}. \tag{3.5}$$

Elements of $\mathbb{R}_*^{\times_{\alpha\in D} r_\alpha}$ are called *full-rank core tensors* of shape $\mathfrak{r}$.

Remark 3.2.1 Since $\mathrm{rank}\mathcal{M}_\alpha(C) = r_\alpha$ if and only if $\mathcal{M}_\alpha(C)\mathcal{M}_\alpha(C)^T \in \mathrm{GL}(\mathbb{R}^{r_\alpha})$, and the determinant is a continuous function, $\mathbb{R}_*^{\times_{\alpha\in D} r_\alpha}$ is an open subset of $\mathbb{R}^{\times_{\alpha\in D} r_\alpha}$, hence a smooth finite-dimensional manifold in its own right. In the scalar case $r_\alpha = 1$ for all α, we recover $\mathbb{R}_* = \mathbb{R} \setminus \{0\} = \mathrm{GL}(\mathbb{R})$.

3.3 The Tucker format

Minimal subspaces: first appearance

We introduce minimal subspaces here in their algebraic form; the full theory (existence, uniqueness, lattice properties, best approximation) is developed in Chapter 4.

Definition 3.3.1 Let $\mathbf{v} \in \mathbf{V}_D = {}_a\bigotimes_{\alpha\in D} V_\alpha$ and $\alpha \in D$. Under the identification $\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$, the *minimal subspace* of $\mathbf{v}$ in direction α is the unique subspace $U_\alpha^{\min}(\mathbf{v}) \subset V_\alpha$ of smallest dimension such that $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a \mathbf{V}_\alpha^{[\alpha]}$. The *$\alpha$-rank* of $\mathbf{v}$ is $r_\alpha(\mathbf{v}) := \dim U_\alpha^{\min}(\mathbf{v})$.

The key properties — existence, uniqueness, and the symmetry $\dim U_\alpha^{\min}(\mathbf{v}) = \dim U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$ — are established in Chapter 4. Here we proceed taking these properties for granted.

The Tucker format and the set $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$

Definition 3.3.2 Let $\mathfrak{r} = (r_\alpha)_{\alpha\in D} \in \mathbb{N}_0^d$ and let $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$ for each $\alpha \in D$. A tensor $\mathbf{v} \in \mathbf{V}_D$ is said to be in *Tucker format* with subspace tuple $(U_\alpha)_{\alpha\in D}$ if

$$\mathbf{v} \in {}_a\bigotimes_{\alpha\in D} U_\alpha.$$

The *Tucker rank* of $\mathbf{v}$ is the tuple $\mathfrak{r}(\mathbf{v}) = (r_\alpha(\mathbf{v}))_{\alpha\in D}$, and the set of tensors with fixed Tucker rank $\mathfrak{r}$ is

$$\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D) := \big\{\mathbf{v} \in \mathbf{V}_D : r_\alpha(\mathbf{v}) = r_\alpha \text{ for all } \alpha \in D\big\}. \tag{3.6}$$

The Tucker format thus represents $\mathbf{v}$ as an element of a subspace $U_1 \otimes \cdots \otimes U_d$ spanned by r_α-dimensional subspaces $U_\alpha \subset V_\alpha$. Unlike the subspaces, the representation is not unique; the set $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ collects all tensors for which the *minimal* such subspaces have dimension exactly r_α.

Admissible Tucker ranks

Definition 3.3.3 A tuple $\mathfrak{r} = (r_\alpha)_{\alpha\in D} \in \mathbb{N}_0^d$ is an *admissible Tucker rank* for $\mathbf{V}_D$ if there exists $\mathbf{v} \in \mathbf{V}_D$ with $r_\alpha(\mathbf{v}) = r_\alpha$ for all $\alpha \in D$. The set of all admissible Tucker ranks is

$$\mathcal{AD}(\mathbf{V}_D) := \big\{(r_\alpha(\mathbf{v}))_{\alpha\in D} \in \mathbb{N}_0^d : \mathbf{v} \in \mathbf{V}_D\big\}. \tag{3.7}$$

The zero tuple $\mathbf{0} = (0, \ldots, 0)$ is always admissible (corresponding to $\mathbf{v} = 0$). The tuple $\mathbf{1} = (1, \ldots, 1)$ is admissible if and only if $\dim V_\alpha \geq 1$ for all α. Since the r_α are bounded above by $\dim V_\alpha$ (in finite dimensions), $\mathcal{AD}(\mathbf{V}_D)$ is a finite set when all V_α are finite-dimensional.

The following decomposition is one of the foundational structural facts of the whole theory.

Proposition 3.3.1 *The algebraic tensor space decomposes as a disjoint union:*

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D). \tag{3.8}$$

Proof. Since every $\mathbf{v} \in \mathbf{V}_D$ has a well-defined Tucker rank tuple (Chapter 4), the sets $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ cover $\mathbf{V}_D$; they are disjoint by definition of the rank. $\square$

The decomposition (3.8) is the algebraic precursor of the manifold decomposition established in Chapter 8: each component $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ will be shown to carry the structure of an infinite-dimensional $\mathcal{C}^\infty$-Banach manifold.

Explicit representation of Tucker tensors

The following lemma, which combines the definition of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ with the properties of minimal subspaces, gives three equivalent characterisations of Tucker tensors. It is the algebraic heart of the Tucker manifold construction.

Lemma 3.3.1 *Let* $\mathbf{V}_D = {}_a\bigotimes_{\alpha \in D} V_\alpha$ *and* $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ *with* $\mathfrak{r} \neq \mathbf{0}$. *For* $\mathbf{v} \in \mathbf{V}_D$*, the following conditions are equivalent.*

(a) $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.

(b) (Tucker representation) *For each* $\alpha \in D$ *there exists a basis* $\{u_{i_\alpha}^{(\alpha)} : 1 \leq i_\alpha \leq r_\alpha\}$ *of* $U_\alpha^{\min}(\mathbf{v})$ *and a unique full-rank core tensor* $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$*, once the bases are fixed, such that*

$$\mathbf{v} = \sum_{\substack{1 \leq i_\alpha \leq r_\alpha \\ \alpha \in D}} C^{(D)}_{(i_\alpha)_{\alpha \in D}} \bigotimes_{\alpha \in D} u_{i_\alpha}^{(\alpha)}. \tag{3.9}$$

(c) (Bilinear form) *For each* $\alpha \in D$ *there exist linearly independent vectors* $\{u_{i_\alpha}^{(\alpha)}\}_{i_\alpha=1}^{r_\alpha}$ *in* V_α *and linearly independent vectors* $\{\mathbf{U}_{i_\alpha}^{(\alpha)}\}_{i_\alpha=1}^{r_\alpha}$ *in* $\mathbf{V}_\alpha^{[\alpha]}$ *such that*

$$\mathbf{v} = \sum_{i_\alpha=1}^{r_\alpha} u_{i_\alpha}^{(\alpha)} \otimes \mathbf{U}_{i_\alpha}^{(\alpha)}. \tag{3.10}$$

Moreover, when (3.9) *holds, the vectors* $\mathbf{U}_{i_\alpha}^{(\alpha)}$ *are given by*

$$\mathbf{U}_{i_\alpha}^{(\alpha)} = \sum_{\substack{1 \le i_\beta \le r_\beta \\ \beta \in D \setminus \{\alpha\}}} C^{(D)}_{i_\alpha,(i_\beta)_{\beta \neq \alpha}} \bigotimes_{\beta \neq \alpha} u_{i_\beta}^{(\beta)}. \tag{3.11}$$

Proof. **(a) $\Leftrightarrow$ (c).** If (a) holds, then by Proposition 4.2.1 (Chapter 4), $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$ with $\dim U_\alpha^{\min}(\mathbf{v}) = \dim U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v}) = r_\alpha$, which gives (c). Conversely, (c) implies $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ for all α, hence (a).

(b) $\Rightarrow$ (c). Immediate.

(c) $\Rightarrow$ (b). From (c), one has $U_\alpha^{\min}(\mathbf{v}) = \operatorname{span}\{u_{i_\alpha}^{(\alpha)}\}$ for each α, and since $\mathbf{v} \in {}_a\bigotimes_{\alpha \in D} U_\alpha^{\min}(\mathbf{v})$, there exists $C^{(D)} \in \mathbb{R}^{\times_{\alpha \in D} r_\alpha}$ satisfying (3.9). From (3.11) and the linear independence of the $\mathbf{U}_{i_\alpha}^{(\alpha)}$, one deduces $\operatorname{rank}\mathcal{M}_\alpha(C^{(D)}) = r_\alpha$ for all α, so $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$. $\square$

Remark 3.3.1 From Lemma 3.3.1 and Remark 3.2.1, once bases of the minimal subspaces $U_\alpha^{\min}(\mathbf{v})$ are fixed ($\alpha \in D$), there is a natural identification

$$\mathfrak{M}_{\mathfrak{r}}\Big(\Big({}_a \bigotimes_{\alpha \in D} U_\alpha^{\min}(\mathbf{v})\Big) \xrightarrow{\sim} \mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}, \qquad \mathbf{u} \mapsto E^{(D)}, \tag{3.12}$$

where $E^{(D)}$ is the coordinate array of $\mathbf{u}$ in the chosen bases. This identification is a diffeomorphism (both directions are polynomial), and it is the local model for the Tucker manifold chart; see Section 8.2.

The Tucker format as a flat case of tree-based formats

The Tucker format corresponds to the *shallowest* possible hierarchical structure: the dimension partition tree T_D^{Tucker} of depth 1, whose root is D and whose children are the singletons $\{1\}, \ldots, \{d\}$:

$$T_D^{\text{Tucker}} = \big\{D, \{1\}, \{2\}, \ldots, \{d\}\big\}, \qquad S(D) = \big\{\{1\}, \ldots, \{d\}\big\} = \mathcal{L}(T_D)[T_D^{\text{Tucker}}].$$

In Chapter 6 we will define tree-based formats for arbitrary dimension partition trees T_D and show that $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) = \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D^{\text{Tucker}})$.

3.4 The Hilbert–Schmidt norm as a tensor product norm

The Hilbert–Schmidt operators provide the most natural bridge between the abstract tensor framework of this chapter and the concrete infinite-dimensional setting of Part **??**. This section shows that the Hilbert–Schmidt norm is a cross-norm on $H \otimes H^*$, computes it explicitly for functions in L^2, and connects the Hilbert–Schmidt rank with the Tucker rank.

The Hilbert tensor product

Let H be a separable complex Hilbert space with inner product $\langle\cdot,\cdot\rangle$ and dual H^* (Riesz isomorphism $J : H \to H^*$, $J(v) = \langle v, \cdot\rangle$). The *Hilbert tensor norm* on $H \otimes_a H^*$ is defined for elementary tensors by

$$\|u \otimes \varphi\|_H := \|u\|_H \|\varphi\|_{H^*}, \tag{3.13}$$

and extended to all finite sums by

$$\Big\|\sum_k u_k \otimes \varphi_k\Big\|_H^2 := \sum_{j,k} \varphi_j(u_k)\,\overline{\varphi_k(u_j)} = \mathrm{Tr}\big(([\varphi_j(u_k)]_{jk})^*[\varphi_j(u_k)]_{jk}\big). \tag{3.14}$$

The completion of $(H \otimes_a H^*, \|\cdot\|_H)$ is the space $\mathcal{T}_2(H)$ of Hilbert–Schmidt operators, with $\mathcal{T}_2(H) \cong H \otimes_H H^*$ isometrically (Theorem C.5.1 in Appendix C).

Proposition 3.4.1 (Hilbert tensor norm is a crossnorm satisfying (Inj)) *The Hilbert tensor norm $\|\cdot\|_H$ on $H \otimes_a H^*$ satisfies:*

(i) (Cross)*: $\|u \otimes J(v)\|_H = \|u\|_H \|v\|_H$ for all $u, v \in H$.*
(ii) (Inj)*: $\|\cdot\|_\varepsilon \le \|\cdot\|_H$.*
(iii) $\|\cdot\|_H \le \|\cdot\|_\pi$ *(dominated by the projective norm).*
(iv) *The Grothendieck inequality gives $\|\cdot\|_\pi \le K_G \|\cdot\|_\varepsilon$ on $H \otimes_a H^*$, so all three norms are equivalent with constant $K_G \approx 1.78$.*

Proof. Part (i) is immediate from (3.13). Part (ii): for any $\phi \in H^*$ with $\|\phi\|_{H^*} \le 1$ and $\psi \in (H^*)^*$ with $\|\psi\|_{(H^*)^*} \le 1$, identifying $\psi = J^{-1}(\psi) \in H$ (using J), one has $|(\phi \otimes \psi)(\mathbf{v})| = |\phi((\mathrm{Id} \otimes \psi)(\mathbf{v}))| \le \|\mathbf{v}\|_H$ by the Cauchy–Schwarz inequality in $\mathcal{T}_2(H)$, giving $\|\mathbf{v}\|_\varepsilon \le \|\mathbf{v}\|_H$. Part (iii) is standard: $\|\cdot\|_H \le \|\cdot\|_\pi$ for any reasonable crossnorm. Part (iv) is Grothendieck's theorem (Remark 3.4.1). □

Hilbert–Schmidt rank equals Tucker rank

Proposition 3.4.2 (Hilbert–Schmidt rank and Tucker rank) *Let $A \in \mathcal{T}_2(H)$ and identify A with a tensor $\mathbf{A} \in H \otimes_H H^*$. Then $r_1(\mathbf{A}) = r_2(\mathbf{A}) =$* rank$(A)$*, and the SVD $A = \sum_{k=1}^r s_k v_k \otimes J(u_k)$ is the Tucker decomposition with minimal subspaces $U_1^{\min}(\mathbf{A}) = \overline{\mathrm{Im}(A^*)} \subset H$ and $U_2^{\min}(\mathbf{A}) = J(\overline{\mathrm{Im}(A)}) \subset H^*$.*

Proof. Under the identification $\mathcal{T}_2(H) \cong H \otimes_H H^*$, the contraction map $T_\varphi : H \otimes_H H^* \to H^*$, $T_\varphi(\mathbf{A}) = (\varphi \otimes \mathrm{id}_{H^*})(\mathbf{A}) = \varphi(A(\cdot))$ has range $J(\overline{\mathrm{Im}(A)})$ when φ ranges over H^*. Hence $U_2^{\min}(\mathbf{A}) = J(\overline{\mathrm{Im}(A)})$, and similarly $U_1^{\min}(\mathbf{A}) = \overline{\mathrm{Im}(A^*)}$. □

Explicit computation in L^2

Example 3.4.1 (Hilbert–Schmidt norm of a separable kernel) Let $H = L^2([0,1])$ and $H^* \cong H$ via J. Consider the rank-r kernel $K(x,y) = \sum_{k=1}^{r} \lambda_k \phi_k(x)\psi_k(y)$, viewed as an element of $L^2([0,1]^2) \cong L^2([0,1]) \otimes_H L^2([0,1])$. The associated Hilbert–Schmidt operator $T_K : L^2 \to L^2$, $T_K f = \int_0^1 K(\cdot, y) f(y)\, dy$, has:

- *Hilbert–Schmidt norm:* $\|T_K\|_{\mathcal{T}_2}^2 = \|K\|_{L^2([0,1]^2)}^2 = \int_0^1 \int_0^1 |K(x,y)|^2 \, dx\, dy$.
- *Tucker rank:* $r_1(K) = r_2(K) = r$ (if $\{\phi_k\}$, $\{\psi_k\}$ are linearly independent).
- *Minimal subspaces:* $U_1^{\min}(K) = \operatorname{span}\{\phi_1, \ldots, \phi_r\}$ and $U_2^{\min}(K) = \operatorname{span}\{\psi_1, \ldots, \psi_r\}$ in $L^2([0,1])$.

For the specific case $K(x,y) = xy$ (rank 1 kernel): $\|K\|_{\mathcal{T}_2}^2 = \int_0^1 x^2\, dx \int_0^1 y^2\, dy = \frac{1}{9}$, so $\|K\|_{\mathcal{T}_2} = \frac{1}{3}$. The Tucker rank is $(1,1)$ and the Tucker decomposition is $K = \phi \otimes J(\psi)$ with $\phi(x) = x$, $\psi(y) = y$.

Remark 3.4.1 (The Hilbert tensor norm in the real case) When $\mathbb{K} = \mathbb{R}$ and $H = \mathbb{R}^n$, the Hilbert–Schmidt operators on $\mathbb{R}^n$ are exactly the $n \times n$ matrices with the Frobenius norm $\|A\|_F = (\sum_{ij} A_{ij}^2)^{1/2}$. The identification $\mathcal{T}_2(\mathbb{R}^n) \cong \mathbb{R}^n \otimes_H \mathbb{R}^n$ (using $H^* \cong H$ since J is the identity in the real self-dual case) gives $\|A\|_F = \|A\|_H$, and the Tucker rank of A viewed as a tensor in $\mathbb{R}^n \otimes \mathbb{R}^n$ equals $\operatorname{rank}(A)$ as a matrix. This is the finite-dimensional instance of Proposition 3.4.2.

3.5 The tensor product map and its differential

When the component spaces are normed, the tensor product map inherits both topological and differential properties that underpin the manifold constructions in Part III.

Definition 3.5.1 Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$. The *tensor product map*

$$\bigotimes : \prod_{\alpha \in D} (V_\alpha, \|\cdot\|_\alpha) \longrightarrow (\mathbf{V}_D, \|\cdot\|_D), \qquad (v_\alpha)_{\alpha \in D} \mapsto \bigotimes_{\alpha \in D} v_\alpha, \tag{3.15}$$

(where the product is equipped with the max norm $\|(v_\alpha)\|_\times = \max_\alpha \|v_\alpha\|_\alpha$) is said to be *continuous* if $\|\bigotimes_\alpha v_\alpha\|_D \le C \prod_\alpha \|v_\alpha\|_\alpha$ for some constant $C > 0$.

Continuity of the tensor product map is equivalent to $\|\bigotimes\| < \infty$; it holds whenever $\|\cdot\|_D$ is a crossnorm (Definition 4.5.1) on $\mathbf{V}_D$.

Proposition 3.5.1 *Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$ such that the tensor product map* (3.15) *is continuous. Then:*

(i) *The map* (3.15) *is $\mathcal{C}^\infty$-Fréchet differentiable, with differential*

$$D\bigotimes(v_1,\ldots,v_d)(w_1,\ldots,w_d)=\sum_{\alpha\in D}v_1\otimes\cdots\otimes v_{\alpha-1}\otimes w_\alpha\otimes v_{\alpha+1}\otimes\cdots\otimes v_d. \tag{3.16}$$

(ii) *The tensor product of bounded linear maps,*

$$\bigotimes:\prod_{\alpha\in D}\mathcal{L}(U_\alpha,V_\alpha)\longrightarrow\mathcal{L}\Big({}_a\bigotimes_{\alpha\in D}U_\alpha,\,\mathbf{V}_D\Big),\quad (A_\alpha)\mapsto\bigotimes_{\alpha\in D}A_\alpha,$$

is also continuous and $\mathcal{C}^\infty$-Fréchet differentiable for any finite-dimensional subspaces $U_\alpha\subset V_\alpha$.

Proof. Part (i) follows directly from Proposition 2.5.1, since $\bigotimes$ is a continuous multilinear map. The formula for the differential follows by the Leibniz rule for multilinear maps. For (ii), one checks boundedness of the map using the continuity of (3.15):

$$\|\bigotimes_\alpha A_\alpha\|_{\mathbf{V}_D\leftarrow\bigotimes U_\alpha}\le C\prod_\alpha\|A_\alpha\|_{V_\alpha\leftarrow U_\alpha},$$

where C is the constant in Definition 3.5.1, and the finite-dimensionality of U_α ensures $L(U_\alpha,V_\alpha)=\mathcal{L}(U_\alpha,V_\alpha)$. Fréchet differentiability then follows again from Proposition 2.5.1. □

Remark 3.5.1 Proposition 3.5.1 is the reason why the Tucker manifold $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ is $\mathcal{C}^\infty$ (not merely continuous): all coordinate change maps between charts are built from compositions of projections and the tensor product map, all of which are $\mathcal{C}^\infty$-Fréchet differentiable. See Section 8.3 for the precise argument.

Banach tensor spaces

When a norm is fixed on $\mathbf{V}_D$, its completion is the natural ambient space.

Definition 3.5.2 Let $(V_\alpha,\|\cdot\|_\alpha)$ be normed spaces and $\|\cdot\|_D$ a norm on $\mathbf{V}_D={}_a\bigotimes_{\alpha\in D}V_\alpha$. The *Banach tensor space* associated to this data is the completion

$$\mathbf{V}_{D,\|\cdot\|_D}:={}_{\|\cdot\|_D}\bigotimes_{\alpha\in D}V_\alpha:=\overline{\mathbf{V}_D}^{\|\cdot\|_D}.$$

If $\mathbf{V}_{D,\|\cdot\|_D}$ is a Hilbert space, it is called a *Hilbert tensor space.*

Example 3.5.1 For $I_\alpha\subset\mathbb{R}$ $(\alpha\in D)$ and $1\le p<\infty$, the Lebesgue spaces $L^p(I_\alpha)$ satisfy

$$L^p\Big(\prod_{\alpha\in D}I_\alpha\Big)={}_{\|\cdot\|_{0,p}}\bigotimes_{\alpha\in D}L^p(I_\alpha),$$

where $\|\cdot\|_{0,p}$ denotes the standard L^p norm on the product domain. For $p=2$ this is a Hilbert tensor space.

Example 3.5.2 For Sobolev spaces $H^{N,p}(I_\alpha)$ with $1 < p < \infty$ and $\mathbf{I} = \prod_\alpha I_\alpha$:

$$H^{N,p}(\mathbf{I}) = {}_{\|\cdot\|_{N,p}} \bigotimes_{\alpha \in D} H^{N,p}(I_\alpha).$$

This is reflexive and separable. For $p = 2$ it is a Hilbert tensor space. These spaces arise naturally in the numerical treatment of high-dimensional PDEs; see Section 4.5 and Section 7.5.

Remark 3.5.2 The algebraic tensor space $\mathbf{V}_D$ is dense in $\mathbf{V}_{D,\|\cdot\|_D}$ by construction. The Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ lives inside the algebraic space, but its closure (as a subset of $\mathbf{V}_{D,\|\cdot\|_D}$) and its topology as a manifold are determined by the norm $\|\cdot\|_D$ and, in particular, by whether $\|\cdot\|_D$ is a crossnorm. This interplay between algebraic and topological structure is a recurring theme throughout the monograph.

Chapter 4
Minimal Subspaces in Tensor Representations

In Chapter 3 we introduced the Tucker format and defined the minimal subspace $U_\alpha^{\min}(\mathbf{v})$ informally, taking its existence and properties for granted. This chapter provides the complete theory: a discussion of reflexivity and best-approximation hypotheses (Section 4.1), existence and uniqueness of minimal subspaces (Section 4.2), the lattice structure and inclusion under refinement of partitions (Section 4.3), admissible ranks and the disjoint decomposition of $\mathbf{V}_D$ (Section 4.4), existence of best approximations (Section 4.5), and worked examples (Section 4.6). The results in this chapter are taken from Falcó and Hackbusch [2012], expanded with complete proofs and additional examples.

4.1 Reflexivity and best approximations

The main existence result of this chapter (Theorem 4.5.2) requires $\mathbf{V}_{D,\|\cdot\|_D}$ to be *reflexive*. This section explains why, and verifies reflexivity for the standard tensor spaces.

Theorem 4.1.1 (James, James [1957]) *A Banach space X is reflexive if and only if every $f \in X^*$ attains its norm on $B_X = \{x : \|x\| \leq 1\}$.*

Proof. The "only if" direction follows from the Krein–Milman theorem (the unit ball of a reflexive space is weakly compact). The "if" direction is proved in James [1957]; a streamlined proof is in Megginson [1998]. □

Remark 4.1.1 (James's theorem and tensor approximation) If $\mathbf{V}_{D,\|\cdot\|_D}$ is not reflexive there exists $F \in (\mathbf{V}_{D,\|\cdot\|_D})^*$ not attaining its norm, and one can construct $\mathbf{u} \in \mathbf{V}_{D,\|\cdot\|_D}$ whose best rank-one approximation does not exist. For instance, in $L^1([0,1]) \otimes L^1([0,1])$ (non-reflexive), best rank-r approximations may fail to exist. This is why Theorem 4.5.2 requires reflexivity.

Proposition 4.1.1 (Reflexivity of tensor products) *Let $(V_\alpha, \|\cdot\|_\alpha)$ be Banach spaces with a reasonable crossnorm $\|\cdot\|_D$ on $\mathbf{V}_D$.*

(i) *If all* V_α *are Hilbert spaces and* $\|\cdot\|_D$ *is the Hilbert tensor norm, then* $\mathbf{V}_{D,\|\cdot\|_D}$ *is reflexive.*
(ii) *If all* $V_\alpha = L^p(I_\alpha)$ *with* $1 < p < \infty$*, then* $\mathbf{V}_{D,\|\cdot\|_D} \cong L^p(\prod_\alpha I_\alpha)$ *is reflexive.*
(iii) *With the projective norm* $\|\cdot\|_\pi$*,* $\mathbf{V}_{D,\|\cdot\|_D}$ *may fail to be reflexive even when all* V_α *are reflexive.*

Proof. Part (i): the Hilbert tensor norm makes $\mathbf{V}_{D,\|\cdot\|_D}$ a Hilbert space. Part (ii): Fubini isomorphism and reflexivity of L^p for $1 < p < \infty$. Part (iii): $\ell^2 \otimes_\pi \ell^2$ contains an isomorphic copy of ℓ^1; see [Ryan, 2002, Example 2.19]. □

Proposition 4.1.2 (Dual bound under (Inj)) *If* $\|\cdot\|_D$ *satisfies* (Inj)*, then for any* $\varphi_\alpha \in V_\alpha^*$*:*

$$\|\varphi_1 \otimes \cdots \otimes \varphi_d\|_{(\mathbf{V}_{D,\|\cdot\|_D})^*} \le \prod_{\alpha \in D} \|\varphi_\alpha\|_{V_\alpha^*}. \tag{4.1}$$

Proof. Follows directly from the definition of the injective norm and (Inj): $|(\bigotimes_\alpha \varphi_\alpha)(\mathbf{v})| \le \|\mathbf{v}\|_\varepsilon \cdot \prod_\alpha \|\varphi_\alpha\| \le \|\mathbf{v}\|_D \cdot \prod_\alpha \|\varphi_\alpha\|$. □

4.2 Existence and uniqueness of minimal subspaces

The case $d = 2$: Schmidt decomposition

The case of two factors $\mathbf{V}_D = V_1 \otimes_a V_2$ is classical and serves as the inductive base. The key observation is a lattice property of tensor subspaces.

Lemma 4.2.1 (Intersection lemma) *Let* X_i, Y_i *be subspaces of* V_i *for* $i = 1, 2$*. Then*

$$(X_1 \otimes_a X_2) \cap (Y_1 \otimes_a Y_2) = (X_1 \cap Y_1) \otimes_a (X_2 \cap Y_2). \tag{4.2}$$

Proof. The inclusion $(X_1 \cap Y_1) \otimes_a (X_2 \cap Y_2) \subseteq (X_1 \otimes_a X_2) \cap (Y_1 \otimes_a Y_2)$ is immediate. For the reverse inclusion, let $\mathbf{v} \in (X_1 \otimes_a X_2) \cap (Y_1 \otimes_a Y_2)$ with representations

$$\mathbf{v} = \sum_{\nu=1}^{n_x} x_\nu^{(1)} \otimes x_\nu^{(2)} = \sum_{\nu=1}^{n_y} y_\nu^{(1)} \otimes y_\nu^{(2)}, \quad x_\nu^{(i)} \in X_i,\ y_\nu^{(i)} \in Y_i.$$

By adding terms with zero coefficient, assume $n_x = n_y = n$ and that $\{x_\nu^{(1)}\}$ and $\{y_\nu^{(1)}\}$ are each linearly independent (passing to a minimal representation). The two representations yield the same matrix $M_1(\mathbf{v})$ under any identification with a bilinear form, so $\operatorname{span}\{x_\nu^{(1)}\} = \operatorname{span}\{y_\nu^{(1)}\} \subset X_1 \cap Y_1$. A symmetric argument for the second factor gives $\operatorname{span}\{x_\nu^{(2)}\} \subset X_2 \cap Y_2$, so $\mathbf{v} \in (X_1 \cap Y_1) \otimes_a (X_2 \cap Y_2)$. □

Proposition 4.2.1 (Minimal subspace: case $d = 2$) *Let* $\mathbf{v} \in V_1 \otimes_a V_2$*. There exist unique subspaces* $U_i \subset V_i$ *of finite dimension such that:*

(i) $\mathbf{v} \in U_1 \otimes_a U_2$.
(ii) *If* $\mathbf{v} \in X_1 \otimes_a X_2$ *for any subspaces* X_i*, then* $U_i \subset X_i$.

Moreover, $\dim U_1 = \dim U_2$.

Proof. **Existence.** Fix any representation $\mathbf{v} = \sum_\nu v_\nu^{(1)} \otimes v_\nu^{(2)}$ and set $U_i = \operatorname{span}\{v_\nu^{(i)}\}$. By passing to a representation with linearly independent $v_\nu^{(1)}$, one can also take $v_\nu^{(2)}$ linearly independent (otherwise row reduce). Then $\dim U_1 = \dim U_2$ (both equal the rank of the bilinear form).

Minimality. If $\mathbf{v} \in X_1 \otimes_a X_2$, then by the intersection lemma, $\mathbf{v} \in (U_1 \cap X_1) \otimes_a (U_2 \cap X_2)$, which forces $U_i \subset X_i$ by minimality of U_i.

Uniqueness. If U_i and U_i' both satisfy (i)–(ii), minimality of U_i applied to U_i' gives $U_i \subset U_i'$, and vice versa. ◻

The general case: existence and uniqueness

Theorem 4.2.1 (Existence and uniqueness of minimal subspaces) *Let* $D = \{1, \dots, d\}$*,* $\mathbf{V}_D = {}_a\bigotimes_{\alpha \in D} V_\alpha$*, and* $\mathbf{v} \in \mathbf{V}_D$*. For each* $\alpha \in D$*, under the identification* $\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$*, there exists a unique subspace* $U_\alpha^{\min}(\mathbf{v}) \subset V_\alpha$ *of finite dimension* $r_\alpha < \infty$ *such that:*

(a) $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a \mathbf{V}_\alpha^{[\alpha]}$.
(b) *If* $\mathbf{v} \in U_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ *for any subspace* $U_\alpha \subset V_\alpha$*, then* $U_\alpha^{\min}(\mathbf{v}) \subset U_\alpha$.

Furthermore, there exists a unique subspace $U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v}) \subset \mathbf{V}_\alpha^{[\alpha]}$ *of dimension* r_α *such that* $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$.

Proof. Apply Proposition 4.2.1 to the two-factor product $V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$. The subspace $U_\alpha^{\min}(\mathbf{v})$ is the minimal subspace in the V_α-factor. Uniqueness follows from the same argument as in $d = 2$. ◻

Definition 4.2.1 The subspace $U_\alpha^{\min}(\mathbf{v})$ of Theorem 4.2.1 is called the *minimal subspace* of $\mathbf{v}$ in direction α. The integer $r_\alpha(\mathbf{v}) := \dim U_\alpha^{\min}(\mathbf{v})$ is the *α-rank* of $\mathbf{v}$.

Characterisation via contractions

The following characterisation, which is central to both the proof of Chapter 8 and the tree-based theory of Chapter 6, expresses $U_\alpha^{\min}(\mathbf{v})$ as the range of a family of contractions.

Theorem 4.2.2 (Contraction characterisation) *Let* $\mathbf{v} \in \mathbf{V}_D$ *and* $\alpha \in D$*. Then*

$$U_\alpha^{\min}(\mathbf{v}) = \operatorname{span}\big\{(\mathrm{id}_{V_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in (\mathbf{V}_\alpha^{[\alpha]})^*\big\}, \tag{4.3}$$

where $(\mathrm{id}_{V_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) \in V_\alpha$ *is the contraction of* $\mathbf{v}$ *with the functional* $\varphi^{[\alpha]}$ *in all directions except* α.

Proof. Write $\mathbf{v} = \sum_\nu u_\nu \otimes \mathbf{U}_\nu$ with $u_\nu \in V_\alpha$ and $\mathbf{U}_\nu \in \mathbf{V}_\alpha^{[\alpha]}$. For $\varphi \in (\mathbf{V}_\alpha^{[\alpha]})^*$: $(\mathrm{id}_{V_\alpha} \otimes \varphi)(\mathbf{v}) = \sum_\nu \varphi(\mathbf{U}_\nu) u_\nu \in \operatorname{span}\{u_\nu\} = U_\alpha^{\min}(\mathbf{v})$. Conversely, by taking $\varphi = \varphi_\nu$ dual to a basis of $U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$, one recovers a basis of $U_\alpha^{\min}(\mathbf{v})$, so equality holds. □

Remark 4.2.1 In Theorem 4.2.2, one can replace $(\mathbf{V}_\alpha^{[\alpha]})^*$ by $(U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v}))^*$ without changing the span, since $\mathbf{v} \in V_\alpha \otimes_a U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$. This gives the tighter formula

$$U_\alpha^{\min}(\mathbf{v}) = \operatorname{span}\big\{(\mathrm{id}_{V_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in (U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v}))^*\big\}, \tag{4.4}$$

which is often more useful in practice.

4.3 Lattice properties and inclusion under partitions

Monotonicity

Proposition 4.3.1 (Monotonicity of minimal subspaces) *Let* $\mathbf{v} \in \mathbf{V}_D$ *and* $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$ *for each* $\alpha \in D$. *Then:*

$$\mathbf{v} \in {}_a\bigotimes_{\alpha \in D} U_\alpha \quad \Longrightarrow \quad U_\alpha^{\min}(\mathbf{v}) \subset U_\alpha \quad \forall\, \alpha \in D.$$

Proof. Under $\mathbf{v} \in {}_a\bigotimes_{\alpha\in D} U_\alpha$, the identification $\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ gives $\mathbf{v} \in U_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$, so the minimality of $U_\alpha^{\min}(\mathbf{v})$ (Theorem 4.2.1(b)) yields $U_\alpha^{\min}(\mathbf{v}) \subset U_\alpha$. □

Inclusion under partition refinement

The following proposition is the key algebraic fact underlying the tree-based format: it shows that minimal subspaces at a coarser level of a tree are contained in tensor products of minimal subspaces at finer levels.

Proposition 4.3.2 (Inclusion under refinement) *Let* $\mathbf{v} \in \mathbf{V}_D$ *and let* $\alpha \subset D$ *with* $\#(\alpha) \geq 2$. *Let* $\mathcal{P}_\alpha$ *be a non-trivial partition of* α. *Then*

$$U_\alpha^{\min}(\mathbf{v}) \subset {}_a\bigotimes_{\beta \in \mathcal{P}_\alpha} U_\beta^{\min}(\mathbf{v}). \tag{4.5}$$

Proof. From the canonical identification $\mathbf{V}_D \cong \mathbf{V}_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ and Theorem 4.2.2,

$$U_\alpha^{\min}(\mathbf{v}) = \operatorname{span}\big\{(\mathrm{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in (U_{\alpha^c}^{\min}(\mathbf{v}))^*\big\}.$$

For each such element $w_\alpha = (\mathrm{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) \in \mathbf{V}_\alpha$, we apply Proposition 4.3.1 to the tensor space $\mathbf{V}_\alpha \cong {}_a\bigotimes_{\beta\in\mathcal{P}_\alpha} \mathbf{V}_\beta$. Since $w_\alpha \in U_\alpha^{\min}(\mathbf{v})$ and $\mathbf{V}_\alpha = {}_a\bigotimes_{\beta\in\mathcal{P}_\alpha} \mathbf{V}_\beta$, one has $w_\alpha \in {}_a\bigotimes_{\beta\in\mathcal{P}_\alpha} U_\beta^{\min}(w_\alpha)$. Moreover, from the minimality of $U_\beta^{\min}(\mathbf{v})$ and the relation between contractions at different levels, $U_\beta^{\min}(w_\alpha) \subset U_\beta^{\min}(\mathbf{v})$ for each $\beta \in \mathcal{P}_\alpha$. Hence $w_\alpha \in {}_a\bigotimes_{\beta\in\mathcal{P}_\alpha} U_\beta^{\min}(\mathbf{v})$, and taking the span over all such w_α gives (4.5). □

Remark 4.3.1 Proposition 4.3.2 applied iteratively at every internal vertex $\alpha \in T_D$ of a dimension partition tree (Chapter 5) gives the fundamental compatibility of minimal subspaces with the tree structure:

$$U_\alpha^{\min}(\mathbf{v}) \subset {}_a\bigotimes_{\beta\in S(\alpha)} U_\beta^{\min}(\mathbf{v}) \quad \forall\, \alpha \in T_D \setminus \mathcal{L}(T_D). \tag{4.6}$$

This is the algebraic foundation of the tree-based format in Chapter 6.

New characterisation of minimal subspaces

The following result, proved in Falcó et al. [2021] and used in the construction of tree-based tensor formats, gives a characterisation of $U_\beta^{\min}(\mathbf{v})$ via contractions of $U_\alpha^{\min}(\mathbf{v})$ for $\alpha \supset \beta$.

Proposition 4.3.3 *Let $\mathbf{v} \in \mathbf{V}_D$, $\alpha \subset D$ with $\#(\alpha) \geq 2$, and $\mathcal{P}_\alpha$ a non-trivial partition of α. Suppose that $\mathbf{V}_\alpha$ and $\mathbf{V}_\beta$ for $\beta \in \mathcal{P}_\alpha$ are normed spaces. Then for each $\beta \in \mathcal{P}_\alpha$:*

$$U_\beta^{\min}(\mathbf{v}) = \operatorname{span}\big\{(\mathrm{id}_{\mathbf{V}_\beta} \otimes \varphi^{(\alpha\setminus\beta)})(\mathbf{v}_\alpha) : \mathbf{v}_\alpha \in U_\alpha^{\min}(\mathbf{v}),\ \varphi^{(\alpha\setminus\beta)} \in \big({}_a\bigotimes_{\gamma\in\mathcal{P}_\alpha\setminus\{\beta\}} \mathbf{V}_\alpha[\gamma]\big)^*\big\}. \tag{4.7}$$

Proof. Applying Theorem 4.2.2 twice — first for the full index set D and then for α — one obtains two nested contraction characterisations. The formula (4.7) follows by composing these and using that $\mathbf{v} \in \mathbf{V}_\alpha \otimes_a U_{\alpha^c}^{\min}(\mathbf{v})$ allows the replacement of the ambient dual by that of $U_{\alpha^c}^{\min}(\mathbf{v})$. □

4.4 Admissible ranks and the disjoint decomposition

Admissible Tucker ranks

Recall from Chapter 3 (Definition 3.3.3) that the set of admissible Tucker ranks is

$$\mathcal{AD}(\mathbf{V}_D) = \left\{ (r_\alpha(\mathbf{v}))_{\alpha \in D} \in \mathbb{N}_0^d : \mathbf{v} \in \mathbf{V}_D \right\}.$$

The following proposition characterises which tuples are admissible.

Proposition 4.4.1 *A tuple $\mathfrak{r} = (r_\alpha)_{\alpha \in D} \in \mathbb{N}_0^d$ belongs to $\mathcal{AD}(\mathbf{V}_D)$ if and only if there exist subspaces $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$ and a tensor $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}({}_a\bigotimes_{\alpha \in D} U_\alpha)$, i.e., a tensor whose α-unfolding has rank exactly r_α for all $\alpha \in D$.*

Proposition 4.4.2 *For $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$:*

(i) $r_\alpha \leq \dim V_\alpha$ *for all* $\alpha \in D$.
(ii) $r_\alpha = r_{\alpha^c} := \dim U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$ *for all* $\alpha \in D$ *(symmetry of marginal ranks).*
(iii) $r_\alpha \leq \prod_{\beta \neq \alpha} r_\beta$ *for all* $\alpha \in D$.

Proof. Statements (i) and (iii) follow from the representation formula (3.9) and rank properties of multi-index arrays. Statement (ii) follows from Proposition 4.2.1 applied to $V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$. □

The disjoint decomposition

Theorem 4.2.1 immediately gives the following structural result.

Theorem 4.4.1 *The algebraic tensor space $\mathbf{V}_D$ admits the disjoint decomposition*

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D). \tag{4.8}$$

Each $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is non-empty for $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$. The zero tensor $\mathbf{0}$ belongs to $\mathfrak{M}_{\mathbf{0}}(\mathbf{V}_D)$ and each elementary tensor $\bigotimes_\alpha v_\alpha$ (with all $v_\alpha \neq 0$) belongs to $\mathfrak{M}_{\mathbf{1}}(\mathbf{V}_D)$.

Remark 4.4.1 (Topology of the decomposition) While $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is algebraically well-defined, its topological structure is non-trivial. The key result — that each $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a $\mathcal{C}^\infty$-Banach manifold immersed in the ambient Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$ — is established in Chapter 8. The immersion requires the condition (Inj) on the norm $\|\cdot\|_D$ (Definition 4.5.1).

Stratification by rank

Definition 4.4.1 For $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$, the set of tensors with Tucker rank *bounded by* $\mathfrak{r}$ is

$$\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D) := \bigsqcup_{\substack{\mathfrak{s} \in \mathcal{AD}(\mathbf{V}_D) \\ \mathfrak{s} \leq \mathfrak{r}}} \mathfrak{M}_{\mathfrak{s}}(\mathbf{V}_D) = \big\{ \mathbf{v} \in \mathbf{V}_D : r_\alpha(\mathbf{v}) \leq r_\alpha \ \forall\, \alpha \in D \big\}, \tag{4.9}$$

where $\mathfrak{s} \leq \mathfrak{r}$ means $s_\alpha \leq r_\alpha$ for all $\alpha \in D$.

The set $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ is not a linear subspace (it is not closed under addition) but it is closed under scalar multiplication. The key property for applications — that best approximations from $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ exist in Banach tensor spaces — is proved in Section 4.5.

4.5 Best approximations in Banach tensor spaces

In numerical applications, one needs to approximate a given tensor $\mathbf{v} \in \mathbf{V}_{D,\|\cdot\|_D}$ by a tensor of bounded rank. The existence of such a best approximation is non-trivial: the set $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ is not convex, so general theorems about proximinal sets do not apply. The proof uses the weak closedness of $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$.

Crossnorms and the injective norm condition

Definition 4.5.1 Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and $\|\cdot\|_D$ a norm on $\mathbf{V}_D$. We say $\|\cdot\|_D$ is a *crossnorm* if

$$\Big\| \bigotimes_{\alpha \in D} v_\alpha \Big\|_D = \prod_{\alpha \in D} \|v_\alpha\|_\alpha \quad \forall\, v_\alpha \in V_\alpha. \tag{4.10}$$

The *injective norm* $\|\cdot\|_\varepsilon$ on $\mathbf{V}_D$ is defined by

$$\|\mathbf{v}\|_\varepsilon := \sup\big\{ |(\bigotimes_{\alpha \in D} \varphi_\alpha)(\mathbf{v})| : \varphi_\alpha \in V_\alpha^*,\ \|\varphi_\alpha\|_{V_\alpha^*} \leq 1 \big\}.$$

We say $\|\cdot\|_D$ satisfies condition (Inj) if $\|\mathbf{v}\|_D \geq \|\mathbf{v}\|_\varepsilon$ for all $\mathbf{v} \in \mathbf{V}_D$.

A crossnorm satisfies (Inj) if and only if it is a *reasonable crossnorm* in the sense that its dual norm is also a crossnorm. The injective norm is the smallest reasonable crossnorm (see Hackbusch [2019], Chapter 4).

Remark 4.5.1 The L^p norm on $L^p(\prod_\alpha I_\alpha)$ is a reasonable crossnorm for $1 \leq p \leq \infty$ (see Hackbusch [2019], Example 4.72). The Sobolev norm $\|\cdot\|_{(0,1),p}$ on

$H^{1,p}(I_1) \otimes_a H^{1,p}(I_2)$ is a crossnorm (Example 4.42 in Hackbusch [2019]), hence satisfies (Inj).

Weak closedness of bounded-rank sets

Theorem 4.5.1 (Weak closedness, [Falcó and Hackbusch, 2012, Theorem 5.1]) *Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$ satisfying condition* (Inj). *Then the set $\mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D)$ is weakly closed in $\mathbf{V}_{D,\|\cdot\|_D}$.*

Sketch. Let $(\mathbf{v}_n) \subset \mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D)$ converge weakly to $\mathbf{v} \in \mathbf{V}_{D,\|\cdot\|_D}$. We must show $\mathbf{v} \in \mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D)$, i.e., $r_\alpha(\mathbf{v}) \leq r_\alpha$ for all $\alpha \in D$.

Fix $\alpha \in D$ and consider the α-unfolding map $\mathcal{M}_\alpha : \mathbf{V}_{D,\|\cdot\|_D} \to \mathcal{L}((V_\alpha)^*, \mathbf{V}_\alpha^{[\alpha]})$ defined by $\mathcal{M}_\alpha(\mathbf{v})(\varphi) = (\varphi \otimes \mathrm{id})(\mathbf{v})$. Since $r_\alpha(\mathbf{v}_n) \leq r_\alpha$, we have $\mathrm{rank}\mathcal{M}_\alpha(\mathbf{v}_n) \leq r_\alpha$ for all n. The condition (Inj) implies that $\mathcal{M}_\alpha$ is continuous for the weak topology on $\mathbf{V}_{D,\|\cdot\|_D}$ and the weak operator topology on $\mathcal{L}((V_\alpha)^*, \mathbf{V}_\alpha^{[\alpha]})$. Since the set of operators of rank $\leq r_\alpha$ is closed in the weak operator topology (it is a closed variety), the limit $\mathcal{M}_\alpha(\mathbf{v})$ has rank $\leq r_\alpha$, i.e., $r_\alpha(\mathbf{v}) \leq r_\alpha$. □

Existence of best approximations

Theorem 4.5.2 (Best approximation, [Falcó and Hackbusch, 2012, Theorem 5.3]) *Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces, let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$ satisfying* (Inj), *and assume that $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive. Then for every $\mathbf{u} \in \mathbf{V}_{D,\|\cdot\|_D}$ and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$, there exists $\mathbf{v}_* \in \mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})$ such that*

$$\|\mathbf{u} - \mathbf{v}_*\|_D = \inf_{\mathbf{v} \in \mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})} \|\mathbf{u} - \mathbf{v}\|_D. \tag{4.11}$$

Proof. Since $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive, any bounded sequence has a weakly convergent subsequence (Eberlein–Šmulian theorem). Let $(\mathbf{v}_n) \subset \mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})$ be a minimising sequence for the infimum in (4.11). The sequence $(\mathbf{v}_n)$ is bounded (the infimum is attained or approached by a bounded set), so by reflexivity there is a weakly convergent subsequence $\mathbf{v}_{n_k} \rightharpoonup \mathbf{v}_*$. By Theorem 4.5.1, $\mathbf{v}_* \in \mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})$. Since the norm is weakly lower semicontinuous, $\|\mathbf{u} - \mathbf{v}_*\|_D \leq \liminf_k \|\mathbf{u} - \mathbf{v}_{n_k}\|_D = \inf$, so $\mathbf{v}_*$ is a best approximation. □

Corollary 4.5.1 *For $I_\alpha \subset \mathbb{R}$ $(\alpha \in D)$ and $1 < p < \infty$, the Banach tensor space $L^p(\prod_\alpha I_\alpha)$ is reflexive and its norm satisfies* (Inj) *(Example 4.5.1). Hence every $f \in L^p(\prod_\alpha I_\alpha)$ has a best Tucker approximation of any prescribed rank.*

Corollary 4.5.2 *For $1 < p < \infty$, $N \geq 1$, and bounded intervals I_α:*

(i) *$H^{N,p}(\prod_\alpha I_\alpha)$ with the standard Sobolev norm is reflexive and satisfies* (Inj).

(ii) *The space* $H^{1,p}(I_1) \otimes_{\|\cdot\|_{(0,1),p}} H^{1,p}(I_2)$ *with the partial-derivative crossnorm satisfies* (Inj).

In both cases, best Tucker approximations of any rank exist.

Example 4.5.1 Consider $d = 2$, $V_1 = H^1(0,1)$, $V_2 = H^1(0,1)$, and the ambient space $V_1 \otimes_{H^1} V_2$ with the norm $\|f\|_{H^1}^2 = \|f\|_{L^2}^2 + \|\partial_{x_2} f\|_{L^2}^2$. This Banach tensor space is reflexive and satisfies (Inj), so every $f \in V_1 \otimes_{H^1} V_2$ has a best approximation of rank $\leq r$ for any $r \geq 1$. This setting arises in parametric PDEs where x_1 is the spatial variable and x_2 is the stochastic parameter.

Extension to intersections

In many applications (including the tree-based format), the ambient space is an intersection of Banach tensor spaces, each with its own norm.

Theorem 4.5.3 (Best approximation in intersections, [Falcó and Hackbusch, 2012, Theorem 5.6]) *Let* $\mathbf{V}^{(1)}_{D,\|\cdot\|_D}, \ldots, \mathbf{V}^{(k)}_{D,\|\cdot\|_D}$ *be reflexive Banach tensor spaces over the same algebraic tensor space* $\mathbf{V}_D$*, each with a norm satisfying* (Inj)*. Equip the intersection* $Z = \mathbf{V}^{(1)}_{D,\|\cdot\|_D} \cap \cdots \cap \mathbf{V}^{(k)}_{D,\|\cdot\|_D}$ *with the norm* $\|\mathbf{v}\|_Z = \max_i \|\mathbf{v}\|_i$*. If* Z *is reflexive (which holds, e.g., when* $k = 2$ *and both norms are reflexive), then every* $\mathbf{u} \in Z$ *has a best approximation in* $\mathfrak{M}_{\leq \mathfrak{r}}(Z)$*.*

4.6 Worked examples

Example 4.6.1 (Gaussian kernel in $L^2([0,1]^2)$*)* Let $f(x_1, x_2) = e^{-\gamma(x_1 - x_2)^2}$, $\gamma > 0$, in $L^2([0,1]^2)$. Viewed as a Hilbert–Schmidt operator $T_f : L^2([0,1]) \to L^2([0,1])$, the spectral decomposition gives

$$f = \sum_{k=1}^{\infty} \lambda_k \, \phi_k \otimes \phi_k,$$

so $r_1(f) = r_2(f) = \infty$. The rank-r truncation $f_r = \sum_{k=1}^{r} \lambda_k \, \phi_k \otimes \phi_k$ is the best Tucker-(r, r) approximation (Eckart–Young), with error $\|f - f_r\|^2 = \sum_{k>r} \lambda_k^2$. For the Gaussian kernel the eigenvalues decay super-algebraically: $\lambda_k \sim Ce^{-c\sqrt{k}}$, so achieving tolerance ε requires only $r = O(\log(1/\varepsilon)^2)$ terms.

Example 4.6.2 (Parametric elliptic PDE) Let $\Gamma = [-1,1]^s$ and consider the family of elliptic problems

$$-\frac{\partial}{\partial x}\Big[a(x, \mathbf{y}) \, \frac{\partial u}{\partial x}(x, \mathbf{y})\Big] = f(x), \quad x \in (0,1), \ \mathbf{y} \in \Gamma, \tag{4.12}$$

with $a(x, \mathbf{y}) = a_0(x) + \sum_{j=1}^{s} y_j a_j(x) \geq a_{\min} > 0$. The solution $\mathbf{y} \mapsto u(\cdot, \mathbf{y})$ defines an element $\mathbf{u} \in H_0^1(0,1) \otimes L^2(\Gamma)$. The minimal subspace

$$U_x^{\min}(\mathbf{u}) = \overline{\operatorname{span}\{u(\cdot, \mathbf{y}) : \mathbf{y} \in \Gamma\}} \subset H_0^1(0,1)$$

has dimension equal to the effective rank of the solution manifold. By Corollary 4.5.2, a best Tucker-(r, r) approximation $\mathbf{u}_r = \sum_{k=1}^{r} \lambda_k v_k^{(x)} \otimes v_k^{(\mathbf{y})}$ exists, with $\|\mathbf{u} - \mathbf{u}_r\|^2 = \sum_{k>r} \lambda_k^2$. When a is analytic in $\mathbf{y}$, the singular values decay exponentially and low-rank approximations are highly accurate Nouy [2017].

4.7 Historical notes and comparison with the literature

The notion of minimal subspace was introduced in Falcó and Hackbusch [2012]. It generalises to infinite dimensions the classical concept of matrix rank: for $d = 2$ and finite-dimensional V_α, the α-rank $r_\alpha(\mathbf{v}) = \operatorname{rank}\mathcal{M}_\alpha(\mathbf{v})$ coincides with the classical matrix rank of the α-unfolding, introduced by Hitchcock Hitchcock [1927] in 1927. The higher-order version for finite-dimensional spaces was studied by Tucker Tucker [1966], De Lathauwer et al. De Lathauwer et al. [2000], and Kolda–Bader Kolda and Bader [2009].

The infinite-dimensional theory of Falcó and Hackbusch [2012] is motivated by the numerical treatment of high-dimensional PDEs, where $V_\alpha = L^2(\mathbb{R}^3)$ or Sobolev spaces, and the ambient tensor space is infinite-dimensional. The existence of best approximations (Theorem 4.5.2) removes a fundamental obstacle in the convergence analysis of alternating least-squares and related iterative methods.

The weak closedness argument of Theorem 4.5.1 is modelled on Uschmajew's result for the case of the spectral norm on finite-dimensional spaces Uschmajew and Vandereycken [2013a]. The innovation of Falcó and Hackbusch [2012] is the extension to general reflexive Banach spaces under the crossnorm condition (Inj), which is the natural condition for tensor product norms.

The connection between minimal subspaces and the geometry of Tucker manifolds is developed in Chapter 8, where the rank map $\varrho : \mathbf{V}_D \to \prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ defined by $\varrho(\mathbf{v}) = (U_\alpha^{\min}(\mathbf{v}))_{\alpha \in D}$ plays a central role in the construction of local charts.

Part II
Tree-Based Tensor Formats

Chapter 5
Dimension Partition Trees

The Tucker format developed in Chapters 3 and 4 corresponds to the shallowest possible hierarchical structure: all d factor spaces are placed at the same level below the root. Richer formats arise by organising the dimensions into a *hierarchy*, encoded by a rooted tree. This chapter introduces dimension partition trees, establishes their basic properties, and presents the principal examples (Tucker, Hierarchical Tucker, Tensor Train). The material prepares the ground for the algebraic (Chapter 6) and topological (Chapter 7) development of tree-based tensor formats.

Throughout, $D = \{1, \ldots, d\}$ with $d \geq 2$.

5.1 Definition and basic properties

Definition 5.1.1 (Dimension partition tree) A finite rooted tree T_D is called a *dimension partition tree of D* if:

(a) Every vertex $\alpha \in T_D$ is a non-empty subset of D.
(b) The root of T_D is D itself.
(c) Every non-leaf vertex $\alpha \in T_D$ (i.e., every α with $\#(\alpha) \geq 2$) has at least two children, and the set of children of α, denoted $S(\alpha)$, forms a *non-trivial partition* of α (i.e., a partition into at least two non-empty subsets whose union is α).
(d) Every vertex α with $\#(\alpha) = 1$ has no children; such vertices are called *leaves*.

The set of all leaves of T_D is denoted $\mathcal{L}(T_D)$. Condition (c) forces every non-leaf to have at least two children, so the tree is never a path of singletons. The following observation is immediate from the definition.

Proposition 5.1.1 *For any dimension partition tree T_D, the set of leaves coincides with the set of singletons of D:*

$$\mathcal{L}(T_D) = \big\{\{j\} : j \in D\big\}.$$

In particular, $\mathcal{L}(T_D)$ is the trivial partition of D into singletons.

Proof. By (d), leaves are exactly the vertices with $\#(\alpha) = 1$, i.e., the singletons $\{j\}$ for $j \in D$. Since the root is D and each non-leaf splits into a partition of its label set, every $j \in D$ is eventually reached as a leaf by iterating the partition. $\square$

Levels, depth, and level-partitions

Definition 5.1.2 The *level* of a vertex $\alpha \in T_D$ is defined recursively by

$$\mathrm{level}(D) = 0, \qquad \mathrm{level}(\beta) = \mathrm{level}(\alpha) + 1 \quad \text{for } \beta \in S(\alpha).$$

The *depth* of the tree is

$$\mathrm{depth}(T_D) := \max_{\alpha \in T_D} \mathrm{level}(\alpha). \tag{5.1}$$

Since leaves are the only vertices with $\#(\alpha) = 1$, and every path from root to leaf passes through a strictly decreasing chain of subset sizes, the depth satisfies $1 \leq \mathrm{depth}(T_D) \leq d - 1$.

Definition 5.1.3 For $1 \leq k \leq \mathrm{depth}(T_D)$, the *$k$-th level partition* of D induced by T_D is

$$\mathcal{P}_k(T_D) := \big\{\alpha \in T_D : \mathrm{level}(\alpha) = k\big\}. \tag{5.2}$$

By convention, $\mathcal{P}_0(T_D) = \{D\}$.

Proposition 5.1.2 *For each $1 \leq k \leq \mathrm{depth}(T_D)$, the collection $\mathcal{P}_k(T_D)$ is a partition of D. Moreover:*

(i) *Every element of $\mathcal{P}_k(T_D)$ is a subset of some element of $\mathcal{P}_{k-1}(T_D)$ (refinement property).*
(ii) *$\mathcal{P}_{\mathrm{depth}(T_D)}(T_D) = \mathcal{L}(T_D)$, the trivial partition of D.*

Proof. The elements of $\mathcal{P}_k(T_D)$ are precisely the vertex labels at level k. Since each such vertex α is the child of a unique parent $\beta \in \mathcal{P}_{k-1}(T_D)$ satisfying $\alpha \subset \beta$, and since $S(\beta)$ partitions β, the elements of $\mathcal{P}_k(T_D)$ at level k partition D by induction on k. Parts (i) and (ii) follow directly. $\square$

The sequence of partitions $\mathcal{P}_0(T_D) \succ \mathcal{P}_1(T_D) \succ \cdots \succ \mathcal{P}_{\mathrm{depth}(T_D)}(T_D) = \mathcal{L}(T_D)$ (where $\mathcal{P} \succ \mathcal{Q}$ means $\mathcal{Q}$ refines $\mathcal{P}$) captures the hierarchical structure of the tree. This is the central object connecting dimension partition trees to tensor spaces: for each level k, the identification $\mathbf{V}_D \cong \bigotimes_{\alpha \in \mathcal{P}_k(T_D)} \mathbf{V}_\alpha$ (Equation (3.4)) allows one to view $\mathbf{v} \in \mathbf{V}_D$ as a Tucker tensor at level k.

5.2 Special trees and standard formats

The following three families of dimension partition trees correspond to the three most widely used tensor formats.

The Tucker tree

Example 5.2.1 (Tucker tree) The *Tucker tree* T_D^{Tucker} over D has depth 1 and satisfies

$$T_D^{\mathrm{Tucker}} = \big\{D,\, \{1\},\, \{2\},\, \dots,\, \{d\}\big\}, \qquad S(D) = \mathcal{L}(T_D)[T_D^{\mathrm{Tucker}}] = \big\{\{j\} : j \in D\big\}.$$

The only non-trivial partition step is $D \to \{\{1\}, \dots, \{d\}\}$.

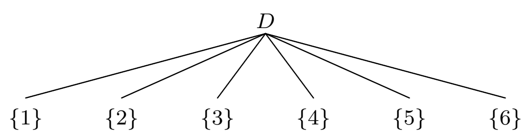

Fig. 5.1 Tucker tree over $D = \{1, 2, 3, 4, 5, 6\}$: depth 1, one level partition $\mathcal{P}_1 = \mathcal{L}(T_D)$.

The Tucker tree is the unique tree of depth 1 over D. The corresponding tensor format is exactly the Tucker format of Chapter 3: $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D^{\mathrm{Tucker}}) = \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.

Binary trees and the Hierarchical Tucker format

Definition 5.2.1 A dimension partition tree T_D is *binary* if every non-leaf vertex has exactly two children: $\#(S(\alpha)) = 2$ for all $\alpha \in T_D$ with $\#(\alpha) \geq 2$.

Binary trees are the natural hierarchical structure for the *Hierarchical Tucker* (HT) format introduced by Hackbusch and Kühn Hackbusch and Kühn [2009].

Example 5.2.2 (Hierarchical Tucker tree) For $D = \{1, 2, 3, 4\}$, the balanced binary HT tree is

$$T_D^{\mathrm{HT}} = \big\{D,\, \{1,2\},\, \{3,4\},\, \{1\},\, \{2\},\, \{3\},\, \{4\}\big\},$$

with $S(D) = \{\{1, 2\}, \{3, 4\}\}$, $S(\{1, 2\}) = \{\{1\}, \{2\}\}$, $S(\{3, 4\}) = \{\{3\}, \{4\}\}$. Here $\mathrm{depth}(T_D^{\mathrm{HT}}) = 2$, with level partitions $\mathcal{P}_1 = \{\{1, 2\}, \{3, 4\}\}$ and $\mathcal{P}_2 = \mathcal{L}(T_D) = \{\{1\}, \{2\}, \{3\}, \{4\}\}$.

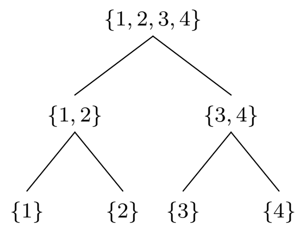

Fig. 5.2 Balanced binary tree over $D = \{1,2,3,4\}$: depth 2, corresponding to the HT format.

Example 5.2.3 (Non-binary tree) For $D = \{1,2,3,4,5,6\}$:

$$T_D = \big\{D,\ \{1,2,3\},\ \{4,5\},\ \{6\},\ \{1\},\ \{2,3\},\ \{4\},\ \{5\},\ \{2\},\ \{3\}\big\},$$

where $S(D) = \{\{1,2,3\},\{4,5\},\{6\}\}$ (three children), $S(\{1,2,3\}) = \{\{1\},\{2,3\}\}$, $S(\{2,3\}) = \{\{2\},\{3\}\}$, $S(\{4,5\}) = \{\{4\},\{5\}\}$. Here $\operatorname{depth}(T_D) = 3$, with $\mathcal{P}_1 = \{\{1,2,3\},\{4,5\},\{6\}\}$, $\mathcal{P}_2 = \{\{1\},\{2,3\},\{4\},\{5\},\{6\}\}$, $\mathcal{P}_3 = \mathcal{L}(T_D)$.

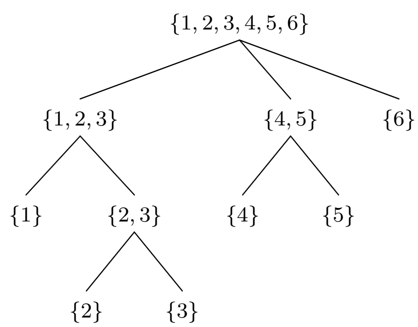

Fig. 5.3 Non-binary dimension partition tree over $D = \{1,2,3,4,5,6\}$: depth 3.

The Tensor Train tree

Example 5.2.4 (Tensor Train tree) For $D = \{1,\dots,d\}$, $d \geq 2$, the *Tensor Train* (TT) tree Oseledets [2011] is the linear (chain) binary tree

$$T_D^{\mathrm{TT}} = \{D\} \cup \big\{\{j\} : 1 \leq j \leq d\big\} \cup \big\{\{1,\dots,j\} : 2 \leq j \leq d-1\big\},$$

with $S(\{1,\dots,j\}) = \{\{1,\dots,j-1\},\{j\}\}$ for $j \geq 2$. Here $\operatorname{depth}(T_D^{\mathrm{TT}}) = d-1$. The level partitions are:

$$\mathcal{P}_k(T_D^{\mathrm{TT}}) = \big\{\{1,\dots,d-k\},\ \{d-k+1\},\ \dots,\ \{d\}\big\}, \quad 1 \leq k \leq d-1.$$

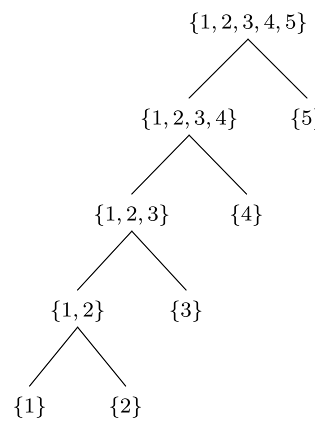

Fig. 5.4 Tensor Train tree over $D = \{1, 2, 3, 4, 5\}$: depth 4, linear binary structure.

Remark 5.2.1 The TT format is a special case of the HT format: T_D^{TT} is a binary tree with maximal depth $d-1$ and a linear (path) structure, whereas HT trees are typically balanced with depth $\lceil \log_2 d \rceil$. This is reflected in their computational cost: TT has storage $O(dr^2 n)$ while balanced HT has $O(dr^3)$, with r the maximal rank and $n = \max_j \dim V_j$ Hackbusch and Kühn [2009], Oseledets [2011].

5.3 Graph-theoretic properties

We record here the combinatorial facts about dimension partition trees that will be used in Chapters 6 and 9.

Proposition 5.3.1 *Let T_D be a dimension partition tree over $D = \{1, \ldots, d\}$. Then:*

(i) $\#(\mathcal{L}(T_D)) = d$. *The number of internal (non-leaf) vertices satisfies* $\#(T_D \setminus \mathcal{L}(T_D)) \leq d-1$.
(ii) *If T_D is binary, then $\#(T_D) = 2d-1$, with exactly $d-1$ internal vertices.*
(iii) $\mathrm{depth}(T_D) \leq d-1$, *with equality if and only if $T_D = T_D^{\mathrm{TT}}$ (up to re-labelling).*

Proof. Part (i): every leaf is a singleton $\{j\}$ for a unique $j \in D$, so $\#(\mathcal{L}(T_D)) = d$. The bound on internal vertices follows since each internal vertex contributes at least two children which are eventually singletons. Part (ii): in a binary tree with d leaves, the number of internal nodes is exactly $d-1$, giving $2d-1$ total vertices. Part (iii): the depth is maximised when each step reduces the largest set by exactly one element, giving the TT structure. ☐

Proposition 5.3.2 (Recursive subtree structure) *Let T_D be a dimension partition tree over D and $\alpha \in T_D$ with $\#(\alpha) \geq 2$. Then the restriction*

$$T_D|_\alpha := \{\beta \in T_D : \beta \subseteq \alpha\}$$

is itself a dimension partition tree over α (with root α).

This recursive structure is fundamental to the inductive proofs in Chapter 6: properties of tree-based tensor formats are proved by induction on $\mathrm{depth}(T_D)$, reducing at each step to the subtrees $T_D|_\beta$ for $\beta \in S(D)$.

5.4 The intersection representation

The central structural theorem of the tree-based theory expresses the set of tree-based tensors with fixed rank as an intersection of Tucker manifolds — one at each level of the tree. This is the key result connecting the tree structure to the geometry of Chapters 9.

For each level partition $\mathcal{P}_k(T_D)$, recall from Section 3.3 that the identification $\mathbf{V}_D \cong \bigotimes_{\alpha\in\mathcal{P}_k(T_D)} \mathbf{V}_\alpha$ allows one to define the Tucker set

$$\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)) := \left\{\mathbf{v} \in \mathbf{V}_D : \dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha \text{ for all } \alpha \in \mathcal{P}_k(T_D)\right\}, \quad (5.3)$$

where $\mathfrak{r}_k = (r_\alpha)_{\alpha\in\mathcal{P}_k(T_D)}$.

Theorem 5.4.1 (Intersection representation, [Falcó et al., 2023, Theorem 3.12]) *Let T_D be a dimension partition tree over D and $\mathfrak{r} = (r_\alpha)_{\alpha\in T_D} \in \mathcal{AD}(\mathbf{V}_D, T_D)$. Then*

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) = \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)), \quad (5.4)$$

where $\mathfrak{r}_k = (r_\alpha)_{\alpha\in\mathcal{P}_k(T_D)}$.

Sketch. The inclusion $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \subseteq \bigcap_k \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ is immediate: if $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ for all $\alpha \in T_D$, then in particular for all $\alpha \in \mathcal{P}_k(T_D)$ at each level k. For the reverse inclusion, one uses Proposition 4.3.2 inductively: from $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ at the leaf level ($k = \mathrm{depth}(T_D)$) and the compatibility condition $U_\alpha^{\min}(\mathbf{v}) \subset {}_a\bigotimes_{\beta\in S(\alpha)} U_\beta^{\min}(\mathbf{v})$, one deduces $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ at all intermediate levels. See Falcó et al. [2023] for the complete proof. $\square$

Remark 5.4.1 For $T_D = T_D^{\mathrm{Tucker}}$ (depth 1), the intersection (5.4) reduces to a single factor: $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D^{\mathrm{Tucker}}) = \mathfrak{M}_{\mathfrak{r}_1}(\mathbf{V}_D, \mathcal{L}(T_D)) = \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$, recovering the Tucker format.

Example 5.4.1 For $D = \{1, 2, 3, 4\}$ and the balanced HT tree T_D^{HT} of Example 5.2.2 with $\mathfrak{r} = (r_D, r_{\{1,2\}}, r_{\{3,4\}}, r_{\{1\}}, r_{\{2\}}, r_{\{3\}}, r_{\{4\}})$:

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D^{\mathrm{HT}}) = \mathfrak{M}_{\mathfrak{r}_1}(\mathbf{V}_D, \mathcal{P}_1) \cap \mathfrak{M}_{\mathfrak{r}_2}(\mathbf{V}_D, \mathcal{P}_2),$$

where $\mathcal{P}_1 = \{\{1,2\},\{3,4\}\}$ and $\mathcal{P}_2 = \mathcal{L}(T_D)$. Concretely:

$$\mathfrak{M}_{\mathfrak{r}_1}(\mathbf{V}_D, \mathcal{P}_1) = \big\{\mathbf{v} \in \mathbf{V}_D : \dim U_{\{1,2\}}^{\min}(\mathbf{v}) = r_{\{1,2\}},\ \dim U_{\{3,4\}}^{\min}(\mathbf{v}) = r_{\{3,4\}}\big\},$$
$$\mathfrak{M}_{\mathfrak{r}_2}(\mathbf{V}_D, \mathcal{P}_2) = \big\{\mathbf{v} \in \mathbf{V}_D : \dim U_{\{j\}}^{\min}(\mathbf{v}) = r_{\{j\}},\ j = 1,2,3,4\big\}.$$

The intersection captures tensors that are simultaneously Tucker at the level of pairs ($\{1,2\},\{3,4\}$) and at the level of singletons.

Remark 5.4.2 (Degenerate case) Not all intersections define distinct sets. If for some internal vertex $\alpha \in T_D$ the rank r_α equals $\prod_{\beta \in S(\alpha)} r_\beta$, the rank constraint at α is automatically satisfied whenever the constraints at the children are. In such cases one of the Tucker factors in (5.4) is redundant. This situation is analysed in detail in [Falcó et al., 2023, Theorem 3.12].

5.5 Concrete comparisons: Tucker, HT, and TT for $d = 3, 4, 5$

This section works through the three standard formats in detail for small values of d, comparing storage costs, rank constraints, and the inclusion relations between the corresponding tensor sets. Throughout, $V_\alpha = \mathbb{R}^n$ for all α (so $n_\alpha = n$) and r denotes a uniform rank bound $r_\alpha = r$ for all $\alpha \in T_D \setminus \{D\}$.

The case $d = 3$

For $D = \{1,2,3\}$ there are exactly two possible tree structures:

Tucker tree (depth(=)1).

The only tree of depth 1 is $T_D^{\text{Tucker}} = \{D, \{1\}, \{2\}, \{3\}\}$ with $S(D) = \{\{1\},\{2\},\{3\}\}$. A Tucker tensor $\mathbf{v} \in \mathfrak{M}_{(r,r,r)}(\mathbb{R}^n \otimes \mathbb{R}^n \otimes \mathbb{R}^n)$ has the form

$$\mathbf{v} = \sum_{i_1,i_2,i_3=1}^{r} C_{i_1 i_2 i_3}\, u_{i_1}^{(1)} \otimes u_{i_2}^{(2)} \otimes u_{i_3}^{(3)}, \quad C \in \mathbb{R}_*^{r\times r\times r}.$$

Storage: $3rn$ (basis vectors) $+r^3$ (core) $= 3rn + r^3$.

TT tree (depth(=)2).

The only binary tree of depth 2 is $T_D^{\text{TT}} = \{D, \{1,2\}, \{1\}, \{2\}, \{3\}\}$ with $S(D) = \{\{1,2\},\{3\}\}$ and $S(\{1,2\}) = \{\{1\},\{2\}\}$. A TT tensor $\mathbf{v} \in \mathcal{FT}_{(r,r,r,r)}(\mathbb{R}^n \otimes \mathbb{R}^n \otimes$

$\mathbb{R}^n, T_D^{\mathrm{TT}})$ has the form

$$\mathbf{v} = \sum_{i_1,i_2,i_3=1}^{n} \sum_{\alpha_1,\alpha_2=1}^{r} G^{(1)}_{i_1,\alpha_1} G^{(2)}_{\alpha_1,i_2,\alpha_2} G^{(3)}_{\alpha_2,i_3}\, e_{i_1} \otimes e_{i_2} \otimes e_{i_3},$$

with *TT cores* $G^{(1)} \in \mathbb{R}^{n\times r}$, $G^{(2)} \in \mathbb{R}^{r\times n\times r}$, $G^{(3)} \in \mathbb{R}^{r\times n}$. Storage: $rn + r^2n + rn = 2rn + r^2n$.

Inclusion and storage comparison.

By Theorem 5.4.1, every TT tensor is also a Tucker tensor (with Tucker rank $\leq (r^2, r, r)$ in general), but not conversely. The storage comparison:

Format	Tree depth	Storage ($d = 3$, uniform rank r)
Tucker	1	$3rn + r^3$
TT	2	$2rn + r^2n$

For $r \ll n$, Tucker storage is $O(rn)$ while TT storage is $O(r^2n)$: Tucker is cheaper for $d = 3$ because the core is only r^3 (small when r is small), while TT pays r^2n for the middle core.

The case $d = 4$

For $D = \{1, 2, 3, 4\}$ the three standard formats are:

Tucker tree (depth(=)1).

$S(D) = \{\{1\}, \{2\}, \{3\}, \{4\}\}$. Storage: $4rn + r^4$.

Balanced HT tree (depth(=)2).

From Example 5.2.2: $S(D) = \{\{1, 2\}, \{3, 4\}\}$, each split further into singletons. An HT tensor has the form

$$\mathbf{v} = \sum_{k=1}^{r_{12}} \sum_{l=1}^{r_{34}} C_{kl}\, \mathbf{u}_k^{(12)} \otimes \mathbf{u}_l^{(34)},$$

where $\mathbf{u}_k^{(12)} \in V_1 \otimes V_2$ are themselves Tucker tensors:

$$\mathbf{u}_k^{(12)} = \sum_{i_1, i_2=1}^{r} B_{k,i_1,i_2}^{(12)} \, u_{i_1}^{(1)} \otimes u_{i_2}^{(2)},$$

and similarly for $\mathbf{u}_l^{(34)}$. With uniform rank r at all levels: storage $= 4rn + 2r^2 r + r^2 = 4rn + 2r^3 + r^2$.

TT tree (depth(=)3).

T_D^{TT} for $d = 4$: $S(D) = \{\{1,2,3\},\{4\}\}$, $S(\{1,2,3\}) = \{\{1,2\},\{3\}\}$, $S(\{1,2\}) = \{\{1\},\{2\}\}$. Storage: $rn + r^2 n + r^2 n + rn = 2rn \cdot 2 + 2r^2 n = 4rn + 2r^2 n$.

Storage comparison ($d = 4$).

Format	Tree depth	Storage ($d = 4$)	Leading term
Tucker	1	$4rn + r^4$	r^4 (exponential in d)
HT (balanced)	2	$4rn + 2r^3 + r^2$	r^3
TT	3	$4rn + 2r^2 n$	$r^2 n$

For $r \ll n$: Tucker $\gg$ HT $\gg$ TT in storage. The Tucker core r^4 grows exponentially in d, while HT and TT have polynomial storage in r.

The case $d = 5$

Example 5.5.1 (Three trees for $d = 5$) For $D = \{1,2,3,4,5\}$, the three standard formats correspond to:

Tucker (depth(=)1).

$S(D) = \{\{1\},\{2\},\{3\},\{4\},\{5\}\}$. Storage: $5rn + r^5$.

Balanced HT (depth(=)3).

One natural binary tree splits as: $S(D) = \{\{1,2,3\},\{4,5\}\}$, $S(\{1,2,3\}) = \{\{1,2\},\{3\}\}$, $S(\{1,2\}) = \{\{1\},\{2\}\}$, $S(\{4,5\}) = \{\{4\},\{5\}\}$. Storage (uniform rank r): $5rn + 4r^3 + r^2$.

TT (depth(=)4).

$S(\{1,\dots,j\}) = \{\{1,\dots,j-1\},\{j\}\}$ for $j = 5,4,3,2$. Storage: $2rn + 3r^2n$.

Remark 5.5.1 (The curse of dimensionality and its mitigation) The fundamental issue is that for a full tensor $\mathbf{v} \in \bigotimes_{j=1}^{d} \mathbb{R}^n$, storage requires n^d numbers — exponential in d. The three formats address this differently:

- *Tucker*: storage $drn + r^d$. The factor r^d is still exponential in d, so Tucker is practical only for small d (typically $d \le 6$) or when $r = 1$ (separable tensors).
- *HT*: storage $O(dr^3)$. The core tensors at each internal node have size $r\times r\times r$, independent of n. For balanced binary trees with d leaves and depth $\lceil \log_2 d \rceil$, the total cost is $O(d(rn + r^3))$.
- *TT*: storage $O(dr^2n)$. The dependence on n is linear and the dependence on r is quadratic, but the cost is independent of d up to a linear factor. TT is the most efficient for large d.

The choice of tree reflects the entanglement/correlation structure of the problem: problems with local interactions (e.g., 1D quantum chains, tridiagonal systems) are efficiently represented by TT; problems with hierarchical correlations (e.g., multi-scale PDEs, wavelet bases) are better suited to HT.

Inclusion relations

The intersection representation (Theorem 5.4.1) implies the following inclusion chain for the same dimension d and rank bound r, comparing formats by number of constraints:

Proposition 5.5.1 (Inclusion of tensor sets) *Let T_D and T'_D be two dimension partition trees over D with* $\operatorname{depth}(T_D) \le \operatorname{depth}(T'_D)$. *If every level partition of T'_D refines the corresponding level partition of T_D, then*

$$\mathcal{FT}_{\mathfrak{r}'}(\mathbf{V}_D, T'_D) \subseteq \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D),$$

where $\mathfrak{r}'$ and $\mathfrak{r}$ are related by the rank bounds at the corresponding nodes.

Proof. By Theorem 5.4.1, $\mathcal{FT}_{\mathfrak{r}'}(\mathbf{V}_D, T'_D)$ is an intersection over more levels than $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, hence a subset. □

For fixed d and uniform rank r, the inclusion chain is:

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D^{\mathrm{TT}}) \subseteq \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D^{\mathrm{HT}}) \subseteq \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) = \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D^{\mathrm{Tucker}}), \tag{5.5}$$

since T_D^{TT} has the most levels (deepest tree) and T_D^{Tucker} has the fewest (shallowest). Each inclusion is in general strict: a Tucker tensor need not satisfy the additional structural constraints of HT or TT.

5.6 Historical notes

Dimension partition trees formalise the hierarchical structure of tensor representations that had appeared in various forms in the literature. The HT format was introduced by Hackbusch and Kühn in Hackbusch and Kühn [2009] using a binary tree structure, with the specific aim of reducing the exponential storage of full Tucker tensors to $O(dr^3)$. The TT format, independently introduced by Vidal Vidal [2003] in quantum information (under the name *matrix product states*) and by Oseledets Oseledets [2011] in numerical mathematics (as *TT decomposition*), corresponds to the linear chain binary tree.

The general notion of a dimension partition tree, unifying these formats under a common combinatorial framework, was introduced in Falcó et al. [2021]. The intersection representation (Theorem 5.4.1) was established in Falcó et al. [2023] and is the key to the geometric theory of Chapter 9: it reduces the analysis of the tree-based manifold $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ to the intersection of Tucker manifolds, each of which has the structure established in Chapter 8.

The connection between dimension partition trees and quantum physics is deeper than a mere formal analogy. In quantum information, a tensor network is a collection of tensors connected by contraction along shared indices, and the tree structure encodes the entanglement structure of a quantum state. The area law for entanglement entropy in one-dimensional quantum systems Vidal [2003] is the physical motivation for the TT format; balanced binary trees encode area laws in higher-dimensional systems. This connection is discussed further in Chapter 13.

Chapter 6
Algebraic Tree-Based Tensor Formats

This chapter develops the algebraic theory of tree-based tensor formats. Given a dimension partition tree T_D over D and a tensor space $\mathbf{V}_D = {}_a\bigotimes_{j\in D} V_j$, we associate to each vertex $\alpha \in T_D$ a subspace $\mathbf{V}_\alpha$ and a minimal subspace $U_\alpha^{\min}(\mathbf{v})$, define the tree-based rank and the sets $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ and $\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D)$, and establish the central characterisation theorem (Theorem 6.4.1) expressing every tensor of fixed tree-based rank in terms of a hierarchical system of *transfer tensors*. The chapter closes with a discussion of the special cases Tucker, HT, and TT.

Throughout, T_D is a dimension partition tree over $D = \{1, \ldots, d\}$ with $d \geq 2$, and V_j $(j \in D)$ are vector spaces over $\mathbb{K}$.

6.1 Tree-based tensor spaces

Associated spaces at each vertex

For each $\alpha \in T_D$ we associate a vector space $\mathbf{V}_\alpha$ defined recursively from the leaves upward:

- For leaves $\{j\} \in \mathcal{L}(T_D)$: $\mathbf{V}_{\{j\}} = V_j$.
- For internal vertices $\alpha \in T_D \setminus \mathcal{L}(T_D)$: $\mathbf{V}_\alpha = {}_a\bigotimes_{\beta\in S(\alpha)} \mathbf{V}_\beta$.

Since $S(\alpha)$ partitions α, the canonical identification $\mathbf{V}_\alpha = {}_a\bigotimes_{j\in\alpha} V_j$ holds at every vertex by the associativity of Equation (3.4). In particular, $\mathbf{V}_D = \mathbf{V}_D$.

Definition 6.1.1 The *tensor space in tree-based format* associated to $(\mathbf{V}_D, T_D)$ is the pair $(\mathbf{V}_D, T_D)$ together with the collection of spaces $\{\mathbf{V}_\alpha\}_{\alpha\in T_D\setminus\{D\}}$.

Minimal subspaces compatible with the tree

Recall from Proposition 4.3.2 that for any $\alpha \in T_D \setminus \mathcal{L}(T_D)$ and any $\mathbf{v} \in \mathbf{V}_D$:

$$U_\alpha^{\min}(\mathbf{v}) \subset {}_a \bigotimes_{\beta \in S(\alpha)} U_\beta^{\min}(\mathbf{v}) \subset \mathbf{V}_\alpha. \tag{6.1}$$

This compatibility is the algebraic backbone of the tree-based format: the minimal subspace at each internal vertex is contained in the tensor product of the minimal subspaces of its children.

6.2 Tree-based rank and admissible tuples

Definition 6.2.1 (Tree-based rank) For $\mathbf{v} \in \mathbf{V}_D$, its *tree-based rank with respect to* T_D is the tuple

$$\operatorname{rank}_{T_D}(\mathbf{v}) := \left(\dim U_\alpha^{\min}(\mathbf{v})\right)_{\alpha \in T_D} \in \mathbb{N}_0^{\#(T_D)}. \tag{6.2}$$

Definition 6.2.2 (Admissible tuples) A tuple $\mathfrak{r} = (r_\alpha)_{\alpha \in T_D} \in \mathbb{N}^{\#(T_D)}$ is *admissible for* $(\mathbf{V}_D, T_D)$ if there exists $\mathbf{v} \in \mathbf{V}_D$ with $\operatorname{rank}_{T_D}(\mathbf{v}) = \mathfrak{r}$. The set of all admissible tuples is

$$\mathcal{AD}(\mathbf{V}_D, T_D) := \left\{\operatorname{rank}_{T_D}(\mathbf{v}) : \mathbf{v} \in \mathbf{V}_D\right\}. \tag{6.3}$$

Remark 6.2.1 Admissible tuples satisfy the compatibility constraints inherited from Proposition 4.4.2 and Equation (6.1). Specifically, for each $\alpha \in T_D \setminus \mathcal{L}(T_D)$:

$$r_\alpha \leq \prod_{\beta \in S(\alpha)} r_\beta. \tag{6.4}$$

Equality in (6.4) is the generic case; strict inequality indicates that the tensor has additional structure at vertex α.

Sets of tensors with fixed and bounded tree-based rank

Definition 6.2.3 Let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$. Define:

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) := \left\{\mathbf{v} \in \mathbf{V}_D : \dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha \text{ for all } \alpha \in T_D\right\}, \tag{6.5}$$

$$\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D) := \left\{\mathbf{v} \in \mathbf{V}_D : \dim U_\alpha^{\min}(\mathbf{v}) \leq r_\alpha \text{ for all } \alpha \in T_D\right\}. \tag{6.6}$$

The relation between these two sets is:

$$\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D) = \bigsqcup_{\substack{\mathfrak{s} \in \mathcal{AD}(\mathbf{V}_D, T_D) \\ \mathfrak{s} \leq \mathfrak{r}}} \mathcal{FT}_{\mathfrak{s}}(\mathbf{V}_D, T_D). \tag{6.7}$$

Remark 6.2.2 (Tucker as special case) For $T_D = T_D^{\text{Tucker}}$ (depth 1, $S(D) = \mathcal{L}(T_D)$), the tree-based rank reduces to the Tucker rank: $\text{rank}_{T_D^{\text{Tucker}}}(\mathbf{v}) = (r_D, r_{\{1\}}, \ldots, r_{\{d\}})$. Since $r_D = 1$ for all non-zero $\mathbf{v}$ (the root has a single rank), the effective tuple is $(r_{\{1\}}, \ldots, r_{\{d\}})$ and $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D^{\text{Tucker}}) = \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.

6.3 Minimal subspaces compatible with a tree: new characterisation

The following proposition, proved in Falcó et al. [2021], gives a characterisation of $U_\beta^{\min}(\mathbf{v})$ for an internal vertex β in terms of contractions of $U_\alpha^{\min}(\mathbf{v})$ for its parent $\alpha = \text{parent}(\beta)$. This is the tree-based analogue of the contraction characterisation Theorem 4.2.2.

Proposition 6.3.1 (Tree-based characterisation of minimal subspaces, [Falcó et al., 2021, Proposition 2.3]) *Let $\mathbf{v} \in \mathbf{V}_D$, $\alpha \in T_D \setminus \mathcal{L}(T_D)$ with $\#(S(\alpha)) \geq 2$, and assume that $\mathbf{V}_\alpha$ and $\mathbf{V}_\beta$ for $\beta \in S(\alpha)$ are normed spaces. Then for each $\beta \in S(\alpha)$:*

$$U_\beta^{\min}(\mathbf{v}) = \text{span}\Big\{\big(\text{id}_{\mathbf{V}_\beta} \otimes \varphi^{(\alpha\setminus\beta)}\big)(\mathbf{v}_\alpha) : \mathbf{v}_\alpha \in U_\alpha^{\min}(\mathbf{v}),\ \varphi^{(\alpha\setminus\beta)} \in \Big({}_a \bigotimes_{\gamma \in S(\alpha)\setminus\{\beta\}} \mathbf{V}_\alpha[\gamma]\Big)^*\Big\}. \tag{6.8}$$

Proof. This follows directly from Proposition 4.3.3 applied at level k of the tree, with α playing the role of the block and $S(\alpha)$ playing the role of the partition $\mathcal{P}_\alpha$. □

6.4 The transfer tensor representation

The main result of this section is the characterisation of every $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ by a system of *transfer tensors* $\{C^{(\alpha)}\}_{\alpha \in T_D}$, one at each vertex. This is the algebraic heart of the tree-based format and the starting point for all computational algorithms (ALS, DMRG, etc.).

Transfer tensors and the representation formula

Definition 6.4.1 Let $\alpha \in T_D$ with $\#(S(\alpha)) \geq 2$ and let $\mathfrak{r}_\alpha = (r_\beta)_{\beta \in S(\alpha)}$. A *transfer tensor at vertex α* is an array

$$C^{(\alpha)} \in \mathbb{R}^{r_\alpha \times \prod_{\beta \in S(\alpha)} r_\beta},$$

where we view it as a matrix with r_α rows and $\prod_{\beta \in S(\alpha)} r_\beta$ columns (indexed by the multi-index $(i_\beta)_{\beta \in S(\alpha)}$). The transfer tensor $C^{(\alpha)}$ is said to be *full-rank* if $\operatorname{rank}\mathcal{M}_\alpha(C^{(\alpha)}) = r_\alpha$.

For the root vertex D, the transfer tensor $C^{(D)}$ plays a special role:

Definition 6.4.2 The *root transfer tensor* is

$$C^{(D)} \in \mathbb{R}^{\times_{\alpha \in S(D)} r_\alpha},$$

a multi-dimensional array of shape $\prod_{\alpha \in S(D)} r_\alpha$ (no "row" index since the root has no parent). The full-rank condition at the root is $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha \in S(D)} r_\alpha}$, the open set of Definition 3.2.2.

The main characterisation theorem

Theorem 6.4.1 (Transfer tensor characterisation, [Falcó et al., 2021, Theorem 2.6]) *Let T_D be a dimension partition tree over D with $\operatorname{depth}(T_D) = L$, and let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$ with $\mathfrak{r} \neq \mathbf{0}$. For $\mathbf{v} \in \mathbf{V}_D$, the following statements are equivalent.*

(a) $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$.

(b) *Given bases $\{u_{i_k}^{(k)} : 1 \leq i_k \leq r_k\}$ of $U_k^{\min}(\mathbf{v})$ for each $k \in \mathcal{L}(T_D)$, there exist:*

- *A unique root transfer tensor $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha \in S(D)} r_\alpha}$ such that*

$$\mathbf{v} = \sum_{\substack{1 \leq i_\alpha \leq r_\alpha \\ \alpha \in S(D)}} C^{(D)}_{(i_\alpha)_{\alpha \in S(D)}} \bigotimes_{\alpha \in S(D)} \mathbf{u}_{i_\alpha}^{(\alpha)}, \tag{6.9}$$

- *For each internal vertex $\mu \in T_D \setminus (\{D\} \cup \mathcal{L}(T_D))$ with $S(\mu) \neq \emptyset$, a unique transfer tensor $C^{(\mu)} \in \mathbb{R}^{r_\mu \times \prod_{\beta \in S(\mu)} r_\beta}$ with $\operatorname{rank}\mathcal{M}_\mu(C^{(\mu)}) = r_\mu$, and a basis $\{\mathbf{u}_{i_\mu}^{(\mu)} : 1 \leq i_\mu \leq r_\mu\}$ of $U_\mu^{\min}(\mathbf{v})$ given by:*

$$\mathbf{u}_{i_\mu}^{(\mu)} = \sum_{\substack{1 \leq i_\beta \leq r_\beta \\ \beta \in S(\mu)}} C^{(\mu)}_{i_\mu, (i_\beta)_{\beta \in S(\mu)}} \bigotimes_{\beta \in S(\mu)} \mathbf{u}_{i_\beta}^{(\beta)}, \quad 1 \leq i_\mu \leq r_\mu. \tag{6.10}$$

Proof. **(b) $\Rightarrow$ (a).** Clear: the representation gives $\dim U_\alpha^{\min}(\mathbf{v}) \leq r_\alpha$ at each level, and the full-rank conditions force equality.

(a) $\Rightarrow$ (b). We construct the transfer tensors level by level from the leaves upward.

Step 1 (Root). Since $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ and $\mathbf{v} \in {}_a\bigotimes_{\alpha \in S(D)} U_\alpha^{\min}(\mathbf{v})$ (by the compatibility condition (6.1)), there exists a unique $C^{(D)} \in \mathbb{R}^{\times_{\alpha \in S(D)} r_\alpha}$ such that (6.9) holds (choosing any bases of the $U_\alpha^{\min}(\mathbf{v})$). The full-rank condition $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha\in S(D)} r_\alpha}$ follows from $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ for $\alpha \in S(D)$: the rank $\operatorname{rank}\mathcal{M}_\alpha(C^{(D)}) = r_\alpha$ for each α because otherwise one could find a subspace of $U_\alpha^{\min}(\mathbf{v})$ of smaller dimension still containing $\mathbf{v}$, contradicting minimality.

Step 2 (Internal vertices). For each $\mu \in T_D \setminus (\{D\} \cup \mathcal{L}(T_D))$ with $S(\mu) \neq \emptyset$, the inclusion $U_\mu^{\min}(\mathbf{v}) \subset {}_a\bigotimes_{\beta\in S(\mu)} U_\beta^{\min}(\mathbf{v})$ (Proposition 4.3.2) gives the existence of $C^{(\mu)}$ satisfying (6.10). The full-rank condition $\operatorname{rank}\mathcal{M}_\mu(C^{(\mu)}) = r_\mu$ follows from the linear independence of $\{\mathbf{u}_{i_\mu}^{(\mu)}\}$ (as a basis of $U_\mu^{\min}(\mathbf{v})$). ☐

Remark 6.4.1 (Uniqueness up to gauge) The transfer tensors $\{C^{(\alpha)}\}$ are uniquely determined once the bases $\{u_{i_k}^{(k)}\}_{k\in\mathcal{L}(T_D)}$ are fixed. Changing the bases introduces a *gauge transformation*: if $\{(\tilde{u}_{i_k}^{(k)})\}$ is another set of bases related by invertible matrices $G_k \in \mathrm{GL}(\mathbb{R}^{r_k})$ (via $\tilde{u}_{i_k}^{(k)} = \sum_j (G_k)_{ij} u_j^{(k)}$), then the transfer tensors transform by contracting with the corresponding G_k. The freedom to choose orthonormal bases (when the V_j are Hilbert spaces) is exploited in the HOSVD and related algorithms.

Recursive computation

Equation (6.10) has a natural recursive interpretation: the basis vectors of $U_\mu^{\min}(\mathbf{v})$ are built bottom-up from the leaf bases, level by level, using the transfer tensors as "gluing" coefficients. The full representation can be written compactly as:

$$\mathbf{v} = \sum_{(i_j)_{j\in D}} \Big[\text{network contraction of } \{C^{(\alpha)}\}_{\alpha\in T_D}\Big]_{(i_j)_{j\in D}} \bigotimes_{j\in D} u_{i_j}^{(\{j\})}, \tag{6.11}$$

where the "network contraction" is the tensor network obtained by connecting the transfer tensors along the edges of T_D.

6.5 Special cases

We verify Theorem 6.4.1 in the three standard formats, recovering known representations.

Tucker format

For $T_D = T_D^{\text{Tucker}}$ (depth 1), $S(D) = \mathcal{L}(T_D) = \{\{j\} : j \in D\}$, and there are no internal vertices other than the root. The only transfer tensor is the root tensor

$C^{(D)}$, and Theorem 6.4.1 reduces to:

$$\mathbf{v} = \sum_{\substack{1\le i_j\le r_j\\ j\in D}} C^{(D)}_{(i_j)_{j\in D}} \bigotimes_{j\in D} u^{(\{j\})}_{i_j}, \quad C^{(D)} \in \mathbb{R}_*^{\times_{j\in D} r_j}.$$

This is exactly the Tucker representation (3.9) of Lemma 3.3.1, with $C^{(D)}$ playing the role of the core tensor.

Hierarchical Tucker format ($d = 4$, balanced binary)

For T_D^{HT} of Example 5.2.2 with $D = \{1,2,3,4\}$, $S(D) = \{\{1,2\},\{3,4\}\}$, and the two internal vertices $\{1,2\}$ and $\{3,4\}$ each have $S(\{i,j\}) = \{\{i\},\{j\}\}$. Theorem 6.4.1 gives:

- Leaf bases: $\{u^{(\{j\})}_{i_j}\}_{i_j=1}^{r_j}$ for $j = 1,2,3,4$.
- Transfer tensors at level 2: $C^{(\{1,2\})} \in \mathbb{R}^{r_{\{1,2\}}\times r_1 r_2}$ and $C^{(\{3,4\})} \in \mathbb{R}^{r_{\{3,4\}}\times r_3 r_4}$, defining bases of $U^{\min}_{\{1,2\}}(\mathbf{v})$ and $U^{\min}_{\{3,4\}}(\mathbf{v})$:

$$\mathbf{u}^{(\{1,2\})}_{i_{12}} = \sum_{i_1=1}^{r_1}\sum_{i_2=1}^{r_2} C^{(\{1,2\})}_{i_{12},i_1 i_2}\, u^{(\{1\})}_{i_1} \otimes u^{(\{2\})}_{i_2}.$$

- Root transfer tensor: $C^{(D)} \in \mathbb{R}_*^{r_{\{1,2\}}\times r_{\{3,4\}}}$, giving:

$$\mathbf{v} = \sum_{i_{12}=1}^{r_{\{1,2\}}}\sum_{i_{34}=1}^{r_{\{3,4\}}} C^{(D)}_{i_{12},i_{34}}\, \mathbf{u}^{(\{1,2\})}_{i_{12}} \otimes \mathbf{u}^{(\{3,4\})}_{i_{34}}.$$

The total storage for this representation is $r_1 r_2 r_{\{1,2\}} + r_3 r_4 r_{\{3,4\}} + r_{\{1,2\}} r_{\{3,4\}} + (r_1+r_2+r_3+r_4)n$, where $n = \max_j \dim V_j$, compared to n^4 for a full tensor.

Tensor Train format

For T_D^{TT} over $D = \{1,\dots,d\}$, with $S(\{1,\dots,k\}) = \{\{1,\dots,k-1\},\{k\}\}$ for $k \ge 2$, Theorem 6.4.1 gives the *TT decomposition*:

$$\mathbf{v} = \sum_{\substack{1\le i_j\le r_j\\ j\in D}} G^{(1)}_{i_1} G^{(2)}_{i_1 i_2} \cdots G^{(d-1)}_{i_{d-2} i_{d-1}} G^{(d)}_{i_{d-1} i_d} \bigotimes_{j\in D} u^{(\{j\})}_{i_j}, \tag{6.12}$$

where $G^{(k)} \in \mathbb{R}^{r_{k-1}\times r_k}$ are the transfer matrices at level k (with $r_0 = r_d = 1$). This is the standard TT format of Oseledets [2011], expressed in terms of minimal subspace bases.

6.6 Disjoint decomposition and rank stratification

Theorem 6.6.1 *The algebraic tensor space* $\mathbf{V}_D$ *admits the disjoint decomposition*

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)} \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D). \tag{6.13}$$

Every non-zero $\mathbf{v} \in \mathbf{V}_D$ *belongs to exactly one stratum* $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ *for* $\mathfrak{r} = \mathrm{rank}_{T_D}(\mathbf{v})$.

Proof. Follows directly from the uniqueness of minimal subspaces (Theorem 4.2.1) and the definition of tree-based rank. □

Proposition 6.6.1 (Relation to Tucker decomposition) *For any* T_D *and* $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$*, the set* $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ *is contained in the Tucker set of rank* $\mathfrak{r}_{\mathrm{leaf}}$*:*

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \subset \mathfrak{M}_{\mathfrak{r}_{\mathrm{leaf}}}(\mathbf{V}_D), \quad \mathfrak{r}_{\mathrm{leaf}} = (r_{\{j\}})_{j \in D}.$$

Moreover, from Theorem 5.4.1:

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) = \mathfrak{M}_{\mathfrak{r}_{\mathrm{leaf}}}(\mathbf{V}_D) \cap \mathfrak{M}_{\mathfrak{r}_1}(\mathbf{V}_D, \mathcal{P}_1(T_D)) \cap \cdots \cap \mathfrak{M}_{\mathfrak{r}_{\mathrm{depth}()-1}}(\mathbf{V}_D, \mathcal{P}_{\mathrm{depth}()-1}(T_D)).$$

The tree-based format imposes additional *rank constraints beyond the Tucker format at each intermediate level of the tree.*

6.7 Historical notes and comparison

The characterisation of tree-based tensors by transfer tensors (Theorem 6.4.1) generalises the Tucker core representation (for depth 1 trees) and the HT representation of Hackbusch–Kühn Hackbusch and Kühn [2009] (for binary trees). The TT case (6.12) was established by Oseledets Oseledets [2011]; the connection to the minimal subspace framework was made explicit in Falcó and Hackbusch [2012].

The general tree-based format, unifying all these cases, was introduced in Falcó et al. [2021]. The key innovation there is the *new characterisation of minimal subspaces* (Proposition 6.3.1), which expresses $U_\beta^{\min}(\mathbf{v})$ via contractions of $U_\alpha^{\min}(\mathbf{v})$ for the parent vertex α. This extends the results of Falcó and Hackbusch [2012] (where only the bilinear splitting $V_\alpha \otimes V_{\alpha^c}$ was used) to the full tree structure.

The gauge freedom (Remark 6.4.1) is the algebraic manifestation of the non-uniqueness of tensor decompositions. In the TT case, canonical gauges (left-orthogonal, right-orthogonal, and mixed-canonical) are used in the DMRG algorithm Holtz et al. [2012]. For general trees, the analogous gauge conditions lead to the HOSVD (Higher-Order SVD) of De Lathauwer et al. De Lathauwer et al. [2000].

Chapter 7
Topological Tree-Based Tensor Spaces

The algebraic theory of Chapter 6 operates entirely within the algebraic tensor space $\mathbf{V}_D$. For applications — numerical solution of PDEs, best approximation, dynamical systems — one needs a *topological* framework: a norm on $\mathbf{V}_D$ (or on its completion) and conditions guaranteeing that the tensor product maps at each level of the tree are continuous. This chapter introduces topological tree-based tensor spaces (Section 7.1), the T_D-continuity condition (Section 7.2), the key identification theorem (Section 7.3), and the existence of best approximations in the tree-based setting (Section 7.5). The main results are drawn from Falcó et al. [2021], which significantly weakens the norm conditions required in Falcó and Hackbusch [2012].

7.1 Topological tree-based tensor spaces

Definition 7.1.1 (Topological tree-based tensor space) Let T_D be a dimension partition tree over D and let $(V_j, \|\cdot\|_j)$ be a normed space for each $j \in D$, with completion $V_{j,\|\cdot\|_j}$ (a Banach space). A *topological tree-based representation* of the ambient Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D} = {}_{\|\cdot\|_D}\bigotimes_{j\in D} V_j$ is a collection $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}_{\alpha\in T_D\setminus\{D\}}$ such that:

(i) For each leaf $\{j\} \in \mathcal{L}(T_D)$: $V_{\{j\},\|\cdot\|_j} = V_{j,\|\cdot\|_j}$.
(ii) For each internal vertex $\alpha \in T_D \setminus \mathcal{L}(T_D)$: $\mathbf{V}_{\alpha,\|\cdot\|_\alpha}$ is a Banach tensor space

$$\mathbf{V}_{\alpha,\|\cdot\|_\alpha} = {}_{\|\cdot\|_\alpha}\bigotimes_{\beta\in S(\alpha)} V_\beta, \tag{7.1}$$

i.e., the completion of the algebraic space $\mathbf{V}_\alpha = {}_a\bigotimes_{\beta\in S(\alpha)} \mathbf{V}_\beta$ under the norm $\|\cdot\|_\alpha$.

The key question is when the norm $\|\cdot\|_\alpha$ on the algebraic tensor space ${}_a\bigotimes_{\beta\in S(\alpha)} \mathbf{V}_\beta$ is *compatible* with completing each factor: that is, when does

$$ {}_{\|\cdot\|_\alpha}\bigotimes_{\beta\in S(\alpha)} V_{\beta,\|\cdot\|_\beta} = {}_{\|\cdot\|_\alpha}\bigotimes_{\beta\in S(\alpha)} V_\beta = {}_{\|\cdot\|_\alpha}\bigotimes_{j\in\alpha} V_j \tag{7.2} $$

hold? The answer is provided by the T_D-continuity condition.

7.2 The T_D-continuity condition

Definition 7.2.1 (T_D-continuous tensor product) Let $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}_{\alpha\in T_D\setminus\{D\}}$ be a topological tree-based representation of $\mathbf{V}_{D,\|\cdot\|_D}$. The tensor product map $\bigotimes$ is *T_D-continuous* if for each internal vertex $\alpha\in T_D\setminus\mathcal{L}(T_D)$, the map

$$ \bigotimes : \prod_{\beta\in S(\alpha)} (V_{\beta,\|\cdot\|_\beta}, \|\cdot\|_\beta) \longrightarrow (\mathbf{V}_{\alpha,\|\cdot\|_\alpha}, \|\cdot\|_\alpha) \tag{7.3} $$

is continuous (equivalently, bounded as a multilinear map).

T_D-continuity is equivalent to requiring that each norm $\|\cdot\|_\alpha$ is a crossnorm on the product $\prod_{\beta\in S(\alpha)} V_{\beta,\|\cdot\|_\beta}$ (or at least bounds it from above up to a constant).

Lemma 7.2.1 ([Hackbusch, 2019, Lemma 4.34]) *Let $(V_j,\|\cdot\|_j)$ be normed spaces and $\|\cdot\|$ a norm on ${}_a\bigotimes_{j=1}^d V_j$ such that the tensor product map $\bigotimes$: $\prod_j (V_{j,\|\cdot\|_j},\|\cdot\|_j) \to ({}_a\bigotimes_j V_{j,\|\cdot\|_j}, \|\cdot\|)$ is continuous. Then*

$$ {}_{\|\cdot\|}\bigotimes_{j=1}^{d} V_{j,\|\cdot\|_j} = {}_{\|\cdot\|}\bigotimes_{j=1}^{d} V_j. $$

7.3 The identification theorem

The following theorem is the topological counterpart of the algebraic associativity: it shows that, under T_D-continuity, completing at each vertex of the tree gives the same result as completing all factors simultaneously.

Theorem 7.3.1 (Identification theorem, [Falcó et al., 2021, Theorem 3.2]) *Let $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}_{\alpha\in T_D\setminus\{D\}}$ be a topological tree-based representation of $\mathbf{V}_{D,\|\cdot\|_D}$ such that the tensor product map $\bigotimes$ is T_D-continuous. Then for each $\alpha\in T_D\setminus\mathcal{L}(T_D)$:*

$$ \mathbf{V}_{\alpha,\|\cdot\|_\alpha} = {}_{\|\cdot\|_\alpha}\bigotimes_{\beta\in S(\alpha)} V_{\beta,\|\cdot\|_\beta} = {}_{\|\cdot\|_\alpha}\bigotimes_{\beta\in S(\alpha)} V_\beta = {}_{\|\cdot\|_\alpha}\bigotimes_{j\in\alpha} V_j. \tag{7.4} $$

Proof. Apply Lemma 7.2.1 at each internal vertex α, using the T_D-continuity of the tensor product map at α. The third equality in (7.4) follows from the algebraic identity $\mathbf{V}_\alpha = {}_a\bigotimes_{j\in\alpha} V_j$ (associativity of tensor products, Proposition 3.1.2). $\square$

Example 7.3.1 (Anisotropic Sobolev space) Consider $D = \{1,2,3\}$, $V_1 = L^p(I_1)$, $V_2 = H^{N,p}(I_2)$, $V_3 = H^{N,p}(I_3)$, and the tree

$$T_D : D \to \big\{\{1\},\{2,3\}\big\} \to \big\{\{1\},\{2\},\{3\}\big\},$$

with norms $\|\cdot\|_{23}$ on $V_2 \otimes_a V_3$ and $\|\cdot\|_{123}$ on V_D. If both tensor product maps are continuous (i.e., $\bigotimes$ is T_D-continuous), then

$${}_{\|\cdot\|_{123}}\bigotimes_{j=1}^{3} V_j = L^p(I_1) \otimes_{\|\cdot\|_{123}} \big(H^{N,p}(I_2) \otimes_{\|\cdot\|_{23}} H^{N,p}(I_3)\big),$$

which is the anisotropic Sobolev space of Figure 7.1.

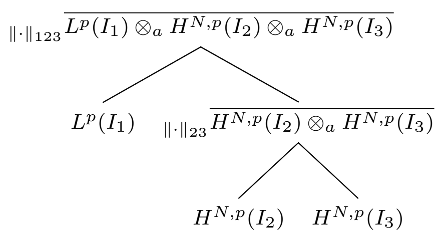

Fig. 7.1 Topological tree-based tensor space for an anisotropic Sobolev product. The intermediate space at vertex $\{2,3\}$ is the Banach completion of $H^{N,p}(I_2) \otimes_a H^{N,p}(I_3)$ under $\|\cdot\|_{23}$.

7.4 The tree-based injective norm condition

For the existence of best approximations, a stronger condition than T_D-continuity is required. The following condition is the tree-based analogue of condition (Inj) from Section 4.5.

Definition 7.4.1 (T_D-injective norm condition) A topological tree-based representation $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}$ satisfies the *T_D-injective norm condition* (Inj)[T_D] if for each $\alpha \in T_D \setminus \mathcal{L}(T_D)$:

$$\|\cdot\|_\alpha \gtrsim \|\cdot\|_{\varepsilon(S(\alpha))}, \tag{7.5}$$

where $\|\cdot\|_{\varepsilon(S(\alpha))}$ is the injective norm on ${}_a\bigotimes_{\beta\in S(\alpha)} V_{\beta,\|\cdot\|_\beta}$.

Proposition 7.4.1 ([Falcó et al., 2021, Proposition 3.3]) *Condition* (Inj)*[T_D] implies T_D-continuity.*

Proof. The injective norm is a reasonable crossnorm (Proposition 4.4.2), so $\|\cdot\|_\alpha \gtrsim \|\cdot\|_{\varepsilon(S(\alpha))}$ implies that $\bigotimes : \prod_{\beta\in S(\alpha)} V_{\beta,\|\cdot\|_\beta} \to \mathbf{V}_{\alpha,\|\cdot\|_\alpha}$ is continuous. $\square$

Remark 7.4.1 The condition (Inj)[T_D] holds for all common tensor product norms arising from L^p and Sobolev spaces. In particular, the L^p norm on $L^p(\prod_j I_j)$ satisfies (Inj)[T_D] for any binary tree over D (this follows from the fact that L^p norms are reasonable crossnorms; see Hackbusch [2019], Example 4.72).

Minimal subspaces in the completion

Under condition (Inj)[T_D], the definition of minimal subspaces can be extended from $\mathbf{V}_D$ to the Banach completion $\mathbf{V}_{D,\|\cdot\|_D}$. This extension is needed for tensors in $\mathbf{V}_{D,\|\cdot\|_D} \setminus \mathbf{V}_D$, which arise naturally as weak limits of sequences in $\mathbf{V}_D$.

Lemma 7.4.1 (Extension of minimal subspaces, [Falcó and Hackbusch, 2012, Lemma 3.8]) *Let* $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}$ *satisfy* (Inj)*[*T_D*]. For* $\mathbf{v} \in \mathbf{V}_{D,\|\cdot\|_D}$ *and* $\alpha \in S(D)$*, define*

$$U_\alpha^{\min}(\mathbf{v}) := \overline{\operatorname{span}\Big\{\overline{(\mathrm{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})}(\mathbf{v}) : \varphi^{[\alpha]} \in {}_a\bigotimes_{\beta\in S(D)\setminus\{\alpha\}} \mathbf{V}_\beta^*\Big\}}^{\|\cdot\|_\alpha}, \qquad (7.6)$$

where $\overline{(\mathrm{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})} \in \mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D}, \mathbf{V}_{\alpha,\|\cdot\|_\alpha})$ *is the continuous extension of the contraction map. Then* $U_\alpha^{\min}(\mathbf{v})$ *is a well-defined closed subspace of* $\mathbf{V}_{\alpha,\|\cdot\|_\alpha}$ *with* $\dim U_\alpha^{\min}(\mathbf{v}) < \infty$*.*

The lemma allows one to define the tree-based rank $\mathrm{rank}_{T_D}(\mathbf{v})$ for any $\mathbf{v} \in \mathbf{V}_{D,\|\cdot\|_D}$, not just for $\mathbf{v} \in \mathbf{V}_D$. This is essential for the best approximation theorem.

7.5 Weak closedness and best approximations

Theorem 7.5.1 (Weak lower semicontinuity of rank, [Falcó et al., 2021, Theorem 3.4]) *Let* $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}$ *satisfy* (Inj)*[*T_D*], and let* $(\mathbf{v}_n) \subset \mathbf{V}_{D,\|\cdot\|_D}$ *be a sequence converging weakly to* $\mathbf{v} \in \mathbf{V}_{D,\|\cdot\|_D}$*. Then for each* $\alpha \in T_D \setminus \{D\}$*:*

$$\dim U_\alpha^{\min}(\mathbf{v}) \le \liminf_{n\to\infty} \dim U_\alpha^{\min}(\mathbf{v}_n). \qquad (7.7)$$

Proof. The proof proceeds in two stages: first for vertices $\alpha \in S(D)$ (children of the root), then by downward induction over the tree.

Stage 1: Vertices in $S(D)$.

Fix $\alpha \in S(D)$. By the extension formula (7.6), for any $\varphi^{[\alpha]} \in {}_a\bigotimes_{\beta\in S(D)\setminus\{\alpha\}} \mathbf{V}_\beta^*$, the operator

$$T_{\varphi^{[\alpha]}} := \overline{(\mathrm{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})} \in \mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D}, \mathbf{V}_{\alpha,\|\cdot\|_\alpha})$$

is bounded, with $\|T_{\varphi^{[\alpha]}}\| \le \|\varphi^{[\alpha]}\|_*$ by condition (Inj)$[T_D]$ (which implies the contraction maps are controlled by the dual functional norms; see the proof of Lemma 7.4.1). Hence weak convergence $\mathbf{v}_n \rightharpoonup \mathbf{v}$ in $\mathbf{V}_{D,\|\cdot\|_D}$ implies

$$T_{\varphi^{[\alpha]}}(\mathbf{v}_n) \rightharpoonup T_{\varphi^{[\alpha]}}(\mathbf{v}) \quad \text{weakly in } \mathbf{V}_{\alpha,\|\cdot\|_\alpha}, \qquad \forall\, \varphi^{[\alpha]}. \tag{7.8}$$

Now suppose for contradiction that $\dim U_\alpha^{\min}(\mathbf{v}) > \liminf_{n\to\infty} \dim U_\alpha^{\min}(\mathbf{v}_n)$. Then there exists a subsequence (still denoted $(\mathbf{v}_n)$) and an integer $m > 0$ such that $\dim U_\alpha^{\min}(\mathbf{v}_n) \le m$ for all n, while $\dim U_\alpha^{\min}(\mathbf{v}) \ge m+1$.

Choose $m+1$ functionals $\varphi_1^{[\alpha]}, \dots, \varphi_{m+1}^{[\alpha]}$ such that the vectors $w_k := T_{\varphi_k^{[\alpha]}}(\mathbf{v}) \in U_\alpha^{\min}(\mathbf{v})$ are linearly independent (possible since $\dim U_\alpha^{\min}(\mathbf{v}) \ge m+1$). By (7.8), $T_{\varphi_k^{[\alpha]}}(\mathbf{v}_n) \rightharpoonup w_k$ for each k.

Since $T_{\varphi_k^{[\alpha]}}(\mathbf{v}_n) \in U_\alpha^{\min}(\mathbf{v}_n)$ and $\dim U_\alpha^{\min}(\mathbf{v}_n) \le m$, the $m+1$ vectors $T_{\varphi_1^{[\alpha]}}(\mathbf{v}_n), \dots, T_{\varphi_{m+1}^{[\alpha]}}(\mathbf{v}_n)$ are linearly dependent for every n. That is, there exist scalars $\lambda_1^n, \dots, \lambda_{m+1}^n$, not all zero, with $\sum_k \lambda_k^n T_{\varphi_k^{[\alpha]}}(\mathbf{v}_n) = 0$. Normalising so that $\max_k |\lambda_k^n| = 1$ and passing to a subsequence, we may assume $\lambda_k^n \to \lambda_k$ for each k, with $\max_k |\lambda_k| = 1$. The weak limit gives $\sum_k \lambda_k w_k = 0$, contradicting the linear independence of $w_1, \dots, w_{m+1}$. Therefore

$$\dim U_\alpha^{\min}(\mathbf{v}) \le \liminf_{n\to\infty} \dim U_\alpha^{\min}(\mathbf{v}_n) \quad \forall\, \alpha \in S(D). \tag{7.9}$$

Stage 2: Downward induction over T_D.

We now extend (7.9) to every $\alpha \in T_D \setminus \{D\}$ by induction on the level $\mathrm{level}(\alpha)$ of the vertex, proceeding from the root downward.

Base case: $\alpha \in S(D)$. This is (7.9).

Inductive step: Let $\alpha \in T_D \setminus (\{D\} \cup \mathcal{L}(T_D))$ and assume that (7.7) holds for α (established at the previous level). Fix $\beta \in S(\alpha)$.

By the tree-based characterisation of minimal subspaces (Proposition 6.3.1), for any $\mathbf{v}_\alpha \in U_\alpha^{\min}(\mathbf{v})$ and any functional $\varphi^{(\alpha\setminus\beta)} \in ({}_a\bigotimes_{\gamma\in S(\alpha)\setminus\{\beta\}} \mathbf{V}_\alpha[\gamma])^*$, the contraction

$$S_{\varphi^{(\alpha\setminus\beta)}}(\mathbf{v}_\alpha) := (\mathrm{id}_{\mathbf{V}_\beta} \otimes \varphi^{(\alpha\setminus\beta)})(\mathbf{v}_\alpha) \in U_\beta^{\min}(\mathbf{v}).$$

Under condition (Inj)$[T_D]$, the composition

$$R_\phi := S_{\varphi^{(\alpha\setminus\beta)}} \circ T_{\varphi^{[\alpha]}} \in \mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D}, \mathbf{V}_\beta)$$

is bounded, and $\mathbf{v}_n \rightharpoonup \mathbf{v}$ implies $R_\phi(\mathbf{v}_n) \rightharpoonup R_\phi(\mathbf{v})$ weakly in $\mathbf{V}_\beta$.

Since $R_\phi(\mathbf{v}_n) \in U_\beta^{\min}(\mathbf{v}_n)$ and $\dim U_\beta^{\min}(\mathbf{v}_n) \le \dim U_\alpha^{\min}(\mathbf{v}_n)$, the same contradiction argument as in Stage 1 applies: if $\dim U_\beta^{\min}(\mathbf{v}) > \liminf_n \dim U_\beta^{\min}(\mathbf{v}_n)$, we find linearly independent weak-limit vectors in $U_\beta^{\min}(\mathbf{v})$ that are linear combinations of weakly convergent sequences confined to a subspace of smaller dimension, yielding a contradiction.

Hence $\dim U_\beta^{\min}(\mathbf{v}) \le \liminf_{n\to\infty} \dim U_\beta^{\min}(\mathbf{v}_n)$ for all $\beta \in S(\alpha)$.

Applying this argument level by level from the root D down to the leaves gives (7.7) for all $\alpha \in T_D \setminus \{D\}$. $\square$

Theorem 7.5.2 (Weak closedness, [Falcó et al., 2021, Theorem 3.5]) *Let* $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}$ *satisfy* (Inj)*[T_D] and let* $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$*. Then* $\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$ *is weakly closed in* $\mathbf{V}_{D,\|\cdot\|_D}$*.*

Proof. Immediate from Theorem 7.5.1: if $\mathbf{v}_n \in \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$ and $\mathbf{v}_n \rightharpoonup \mathbf{v}$, then $\dim U_\alpha^{\min}(\mathbf{v}) \leq \liminf \dim U_\alpha^{\min}(\mathbf{v}_n) \leq r_\alpha$ for all $\alpha \in T_D$, so $\mathbf{v} \in \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$. $\square$

Theorem 7.5.3 (Best approximation, [Falcó et al., 2021, Theorem 3.6]) *Let* $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}$ *satisfy* (Inj)*[T_D] and assume* $\mathbf{V}_{D,\|\cdot\|_D}$ *is reflexive. Then for every* $\mathbf{u} \in \mathbf{V}_{D,\|\cdot\|_D}$ *and* $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$*, there exists* $\mathbf{v}_* \in \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$ *such that*

$$\|\mathbf{u} - \mathbf{v}_*\|_D = \inf_{\mathbf{v} \in \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)} \|\mathbf{u} - \mathbf{v}\|_D. \tag{7.10}$$

Proof. Identical to the proof of Theorem 4.5.2: take a minimising sequence $(\mathbf{v}_n) \subset \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$, extract a weakly convergent subsequence (by reflexivity), and apply Theorem 7.5.2 to conclude the limit is in $\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$. Lower semicontinuity of the norm then gives the minimum. $\square$

Remark 7.5.1 (Improvement over prior results) Theorem 7.5.3 improves the earlier result of Theorem 4.5.2 in two ways:

1. It applies to *any* tree T_D, not just the Tucker tree T_D^{Tucker}.
2. The norm condition (Inj)[T_D] is *weaker* than the condition of Falcó and Hackbusch [2012]: it requires the injective norm bound only *locally* at each vertex, whereas Falcó and Hackbusch [2012] required a global injective norm bound on the full tensor product.

In particular, Theorem 7.5.3 applies to the anisotropic Sobolev space of Example 7.3.1, which is *not* covered by Theorem 4.5.2.

Corollary 7.5.1 *For* $1 < p < \infty$, $L^p(\prod_{j\in D} I_j)$ *with any dimension partition tree* T_D *satisfies the hypotheses of Theorem 7.5.3. Hence every* $f \in L^p(\prod_j I_j)$ *has a best tree-based approximation of any prescribed tree-based rank.*

7.6 Historical notes

The first best approximation results for tensor formats in infinite- dimensional spaces were proved in Falcó and Hackbusch [2012] for the Tucker format, under the assumption that $\|\cdot\|_D \gtrsim \|\cdot\|_\varepsilon$ globally. The extension to tree-based formats under the weaker condition (Inj)[T_D] was achieved in Falcó et al. [2021]. The key improvement is the use of the *local* injective norm bound at each vertex of the tree, combined with the new characterisation of minimal subspaces (Proposition 6.3.1) to propagate weak lower semicontinuity up the tree.

The anisotropic Sobolev space of Example 7.3.1 motivates this improvement: in parametric PDEs where the spatial variable $x_1 \in I_1$ and the parameter $x_2 \in I_2$ have different regularity, the natural norm on $L^p(I_1) \otimes H^{N,p}(I_2)$ is not a global injective norm, but it does satisfy (Inj)$[T_D]$ for the two-level tree. The existence of best approximations in such spaces is crucial for the convergence analysis of greedy algorithms for parametric PDEs (cf. Nouy [2017]).

Part III
Differential Geometry of Tensor Formats

Chapter 8
The Tucker Manifold

This chapter establishes the central geometric result of the monograph: the set $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ of tensors with fixed Tucker rank carries the structure of a $\mathcal{C}^\infty$-Banach manifold, and is an immersed submanifold of the ambient Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$. Section 8.1 introduces the rank map and motivates the manifold structure. Section 8.2 constructs the local charts explicitly. Section 8.3 proves the $\mathcal{C}^\infty$ atlas (Theorem 8.3.1). Section 8.6 develops the fibre bundle structure in detail: local trivialisations, transition functions, and the structure group $G_{\mathfrak{r}} = \prod_\alpha \mathrm{GL}(r_\alpha)\mathbb{R}$. Section 8.4 identifies the tangent space. Section 8.5 proves the immersion theorem (Theorem 8.5.1). Throughout, $D = \{1, \ldots, d\}$, $d \geq 2$, and $(V_\alpha, \|\cdot\|_\alpha)$ are normed spaces.

8.1 The rank map and the decomposition of $\mathbf{V}_D$

Recall from Theorem 4.4.1 that

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D).$$

The map

$$\varrho : \mathbf{V}_D \longrightarrow \bigcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)} \prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha), \qquad \mathbf{v} \mapsto \left(U_\alpha^{\min}(\mathbf{v})\right)_{\alpha \in D}, \tag{8.1}$$

sends each tensor to its tuple of minimal subspaces. For fixed $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$, the restriction

$$\varrho_{\mathfrak{r}} := \varrho\big|_{\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)} : \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \longrightarrow \prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha) \tag{8.2}$$

is surjective: for any tuple $(U_\alpha)_{\alpha\in D}$ with $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$, any full-rank tensor in ${}_a\bigotimes_\alpha U_\alpha$ belongs to $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and maps to (U_α).

The product of Grassmannians $\prod_{\alpha\in D}\mathbb{G}_{r_\alpha}(V_\alpha)$ is a Banach manifold (Theorem 2.3.1 applied factor by factor), and the rank map $\varrho_{\mathfrak{r}}$ is its natural base for the bundle structure of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.

8.2 Local charts

Fix $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$ and $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$. Let $V_{\alpha,\|\cdot\|_\alpha}$ denote the completion of V_α and choose for each $\alpha \in D$ a closed complement $W_\alpha^{\min}(\mathbf{v})$ of $U_\alpha^{\min}(\mathbf{v})$ in $V_{\alpha,\|\cdot\|_\alpha}$:

$$V_{\alpha,\|\cdot\|_\alpha} = U_\alpha^{\min}(\mathbf{v}) \oplus W_\alpha^{\min}(\mathbf{v}).$$

Such a complement exists by Proposition 2.2.2, since $U_\alpha^{\min}(\mathbf{v})$ is finite-dimensional.

Chart domain

Define the *chart domain at* $\mathbf{v}$ by

$$\mathcal{U}(\mathbf{v}) \coloneqq \big\{\mathbf{w} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) : U_\alpha^{\min}(\mathbf{w}) \in \mathbb{G}(W_\alpha^{\min}(\mathbf{v}), V_\alpha),\ \forall \alpha \in D\big\}, \tag{8.3}$$

where $\mathbb{G}(W, X) = \{U \in \mathbb{G}(X) : X = U \oplus W\}$ is the chart domain of the Grassmannian at $U_\alpha^{\min}(\mathbf{v})$ (see Section 2.3).

Chart map

Fix bases $\{u_{i_\alpha}^{(\alpha)}\}_{i_\alpha=1}^{r_\alpha}$ of $U_\alpha^{\min}(\mathbf{v})$ for each $\alpha \in D$. By Lemma 3.3.1, there is a unique $C^{(D)}(\mathbf{v}) \in \mathbb{R}_*^{\times r_\alpha}$ such that $\mathbf{v}$ has the Tucker representation (3.9). For each $\mathbf{w} \in \mathcal{U}(\mathbf{v})$, define the *chart map*

$$\xi_{\mathbf{v}} : \mathcal{U}(\mathbf{v}) \longrightarrow \prod_{\alpha\in D} \mathcal{L}\big(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})\big) \times \mathbb{R}_*^{\times r_\alpha}, \tag{8.4}$$

by sending $\mathbf{w} = (\bigotimes_\alpha \exp(L_\alpha))(\mathbf{u}(E^{(D)}))$ to the pair $((L_\alpha)_{\alpha\in D}, E^{(D)})$, where $(L_\alpha, E^{(D)})$ are uniquely determined by Lemma 8.2.1 below.

Lemma 8.2.1 (Local representation, [Falcó et al., 2019, Lemma 3.7]) *Under the conditions above, for* $\mathbf{w} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ *the following are equivalent:*

(a) $\mathbf{w} \in \mathcal{U}(\mathbf{v})$.
(b) *There exists a unique pair* $((L_\alpha)_{\alpha\in D}, E^{(D)}) \in \prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \times \mathbb{R}_*^{\times r_\alpha}$ *such that*

$$\mathbf{w} = \Big(\bigotimes_{\alpha\in D} \exp(L_\alpha)\Big)(\mathbf{u}(E^{(D)})), \tag{8.5}$$

where $\mathbf{u}(E^{(D)}) \in \mathfrak{M}_{\mathfrak{r}}(()_a \bigotimes_\alpha U_\alpha^{\min}(\mathbf{v}))$ *is the tensor with core* $E^{(D)}$ *in the fixed bases.*

Here $\exp(L_\alpha) = \mathrm{id}_{V_\alpha} + L_\alpha$ by the nilpotency of $L_\alpha \in \mathcal{L}_{(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v}))}(V_\alpha)$ (Proposition 2.2.3), so the representation is polynomial and $\mathcal{C}^\infty$.

Remark 8.2.1 Formula (8.5) has a clean geometric interpretation: $\mathbf{w}$ is obtained from the "reference" tensor $\mathbf{u}(E^{(D)})$ (which lives in the fixed subspaces $U_\alpha^{\min}(\mathbf{v})$) by applying the "rotation" $\bigotimes_\alpha \exp(L_\alpha)$ which moves each subspace $U_\alpha^{\min}(\mathbf{v})$ to $U_\alpha^{\min}(\mathbf{w}) = G(L_\alpha) = \mathrm{span}\{(\mathrm{id} + L_\alpha)(u_{i_\alpha}^{(\alpha)})\}$.

8.3 The $\mathcal{C}^\infty$-Banach manifold structure

Lemma 8.3.1 (Smooth transitions, [Falcó et al., 2019, Lemma 3.8]) *Under the continuity assumption on the tensor product map* (3.15)*, for* $\mathbf{v}, \mathbf{v}' \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ *with* $\mathcal{U}(\mathbf{v}) \cap \mathcal{U}(\mathbf{v}') \neq \emptyset$*, the transition map*

$$\xi_{\mathbf{v}'} \circ \xi_{\mathbf{v}}^{-1} : \xi_{\mathbf{v}}(\mathcal{U}(\mathbf{v}) \cap \mathcal{U}(\mathbf{v}')) \longrightarrow \xi_{\mathbf{v}'}(\mathcal{U}(\mathbf{v}) \cap \mathcal{U}(\mathbf{v}'))$$

is $\mathcal{C}^\infty$*-Fréchet differentiable* (*analytic if the* V_α *are complex Banach spaces*).

Sketch. For $\mathbf{w} \in \mathcal{U}(\mathbf{v}) \cap \mathcal{U}(\mathbf{v}')$, the transition map is

$$(L_\alpha, E^{(D)}) \mapsto \big((\Psi_{\mathbf{v}'}^{(\alpha)} \circ (\Psi_{\mathbf{v}}^{(\alpha)})^{-1})(L_\alpha),\ f(\mathfrak{L}, \mathbf{u}(E^{(D)}))\big),$$

where $\Psi_{\mathbf{v}}^{(\alpha)}$ is the Grassmann chart map at $U_\alpha^{\min}(\mathbf{v})$ (Theorem 2.3.1), and $f(\mathfrak{L}, \mathbf{u}(E^{(D)})) = (\bigotimes_\alpha \exp(L_\alpha - L'_\alpha))(\mathbf{u}(E^{(D)}))$. The first component is analytic (transition map of the Grassmannian). The second is $\mathcal{C}^\infty$ by Proposition 3.5.1 (tensor product of $\mathcal{C}^\infty$ maps) and the fact that the evaluation map is multilinear and continuous. See Falcó et al. [2019] for the complete argument. □

Theorem 8.3.1 ($\mathcal{C}^\infty$ Tucker manifold, [Falcó et al., 2019, Theorem 3.17]) *Let* $(V_\alpha, \|\cdot\|_\alpha)$ *be normed spaces for* $\alpha \in D$ *and let* $\|\cdot\|_D$ *be a norm on* $\mathbf{V}_D$ *such that the tensor product map* (3.15) *is continuous. Then:*

(i) *The collection* $\{(\mathcal{U}(\mathbf{v}), \xi_{\mathbf{v}})\}_{\mathbf{v}\in\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)}$ *is a* $\mathcal{C}^\infty$*-atlas on* $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$*, which is thus a* $\mathcal{C}^\infty$*-Banach manifold modelled on the Banach space*

$$\mathbf{E}_D := \prod_{\alpha\in D} \mathcal{L}\big(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})\big) \times \mathbb{R}^{\times_{\alpha\in D} r_\alpha}. \tag{8.6}$$

(ii) *The triple* $(\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D), \prod_{\alpha\in D} \mathbb{G}_{r_\alpha}(V_\alpha), \varrho_{\mathfrak{r}})$ *is a* $\mathcal{C}^\infty$*-fibre bundle with typical fibre* $\mathbb{R}_*^{\times r_\alpha}$*.*

If the V_α are complex Banach spaces, the manifold is analytic.

Proof. The atlas conditions AT1–AT3 are verified as follows. AT1: $\{\mathcal{U}(\mathbf{v})\}$ covers $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ since every $\mathbf{v}$ belongs to its own chart domain. AT2: $\xi_\mathbf{v}$ is a bijection onto $\prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \times \mathbb{R}_*^{\times r_\alpha}$ by Lemma 8.2.1. AT3: Lemma 8.3.1 gives $\mathcal{C}^\infty$ transitions. The fibre bundle structure follows because $\varrho_\mathfrak{r}$ acts locally as the projection onto $\prod_\alpha \mathbb{G}(W_\alpha^{\min}(\mathbf{v}), V_\alpha)$ (the Grassmannian chart domain), with fibre $\mathbb{R}_*^{\times r_\alpha}$. □

Remark 8.3.1 (Independence from the norm) The manifold structure of $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ is independent of the choice of $\|\cdot\|_D$: the chart maps $\xi_\mathbf{v}$ depend only on the splitting $V_{\alpha,\|\cdot\|_\alpha} = U_\alpha^{\min}(\mathbf{v}) \oplus W_\alpha^{\min}(\mathbf{v})$, and the $\mathcal{C}^\infty$ regularity follows from the continuity of the tensor product map, which holds for any crossnorm. Different norms give equivalent smooth structures.

Corollary 8.3.1 (Hilbert manifold) *If $V_{\alpha,\|\cdot\|_\alpha}$ is a Hilbert space for each $\alpha \in D$, then $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ is a $\mathcal{C}^\infty$-Hilbert manifold modelled on $\prod_\alpha W_\alpha^{r_\alpha} \times \mathbb{R}^{\times r_\alpha}$, where $W_\alpha^{r_\alpha} = (U_\alpha)^\perp \times \cdots$ (each $\mathcal{L}(U_\alpha, W_\alpha) \cong W_\alpha^{r_\alpha}$ via the chosen orthonormal basis of U_α).*

Example 8.3.1 For $1 < p < \infty$, the L^p tensor space $L^p(\prod_\alpha I_\alpha)$ satisfies the hypotheses of Theorem 8.3.1. Hence $\mathfrak{M}_\mathfrak{r}(()_a \bigotimes_\alpha L^p(I_\alpha))$ is a $\mathcal{C}^\infty$-Banach manifold. For $p = 2$ it is a Hilbert manifold.

Example 8.3.2 For $V_1 = H^{1,p}(I_1)$, $V_2 = H^{1,p}(I_2)$, and either the full Sobolev norm $\|\cdot\|_{H^{1,p}}$ or the partial norm $\|\cdot\|_{(0,1),p}$ on $V_1 \otimes_a V_2$, both norms make the tensor product map continuous (see [Hackbusch, 2019, Examples 4.41, 4.42]). Hence $\mathfrak{M}_{(r,r)}(H^{1,p}(I_1) \otimes_a H^{1,p}(I_2))$ is a $\mathcal{C}^\infty$-Banach manifold modelled on $\mathcal{L}(U_1, W_1) \times \mathcal{L}(U_2, W_2) \times \mathrm{GL}(\mathbb{R}^r)$.

8.4 The tangent space

Definition 8.4.1 (Tangent subspace $Z^{(D)}(\mathbf{v})$) For $\mathbf{v} \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$, define the linear subspace

$$Z^{(D)}(\mathbf{v}) \coloneqq {}_a\bigotimes_{\alpha\in D} U_\alpha^{\min}(\mathbf{v}) \oplus \bigoplus_{\alpha\in D} W_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v}) \;\subset\; \mathbf{V}_{D,\|\cdot\|_D}. \tag{8.7}$$

$Z^{(D)}(\mathbf{v})$ is a direct sum of $d+1$ subspaces: the core subspace ${}_a\bigotimes_\alpha U_\alpha^{\min}(\mathbf{v})$ (dimension $\prod r_\alpha$) and d "tangent directions" $W_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$.

Proposition 8.4.1 (Tangent space identification, [Falcó et al., 2019, Proposition 4.13]) *Under the continuity assumption, the tangent map of the inclusion $\mathfrak{i} : \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D) \hookrightarrow \mathbf{V}_{D,\|\cdot\|_D}$ at $\mathbf{v}$ is injective, and*

$$T_\mathbf{v}\mathfrak{i}\big(\mathbb{T}_\mathbf{v}(\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D))\big) = Z^{(D)}(\mathbf{v}).$$

More explicitly, for $(\dot{\mathfrak{L}}, \dot{C}^{(D)}) \in \mathbb{T}_{\mathbf{v}}(\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D))$:

$$T_{\mathbf{v}}\mathfrak{i}(\dot{\mathfrak{L}}, \dot{C}^{(D)}) = \sum_{(i_\alpha)} \dot{C}^{(D)}_{(i_\alpha)} \bigotimes_{\alpha} u^{(\alpha)}_{i_\alpha} + \sum_{\alpha \in D} \sum_{i_\alpha = 1}^{r_\alpha} \dot{u}^{(\alpha)}_{i_\alpha} \otimes \mathbf{U}^{(\alpha)}_{i_\alpha}, \tag{8.8}$$

where $\dot{u}^{(\alpha)}_{i_\alpha} = \dot{L}_\alpha(u^{(\alpha)}_{i_\alpha}) \in W^{\min}_\alpha(\mathbf{v})$ and $\mathbf{U}^{(\alpha)}_{i_\alpha} \in U^{\min}_{D\setminus\{\alpha\}}(\mathbf{v})$ is given by (3.11)*.*

Proof. The formula (8.8) follows by differentiating $(\mathfrak{i} \circ \xi_{\mathbf{v}}^{-1})$ at the origin $(\mathfrak{o}, C^{(D)})$ using Proposition 3.5.1 and the Leibniz rule for tensor products. Injectivity of $T_{\mathbf{v}}\mathfrak{i}$ follows from the linear independence of $\{\bigotimes_\alpha u^{(\alpha)}_{i_\alpha}\}$ (a basis of ${}_a\bigotimes_\alpha U^{\min}_\alpha(\mathbf{v})$) and of $\{\mathbf{U}^{(\alpha)}_{i_\alpha}\}_{i_\alpha}$ for each α (a basis of $U^{\min}_{D\setminus\{\alpha\}}(\mathbf{v})$). See Falcó et al. [2019] for the complete proof. ◻

8.5 Immersion in the ambient Banach space

To show that $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is an immersed submanifold of $\mathbf{V}_{D,\|\cdot\|_D}$, it remains to show that $Z^{(D)}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$, i.e., that $Z^{(D)}(\mathbf{v})$ is a split subspace of the ambient Banach space. This requires condition (Inj).

Lemma 8.5.1 ([Falcó et al., 2019, Lemma 4.13]) *Assume* (Inj) *holds. Let $\beta \in D$ and $W_\beta \in \mathbb{G}(V_{\beta,\|\cdot\|_\beta})$ with $V_{\beta,\|\cdot\|_\beta} = U_\beta \oplus W_\beta$ for some finite-dimensional U_β. Then for every finite-dimensional subspace $U_{[\beta]} \subset \mathbf{V}^{[\beta]}_\alpha$:*

$$W_\beta \otimes_a U_{[\beta]} \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D}).$$

Proof. The proof uses the continuity of the projection $\mathrm{id}_\beta \otimes P_{U_{[\beta]} \oplus W_{[\beta]}}$ on $\mathbf{V}_{D,\|\cdot\|_D}$ (which follows from (Inj) and a norm estimate via [Falcó and Hackbusch, 2012, Lemma 3.18]), and then decomposes $\mathrm{id}_\beta = P_{U_\beta \oplus W_\beta} + P_{W_\beta \oplus U_\beta}$ to construct an explicit bounded projection onto $W_\beta \otimes_a U_{[\beta]}$. See Falcó et al. [2019] for details. ◻

Lemma 8.5.2 ([Falcó et al., 2019, Lemma 4.14]) *Let X be a Banach space and $U, V \in \mathbb{G}(X)$ with $U \cap V = \{0\}$. Then $U \oplus V \in \mathbb{G}(X)$ and $U \cap V \in \mathbb{G}(X)$.*

Proof. Since $U, V \in \mathbb{G}(X)$ there exist $U', V' \in \mathbb{G}(X)$ with $X = U \oplus U' = V \oplus V'$. Then $U \oplus V \oplus (U' \cap V') = (U \cap V') \oplus (V \cap U') \oplus (U' \cap V') = X$, proving $U \oplus V \in \mathbb{G}(X)$. The second claim follows from $X = (U \cap V) \oplus (U \cap V') \oplus (V \cap U') \oplus (U' \cap V')$. ◻

Theorem 8.5.1 (Immersion theorem, [Falcó et al., 2019, Theorem 4.15]) *Assume* (Inj) *holds. Then for each $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$:*

$$Z^{(D)}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D}). \tag{8.9}$$

Consequently, $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is an immersed submanifold of $\mathbf{V}_{D,\|\cdot\|_D}$.

Proof. For each $\alpha \in D$: $W_\alpha^{\min}(\mathbf{v}) \in \mathbb{G}(V_{\alpha,\|\cdot\|_\alpha})$ and $U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$ is finite-dimensional. By Lemma 8.5.1, $W_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$. Also, ${}_a\bigotimes_\alpha U_\alpha^{\min}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$ (finite-dimensional, hence split by Proposition 2.2.2). The $d+1$ summands in (8.7) have pairwise trivial intersection (by the linear independence properties of minimal subspaces and the direct sum structure), so Lemma 8.5.2 applied iteratively gives $Z^{(D)}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$. □

Corollary 8.5.1 (Best projection onto tangent space) *If $\mathbf{V}_{D,\|\cdot\|_D}$ is additionally reflexive, then for any $\mathbf{v} \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ and $\dot{\mathbf{u}} \in \mathbf{V}_{D,\|\cdot\|_D}$, there exists $\dot{\mathbf{v}}_{\text{best}} \in Z^{(D)}(\mathbf{v})$ such that*

$$\|\dot{\mathbf{u}} - \dot{\mathbf{v}}_{\text{best}}\|_D = \min_{\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v})} \|\dot{\mathbf{u}} - \dot{\mathbf{v}}\|_D.$$

Proof. Since $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive, every closed subspace is proximinal (proximinal = best approximations exist). $Z^{(D)}(\mathbf{v})$ is closed (it is a split subspace by Theorem 8.5.1). □

8.6 The Tucker fibre bundle

Theorem 8.3.1(ii) asserts that $(\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D), \prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha), \varrho_\mathfrak{r})$ is a fibre bundle. In this section we develop this structure in detail: we identify the transition functions, the structure group, and the local trivialisations explicitly in terms of the chart maps, and we compare the bundle structure with the quotient description available in finite dimensions.

Local trivialisations

Fix $\mathbf{v} \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ and recall the chart domain $\mathcal{U}(\mathbf{v})$ from (8.3). The rank map $\varrho_\mathfrak{r}$ restricted to $\mathcal{U}(\mathbf{v})$ takes values in

$$\mathcal{B}(\mathbf{v}) := \prod_{\alpha \in D} \mathbb{G}(W_\alpha^{\min}(\mathbf{v}), V_\alpha) \subset \prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha),$$

which is the product of the Grassmannian chart domains at $(U_\alpha^{\min}(\mathbf{v}))_{\alpha \in D}$, an open subset of the base.

Proposition 8.6.1 (Local trivialisation) *For each $\mathbf{v} \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$, the map*

$$\chi_\mathbf{v} : \varrho_\mathfrak{r}^{-1}\big(\mathcal{B}(\mathbf{v})\big) \xrightarrow{\sim} \mathcal{B}(\mathbf{v}) \times F_\mathfrak{r}, \qquad \mathbf{w} \longmapsto \big((U_\alpha^{\min}(\mathbf{w}))_{\alpha \in D},\ E^{(D)}(\mathbf{w})\big), \quad (8.10)$$

where $F_\mathfrak{r} := \mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$ and $E^{(D)}(\mathbf{w})$ is the core tensor of $\mathbf{w}$ in the chart at $\mathbf{v}$ (Lemma 8.2.1), is a $\mathcal{C}^\infty$ diffeomorphism satisfying $\mathrm{pr}_1 \circ \chi_\mathbf{v} = \varrho_\mathfrak{r}$.

Proof. By Lemma 8.2.1, every $\mathbf{w} \in \mathcal{U}(\mathbf{v})$ is uniquely represented by $((L_\alpha)_\alpha, E^{(D)})$, and $\varrho_\mathfrak{r}(\mathbf{w}) = (U_\alpha^{\min}(\mathbf{w}))_\alpha = (G(L_\alpha))_\alpha \in \mathcal{B}(\mathbf{v})$. The Grassmannian chart maps $\Psi_{U_\alpha^{\min}(\mathbf{v}) \oplus W_\alpha^{\min}(\mathbf{v})}$ give a diffeomorphism $\mathcal{B}(\mathbf{v}) \cong \prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v}))$. Under this identification, $\chi_\mathbf{v}$ becomes the chart map $\xi_\mathbf{v}$ restricted to the base component (the L_α part) and the fibre component (the $E^{(D)}$ part). The chart map is a $\mathcal{C}^\infty$ diffeomorphism by Lemma 8.3.1 and Lemma 8.2.1, and $\mathrm{pr}_1 \circ \chi_\mathbf{v} = \varrho_\mathfrak{r}$ is immediate from the definition. □

Transition functions and structure group

Proposition 8.6.2 (Transition functions) *For $\mathbf{v}, \mathbf{v}' \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ with $\mathcal{B}(\mathbf{v}) \cap \mathcal{B}(\mathbf{v}') \neq \emptyset$, the transition function*

$$g_{\mathbf{v}'\mathbf{v}} : \mathcal{B}(\mathbf{v}) \cap \mathcal{B}(\mathbf{v}') \longrightarrow \mathrm{GL}(F_\mathfrak{r}) \tag{8.11}$$

defined by $\chi_{\mathbf{v}'} \circ \chi_\mathbf{v}^{-1}(U, e) = (U, g_{\mathbf{v}'\mathbf{v}}(U) \cdot e)$ is a $\mathcal{C}^\infty$ map taking values in the group

$$G_\mathfrak{r} := \prod_{\alpha \in D} \mathrm{GL}(r_\alpha)\mathbb{R} \subset \mathrm{GL}(F_\mathfrak{r}), \tag{8.12}$$

acting on $F_\mathfrak{r} = \mathbb{R}_*^{\times r_\alpha}$ by simultaneous change of basis: for $g = (g_\alpha)_\alpha \in G_\mathfrak{r}$ and $E^{(D)} \in F_\mathfrak{r}$,

$$(g \cdot E^{(D)})_{i_1 \ldots i_d} = \sum_{j_1, \ldots, j_d} (g_1)_{i_1 j_1} \cdots (g_d)_{i_d j_d} \, E^{(D)}_{j_1 \ldots j_d}. \tag{8.13}$$

Proof. For $\mathbf{w} \in \varrho_\mathfrak{r}^{-1}(\mathcal{B}(\mathbf{v}) \cap \mathcal{B}(\mathbf{v}'))$, write $\chi_\mathbf{v}(\mathbf{w}) = (U, E^{(D)})$ and $\chi_{\mathbf{v}'}(\mathbf{w}) = (U, E'^{(D)})$ (both have the same base component since $\varrho_\mathfrak{r}(\mathbf{w}) = U$ is intrinsic). The two core tensors $E^{(D)}$ and $E'^{(D)}$ represent the same tensor $\mathbf{w}$ in the two chart bases:

$$\mathbf{w} = \sum_{(i_\alpha)} E^{(D)}_{(i_\alpha)} \bigotimes_\alpha \exp(L_\alpha)(u^{(\alpha)}_{i_\alpha}) = \sum_{(i_\alpha)} E'^{(D)}_{(i_\alpha)} \bigotimes_\alpha \exp(L'_\alpha)(u'^{(\alpha)}_{i_\alpha}).$$

Since both $\exp(L_\alpha)(u^{(\alpha)}_{i_\alpha})$ and $\exp(L'_\alpha)(u'^{(\alpha)}_{i_\alpha})$ are bases of $U_\alpha^{\min}(\mathbf{w})$, there exist matrices $g_\alpha \in \mathrm{GL}(r_\alpha)\mathbb{R}$ relating them: $\exp(L_\alpha)(u^{(\alpha)}_{i_\alpha}) = \sum_j (g_\alpha)_{ij} \exp(L'_\alpha)(u'^{(\alpha)}_{j_\alpha})$. Substituting into the two representations gives $E'^{(D)}_{(i_\alpha)} = \sum_{(j_\alpha)} \prod_\alpha (g_\alpha)_{i_\alpha j_\alpha} E^{(D)}_{(j_\alpha)}$, which is exactly (8.13).

The map $U \mapsto g_{\mathbf{v}'\mathbf{v}}(U)$ is $\mathcal{C}^\infty$ because it is the composition of $\mathcal{C}^\infty$ Grassmannian transition maps (Lemma 8.3.1, Step 1). □

Remark 8.6.1 (The bundle is principal over its structure group) Proposition 8.6.2 shows that the Tucker bundle has structure group $G_\mathfrak{r} = \prod_\alpha \mathrm{GL}(r_\alpha)\mathbb{R}$, which is a *finite-dimensional* Lie group regardless of the (possibly infinite-dimensional) component spaces V_α. This is the key reason why the bundle has good properties (e.g., the fibre is a smooth manifold, and the bundle admits a $\mathcal{C}^\infty$ atlas) even in infinite dimensions.

The fibre $F_\mathfrak{r} = \mathbb{R}_*^{\times r_\alpha}$ is the set of arrays of full Tucker rank, i.e., the complement of a closed algebraic variety in $\mathbb{R}^{\times r_\alpha}$. The group $G_\mathfrak{r}$ acts on $F_\mathfrak{r}$ freely (a full-rank array cannot be fixed by a non-trivial gauge transformation) and properly (since $G_\mathfrak{r}$ is finite-dimensional Lie and acts by linear isomorphisms). Therefore the bundle is a *principal* $G_\mathfrak{r}$*-bundle* in the classical sense.

The quotient description and its limitations

In finite dimensions, the structure group $G_\mathfrak{r}$ acts on the total space of frames:

$$\widetilde{\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)} := \big\{(\mathbf{v}, (u_{i_\alpha}^{(\alpha)})_{\alpha,i_\alpha}) : \mathbf{v} \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D),\ \{u_{i_\alpha}^{(\alpha)}\}_{i_\alpha=1}^{r_\alpha} \text{ basis of } U_\alpha^{\min}(\mathbf{v})\big\}, \tag{8.14}$$

and $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D) = \widetilde{\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)}/G_\mathfrak{r}$, giving the quotient description of Koch and Lubich [2007], Uschmajew and Vandereycken [2013b].

In infinite dimensions, this quotient description breaks down for the following reason: the frame bundle $\widetilde{\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)}$ has a natural topology as a product $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D) \times \prod_\alpha \mathrm{GL}(\mathbb{R}^{r_\alpha}, U_\alpha^{\min}(\mathbf{v}))$, but the group $G_\mathfrak{r}$ acts on the *second factor only*, and the projection $\widetilde{\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)} \to \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ is a principal $G_\mathfrak{r}$-bundle over $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ — which is what we are trying to construct. The quotient construction is circular in infinite dimensions.

The explicit atlas approach of Section 8.2 resolves this by working directly with the Grassmannian structure of the base $\prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$, bypassing the group action entirely. The bundle structure is then recovered a posteriori from Propositions 8.6.1 and 8.6.2.

Special cases

Example 8.6.1 ($d = 2$: line bundle and matrix bundle) For $d = 2$:

(i) If $r_1 = r_2 = 1$ (rank-one matrices), the fibre is $F_{(1,1)} = \mathbb{R}^* = \mathbb{R} \setminus \{0\}$ and the structure group is $G_{(1,1)} = \mathbb{R}^* \times \mathbb{R}^*$. The Tucker bundle is a *line bundle* over $\mathbb{G}_1(V_1) \times \mathbb{G}_1(V_2) \cong \mathbb{P}(V_1) \times \mathbb{P}(V_2)$ (the product of projective spaces).

(ii) If $r_1 = r_2 = r \geq 2$ (matrices of rank r), the fibre is $F_{(r,r)} = \mathrm{GL}(r)\mathbb{R} \subset \mathbb{R}^{r\times r}$ and the structure group is $G_{(r,r)} = \mathrm{GL}(r)\mathbb{R} \times \mathrm{GL}(r)\mathbb{R}$, acting by $(g_1, g_2)\cdot M = g_1 M g_2^{-1}$. This recovers the principal bundle description of rank-r matrices as $\mathrm{GL}(r)\mathbb{R} \times \mathrm{GL}(r)\mathbb{R}$-equivariant maps.

Example 8.6.2 (Hilbert case: unitary structure group) If each V_α is a Hilbert space and the bases $\{u_{i_\alpha}^{(\alpha)}\}$ are chosen to be orthonormal, the structure group reduces from $\mathrm{GL}(r_\alpha)\mathbb{R}$ to the orthogonal group $O(r_\alpha)$. In the complex Hilbert case it reduces to $U(r_\alpha)$ (unitary group). This is the structure group used in the Hilbert-space treatments of Koch and Lubich [2007] for MCTDH and is the reason why gauge invariance in quantum chemistry can be phrased in terms of unitary transformations.

8.7 The tensor space as a Banach manifold

Corollary 8.7.1 ([Falcó et al., 2019, Corollary 3.19]) *Under the continuity assumption on the tensor product map, the algebraic tensor space $\mathbf{V}_D$ itself is a $\mathcal{C}^\infty$-Banach manifold (not modelled on a single Banach space), with atlas given by the disjoint union of the Tucker manifold atlases over all $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$:*

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D).$$

8.8 Historical notes and comparison

The $\mathcal{C}^\infty$-Banach manifold structure of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ in infinite-dimensional spaces was established in Falcó et al. [2019]. Prior work had treated only finite-dimensional or Hilbert space settings:

Koch and Lubich Koch and Lubich [2007] proved that the Tucker manifold is a quotient manifold in finite dimensions and used it for the Dirac–Frenkel variational principle for the MCTDH method. Their approach, however, uses the Hilbert structure (metric projection) and does not extend to Banach spaces.

Uschmajew and Vandereycken Uschmajew and Vandereycken [2013a] studied the HT manifold in finite dimensions as a quotient manifold $\mathrm{GL}(r_1) \times \cdots \times \mathrm{GL}(r_d) \backslash \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)/G$ for a gauge group G. The description in Falcó et al. [2019] is fundamentally different: it provides an *explicit atlas* with local charts given by formula (8.5), valid in the infinite-dimensional Banach setting.

The key novelties of Falcó et al. [2019] are:

1. The chart maps $\xi_{\mathbf{v}}$ are given explicitly in terms of the Grassmann chart maps $\Psi_{\mathbf{v}}^{(\alpha)}$, making the geometric structure completely transparent.
2. The immersion theorem (Theorem 8.5.1) holds in general Banach spaces under condition (Inj), without requiring a Hilbert or reflexive structure.
3. The fibre bundle structure identifies the typical fibre as $\mathbb{R}_*^{\times r_\alpha}$, a finite-dimensional smooth manifold (an open subset of a Euclidean space).

Chapter 9
The Tree-Based Manifold

This chapter establishes the geometric counterpart of Chapter 8 for general tree-based tensor formats. The central result is that $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ — when it is a *proper* set of tree-based tensors (Definition 9.1.1) — carries a natural $\mathcal{C}^\infty$-Banach manifold structure that is compatible with the Tucker manifold structures at each level of the tree, and is an immersed submanifold of the ambient Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$.

The key idea, due to Falcó et al. [2023], is that $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) = \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ (Theorem 5.4.1) is an intersection of Tucker manifolds, and its chart system is obtained by taking the intersection of the Tucker chart systems at each level. The resulting charts are natural and compatible with those of each Tucker factor.

Throughout, $(V_\alpha, \|\cdot\|_\alpha)$ are normed spaces, $\|\cdot\|_D$ satisfies (Inj) at each level of T_D, and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$.

9.1 Proper sets of tree-based tensors

Not every $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is a genuine intersection of Tucker manifolds — in degenerate cases it may coincide with a single Tucker manifold. The following definition isolates the non-degenerate case.

Definition 9.1.1 (Proper tree-based set) The set $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is called a *proper set of tree-based tensors* with rank $\mathfrak{r}$ if

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \neq \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)) \quad \text{for all } 1 \leq k \leq \mathrm{depth}(T_D).$$

Example 9.1.1 (Degenerate case) For $D = \{1, \dots, 6\}$ and the tree T_D with $S(D) = \{\{1\}, \{2,3,4,5,6\}\}$ (depth 2):

$$\mathcal{P}_1(T_D) = \big\{\{1\}, \{2,3,4,5,6\}\big\},$$
$$\mathcal{P}_2(T_D) = \mathcal{L}(T_D).$$

Since $\dim U_{\{1\}}^{\min}(\mathbf{v}) = \dim U_{\{2,3,4,5,6\}}^{\min}(\mathbf{v})$ for all $\mathbf{v} \in \mathbf{V}_D$ (Theorem 4.2.1), the constraint at level 1 is automatically satisfied whenever the leaf-level constraints hold. Hence $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) = \mathfrak{M}_{\mathfrak{r}_2}(\mathbf{V}_D, \mathcal{L}(T_D))$ and $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is *not* a proper set. The geometry reduces to the Tucker manifold.

In all examples arising from balanced binary trees (HT format), Tensor Train, and most practical trees with $\operatorname{depth}(T_D) \geq 2$, the set $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is proper.

9.2 The Laplacian-like map and the intersection of model spaces

For each level k of the tree, the Tucker manifold $\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ has model space

$$\mathbf{E}_k := \Big(\bigoplus_{\alpha \in \mathcal{P}_k(T_D)} \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \otimes \operatorname{span}\{\mathbf{id}_{[\alpha]}\} \Big) \times \mathbb{R}^{\times_{\alpha \in \mathcal{P}_k(T_D)} r_\alpha},$$

where $\mathbf{id}_{[\alpha]} = \bigotimes_{\beta \in \mathcal{P}_k(T_D) \setminus \{\alpha\}} \mathrm{id}_{\mathbf{V}_\beta}$ is the identity on all factors except α.

The *Laplacian-like map* Δ_k embeds the product of $\mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v}))$ spaces into $\mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D})$:

$$\Delta_k : \prod_{\alpha \in \mathcal{P}_k(T_D)} \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \longrightarrow \mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D}), \quad (L_\alpha)_\alpha \mapsto \sum_{\alpha \in \mathcal{P}_k(T_D)} L_\alpha \otimes \mathbf{id}_{[\alpha]}. \tag{9.1}$$

Proposition 9.2.1 ([Falcó et al., 2023, Proposition 4.5]) *For* $(L_\alpha)_{\alpha \in \mathcal{P}_k(T_D)} \in \prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v}))$*:*

$$\exp\big(\Delta_k((L_\alpha)_\alpha)\big) = \bigotimes_{\alpha \in \mathcal{P}_k(T_D)} \exp(L_\alpha). \tag{9.2}$$

Proof. Put $L = \sum_\alpha L_\alpha \otimes \mathbf{id}_{[\alpha]}$. For $\alpha \neq \beta$: $(L_\alpha \otimes \mathbf{id}_{[\alpha]}) \circ (L_\beta \otimes \mathbf{id}_{[\beta]}) = L_\alpha \otimes L_\beta \otimes \mathbf{id}_{[\alpha,\beta]}$, and by the nilpotency of L_α (Proposition 2.2.3), $\exp(L_\alpha \otimes \mathbf{id}_{[\alpha]}) = \exp(L_\alpha) \otimes \mathbf{id}_{[\alpha]}$. Since the summands commute, $\exp(L) = \bigodot_\alpha \exp(L_\alpha \otimes \mathbf{id}_{[\alpha]}) = \bigotimes_\alpha \exp(L_\alpha)$. □

The critical consequence is that the local representation $\mathbf{w} = \exp(L)(\mathbf{u})$ for $L \in \mathbf{E}_k$ is *independent of* k: for any k and any $L \in \mathbf{E}_k$ with the same underlying tensor $L = \Delta_k((L_\alpha)_\alpha)$, the formula $\mathbf{w} = (\bigotimes_\alpha \exp(L_\alpha))(\mathbf{u})$ gives the same $\mathbf{w}$.

9.3 Charts for the tree-based manifold

Intersection of chart domains

For $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, define the *tree-based chart domain*:

$$\mathcal{W}(\mathbf{v}) \coloneqq \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathcal{U}^{(k)}(\mathbf{v}), \tag{9.3}$$

where $\mathcal{U}^{(k)}(\mathbf{v})$ is the chart domain of $\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ at $\mathbf{v}$ (Section 8.2).

Model space

Define the *tree-based model space*:

$$\mathbf{E}(\mathbf{v}) \coloneqq \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathbf{E}_k \subset \mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D}), \tag{9.4}$$

and the *tree-based core set*:

$$\mathcal{O}(\mathbf{v}) \coloneqq \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathfrak{M}_{\mathfrak{r}_k}()\Big)\big({}_a \bigotimes_{\alpha \in \mathcal{P}_k(T_D)} U_\alpha^{\min}(\mathbf{v})\Big).$$

Lemma 9.3.1 ([Falcó et al., 2023, Lemma 4.6]) *Under condition* (Inj) *at each level,* $\mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v})$ *is an open subset of the Banach space* $\mathbf{E}(\mathbf{v}) \times {}_a \bigotimes_{\alpha \in \mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$.

Proof. We show separately that $\mathcal{O}(\mathbf{v})$ is open in ${}_a \bigotimes_{\alpha \in \mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$ and that $\mathbf{E}(\mathbf{v})$ is a Banach space.

Step 1: $\mathcal{O}(\mathbf{v})$ is open.

Fix $k \in \{1, \ldots, \mathrm{depth}(T_D)\}$. The space $A_k \coloneqq {}_a \bigotimes_{\alpha \in \mathcal{P}_k(T_D)} U_\alpha^{\min}(\mathbf{v})$ is finite-dimensional, since each $U_\alpha^{\min}(\mathbf{v})$ is finite-dimensional (Theorem 4.2.1). The set $\mathcal{O}_k \coloneqq \mathfrak{M}_{\mathfrak{r}_k}(()A_k)$ consists of tensors in A_k with α-ranks equal to r_α for all $\alpha \in \mathcal{P}_k(T_D)$. Its complement is the zero set of the rank-r_α minors of the unfolding matrices; since these are polynomial conditions on the coordinates of A_k, the complement is a closed algebraic variety. Hence $\mathcal{O}_k$ is open in A_k.

Now, all spaces A_k are identified with $A_1 = {}_a \bigotimes_{\alpha \in \mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$ via the canonical isomorphisms induced by the tree structure (each $\mathbf{V}_\alpha[\alpha]$ for $\alpha \in \mathcal{P}_k(T_D)$ is a subspace of $\mathbf{V}_\alpha[\beta]$ for the appropriate $\beta \in \mathcal{P}_1(T_D)$). Under this identification, $\mathcal{O}_k$ becomes an open subset of A_1, and $\mathcal{O}(\mathbf{v}) = \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathcal{O}_k$ is a finite intersection of open sets in the finite-dimensional space A_1, hence open.

Step 2: $\mathbf{E}(\mathbf{v})$ is a Banach space.

For each level k, the model space $\mathbf{E}_k$ embeds into $\mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D})$ via the Laplacian map Δ_k (equation (9.1)). The image is:

$$\Delta_k\Big(\prod_{\alpha\in\mathcal{P}_k(T_D)}\mathcal{L}\big(U_\alpha^{\min}(\mathbf{v}),W_\alpha^{\min}(\mathbf{v})\big)\Big)=\Big\{\sum_{\alpha\in\mathcal{P}_k(T_D)}L_\alpha\otimes\mathbf{id}_{[\alpha]}:L_\alpha\in\mathcal{L}\big(U_\alpha^{\min}(\mathbf{v}),W_\alpha^{\min}(\mathbf{v})\big)\Big\},$$

which is a closed subspace of $\mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D})$: it is the image of a finite-dimensional Banach space under a bounded linear map, hence finite-dimensional and therefore closed. Therefore $\mathbf{E}_k\subset\mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D})$ is closed.

The intersection $\mathbf{E}(\mathbf{v})=\bigcap_{k=1}^{\mathrm{depth}(T_D)}\mathbf{E}_k$ is a finite intersection of closed subspaces of $\mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D})$, hence closed, and therefore a Banach space in its own right.

Conclusion.

Since $\mathbf{E}(\mathbf{v})$ is a Banach space and $\mathcal{O}(\mathbf{v})$ is open in the finite-dimensional space A_1, the product $\mathbf{E}(\mathbf{v})\times\mathcal{O}(\mathbf{v})$ is open in $\mathbf{E}(\mathbf{v})\times A_1=\mathbf{E}(\mathbf{v})\times{}_a\bigotimes_{\alpha\in\mathcal{P}_1(T_D)}U_\alpha^{\min}(\mathbf{v})$. $\square$

Tree-based chart map

Since $L\in\mathbf{E}(\mathbf{v})$ satisfies $L\in\mathbf{E}_k$ for all k, the formula $\mathbf{w}=\exp(L)(\mathbf{u})$ is well-defined and independent of the level index k by Proposition 9.2.1. This allows us to define:

Definition 9.3.1 (Tree-based chart) For $\mathbf{v}\in\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)$, the *tree-based chart map*

$$\boldsymbol{\xi}_{\mathbf{v}}:\mathcal{W}(\mathbf{v})\longrightarrow\mathbf{E}(\mathbf{v})\times\mathcal{O}(\mathbf{v}) \tag{9.5}$$

is defined by $\boldsymbol{\xi}_{\mathbf{v}}(\mathbf{w})=(L,\mathbf{u})$ where $\mathbf{w}=\exp(L)(\mathbf{u})$, with $L\in\mathbf{E}(\mathbf{v})$ and $\mathbf{u}\in\mathcal{O}(\mathbf{v})$ uniquely determined by any Tucker chart $\xi_{\mathbf{v}}^{(k)}$.

9.4 The $\mathcal{C}^\infty$-Banach manifold structure

Theorem 9.4.1 ($\mathcal{C}^\infty$ tree-based manifold, [Falcó et al., 2023, Theorem 4.7]) *Let T_D be a dimension partition tree over D with* $\mathrm{depth}(T_D)\geq 2$, *and let $\mathfrak{r}\in\mathcal{AD}(\mathbf{V}_D,T_D)$ be such that $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)$ is a proper set of tree-based tensors. Assume* (Inj) *holds at each level of T_D. Then:*

(i) *The collection $\{(\mathcal{W}(\mathbf{v}),\boldsymbol{\xi}_{\mathbf{v}})\}_{\mathbf{v}\in\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)}$ is a $\mathcal{C}^\infty$-atlas on $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)$, making it a $\mathcal{C}^\infty$-Banach manifold modelled on $\mathbf{E}(\mathbf{v})\times{}_a\bigotimes_{\alpha\in\mathcal{P}_1(T_D)}U_\alpha^{\min}(\mathbf{v})$.*
(ii) *The atlas is compatible with the Tucker manifold atlas at each level: the inclusion*

$$\mathfrak{i}_k : \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \hookrightarrow \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$$

is a $\mathcal{C}^\infty$*-immersion for each* $1 \le k \le \mathrm{depth}(T_D)$.

Proof. We verify the atlas conditions AT1–AT3 for the collection $\{(\mathcal{W}(\mathbf{v}), \boldsymbol{\xi}_{\mathbf{v}})\}_{\mathbf{v}\in\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)}$ and then establish the compatibility with the Tucker atlases.

AT1: Coverage.

For each $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, we have $\mathbf{v} \in \mathcal{W}(\mathbf{v})$ since $\mathbf{v}$ belongs to every Tucker chart domain $\mathcal{U}^{(k)}(\mathbf{v})$ (each Tucker chart is centred at $\mathbf{v}$ by definition). Hence $\bigcup_{\mathbf{v}\in\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)} \mathcal{W}(\mathbf{v}) = \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$.

AT2: $\boldsymbol{\xi}_{\mathbf{v}}$ is a bijection.

For $\mathbf{w} \in \mathcal{W}(\mathbf{v})$, we have $\mathbf{w} \in \mathcal{U}^{(k)}(\mathbf{v})$ for all k. By Lemma 8.2.1 applied at level k (Theorem 8.3.1), there exists a unique pair $(L^{(k)}, \mathbf{u}^{(k)})$ with $L^{(k)} \in \mathbf{E}_k$ and $\mathbf{u}^{(k)} \in \mathcal{O}_k$ such that $\mathbf{w} = \exp(L^{(k)})(\mathbf{u}^{(k)})$.

By Proposition 9.2.1, the map $\exp(L)$ for $L \in \mathbf{E}_k$ acts as $\bigotimes_{\alpha\in\mathcal{P}_k(T_D)} \exp(L_\alpha)$. Since the representation is independent of the level k used (the same tensor $\mathbf{w}$ is represented), we get $L^{(k)} = L^{(k')}$ and $\mathbf{u}^{(k)} = \mathbf{u}^{(k')}$ for all k, k'. In particular, $L := L^{(1)} \in \bigcap_k \mathbf{E}_k = \mathbf{E}(\mathbf{v})$ and $\mathbf{u} := \mathbf{u}^{(1)} \in \bigcap_k \mathcal{O}_k = \mathcal{O}(\mathbf{v})$.

The map $\boldsymbol{\xi}_{\mathbf{v}}(\mathbf{w}) = (L, \mathbf{u})$ is therefore well-defined and injective (since $(L, \mathbf{u})$ uniquely determines $\mathbf{w} = \exp(L)(\mathbf{u})$). It is surjective onto $\mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v})$ by the converse: for any $(L, \mathbf{u}) \in \mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v})$, the tensor $\mathbf{w} = \exp(L)(\mathbf{u})$ belongs to every $\mathcal{U}^{(k)}(\mathbf{v})$ (since $L \in \mathbf{E}_k$ and $\mathbf{u} \in \mathcal{O}_k$ for all k), so $\mathbf{w} \in \mathcal{W}(\mathbf{v})$.

AT3: Smooth transitions.

Let $\mathbf{v}, \mathbf{v}' \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ with $\mathcal{W}(\mathbf{v}) \cap \mathcal{W}(\mathbf{v}') \neq \emptyset$. We must show that $\boldsymbol{\xi}_{\mathbf{v}'} \circ \boldsymbol{\xi}_{\mathbf{v}}^{-1}$ is $\mathcal{C}^\infty$.

For any $\mathbf{w} \in \mathcal{W}(\mathbf{v}) \cap \mathcal{W}(\mathbf{v}')$, pick any level k. Since $\mathbf{w} \in \mathcal{U}^{(k)}(\mathbf{v}) \cap \mathcal{U}^{(k)}(\mathbf{v}')$, the transition map

$$\xi^{(k)}_{\mathbf{v}'} \circ (\xi^{(k)}_{\mathbf{v}})^{-1}$$

is $\mathcal{C}^\infty$ by Lemma 8.3.1 (applied to the Tucker manifold $\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$). The tree-based chart maps satisfy $\boldsymbol{\xi}_{\mathbf{v}} = \xi^{(k)}_{\mathbf{v}}|_{\mathcal{W}(\mathbf{v})}$ for every k (by the uniqueness proved in AT2). Therefore

$$\boldsymbol{\xi}_{\mathbf{v}'} \circ \boldsymbol{\xi}_{\mathbf{v}}^{-1} = \xi^{(k)}_{\mathbf{v}'} \circ (\xi^{(k)}_{\mathbf{v}})^{-1}\big|_{\boldsymbol{\xi}_{\mathbf{v}}(\mathcal{W}(\mathbf{v})\cap\mathcal{W}(\mathbf{v}'))},$$

which is the restriction of a $\mathcal{C}^\infty$ map to an open subset, hence $\mathcal{C}^\infty$.

Part (ii): Compatibility with Tucker atlases.

For each k, the inclusion $\mathfrak{i}_k : \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \hookrightarrow \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ satisfies, in charts,

$$\xi^{(k)}_{\mathbf{v}} \circ \mathfrak{i}_k \circ \boldsymbol{\xi}_{\mathbf{v}}^{-1} = \xi^{(k)}_{\mathbf{v}} \circ (\xi^{(k)}_{\mathbf{v}})^{-1}\big|_{\mathbf{E}(\mathbf{v})\times\mathcal{O}(\mathbf{v})} = \text{inclusion } \mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v}) \hookrightarrow \mathbf{E}_k \times \mathcal{O}_k,$$

which is a bounded linear injection (hence $\mathcal{C}^\infty$ and an immersion). Since this holds for every choice of charts and every k, $\mathfrak{i}_k$ is a $\mathcal{C}^\infty$-immersion.

Manifold structure.

The atlas $\{(\mathcal{W}(\mathbf{v}),\boldsymbol{\xi}_{\mathbf{v}})\}_{\mathbf{v}\in\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)}$ satisfies AT1–AT3, so $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)$ is a $\mathcal{C}^\infty$-Banach manifold. The model space is $\mathbf{E}(\mathbf{v})\times\mathcal{O}(\mathbf{v})$, which is open in $\mathbf{E}(\mathbf{v})\times {}_a\bigotimes_{\alpha\in\mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$ by Lemma 9.3.1, so $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)$ is modelled on this Banach space. □

9.5 Tangent space of the tree-based manifold

Proposition 9.5.1 (Tangent space) *Under the assumptions of Theorem 9.4.1, for* $\mathbf{v}\in\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)$*:*

$$T_{\mathbf{v}}\mathfrak{i}\big(\mathbb{T}_{\mathbf{v}}(\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D))\big)=\bigcap_{k=1}^{\operatorname{depth}(T_D)} T_{\mathbf{v}}\mathfrak{i}_k\big(\mathbb{T}_{\mathbf{v}}(\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D,\mathcal{P}_k(T_D)))\big)=\bigcap_{k=1}^{\operatorname{depth}(T_D)} Z^{(D)}([)\mathcal{P}_k(T_D)]\mathbf{v}, \tag{9.6}$$

where $Z^{(D)}([)\mathcal{P}]\mathbf{v}$ *is the tangent subspace* (8.7) *for the partition* $\mathcal{P}$*.*

Proof. The model space of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)$ is $\mathbf{E}(\mathbf{v})\times\mathcal{O}_1$, where $\mathbf{E}(\mathbf{v})=\bigcap_k\mathbf{E}_k$. Since $T_{\mathbf{v}}\mathfrak{i}_k$ is injective (Proposition 8.4.1), the tangent map $T_{\mathbf{v}}\mathfrak{i}$ is also injective and its image is $\bigcap_k T_{\mathbf{v}}\mathfrak{i}_k(\mathbb{T}_{\mathbf{v}}\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D,\mathcal{P}_k()))$. The last equality follows from the explicit formula (8.8) for each level. □

9.6 Immersion in the ambient Banach space

Theorem 9.6.1 (Immersion of the tree-based manifold, [Falcó et al., 2023, Corollary 4.9]) *Under the assumptions of Theorem 9.4.1:*

$$T_{\mathbf{v}}\mathfrak{i}\big(\mathbb{T}_{\mathbf{v}}(\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D))\big)\in\mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})\quad \textit{for all } \mathbf{v}\in\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D). \tag{9.7}$$

Consequently, $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)$ *is an immersed submanifold of* $\mathbf{V}_{D,\|\cdot\|_D}$*.*

Proof. From Theorem 8.5.1 applied at each level, $Z^{(D)}([)\mathcal{P}_k(])\mathbf{v}\in\mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$ for $1\le k\le\operatorname{depth}(T_D)$. By Proposition 9.5.1, the tangent space of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)$ is $\bigcap_k Z^{(D)}([)\mathcal{P}_k(])\mathbf{v}$. Applying Lemma 8.5.2 iteratively (intersection of split subspaces is split), we obtain $\bigcap_k Z^{(D)}([)\mathcal{P}_k(])\mathbf{v}\in\mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$. □

Remark 9.6.1 (Embedded vs. immersed) Theorem 9.6.1 shows that $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)$ is an *immersed* submanifold of $\mathbf{V}_{D,\|\cdot\|_D}$. Moreover, Theorem 9.4.1(ii) shows it is an *embedded* submanifold of each Tucker manifold $\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D,\mathcal{P}_k(T_D))$. Whether $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)$ is embedded in $\mathbf{V}_{D,\|\cdot\|_D}$ (i.e., the manifold topology coincides with the subspace topology) is a more delicate question related to the global topology discussed in Section 9.8.

9.7 Comparison with alternative descriptions

The geometric description of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ in Falcó et al. [2023] differs from earlier approaches in several respects.

Uschmajew–Vandereycken Uschmajew and Vandereycken [2013a].

For the HT format in finite-dimensional spaces, they construct $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ as a quotient manifold $P\backslash\big(\mathrm{GL}(r_\alpha) \times \cdots\big)$ for a gauge group P. This gives a finite-dimensional manifold and is well-suited for numerical algorithms (Riemannian gradient methods). However, it does not extend to infinite-dimensional spaces and does not make the compatibility with Tucker manifolds at each level explicit.

Holtz–Rohwedder–Schneider Holtz et al. [2012].

For the TT format in finite dimensions, a similar quotient construction is used. The connection to the Tucker manifold at each level is not made explicit.

Falcó–Hackbusch–Nouy (2015 preprint Falcó et al. [2015]).

An earlier version of the atlas in Falcó et al. [2023] was proposed using a different chart system. The description in Falcó et al. [2023] is more natural because the chart maps are directly inherited from the Tucker manifold charts via the Laplacian map (Proposition 9.2.1).

Key novelties of Falcó et al. [2023].

1. The atlas $\{(\mathcal{W}(\mathbf{v}), \boldsymbol{\xi}_{\mathbf{v}})\}$ is *natural*: it is the restriction of the Tucker atlases to the intersection $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, without any additional construction.
2. The *compatibility* with Tucker manifold structures at each level is built into the definition, making the geometric relationships transparent.
3. The description holds in *infinite-dimensional Banach spaces* under the mild condition (Inj).
4. The tangent space formula (9.6) is explicit and computable.

9.8 Topology and connected components

Proposition 9.8.1 *For the Tucker manifold, $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is connected for any $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$. This follows because $\prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ is connected (each $\mathbb{G}_r(V)$ is connected as a Banach manifold) and the fibre $\mathbb{R}_*^{\times r_\alpha}$ is connected.*

For general tree-based formats, the connectedness of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is an open problem in general. The following is known:

Proposition 9.8.2 (i) *For the TT format and binary HT formats in* finite dimensions*, $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is connected (see Uschmajew and Vandereycken [2013a], Holtz et al. [2012]).*
(ii) *For general T_D and infinite-dimensional spaces, connectedness of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is an open question.*

Remark 9.8.1 (Stratification) The decomposition $\mathbf{V}_D = \bigsqcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)} \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ (Theorem 6.6.1) is a stratification of $\mathbf{V}_D$ by tree-based rank. Each stratum $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is an immersed manifold, and $\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D) = \overline{\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)}^{\text{weak}}$ is the weak closure. The boundary strata $\partial \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D) = \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D) \setminus \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ consist of tensors with strictly smaller rank. This stratification is the tensor-format analogue of the rank stratification of matrices.

9.9 Historical notes

The geometric structure of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ as a Banach manifold was established in Falcó et al. [2023]. The fundamental observation — that the tree-based manifold is an intersection of Tucker manifolds and that its atlas can be obtained by intersecting the Tucker atlases — was made possible by the reformulation of the Tucker manifold charts using the Laplacian-like map Δ_k (Proposition 9.2.1).

The compatibility condition between the tree-based atlas and the Tucker atlases at each level (Theorem 9.4.1(ii)) is the key property that distinguishes the description of Falcó et al. [2023] from earlier work: it allows one to apply the Tucker manifold geometry (tangent spaces, projections, Dirac–Frenkel principle) at each level of the tree independently, and then combine the results.

The Dirac–Frenkel variational principle for tree-based formats (Chapter 10) exploits this compatibility in an essential way: the reduced ODE on $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is obtained by projecting the vector field onto the tangent space $\bigcap_k Z^{(D)}([])\mathcal{P}_k([])\mathbf{v}$, and the existence of this projection (Corollary 8.5.1 applied at each level) follows from the fact that each $Z^{(D)}([])\mathcal{P}_k([])\mathbf{v}$ is a split subspace of $\mathbf{V}_{D,\|\cdot\|_D}$.

Part IV
Variational Principles and Applications

Chapter 10
The Dirac–Frenkel Variational Principle

The geometric framework developed in Chapters 8 and 9 has a direct dynamical application: it provides the rigorous foundation for *model reduction* of differential equations on tensor Banach spaces. Given an abstract Cauchy problem on $\mathbf{V}_{D,\|\cdot\|_D}$, one looks for an approximate solution that evolves on the manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (or $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$). The natural way to constrain the dynamics to the manifold is the *Dirac–Frenkel variational principle*: at each time t, the velocity $\dot{\mathbf{v}}_r(t)$ is chosen as the best approximation of the vector field $\mathbf{F}(t, \mathbf{v}_r(t))$ in the tangent space $Z^{(D)}(\mathbf{v}_r(t))$.

This principle was originally proposed by Dirac and Frenkel in 1930 for the time-dependent Schrödinger equation in Hilbert spaces. The present chapter extends it to general Banach tensor spaces, establishing the existence of the reduced ODE and characterising its solutions. Sections 10.1–10.2 set up the abstract framework. Section 10.4 proves the Tucker version (Theorems 5.2 and 5.4 of Falcó et al. [2019]). Section 10.6 extends to the tree-based setting (Falcó et al. [2023]).

10.1 The abstract Cauchy problem on tensor Banach spaces

Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces for $\alpha \in D$, and let $\mathbf{V}_{D,\|\cdot\|_D} = {}_{\|\cdot\|_D}\bigotimes_{\alpha\in D} V_\alpha$ be a reflexive Banach tensor space satisfying condition (Inj). Consider the abstract Cauchy problem:

$$\dot{\mathbf{u}}(t) = \mathbf{F}(t, \mathbf{u}(t)), \quad t \geq 0, \tag{10.1}$$

$$\mathbf{u}(0) = \mathbf{u}_0, \tag{10.2}$$

where $\mathbf{u}_0 \neq \mathbf{0}$, and $\mathbf{F} : [0, \infty) \times \mathbf{V}_{D,\|\cdot\|_D} \to \mathbf{V}_{D,\|\cdot\|_D}$ satisfies the usual conditions for existence and uniqueness of solutions (e.g., Lipschitz continuity in the second argument).

The goal is to approximate $\mathbf{u}(t)$ by a *differentiable curve* $t \mapsto \mathbf{v}_r(t)$ lying in $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ for some fixed $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$, with initial condition $\mathbf{v}_r(0) = \mathbf{v}_0 \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ chosen as an approximation of $\mathbf{u}_0$.

Remark 10.1.1 By Theorem 4.5.2, a best approximation $\mathbf{v}_0 \in \mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D)$ of $\mathbf{u}_0$ exists when $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive. The initial condition $\mathbf{v}_0 \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (with exact rank $\mathfrak{r}$) can always be arranged by taking $\mathbf{v}_0 \in \mathfrak{M}_{\mathfrak{r}'}(\mathbf{V}_D)$ for some $\mathfrak{r}' \leq \mathfrak{r}$ and adjusting the rank.

10.2 The principle in Hilbert spaces

When $\mathbf{V}_{D,\|\cdot\|_D}$ is a Hilbert space with inner product $\langle\cdot,\cdot\rangle_D$, the constraint to $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ takes the classical form. Since $Z^{(D)}(\mathbf{v}_r(t)) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$ (Theorem 8.5.1), the orthogonal projection $\mathcal{P}_{\mathbf{v}_r(t)}$ onto $Z^{(D)}(\mathbf{v}_r(t))$ is well-defined and the *Dirac–Frenkel equation* reads:

$$\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}\big(\mathbf{F}(t, \mathbf{v}_r(t))\big), \tag{10.3}$$

which is equivalent to the variational condition:

$$\langle \dot{\mathbf{v}}_r(t) - \mathbf{F}(t, \mathbf{v}_r(t)),\, \dot{\mathbf{v}}\rangle_D = 0 \quad \text{for all } \dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t)). \tag{10.4}$$

The MCTDH (Multi-Configuration Time-Dependent Hartree) method of quantum dynamics Lubich [2008] is precisely (10.3) with $\mathbf{V}_{D,\|\cdot\|_D} = L^2(\mathbb{R}^{3d})$ and $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ the Hartree manifold ($\mathfrak{r} = (1, \ldots, 1)$).

10.3 Projections in Banach spaces

In a general Banach space, there is no canonical inner product and hence no canonical projection. Two notions replace the orthogonal projection.

Metric projection

Definition 10.3.1 For $\mathbf{v}_r(t) \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and $t \in I$, the *metric projection* $\mathcal{P}_{\mathbf{v}_r(t)}$ maps $\dot{\mathbf{u}} \in \mathbf{V}_{D,\|\cdot\|_D}$ to the nearest point in $Z^{(D)}(\mathbf{v}_r(t))$:

$$\mathcal{P}_{\mathbf{v}_r(t)}(\dot{\mathbf{u}}) \coloneqq \operatorname*{arg\,min}_{\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))} \|\dot{\mathbf{u}} - \dot{\mathbf{v}}\|_D.$$

By Corollary 8.5.1, when $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive the metric projection exists. When $\mathbf{V}_{D,\|\cdot\|_D}$ is strictly convex it is unique (since $Z^{(D)}(\mathbf{v}_r(t))$ is convex). The metric projection is, in general, a *nonlinear* operator on Banach spaces.

Normalised duality mapping

The duality mapping $J : \mathbf{V}_{D,\|\cdot\|_D} \to 2^{\mathbf{V}^*_{D,\|\cdot\|_D}}$ is defined by

$$J(\mathbf{u}) \coloneqq \{ f \in \mathbf{V}^*_{D,\|\cdot\|_D} : \langle \mathbf{u}, f \rangle = \|\mathbf{u}\|_D^2 = \|f\|_*^2 \}.$$

It is single-valued when $\mathbf{V}_{D,\|\cdot\|_D}$ is smooth (Definition 2.1.3), and uniformly continuous on bounded sets when $\mathbf{V}_{D,\|\cdot\|_D}$ is uniformly smooth. In a Hilbert space, J coincides with the Riesz isomorphism.

Theorem 10.3.1 (Metric projection characterisation, [Falcó et al., 2019, Theorem 5.2]) *Let $\mathbf{V}_{D,\|\cdot\|_D}$ be reflexive and strictly convex, and assume* (Inj). *For $\mathbf{v}_r(t) \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and fixed $t \in I$, the following are equivalent:*

(a) $\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}(\mathbf{F}(t, \mathbf{v}_r(t)))$.
(b) $\langle \dot{\mathbf{v}}_r(t) - \dot{\mathbf{v}},\, J(\mathbf{F}(t, \mathbf{v}_r(t)) - \dot{\mathbf{v}}_r(t)) \rangle \geq 0$ *for all* $\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))$.
(c) $\langle \dot{\mathbf{v}},\, J(\mathbf{F}(t, \mathbf{v}_r(t)) - \dot{\mathbf{v}}_r(t)) \rangle = 0$ *for all* $\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))$.

Proof. This is a direct consequence of the classical characterisation of the metric projection onto a closed convex set in a reflexive, strictly convex Banach space, combined with [Alber, 1996, Proposition 2.10]. Condition (c) recovers the Hilbert-space variational form: when J is the Riesz map, (c) becomes the orthogonality condition (10.4). □

Generalised projection

When $\mathbf{V}_{D,\|\cdot\|_D}$ is additionally smooth, an alternative projection is available. Define $\phi : \mathbf{V}_{D,\|\cdot\|_D} \times \mathbf{V}_{D,\|\cdot\|_D} \to \mathbb{R}$ by

$$\phi(\mathbf{u}, \mathbf{v}) = \|\mathbf{u}\|_D^2 - 2\langle \mathbf{u}, J(\mathbf{v}) \rangle + \|\mathbf{v}\|_D^2.$$

The *generalised projection* $\Pi_{\mathbf{v}_r(t)}$ minimises ϕ over $Z^{(D)}(\mathbf{v}_r(t))$:

$$\Pi_{\mathbf{v}_r(t)}(\dot{\mathbf{u}}) \coloneqq \operatorname*{arg\,min}_{\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))} \phi(\dot{\mathbf{v}}, \dot{\mathbf{u}}).$$

It coincides with the metric projection when $\mathbf{V}_{D,\|\cdot\|_D}$ is a Hilbert space.

10.4 The Dirac–Frenkel principle for the Tucker manifold

Theorem 10.4.1 (Dirac–Frenkel for Tucker, [Falcó et al., 2019, Theorems 5.2, 5.4]) *Let $\mathbf{V}_{D,\|\cdot\|_D}$ be reflexive and strictly convex, and assume* (Inj). *Let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$, and let $\mathbf{v}_0 \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.*

(Existence) *There exists a differentiable curve $t \mapsto \mathbf{v}_r(t)$ in $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$, defined for t in a neighbourhood of 0, satisfying*

$$\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}\big(\mathbf{F}(t, \mathbf{v}_r(t))\big), \qquad \mathbf{v}_r(0) = \mathbf{v}_0. \tag{10.5}$$

(Characterisation) *The curve $t \mapsto \mathbf{v}_r(t)$ satisfies* (10.5) *if and only if $\dot{\mathbf{v}}_r(t) \in Z^{(D)}(\mathbf{v}_r(t))$ and*

$$\|\dot{\mathbf{v}}_r(t) - \mathbf{F}(t, \mathbf{v}_r(t))\|_D = \min_{\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))} \|\dot{\mathbf{v}} - \mathbf{F}(t, \mathbf{v}_r(t))\|_D. \tag{10.6}$$

Sketch. **Existence.** The reduced ODE (10.5) is an ODE on the Banach manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$. By Theorem 8.5.1, $Z^{(D)}(\mathbf{v}_r(t))$ is a split subspace of $\mathbf{V}_{D,\|\cdot\|_D}$ for each $\mathbf{v}_r(t) \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$. The right-hand side of (10.5) depends smoothly on $\mathbf{v}_r(t)$ (the metric projection onto a smoothly-varying split subspace is smooth), so local existence and uniqueness follow from the ODE existence theorem on Banach manifolds [Lang, 1999, Theorem 4.1.5].

Characterisation. Since $Z^{(D)}(\mathbf{v}_r(t))$ is closed and convex, Corollary 8.5.1 gives existence of the minimum in (10.6). The equivalence with the metric projection condition (10.5) follows from the uniqueness (strict convexity) and Theorem 10.3.1. □

Remark 10.4.1 (Kinematic equations) In local chart coordinates $(L_\alpha, C^{(D)})$ at $\mathbf{v}_r(t)$, the reduced ODE (10.5) takes the form of a system of coupled equations for the subspace parameters $L_\alpha(t)$ and the core tensor $C^{(D)}(t)$. The tangent formula (8.8) makes this explicit:

$$\dot{\mathbf{v}}_r(t) = \sum_{(i_\alpha)} \dot{C}^{(D)}_{(i_\alpha)} \bigotimes_\alpha u^{(\alpha)}_{i_\alpha} + \sum_\alpha \sum_{i_\alpha} \dot{u}^{(\alpha)}_{i_\alpha} \otimes \mathbf{U}^{(\alpha)}_{i_\alpha}.$$

The Dirac–Frenkel condition (10.6) then yields coupled evolution equations for the basis vectors $u^{(\alpha)}_{i_\alpha}(t) \in V_\alpha$ and the core tensor $C^{(D)}(t)$, which generalise the MCTDH equations to arbitrary Tucker ranks in Banach spaces. In the Hilbert case, imposing the tangential gauge $\langle \dot{u}^{(\alpha)}_i, u^{(\alpha)}_j \rangle = 0$, the closed system reads:

$$\dot{C}^{(D)}_{(j_\alpha)}(t) = \Big\langle \mathbf{F}(t, \mathbf{v}_r(t)), \bigotimes_\alpha u^{(\alpha)}_{j_\alpha}(t) \Big\rangle_D, \tag{10.7a}$$

$$\dot{u}^{(\alpha)}_{j_\alpha}(t) = P^{(\alpha)}_W(t) \big[M^{(\alpha)}(t)^{-1}\big]_{j_\alpha \cdot} \sum_{i_\alpha} F^{(\alpha)}_{i_\alpha \cdot}(t)\, u^{(\alpha)}_{i_\alpha}(t), \tag{10.7b}$$

where $M^{(\alpha)}_{ij}(t) = \langle \mathbf{U}^{(\alpha)}_i(t), \mathbf{U}^{(\alpha)}_j(t) \rangle_{[\alpha]}$ is the mass matrix and $F^{(\alpha)}_{ij}(t)$ the mean-field matrix (10.9).

The Hartree manifold

Example 10.4.1 (Time-dependent Hartree method) For $\mathfrak{r} = \mathbf{1} = (1, \ldots, 1)$, the Tucker manifold $\mathfrak{M}_{\mathbf{1}}(\mathbf{V}_D)$ consists of elementary tensors $\mathbf{v} = \lambda \bigotimes_\alpha v_\alpha$ ($v_\alpha \in V_\alpha$, $\lambda \in \mathbb{R}$), called the *Hartree manifold* Lubich [2008]. The Dirac–Frenkel ODE (10.5) on the Hartree manifold gives the *time-dependent Hartree equations*. In a Hilbert space with $\|v_\alpha\|_\alpha = 1$ (unit sphere constraint):

$$\dot{\lambda}(t) = \langle \bigotimes_\alpha v_\alpha(t), \mathbf{F}(t, \mathbf{v}_r(t)) \rangle_D,$$

$$\lambda(t)\, \dot{v}_\alpha(t) = P_{W_\alpha(t)}\big[(\mathrm{id}_\alpha \otimes \langle \cdot, \bigotimes_{\beta \neq \alpha} v_\beta(t) \rangle)\mathbf{F}(t, \mathbf{v}_r(t))\big],$$

where $P_{W_\alpha(t)}$ is the orthogonal projection onto the complement of $\mathrm{span}\{v_\alpha(t)\}$ in V_α. This is the system solved numerically by MCTDH.

10.5 Kinematic equations for the Tucker manifold

Setup

Fix $\mathbf{v}_r(t) \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and orthonormal bases $\{u^{(\alpha)}_{i_\alpha}(t)\}_{i_\alpha=1}^{r_\alpha}$ of $U_\alpha^{\min}(\mathbf{v}_r(t))$ for each $\alpha \in D$. Define the *mean-field vectors*

$$\mathbf{U}^{(\alpha)}_{i_\alpha}(t) := \sum_{(i_\beta)_{\beta \neq \alpha}} C^{(D)}_{(i_\alpha, (i_\beta)_{\beta \neq \alpha})}(t) \bigotimes_{\beta \neq \alpha} u^{(\beta)}_{i_\beta}(t) \in U^{\min}_{D \setminus \{\alpha\}}(\mathbf{v}_r(t)), \tag{10.8}$$

and the *mean-field matrix*

$$F^{(\alpha)}_{i_\alpha j_\alpha}(t) := \big\langle \mathbf{U}^{(\alpha)}_{i_\alpha}(t),\ \big(\mathrm{id}_\alpha \otimes T^{[\alpha]}_\psi\big)(\mathbf{F}(t, \mathbf{v}_r(t)))\big|_{\psi = u^{(\alpha)}_{j_\alpha}(t)} \big\rangle_{[\alpha]}, \tag{10.9}$$

where $T^{[\alpha]}_\psi(\mathbf{w}) := (\psi \otimes \mathrm{id}_{[\alpha]})(\mathbf{w})$ is the contraction with $\psi \in V_\alpha^*$.

Tangent decomposition

By Proposition 8.4.1, every velocity $\dot{\mathbf{v}}_r(t) \in Z^{(D)}(\mathbf{v}_r(t))$ decomposes as

$$\dot{\mathbf{v}}_r(t) = \sum_{(i_\alpha)} \dot{C}^{(D)}_{(i_\alpha)}(t) \bigotimes_\alpha u^{(\alpha)}_{i_\alpha}(t) + \sum_{\alpha \in D} \sum_{i_\alpha} \dot{u}^{(\alpha)}_{i_\alpha}(t) \otimes \mathbf{U}^{(\alpha)}_{i_\alpha}(t), \tag{10.10}$$

with $\dot{u}^{(\alpha)}_{i_\alpha}(t) \in W^{\min}_\alpha(\mathbf{v}_r(t))$. Imposing the *tangential constraint*

$$\langle \dot{u}_{i_\alpha}^{(\alpha)}(t), u_{j_\alpha}^{(\alpha)}(t)\rangle_\alpha = 0 \quad \forall\, i_\alpha, j_\alpha,\ \alpha \in D, \tag{10.11}$$

removes the gauge freedom. In the Hilbert case, the closed kinematic system is (10.7) with mean-field matrix

$$F_{i_\alpha j_\alpha}^{(\alpha)}(t) = \big\langle \mathbf{U}_{i_\alpha}^{(\alpha)}(t), (\mathrm{id}_\alpha \otimes \langle u_{j_\alpha}^{(\alpha)}(t), \cdot\rangle_\alpha)\mathbf{F}(t, \mathbf{v}_r(t))\big\rangle_{[\alpha]}. \tag{10.12}$$

Remark 10.5.1 (Banach case: duality mapping) In a non-Hilbert Banach space, the inner products in (10.7) are replaced by the duality pairing $\langle \cdot, J(\cdot)\rangle$. The mass matrix $M_{ij}^{(\alpha)} = \langle \mathbf{U}_i^{(\alpha)}, \mathbf{U}_j^{(\alpha)}\rangle_{[\alpha]}$ becomes the *duality mass matrix* $M_{ij}^{(\alpha)} = \langle \mathbf{U}_i^{(\alpha)}, J(\mathbf{U}_j^{(\alpha)})\rangle_{[\alpha]}$, still invertible (by Proposition 8.4.1) but no longer symmetric.

10.6 Extension to tree-based formats

The extension of Theorem 10.4.1 to $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is immediate given the immersion result of Chapter 9.

Theorem 10.6.1 (Dirac–Frenkel for tree-based formats, [Falcó et al., 2023, Section 5]) *Let T_D be a dimension partition tree, $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$ such that $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is a proper set of tree-based tensors. Assume $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive and strictly convex, and that* (Inj) *holds at each level of T_D. Let $\mathbf{v}_0 \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$.*

(Existence) *There exists a differentiable curve $t \mapsto \mathbf{v}_r(t)$ in $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ satisfying*

$$\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}\big(\mathbf{F}(t, \mathbf{v}_r(t))\big), \qquad \mathbf{v}_r(0) = \mathbf{v}_0, \tag{10.13}$$

where $\mathcal{P}_{\mathbf{v}_r(t)}$ is the metric projection onto $T_{\mathbf{v}_r(t)}\mathfrak{i}(\mathbb{T}_{\mathbf{v}_r(t)}(\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D))) = \bigcap_k Z^{(D)}([]\mathcal{P}_k(T_D)]\mathbf{v}_r(t)$.

(Characterisation) *The curve satisfies* (10.13) *if and only if $\dot{\mathbf{v}}_r(t) \in \bigcap_k Z^{(D)}([]\mathcal{P}_k(T_D)]\mathbf{v}_r(t)$ and*

$$\|\dot{\mathbf{v}}_r(t) - \mathbf{F}(t, \mathbf{v}_r(t))\|_D = \min_{\dot{\mathbf{v}} \in \bigcap_k Z^{(D)}([]\mathcal{P}_k([])\mathbf{v}_r(t)} \|\dot{\mathbf{v}} - \mathbf{F}(t, \mathbf{v}_r(t))\|_D. \tag{10.14}$$

Proof. The proof is identical to that of Theorem 10.4.1, with $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ replaced by $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ and $Z^{(D)}(\mathbf{v}_r(t))$ replaced by $\bigcap_k Z^{(D)}([]\mathcal{P}_k([])\mathbf{v}_r(t)$. The key inputs are: (i) $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is a $\mathcal{C}^\infty$-Banach manifold (Theorem 9.4.1); (ii) its tangent space is a split subspace of $\mathbf{V}_{D,\|\cdot\|_D}$ (Theorem 9.6.1); and (iii) the metric projection onto a split subspace in a reflexive strictly convex space is well-defined and unique. □

Remark 10.6.1 (Algorithmic interpretation) Theorem 10.6.1 provides the mathematical foundation for a class of algorithms that integrate tensor-format

ODEs by evolving the transfer tensors $\{C^{(\alpha)}\}_{\alpha\in T_D}$ and the leaf basis vectors $\{u^{(j)}_{i_j}\}_{j\in\mathcal{L}(T_D)}$ along the Dirac–Frenkel trajectories. For the TT format, this is the starting point of the *projector-splitting integrator* of Lubich, Oseledets, and Vandereycken Lubich et al. [2015]. For general tree-based formats, the corresponding algorithm was introduced by Ceruti, Lubich, and Walach Ceruti et al. [2021].

10.7 Worked examples

This section illustrates the Dirac–Frenkel principle through two concrete problems: a linear diffusion equation and a bipartite quantum system. Both examples show explicitly how the abstract ODE on $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ reduces to a finite system of ODEs.

Example 1: The heat equation on $L^2([0,1]^2)$

Example 10.7.1 (Low-rank dynamics for the heat equation) Let $\mathbf{V}_{D,\|\cdot\|_D} = L^2([0,1]^2)$ with $V_1 = V_2 = L^2([0,1])$ and the Hilbert tensor norm. Consider the heat equation

$$\partial_t u(x_1,x_2,t) = \Delta u(x_1,x_2,t), \quad (x_1,x_2)\in[0,1]^2, \quad t>0, \tag{10.15}$$

with homogeneous Dirichlet boundary conditions and initial condition $u(\cdot,\cdot,0) = u_0 \in \mathfrak{M}_{(r,r)}(L^2([0,1]^2))$. The vector field is $\mathbf{F}(\mathbf{v}) = \Delta\mathbf{v}$, where $\Delta = \partial^2_{x_1} + \partial^2_{x_2}$.

Tucker rank evolution.

The exact solution $u(t) = e^{t\Delta}u_0$ preserves the Tucker rank (r,r) when the Laplacian acts separately on each factor. More precisely, writing $u_0 = \sum_{k=1}^r \lambda_k\phi_k\otimes\psi_k$ in Tucker format, the exact solution is

$$u(t) = \sum_{k=1}^r\sum_{j=1}^r \lambda_k\langle e^{t\partial^2_{x_1}}\phi_k,\, e_j\rangle\langle e^{t\partial^2_{x_2}}\psi_k,\, f_j\rangle e_j\otimes f_j,$$

where $\{e_j\}$ and $\{f_j\}$ are eigenbases of $\partial^2_{x_1}$ and $\partial^2_{x_2}$. This shows that the Tucker rank of $u(t)$ can grow to r^2 in finite time, even if u_0 has rank r. The Dirac–Frenkel principle projects this growth back onto the manifold $\mathfrak{M}_{(r,r)}(L^2)$, keeping the rank fixed.

Kinematic equations in the eigenbasis.

Choose $u_i^{(1)}(t) = e_{\sigma_1(i)}$ and $u_j^{(2)}(t) = f_{\sigma_2(j)}$ as (time-independent) eigenbases of the respective Laplacians, with $-\partial_{x_\alpha}^2 e_k = \mu_k^{(\alpha)} e_k$, $\mu_k^{(\alpha)} = k^2\pi^2$. The core tensor $C^{(D)}(t) \in \mathbb{R}_*^{r\times r}$ then satisfies, by (10.7a) (gauge condition: fixed basis, so $\dot{u}^{(\alpha)} = 0$):

$$\dot{C}_{ij}^{(D)}(t) = \langle \Delta(C^{(D)}(t)\cdot e\otimes f),\, e_i\otimes f_j\rangle = -(\mu_i^{(1)} + \mu_j^{(2)})\, C_{ij}^{(D)}(t), \tag{10.16}$$

giving $C_{ij}^{(D)}(t) = C_{ij}^{(D)}(0)\, e^{-(\mu_i^{(1)}+\mu_j^{(2)})t}$.

Comparison with the exact solution.

The projected solution is $\mathbf{v}_r(t) = \sum_{i,j=1}^r C_{ij}^{(D)}(t) e_i \otimes f_j$. The exact solution in the same eigenbasis has components $u_{ij}(t) = u_{ij}(0)\, e^{-(\mu_i^{(1)}+\mu_j^{(2)})t}$ for all $i, j \geq 1$. Since the initial condition u_0 has Tucker rank r, we have $u_{ij}(0) = 0$ for $\max(i, j) > r$, so the projected and exact solutions agree: $\mathbf{v}_r(t) = u(t)$ for all $t > 0$.

Remark 10.7.1 This agrees with the general principle: when the ambient vector field $\mathbf{F}$ is *tangential* to $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (i.e., $\mathbf{F}(\mathbf{v}) \in Z^{(D)}(\mathbf{v})$ for all $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$), the Dirac–Frenkel projected solution equals the exact solution. For the heat equation with a product Laplacian, this holds when the initial condition has Tucker rank $\leq r$ and the eigenbasis is fixed: the vector field $\Delta\mathbf{v} = (\partial_{x_1}^2 + \partial_{x_2}^2)\mathbf{v}$ stays in the tangent space since both $\partial_{x_\alpha}^2$ act separately on the α-factor.

Non-trivial projection: non-product Laplacian.

A more instructive example is the operator with a coupling term: $\mathbf{F}(\mathbf{v}) = \Delta\mathbf{v} + \varepsilon\,(x_1x_2)\,\mathbf{v}$ for $\varepsilon > 0$. The multiplication operator x_1x_2 has the effect of coupling all modes and increases the Tucker rank of $\mathbf{F}(\mathbf{v})$ beyond r. The Dirac–Frenkel projection then gives non-trivial kinematic equations: both the core $C^{(D)}(t)$ and the basis vectors $u_i^{(\alpha)}(t)$ evolve, with the mean-field matrices $F_{ij}^{(\alpha)}(t)$ encoding the effect of the coupling.

Example 2: Time-dependent Schrödinger equation for a bipartite quantum system

Example 10.7.2 (Two-body Schrödinger equation and entanglement) Let $H_1 = H_2 = L^2(\mathbb{R}^3)$ and $\mathbf{V}_{D,\|\cdot\|_D} = H_1 \otimes_H H_2 = L^2(\mathbb{R}^6)$. Consider the time-dependent Schrödinger equation for a two-particle quantum system:

$$i\,\partial_t \psi(x_1, x_2, t) = \mathcal{H}\,\psi(x_1, x_2, t), \tag{10.17}$$

where $\mathcal{H}$ is the Hamiltonian operator

$$\mathcal{H} = \mathcal{H}_1 \otimes I + I \otimes \mathcal{H}_2 + V_{12}(x_1, x_2), \tag{10.18}$$

with $\mathcal{H}_\alpha = -\frac{1}{2}\Delta_{x_\alpha} + V_\alpha(x_\alpha)$ the single-particle Hamiltonians and V_{12} the interaction potential.

The vector field.

The vector field is $\mathbf{F}(\mathbf{v}) = -i\mathcal{H}\mathbf{v}$, which is a (possibly unbounded) skew-adjoint operator on $L^2(\mathbb{R}^6)$. In the Hilbert-space setting of this example, the metric projection $\mathcal{P}_{\mathbf{v}_r(t)}$ is the orthogonal projection onto $Z^{(D)}(\mathbf{v}_r(t))$ (Hilbert-space Dirac–Frenkel, equation (10.3)).

Tucker rank $r = 1$: the Hartree approximation.

For $\mathfrak{r} = (1, 1)$, the Tucker manifold $\mathfrak{M}_{(1,1)}(L^2(\mathbb{R}^6))$ consists of product states $\psi = \phi_1 \otimes \phi_2$ with $\phi_\alpha \in H_\alpha$. The kinematic equations (10.7) reduce to:

$$i\,\dot{\phi}_1(t) = \mathcal{H}_1\phi_1(t) + \langle \phi_2(t), V_{12}\,\phi_2(t)\rangle_{H_2}\,\phi_1(t), \tag{10.19a}$$

$$i\,\dot{\phi}_2(t) = \mathcal{H}_2\phi_2(t) + \langle \phi_1(t), V_{12}\,\phi_1(t)\rangle_{H_1}\,\phi_2(t), \tag{10.19b}$$

where $\langle \phi_\alpha, V_{12}\,\phi_\alpha\rangle_{H_\alpha}$ denotes the *mean-field potential*: the average of V_{12} with respect to the state of the other particle. These are exactly the time-dependent Hartree equations of Lubich [2008].

Tucker rank $r \geq 2$: the MCTDH equations.

For $\mathfrak{r} = (r, r)$ with $r \geq 2$, the Tucker manifold $\mathfrak{M}_{(r,r)}(L^2(\mathbb{R}^6))$ consists of two-body states with Schmidt rank $\leq r$, which by Theorem 11.1.1 (Chapter 11) are exactly the states with entanglement entropy $S \leq \log r$. The kinematic equations (10.7) give:

Core equation (10.7a):

$$i\,\dot{C}^{(D)}_{ij}(t) = \sum_{k,l=1}^{r} \langle u^{(1)}_k(t) \otimes u^{(2)}_l(t),\, V_{12}\,u^{(1)}_i(t) \otimes u^{(2)}_j(t)\rangle\, C^{(D)}_{kl}(t) + (\mu^{(1)}_i + \mu^{(2)}_j)\, C^{(D)}_{ij}(t), \tag{10.20}$$

where $\mathcal{H}_\alpha u^{(\alpha)}_i = \mu^{(\alpha)}_i u^{(\alpha)}_i$ (eigenvalue equation for the single-particle Hamiltonians; in practice one uses instantaneous eigenstates).

Basis equations (10.7b): for each $\alpha \in \{1, 2\}$ and $j = 1, \ldots, r$,

$$i\,\dot{u}_j^{(\alpha)}(t) = \mathcal{H}_\alpha u_j^{(\alpha)}(t) + P_W^{(\alpha)}(t)\big[M^{(\alpha)}(t)^{-1}\big]_{j\cdot}\sum_{k=1}^{r} F_{k\cdot}^{(\alpha)}(t)\,u_k^{(\alpha)}(t), \qquad (10.21)$$

where $F_{ij}^{(\alpha)}(t) = \langle \mathbf{U}_i^{(\alpha)}(t), V_{12}u_j^{(\alpha)}(t) \otimes \mathbf{U}_i^{(\alpha)}(t)\rangle$ is the mean-field matrix at rank r, and $P_W^{(\alpha)}(t)$ projects onto the orthogonal complement of $U_\alpha^{\min}(\mathbf{v}_r(t))$.

Entanglement and rank.

The Tucker rank r controls the entanglement of the approximate solution: $\mathbf{v}_r(t) \in \mathfrak{M}_{(r,r)}(L^2(\mathbb{R}^6))$ implies $S_1(\mathbf{v}_r(t)) \le \log r$ (Remark 11.3.1 in Chapter 11). The Dirac–Frenkel projection *enforces* this bound throughout the evolution: the entanglement entropy of the projected solution cannot exceed $\log r$ at any time. This is the Banach-space analogue of the entanglement barrier in tensor network simulations, and it gives a rigorous meaning to the informal statement that "MCTDH captures up to r entangled modes".

Comparison: Banach vs. Hilbert case.

If one replaces $L^2(\mathbb{R}^3)$ by $W^{1,p}(\mathbb{R}^3)$ with $p \neq 2$ (relevant for relativistic corrections or p-Laplacian models), the inner product is no longer available and the Hilbert-space projection must be replaced by the metric projection $\mathcal{P}_{\mathbf{v}_r(t)}$ of Definition 10.3.1. The kinematic equations retain the same form (10.7a)–(10.7b) but with $\langle\cdot,\cdot\rangle$ replaced by the duality pairing $\langle\cdot, J(\cdot)\rangle$ and the mass matrix replaced by the duality mass matrix of Remark 10.5.1.

10.8 Historical notes

The Dirac–Frenkel variational principle was proposed independently by Dirac Dirac [1930] and Frenkel Frenkel [1934] in 1930 and 1934 for the time-dependent Schrödinger equation in Hilbert space. The formulation used here — projection of the vector field onto the tangent space of a manifold — is that of McLachlan et al. and Lubich.

Koch and Lubich Koch and Lubich [2007] proved existence and uniqueness of solutions to the Dirac–Frenkel ODE on Tucker manifolds in finite-dimensional Hilbert spaces, and used it to analyse the MCTDH method. The extension to infinite-dimensional Banach spaces (Theorem 10.4.1) was established in Falcó et al. [2019], where the key difficulty — showing that the tangent space $Z^{(D)}(\mathbf{v}_r(t))$ is a split subspace of $\mathbf{V}_{D,\|\cdot\|_D}$, without the Hilbert space structure — was resolved by the immersion theorem (Theorem 8.5.1).

The extension to tree-based formats (Theorem 10.6.1) is the main application result of Falcó et al. [2023]: having established the geometry of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ in Chapter 9, the Dirac–Frenkel principle follows with no additional effort.

Chapter 11
Quantum States as Tensor Formats

The theory developed in Parts I–IV connects naturally to quantum mechanics through a single observation: for a non-zero vector v in a Hilbert tensor space $H = \bigotimes_{j=1}^{d} H_j$, the minimal subspace $U_\alpha^{\min}(v)$ coincides with the support of the reduced density operator of the quantum state associated to v. This chapter makes this connection precise and develops the geometric theory of quantum states from the perspective of tensor formats and minimal subspaces.

Remark 11.0.1 (Notation convention for Part **??***)* In Parts I–IV, tensors in $\mathbf{V}_{D,\|\cdot\|_D}$ are written in boldface $(\mathbf{v}, \mathbf{u}, \mathbf{w})$ to distinguish them from their component vectors $(v_\alpha \in V_\alpha)$. In this part, the primary objects are density operators $\rho \in \mathcal{T}_1^{sa}(H)$ and Hilbert-space vectors $v \in H$; we follow the standard convention of the operator-theory literature and use lightface notation throughout. When a pure state $\rho_v = v \otimes J(v)/\|v\|^2$ is considered as a tensor in $H \cong \bigotimes_j H_j$ and the tensor structure is the explicit focus, the boldface notation $\mathbf{v}$ is used.

The state space of a quantum system on H is the set $\mathcal{D}(H)$ of positive trace-class operators of unit trace (see Appendix C for the relevant operator-theoretic background). Each fixed-rank stratum $\mathcal{D}_r(H)$ carries the structure of a smooth Banach manifold, fibred over the Grassmannian $\mathbb{G}_r(H)$ — precisely the structure of the Tucker manifold of Chapter 8 specialised to the trace-class Banach space $\mathcal{T}_1^{sa}(H)$.

Section 11.1 establishes the fundamental connection between minimal subspaces and partial traces for pure states. Section 11.2 extends to mixed states and identifies the rank stratification of $\mathcal{D}(H)$ with the Tucker rank stratification. Section 11.3 connects the Tucker rank vector with the entanglement structure of multipartite systems. Section 11.4 develops the Banach manifold structure of $\mathcal{D}_r(H)$ as a special case of the Tucker manifold theory. Section 11.5 applies the best-approximation theory of Chapter 4 to quantum state compression. Section 11.6 contains historical notes.

Throughout this chapter, $H = \bigotimes_{j=1}^{d} H_j$ is a Hilbert tensor product of separable complex Hilbert spaces H_j, and $\mathcal{T}_1^{sa}(H)$ is the real Banach space of self-adjoint trace-class operators on H, equipped with the trace norm $\|\cdot\|_1$ (see Appendix C).

11.1 Pure states and minimal subspaces

The pure state associated to a vector

For a non-zero vector $v \in H$, the *pure state associated to* v is

$$\rho_v \coloneqq \frac{v \otimes J(v)}{\|v\|^2} \in \mathcal{D}(H), \tag{11.1}$$

where $J : H \to H^*$ is the Riesz isomorphism and $v \otimes J(v)$ is the rank-one operator $w \mapsto \langle v, w\rangle v/\|v\|^2$. Two vectors v and λv ($\lambda \in \mathbb{C}^*$) give the same pure state. The map $v \mapsto \rho_v$ thus induces a bijection between the projective space $\mathbb{P}(H)$ and the set $\mathcal{D}_1(H)$ of pure states.

The fundamental connection

The following theorem is the cornerstone of this chapter: it identifies the minimal subspaces of v (in the sense of Definition 4.2.1) with the supports of the partial traces of ρ_v.

Theorem 11.1.1 (Minimal subspaces and partial traces) *Let $v \in H = \bigotimes_{j=1}^d H_j$ with $\|v\| = 1$ and let $\rho_v = v \otimes J(v)$ be the associated pure state. For each $\alpha \in D = \{1, \ldots, d\}$, let $\rho_\alpha \coloneqq \mathrm{Tr}_{\alpha^c}(\rho_v) \in \mathcal{T}_1^{sa}(H_\alpha)$ be the partial trace over all factors except α. Then:*

(a) *ρ_α is a positive trace-class operator with* $\mathrm{Tr}(\rho_\alpha) = 1$ *and*

$$\mathrm{supp}(\rho_\alpha) = U_\alpha^{\min}(v). \tag{11.2}$$

(b) $\mathrm{rank}(\rho_\alpha) = r_\alpha(v)$*, the Tucker α-rank of v.*

(c) *The non-zero eigenvalues of ρ_α are $\lambda_k^{(\alpha)}(v) > 0$, $k = 1, \ldots, r_\alpha(v)$, with $\sum_k \lambda_k^{(\alpha)}(v) = 1$. For $d = 2$, $\lambda_k^{(1)}(v) = \lambda_k^{(2)}(v) = s_k(v)^2$ where $s_k(v)$ are the Schmidt coefficients of v.*

Proof. By the definition of $U_\alpha^{\min}(v)$ (Definition 4.2.1), $U_\alpha^{\min}(v)$ is the smallest subspace $U \subset H_\alpha$ such that $v \in U \otimes_H H_{\alpha^c}$. This is exactly the range of the map $T : H_\alpha^* \to H_{\alpha^c}$ defined by $T(\varphi) = (\varphi \otimes \mathrm{id}_{\alpha^c})(v)$, hence $U_\alpha^{\min}(v) = \overline{\mathrm{Im}(T^*)}$ where $T^* : H_{\alpha^c}^* \to H_\alpha$ is the adjoint.

On the other hand, $\rho_\alpha = \mathrm{Tr}_{\alpha^c}(\rho_v)$ is the operator on H_α characterised by $\langle u, \rho_\alpha w\rangle_{H_\alpha} = \langle u \otimes \mathrm{id}_{\alpha^c}, \rho_v(w \otimes \mathrm{id}_{\alpha^c})\rangle$ for all $u, w \in H_\alpha$. A direct computation using $\rho_v = v \otimes J(v)$ gives

$$\rho_\alpha = \mathrm{Tr}_{\alpha^c}(v \otimes J(v)) = TT^* : H_\alpha \to H_\alpha$$

(the Gram operator of T composed with its adjoint). Hence $\mathrm{Im}(\rho_\alpha) = \overline{\mathrm{Im}(TT^*)} = \overline{\mathrm{Im}(T^*)} = U_\alpha^{\min}(v)$, proving (11.2). Part (b) follows since $\mathrm{rank}(TT^*) = \mathrm{rank}(T) = r_\alpha(v)$. Part (c) follows from the spectral theorem for compact operators (Theorem C.4.1): the eigenvalues of $\rho_\alpha = TT^*$ are the squared singular values of T, and $\mathrm{Tr}(\rho_\alpha) = 1$ gives $\sum_k \lambda_k^{(\alpha)} = 1$. □

Remark 11.1.1 (The $d = 2$ case: Schmidt decomposition) For $d = 2$, Theorem 11.1.1 reduces to the classical Schmidt decomposition. Every $v \in H_1 \otimes_H H_2$ can be written uniquely (up to phases) as

$$v = \sum_{k=1}^{r} s_k\, e_k \otimes f_k, \quad s_1 \geq s_2 \geq \cdots \geq s_r > 0,$$

with $\{e_k\}_{k=1}^r \subset H_1$ and $\{f_k\}_{k=1}^r \subset H_2$ orthonormal, $r = r_1(v) = r_2(v)$. Then $U_1^{\min}(v) = \mathrm{span}\{e_k\}$ and $U_2^{\min}(v) = \mathrm{span}\{f_k\}$, and $\rho_1 = \sum_k s_k^2\, e_k \otimes J(e_k)$, $\rho_2 = \sum_k s_k^2\, f_k \otimes J(f_k)$.

Remark 11.1.2 (Tucker format as the natural language for multipartite states) Theorem 11.1.1 shows that for a pure state ρ_v, the Tucker rank vector $(r_\alpha(v))_{\alpha\in D}$ is an intrinsic property of the state: it is determined by the ranks of the reduced density operators $\{\rho_\alpha\}$, which are physical observables. The Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(H)$ is therefore the natural variational class for quantum systems with prescribed entanglement structure $(r_\alpha)_{\alpha\in D}$.

11.2 Mixed states and the rank stratification

Hilbert–Schmidt operators as tensors

By Theorem C.5.1, $\mathcal{T}_2(H) \cong H \otimes_H H^*$ isometrically as Hilbert spaces. Under this identification, a Hilbert–Schmidt operator $A \in \mathcal{T}_2(H)$ of rank r corresponds to a tensor in $H \otimes_H H^*$ of Tucker rank (r, r) with respect to the bipartition $\{H, H^*\}$. The SVD of A,

$$A = \sum_{k=1}^{r} s_k(A)\, v_k \otimes J(u_k),$$

is the Tucker decomposition of this tensor, with minimal subspaces $U_1^{\min}(A) = \overline{\mathrm{Im}(A^*)} \subset H$ and $U_2^{\min}(A) = J(\overline{\mathrm{Im}(A)}) \subset H^*$.

Mixed states of rank r

For a mixed state $\rho \in \mathcal{D}_r(H)$ with spectral decomposition

$$\rho = \sum_{k=1}^{r} \lambda_k \, v_k \otimes J(v_k), \quad \lambda_k > 0, \quad \sum_{k=1}^{r} \lambda_k = 1, \tag{11.3}$$

the support is $\mathrm{supp}(\rho) = \mathrm{span}\{v_k : k = 1, \dots, r\} \in \mathbb{G}_r(H)$. The positive core $\sigma \coloneqq V^* \rho V \in \mathcal{D}_+(\mathbb{C}^r)$ (where $V = [v_1 | \cdots | v_r] : \mathbb{C}^r \to H$ is the column matrix of eigenvectors) is the finite-dimensional density operator on $\mathbb{C}^r$ with eigenvalues $\lambda_1, \dots, \lambda_r$.

Proposition 11.2.1 (Rank stratification as Tucker decomposition) *The decomposition*

$$\mathcal{D}(H) = \bigsqcup_{r=1}^{\infty} \mathcal{D}_r(H) \cup \mathcal{D}_\infty(H) \tag{11.4}$$

is the rank stratification of $\mathcal{D}(H)$ by Tucker rank in the Banach tensor space $(\mathcal{T}_1^{sa}(H), \|\cdot\|_1)$. Specifically, viewing $\rho \in \mathcal{D}_r(H)$ as a tensor in $H \otimes_\pi H^*$ via $\mathcal{T}_1(H) \cong H \otimes_\pi H^*$:

(i) $r_1(\rho) = r_2(\rho) = r = \mathrm{rank}(\rho)$.
(ii) $U_1^{\min}(\rho) = \mathrm{supp}(\rho) \subset H$ *and* $U_2^{\min}(\rho) = J(\mathrm{supp}(\rho)) \subset H^*$.
(iii) *The trace norm* $\|\rho\|_1 = \mathrm{Tr}(\rho) = 1$ *is the projective tensor norm, satisfying* (Inj) *with equality at all density operators* (*Remark C.6.1*).

Proof. Parts (i)–(ii) follow from the identification $\mathcal{T}_1(H) \cong H \otimes_\pi H^*$ and the definition of Tucker rank (Definition 4.4.1): $r_\alpha(\rho)$ is the dimension of $U_\alpha^{\min}(\rho)$, which for $\alpha = 1$ is the support in H and for $\alpha = 2$ is the support in H^* (both have dimension r by (11.3)). Part (iii) is Remark C.6.1. □

Corollary 11.2.1 (Weak lower semicontinuity of quantum rank) *In the trace-norm topology, rank is weakly lower semicontinuous on $\mathcal{D}(H)$: if $\rho_n \in \mathcal{D}_r(H)$ and $\|\rho_n - \rho\|_1 \to 0$, then* $\mathrm{rank}(\rho) \le r$.

Proof. This is Theorem 4.5.1 applied to the Tucker manifold $\mathcal{D}_r(H) \subset \mathcal{T}_1^{sa}(H)$: the trace norm satisfies (Inj) (Remark C.6.1), and $\mathcal{T}_1^{sa}(H)$ is reflexive (it is a closed subspace of the Banach space $\mathcal{T}_1(H)$, which is reflexive when H is finite-dimensional, and the statement holds in infinite dimensions by the same argument with the weak topology on $\mathcal{T}_1^{sa}(H)$; see Bratteli and Robinson [1987]). □

11.3 Entanglement and minimal subspace dimension

Separability and Tucker rank one

For a multipartite system $H = \bigotimes_{j=1}^d H_j$ and a pure state $\rho_v = v \otimes J(v)$, Theorem 11.1.1 gives a complete characterisation of separability:

Proposition 11.3.1 (Separability and Tucker rank) *Let $v \in \bigotimes_{j=1}^d H_j$ with $\|v\| = 1$ and let $\alpha \subset D$ be a subset of parties. The following are equivalent:*

(a) *v is separable across the bipartition $\alpha|\alpha^c$: $v = u \otimes w$ for some $u \in \bigotimes_{j\in\alpha} H_j$, $w \in \bigotimes_{j\in\alpha^c} H_j$.*
(b) $r_\alpha(v) = 1$.
(c) $\operatorname{rank}(\rho_\alpha) = 1$*, i.e., the reduced state $\rho_\alpha = \operatorname{Tr}_{\alpha^c}(\rho_v)$ is a pure state.*
(d) *$U_\alpha^{\min}(v)$ is one-dimensional.*

Proof. (a)⇔(b): by definition of Tucker rank (Definition 4.4.1). (b)⇔(c): Theorem 11.1.1(b). (b)⇔(d): $r_\alpha(v) = \dim U_\alpha^{\min}(v)$. □

Entanglement entropy and minimal subspaces

The *entanglement entropy* of the bipartition $\alpha|\alpha^c$ is the von Neumann entropy of the reduced state:

$$S_\alpha(v) := -\operatorname{Tr}\big(\rho_\alpha \log \rho_\alpha\big) = -\sum_{k=1}^{r_\alpha(v)} \lambda_k^{(\alpha)}(v) \log \lambda_k^{(\alpha)}(v), \tag{11.5}$$

where $\{\lambda_k^{(\alpha)}(v)\}$ are the eigenvalues of ρ_α from Theorem 11.1.1(c). Since $\lambda_k^{(\alpha)} > 0$ and $\sum_k \lambda_k^{(\alpha)} = 1$, we have $0 \le S_\alpha(v) \le \log r_\alpha(v)$, with:

- $S_\alpha(v) = 0$ iff $r_\alpha(v) = 1$ (separable, no entanglement).
- $S_\alpha(v) = \log r_\alpha(v)$ iff all eigenvalues are equal $\lambda_k = 1/r_\alpha(v)$ (maximally entangled).

The Tucker rank $r_\alpha(v)$ is thus the *Schmidt rank* of the bipartition, which upper-bounds the entanglement entropy: $S_\alpha(v) \le \log r_\alpha(v)$.

Remark 11.3.1 (The Tucker manifold as a variational class for entanglement) The Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(H)$ for $H = \bigotimes_j H_j$ and $\mathfrak{r} = (r_\alpha)_{\alpha\in D}$ is the set of pure states v with prescribed bipartite Schmidt ranks. States on $\mathfrak{M}_{\mathfrak{r}}(H)$ have entanglement entropies bounded by $S_\alpha \le \log r_\alpha$ for all bipartitions α. The Dirac–Frenkel principle of Chapter 10, applied on $\mathfrak{M}_{\mathfrak{r}}(H)$, yields a variational dynamics that preserves the entanglement structure $(r_\alpha)_\alpha$ in the sense that the Schmidt ranks remain equal to $\mathfrak{r}$ along the trajectory.

11.4 The Banach manifold structure of $\mathcal{D}_r(H)$

The geometry of $\mathcal{D}_r(H)$ is a special case of the Tucker manifold theory of Chapter 8, applied to the Banach tensor space $(\mathcal{T}_1^{sa}(H), \|\cdot\|_1) \cong (H \otimes_\pi H^*, \|\cdot\|_\pi)$.

The atlas

Fix $\rho_0 \in \mathcal{D}_r(H)$ and let $S_0 = \operatorname{supp}(\rho_0) \in \mathbb{G}_r(H)$. Choose an isometry $V_0 : \mathbb{C}^r \to H$ with $\operatorname{Im}(V_0) = S_0$, and set $P_0 = V_0 V_0^*$ (orthogonal projection onto S_0). For $A \in \mathcal{L}(\mathbb{C}^r, S_0^\perp)$ with $\|A\| < 1$, define the isometry

$$V_A := V_0(I + A^*A)^{-1/2} + \iota A(I + A^*A)^{-1/2} \in \mathcal{L}(\mathbb{C}^r, H), \quad V_A^* V_A = I_{\mathbb{C}^r}, \tag{11.6}$$

where $\iota : S_0^\perp \hookrightarrow H$ is the inclusion. The chart map

$$\Phi_{S_0} : \mathcal{L}(\mathbb{C}^r, S_0^\perp) \times \mathcal{D}_+(\mathbb{C}^r) \longrightarrow \mathcal{D}_r(H), \quad (A, \sigma) \longmapsto V_A \sigma V_A^*, \tag{11.7}$$

is the quantum analogue of the Tucker chart map $\xi_{\mathbf{v}}$ of Section 8.2: the Grassmann variable $A \in \mathcal{L}(\mathbb{C}^r, S_0^\perp)$ encodes the motion of the support (analogous to $L_\alpha \in \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})))$, and the core variable $\sigma \in \mathcal{D}_+(\mathbb{C}^r)$ encodes the positive density on the support (analogous to the core tensor $C^{(D)} \in \mathbb{R}_*^{\times r_\alpha}$).

Theorem 11.4.1 (Banach manifold structure of $\mathcal{D}_r(H)$, [Acevedo and Falcó, 2026, Theorem 3.5]) *For each $r \in \mathbb{N}$, the set $\mathcal{D}_r(H)$ admits a smooth $\mathcal{C}^\infty$-Banach manifold structure in the sense of Lang Lang [1999]. Specifically:*

(i) *For each $\rho_0 \in \mathcal{D}_r(H)$, the map Φ_{S_0} in (11.7) is a $\mathcal{C}^\infty$ diffeomorphism from a neighbourhood of $(0, V_0^* \rho_0 V_0)$ in $\mathcal{L}(\mathbb{C}^r, S_0^\perp) \times \mathcal{D}_+(\mathbb{C}^r)$ onto a neighbourhood of ρ_0 in $\mathcal{D}_r(H)$.*
(ii) *The transition maps between overlapping charts are $\mathcal{C}^\infty$ (analytic over $\mathbb{C}$).*
(iii) *The tangent space at $\rho \in \mathcal{D}_r(H)$ is*

$$T_\rho \mathcal{D}_r(H) = \{\dot\rho \in \mathcal{T}_1^{sa}(H) : (I - P)\dot\rho(I - P) = 0,\ \operatorname{Tr}(\dot\rho) = 0\}, \tag{11.8}$$

where $P = P_{\operatorname{supp}(\rho)}$ is the orthogonal projection onto $\operatorname{supp}(\rho)$. *This is a complemented closed subspace of $\mathcal{T}_1^{sa}(H)$.*
(iv) *The inclusion $\iota : \mathcal{D}_r(H) \hookrightarrow \mathcal{T}_1^{sa}(H)$ is a smooth immersion.*

Proof. We prove each part in turn.

Part (i): The chart map is a $\mathcal{C}^\infty$ diffeomorphism.
Fix $\rho_0 \in \mathcal{D}_r(H)$, $S_0 = \operatorname{supp}(\rho_0)$, and an isometry $V_0 : \mathbb{C}^r \to H$ with $\operatorname{Im}(V_0) = S_0$.

Step 1: Φ_{S_0} is smooth. For $\|A\| < 1$, the operator $I_{\mathbb{C}^r} + A^*A$ is invertible in $\mathcal{L}(\mathbb{C}^r, \mathbb{C}^r)$ (since $\mathbb{C}^r$ is finite-dimensional and $A^*A \geq 0$). The map $A \mapsto (I + A^*A)^{-1/2}$ is real-analytic on operator norm balls, hence $A \mapsto V_A = (V_0 + \iota A)(I + A^*A)^{-1/2}$ is $\mathcal{C}^\infty$ by composition. Since $\sigma \mapsto V_A \sigma V_A^*$ is linear in σ for fixed A, the full map $(A, \sigma) \mapsto V_A \sigma V_A^*$ is $\mathcal{C}^\infty$.

Step 2: Values in $\mathcal{D}_r(H)$. For any (A, σ) in the domain: $V_A \sigma V_A^* \geq 0$ (since $\sigma > 0$ and V_A is an isometry); $\operatorname{Tr}(V_A \sigma V_A^*) = \operatorname{Tr}(\sigma V_A^* V_A) = \operatorname{Tr}(\sigma) = 1$ (since $V_A^* V_A = I_{\mathbb{C}^r}$); and $\operatorname{rank}(V_A \sigma V_A^*) = r$ (since $\sigma \succ 0$ and V_A is injective). Hence $\Phi_{S_0}(A, \sigma) \in \mathcal{D}_r(H)$.

Step 3: Local bijectivity and smooth inverse. At $A = 0$: $V_0 = V_0(I)^{-1/2} = V_0$ and $\Phi_{S_0}(0, \sigma) = V_0 \sigma V_0^*$; in particular $\Phi_{S_0}(0, V_0^* \rho_0 V_0) = V_0(V_0^* \rho_0 V_0)V_0^* = \rho_0$. For ρ' near ρ_0 in $\mathcal{D}_r(H)$, $\mathrm{supp}(\rho')$ is near S_0 in $\mathbb{G}_r(H)$, so there is a unique small A with $\mathrm{Im}(V_A) = \mathrm{supp}(\rho')$. The inverse is $\Phi_{S_0}^{-1}(\rho') = (A(\rho'), V_A^* \rho' V_A)$, where $A(\rho') = \chi_{S_0, S_0^\perp}(\mathrm{supp}(\rho'))$ is the Grassmannian chart map. Since the Grassmannian chart is $\mathcal{C}^\infty$ (Theorem 2.3.1) and $\rho' \mapsto V_A^* \rho' V_A$ is linear in ρ', the inverse is $\mathcal{C}^\infty$.

Part (ii): Smooth transition maps.

Let $\rho_0, \tilde{\rho}_0 \in \mathcal{D}_r(H)$ with chart domains $\mathcal{B}(\rho_0)$ and $\mathcal{B}(\tilde{\rho}_0)$ overlapping. The transition map $\Phi_{\tilde{S}_0}^{-1} \circ \Phi_{S_0}$ is the composition:

$$(A, \sigma) \xmapsto{\Phi_{S_0}} \rho \xmapsto{\Phi_{\tilde{S}_0}^{-1}} (\tilde{A}, \tilde{\sigma}),$$

where $\tilde{A} = \chi_{\tilde{S}_0, \tilde{S}_0^\perp}(\mathrm{supp}(\rho))$ is smooth in ρ (Grassmannian transition map, Theorem 2.3.1), and $\tilde{\sigma} = \tilde{V}_{\tilde{A}}^* \rho \tilde{V}_{\tilde{A}}$ is smooth in both $\tilde{A}$ and ρ. The composition is therefore $\mathcal{C}^\infty$.

Part (iii): Tangent space characterisation.

Let $t \mapsto \rho(t) = \Phi_{S_0}(A(t), \sigma(t))$ be a $\mathcal{C}^1$ curve with $A(0) = 0$ and $\sigma(0) = V_0^* \rho_0 V_0$. Differentiating $\rho(t) = V_{A(t)} \sigma(t) V_{A(t)}^*$ at $t = 0$:

$$\dot{\rho}(0) = \dot{V}_0 \sigma_0 V_0^* + V_0 \dot{\sigma}_0 V_0^* + V_0 \sigma_0 \dot{V}_0^*,$$

where $\dot{V}_0 = \frac{d}{dt} V_{A(t)}|_{t=0}$. Since $V_{A(t)}^* V_{A(t)} = I$ implies $\dot{V}_0^* V_0 + V_0^* \dot{V}_0 = 0$ (so $V_0^* \dot{V}_0$ is skew-Hermitian), and $\mathrm{Im}(V_0) = S_0$ (so $\dot{V}_0 \in \mathcal{L}(\mathbb{C}^r, S_0^\perp)$ by the Grassmannian chart), one computes:

$$(I - P)\dot{\rho}(0)(I - P) = (I - P)\dot{V}_0 \sigma_0 V_0^*(I - P) = 0,$$

since $V_0^*(I - P) = 0$ (the range of V_0 is S_0). Also $\mathrm{Tr}(\dot{\rho}(0)) = \mathrm{Tr}(\dot{\sigma}_0) + 2\mathrm{Re}\,\mathrm{Tr}(\sigma_0 V_0^* \dot{V}_0) = 0$ if $\mathrm{Tr}(\dot{\sigma}_0) = 0$ and $V_0^* \dot{V}_0$ is skew-Hermitian. This shows the inclusion $\dot{\rho}(0) \in T_\rho \mathcal{D}_r(H) \supset$ the set in (11.8). The reverse inclusion follows by constructing explicit curves realising each tangent vector of the given form (Step 3 of the proof in Acevedo and Falcó [2026]).

Part (iv): Smooth immersion.

The differential of $\iota : \mathcal{D}_r(H) \hookrightarrow \mathcal{T}_1^{sa}(H)$ at ρ is the map $d\iota_\rho : T_\rho \mathcal{D}_r(H) \to \mathcal{T}_1^{sa}(H)$, $d\iota_\rho(\dot{\rho}) = \dot{\rho}$, which is injective by definition. The image $T_\rho \mathcal{D}_r(H)$ is a complemented subspace of $\mathcal{T}_1^{sa}(H)$ by the explicit bounded projection $\hat{P}_\rho$ of Proposition 11.4.1 below, satisfying $\|\hat{P}_\rho\| \le 4 < \infty$. Hence $d\iota_\rho$ has closed split image (Proposition 2.2.1), i.e., ι is an immersion. $\square$

The tangent projector

Proposition 11.4.1 (Bounded projector onto $T_\rho\mathcal{D}_r(H)$, [Acevedo and Falcó, 2026, Proposition 3.4]) *Let $\rho \in \mathcal{D}_r(H)$ and $P = P_{\mathrm{supp}(\rho)}$. The map*

$$\hat{P}_\rho : \mathcal{T}_1^{sa}(H) \longrightarrow T_\rho\mathcal{D}_r(H), \quad \hat{P}_\rho(X) \coloneqq PXP + PX(I-P) + (I-P)XP - \frac{\mathrm{Tr}(PXP)}{r}P, \tag{11.9}$$

is a bounded linear projection with $\|\hat{P}_\rho\|_{\mathcal{L}(\mathcal{T}_1^{sa}(H))} \leq 4$, and $\hat{P}_\rho(X) = X$ for all $X \in T_\rho\mathcal{D}_r(H)$.

Proof. Write $X \in \mathcal{T}_1^{sa}(H)$ in block form with respect to $H = S \oplus S^\perp$:

$$X = \begin{pmatrix} X_{11} & X_{12}^* \\ X_{12} & X_{22} \end{pmatrix}, \quad X_{11} \in \mathcal{T}_1^{sa}(S),\ X_{12} \in \mathcal{T}_1(S, S^\perp),\ X_{22} \in \mathcal{T}_1^{sa}(S^\perp).$$

By direct computation using $P^2 = P = P^*$:

$$PXP = \begin{pmatrix} X_{11} & 0 \\ 0 & 0 \end{pmatrix}, \qquad PX(I-P) = \begin{pmatrix} 0 & X_{12}^* \\ 0 & 0 \end{pmatrix},$$
$$(I-P)XP = \begin{pmatrix} 0 & 0 \\ X_{12} & 0 \end{pmatrix}, \qquad \frac{\mathrm{Tr}(PXP)}{r}P = \frac{\mathrm{Tr}(X_{11})}{r}\begin{pmatrix} I_S & 0 \\ 0 & 0 \end{pmatrix}.$$

Hence

$$\hat{P}_\rho(X) = \begin{pmatrix} X_{11} - \frac{\mathrm{Tr}(X_{11})}{r} I_S & X_{12}^* \\ X_{12} & 0 \end{pmatrix}. \tag{11.10}$$

Idempotency: Applying $\hat{P}_\rho$ to (11.10): the (S,S) block is $X_{11} - \frac{\mathrm{Tr}(X_{11})}{r} I_S - \frac{1}{r}\mathrm{Tr}(X_{11} - \frac{\mathrm{Tr}(X_{11})}{r} I_S) I_S = X_{11} - \frac{\mathrm{Tr}(X_{11})}{r} I_S$ (since $\mathrm{Tr}(X_{11} - \frac{\mathrm{Tr}(X_{11})}{r} I_S) = \mathrm{Tr}(X_{11}) - \mathrm{Tr}(X_{11}) = 0$), and the off-diagonal blocks are unchanged. So $\hat{P}_\rho^2 = \hat{P}_\rho$.

Image: From (11.10), $\hat{P}_\rho(X)$ has $(S^\perp, S^\perp)$ block zero and $\mathrm{Tr}(\hat{P}_\rho(X)) = \mathrm{Tr}(X_{11} - \frac{\mathrm{Tr}(X_{11})}{r} I_S) + 0 = 0$. So $\hat{P}_\rho(X) \in T_\rho\mathcal{D}_r(H)$.

$\hat{P}_\rho = \mathrm{Id}$ *on $T_\rho\mathcal{D}_r(H)$:* If $Y \in T_\rho\mathcal{D}_r(H)$, its block form is $Y_{22} = 0$ and $\mathrm{Tr}(Y_{11}) = 0$ (by (11.8)). Then $\frac{\mathrm{Tr}(PYP)}{r} = \frac{\mathrm{Tr}(Y_{11})}{r} = 0$, so $\hat{P}_\rho(Y) = Y$.

Norm bound: Using the ideal property $\|BAC\|_1 \leq \|B\|\|A\|_1\|C\|$:

$$\begin{aligned} \|\hat{P}_\rho(X)\|_1 &\leq \|PXP\|_1 + \|PX(I-P)\|_1 + \|(I-P)XP\|_1 + \tfrac{|\mathrm{Tr}(PXP)|}{r}\|P\|_1 \\ &\leq \|X\|_1 + \|X\|_1 + \|X\|_1 + \tfrac{\|PXP\|_1}{r} \cdot r \\ &\leq 4\|X\|_1, \end{aligned}$$

where we used $|\mathrm{Tr}(PXP)| \leq \|PXP\|_1 \leq \|X\|_1$ and $\|P\|_1 = \mathrm{Tr}(P) = r$. □

Remark 11.4.1 (Comparison with the Tucker projector) The projector $\hat{P}_\rho$ in (11.9) is the quantum analogue of the metric projection $\mathcal{P}_{\mathbf{v}_r}$ onto $Z^{(D)}(\mathbf{v}_r)$ from Chapter 10. The explicit formula in terms of blocks with respect to $H = S \oplus S^\perp$ is the quantum version of the tangent space formula (8.8): the PXP block corresponds to the core variation $\dot{C}^{(D)}$, and the off-diagonal blocks correspond to the subspace velocity $\dot{u}^{(\alpha)}_{i_\alpha} \in W^{\min}_\alpha(\mathbf{v}_r)$. The trace correction $-\frac{\mathrm{Tr}(PXP)}{r}P$ is the additional constraint $\mathrm{Tr}(\dot{\rho}) = 0$ that appears because $\mathcal{D}_r(H)$ lies in the affine hyperplane $\{\mathrm{Tr} = 1\}$, not in the full Banach space.

The fibre bundle structure

The support map $\pi : \mathcal{D}_r(H) \to \mathbb{G}_r(H)$, $\pi(\rho) = \mathrm{supp}(\rho)$, is the quantum analogue of the rank map $\varrho_{\mathfrak{r}}$ of Section 8.3.

Theorem 11.4.2 (Fibre bundle, [Acevedo and Falcó, 2026, Theorem 3.8]) *The support map $\pi : \mathcal{D}_r(H) \to \mathbb{G}_r(H)$ is a smooth fibre bundle with typical fibre $\mathcal{D}_+(\mathbb{C}^r)$, a smooth manifold of real dimension $r^2 - 1$. The structure group is $U(r)$ (the unitary group), acting on $\mathcal{D}_+(\mathbb{C}^r)$ by $U \cdot \sigma = U\sigma U^*$.*

This is the quantum specialisation of Theorem 8.3.1(ii) and Propositions 8.6.1–8.6.2: the structure group reduces from $\mathrm{GL}(r)\mathbb{R}$ to $U(r)$ because the core variable $\sigma \in \mathcal{D}_+(\mathbb{C}^r)$ transforms by unitary conjugation (preserving positivity and trace), rather than by an arbitrary invertible change of basis.

11.5 Best approximation and quantum state compression

The best-approximation theorem (Theorem 4.5.2) applied to the quantum setting gives the mathematical foundation for *low-rank approximation of quantum states.*

Theorem 11.5.1 (Existence of best low-rank approximation) *Let $\rho \in \mathcal{D}(H)$ and $r \in \mathbb{N}$. There exists $\rho_r \in \bigcup_{k=1}^r \mathcal{D}_k(H)$ such that*

$$\|\rho - \rho_r\|_1 = \inf_{\sigma \in \bigcup_{k=1}^r \mathcal{D}_k(H)} \|\rho - \sigma\|_1. \tag{11.11}$$

Proof. The set $\bigcup_{k=1}^r \mathcal{D}_k(H)$ is weakly closed in $\mathcal{T}_1^{sa}(H)$ by Corollary 11.2.1 (weak lower semicontinuity of rank). The trace-class space $\mathcal{T}_1^{sa}(H)$ is reflexive when H is finite-dimensional; in infinite dimensions, the result follows from the explicit formula: the best rank-r approximation of ρ in trace norm is given by the *spectral truncation*

$$\rho_r := \frac{1}{\sum_{k=1}^r \lambda_k} \sum_{k=1}^r \lambda_k\, v_k \otimes J(v_k), \tag{11.12}$$

where $\lambda_1 \geq \lambda_2 \geq \cdots$ are the eigenvalues of ρ in decreasing order and $\{v_k\}$ the corresponding eigenvectors. The error is $\|\rho - \rho_r\|_1 = 2\sum_{k>r} \lambda_k$ (Ky Fan's theorem). □

Remark 11.5.1 (Spectral truncation as Tucker approximation) The spectral truncation (11.12) is the quantum analogue of the HOSVD (Higher-Order SVD) truncation for tensor formats (Algorithm D.2 in Appendix D): in both cases, the best rank-r approximation in the ambient norm is obtained by projecting onto the dominant eigenspaces/singular subspaces. For tensors with $d \geq 3$, the analogue of (11.12) gives a near-best Tucker approximation (quasi-optimality with constant $\sqrt{d}$), not an exact best approximation; see Appendix D.2.

11.6 Historical notes

Schmidt decomposition and entanglement.

The Schmidt decomposition for bipartite systems was introduced by Schmidt (1907) in the context of integral equations, long before its quantum-mechanical interpretation. Its role in quantum entanglement was recognised in the 1990s with the development of quantum information theory. The generalisation to multipartite systems via minimal subspaces is the contribution of Falcó and Hackbusch Falcó and Hackbusch [2012].

Tucker format and quantum many-body physics.

The connection between the Tucker format and tree tensor networks (TTN) was developed in the quantum chemistry and condensed matter literature under the name MCTDH (Multi-Configuration Time-Dependent Hartree) and its matrix product state (MPS/DMRG) variants. White's DMRG algorithm White [1992] is algorithmically the DMRG of Appendix D, implemented for quantum spin chains. The mathematical equivalence between Tucker formats and tensor network states was clarified in Hackbusch [2019].

Banach manifold structure of $\mathcal{D}_r(H)$.

The explicit Grassmann-Banach manifold model for fixed-rank density operators in the trace-norm topology, with the support-core charts and bounded tangent projectors, was developed in the companion paper Acevedo and Falcó [2026]. Earlier work in the Hilbert–Schmidt (T_2) setting includes Koch and Lubich [2007]; the move to the trace-class (T_1) setting is motivated by the operational interpretation of the trace norm as the natural metric for state distinguishability (Remark C.6.1).

Chapter 12
The Dirac–Frenkel Principle for Open Quantum Systems

Chapter 10 established the Dirac–Frenkel variational principle for abstract Cauchy problems on tensor Banach manifolds. Chapter 11 identified the fixed-rank stratum $\mathcal{D}_r(H)$ as a smooth Banach manifold within the trace-class Banach space $\mathcal{T}_1^{sa}(H)$, inheriting all the geometric properties developed in Chapters 8–9. This chapter combines these two lines to obtain the Dirac–Frenkel principle for quantum master equations on $\mathcal{T}_1^{sa}(H)$: the Lindblad (GKSL) equation for open quantum systems.

The Lindblad equation describes the evolution of the state $\rho(t) \in \mathcal{D}(H)$ of a quantum system coupled to an environment. Its generator $\mathcal{L}$ maps $\mathcal{T}_1^{sa}(H)$ to itself and preserves $\mathcal{D}(H)$. Since exact simulation of $\rho(t)$ in infinite-dimensional systems is generally intractable, one seeks a reduced dynamics on the finite-rank manifold $\mathcal{D}_r(H)$, obtained by projecting the Lindblad vector field onto the tangent space of $\mathcal{D}_r(H)$.

Section 12.1 formulates the Lindblad equation as an ODE on $\mathcal{T}_1^{sa}(H)$ and identifies its vector field. Section 12.2 develops the tangent space of $\mathcal{D}_r(H)$ via minimal subspaces and establishes the bounded projector. Section 12.3 states and proves the main well-posedness theorem. Section 12.4 analyses which structural properties (trace, rank, positivity) are preserved by the projected dynamics. Section 12.5 derives the kinematic equations in factorised coordinates, recovering the equations of Chapter 10 as a special case. Section 12.6 contains historical notes.

12.1 The Lindblad equation on $\mathcal{T}_1^{sa}(H)$

The GKSL generator

The most general Markovian master equation for the state $\rho(t) \in \mathcal{D}(H)$ of an open quantum system is the *Gorini–Kossakowski–Sudarshan–Lindblad* (GKSL) equation:

$$\dot{\rho}(t) = \mathcal{L}(\rho(t)) := -i[H, \rho(t)] + \sum_{j \in J} \Big(L_j \rho(t) L_j^* - \tfrac{1}{2}\{L_j^* L_j, \rho(t)\} \Big), \qquad (12.1)$$

where:

- $H = H^* \in \mathcal{L}(H)$ is the *Hamiltonian* (self-adjoint, bounded);
- $\{L_j\}_{j \in J} \subset \mathcal{L}(H)$ are the *Lindblad operators* (noise operators), with J a finite or countable index set;
- $[H, \rho] = H\rho - \rho H$ is the commutator and $\{A, B\} = AB + BA$ is the anti-commutator.

When J is finite and all $L_j \in \mathcal{L}(H)$ are bounded, $\mathcal{L}$ is a bounded operator on $\mathcal{T}_1(H)$. The solution to (12.1) is a strongly continuous one-parameter semi-group $(T_t)_{t \geq 0}$ on $\mathcal{T}_1(H)$ that is *completely positive and trace preserving* (CPTP): $T_t(\mathcal{D}(H)) \subset \mathcal{D}(H)$ for all $t \geq 0$.

Remark 12.1.1 (The Lindblad equation as a tensor Banach ODE) The equation (12.1) is a special case of the abstract Cauchy problem (10.1) of Chapter 10, with ambient Banach space $\mathbf{V}_{D,\|\cdot\|_D} = \mathcal{T}_1^{sa}(H)$ and vector field $\mathbf{F}(t, \rho) = \mathcal{L}(\rho)$ (time-independent in the homogeneous case). The trace norm $\|\cdot\|_1$ plays the role of $\|\cdot\|_D$, and the identification $\mathcal{T}_1^{sa}(H) \cong H \otimes_\pi H^*$ (Remark C.6.1) makes $(\mathcal{T}_1^{sa}(H), \|\cdot\|_1)$ a Banach tensor space with $d = 2$, $V_1 = H$, $V_2 = H^*$, satisfying condition (Inj) with the projective norm.

Assumption 12.1.1 (Regularity of $\mathcal{L}$) Throughout this chapter we assume that:

(R1) $H \in \mathcal{L}(H)$ is bounded and J is finite, so $\mathcal{L} : \mathcal{T}_1^{sa}(H) \to \mathcal{T}_1^{sa}(H)$ is a bounded linear operator.
(R2) The CPTP semigroup $(T_t)_{t \geq 0}$ generated by $\mathcal{L}$ satisfies $T_t(\mathcal{D}(H)) \subset \mathcal{D}(H)$.
(R3) For each $\rho_0 \in \mathcal{D}_r(H)$, the restriction $\mathcal{L}|_{\mathcal{D}_r(H)}$ is locally Lipschitz in $\|\cdot\|_1$.

Under Assumption 12.1.1, the full Lindblad equation has a unique global solution $\rho : [0, \infty) \to \mathcal{D}(H)$.

Remark 12.1.2 (Unbounded generators) Assumption 12.1.1(R1) excludes unbounded Lindblad operators (e.g., position or momentum operators in infinite dimensions). The extension to unbounded generators requires domain-adapted fixed-rank strata and is the subject of Open Problem 13.5.2.

12.2 Tangent space of $\mathcal{D}_r(H)$ via minimal subspaces

The tangent space characterisation of Theorem 11.4.1(iii) takes the following explicit form, connecting with the Tucker tangent space of Chapter 8.

Proposition 12.2.1 (Tangent space decomposition) *Let $\rho \in \mathcal{D}_r(H)$, $S = \operatorname{supp}(\rho)$, $P = P_S$. With respect to the orthogonal decomposition $H = S \oplus S^\perp$, every $\dot{\rho} \in T_\rho \mathcal{D}_r(H)$ has a unique block form*

$$\dot{\rho} = \begin{pmatrix} C & B^* \\ B & 0 \end{pmatrix} \in \mathcal{T}_1^{sa}(H), \quad C \in \mathcal{T}_1^{sa}(S),\ \mathrm{Tr}(C) = 0,\ B \in \mathcal{T}_1(S, S^\perp), \tag{12.2}$$

where $\mathcal{T}_1(S, S^\perp)$ denotes the trace-class operators from S to $S^\perp$ (automatically finite-rank since $\dim S = r$*). The decomposition $T_\rho \mathcal{D}_r(H) \cong \{C \in \mathcal{T}_1^{sa}(S) : \mathrm{Tr}(C) = 0\} \oplus \mathcal{T}_1(S, S^\perp)$ identifies:*

- *The* core direction *C: variation of ρ within fixed support S — the quantum analogue of $\dot{C}^{(D)}$ in* (10.10)*.*
- *The* support direction *B: variation of the support subspace S — the quantum analogue of $\dot{u}_{i_\alpha}^{(\alpha)} \in W_\alpha^{\min}(\mathbf{v}_r)$ in* (10.10)*.*

Proof. This is Lemma 3.1 and Corollary 3.2 of Acevedo and Falcó [2026], proved by differentiating the chart curve $(A(t), \sigma(t)) \mapsto V_{A(t)}\sigma(t)V_{A(t)}^*$ at $t = 0$ in local chart coordinates (11.7). The block structure follows from the orthogonality of $H = S \oplus S^\perp$ and the chain rule, exactly as in the proof of Proposition 8.4.1. □

Remark 12.2.1 (The trace constraint) The condition $\mathrm{Tr}(\dot{\rho}) = 0$ in (12.2) is the infinitesimal form of $\mathrm{Tr}(\rho(t)) = 1$. It corresponds to the fact that $\mathcal{D}_r(H)$ lies in the affine hyperplane $\{\mathrm{Tr} = 1\} \subset \mathcal{T}_1^{sa}(H)$, which is an additional constraint not present in the Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \subset \mathbf{V}_{D,\|\cdot\|_D}$. In the kinematic equations of Section 12.5, this constraint is enforced automatically by the trace-removal term in the projector $\hat{P}_\rho$ (Proposition 11.4.1).

12.3 The Dirac–Frenkel principle for Lindblad dynamics

Definition 12.3.1 (Dirac–Frenkel projected Lindblad dynamics) The *Dirac–Frenkel projected Lindblad dynamics* on $\mathcal{D}_r(H)$ is the ODE

$$\dot{\rho}(t) = \hat{P}_{\rho(t)}\big(\mathcal{L}(\rho(t))\big), \qquad \rho(0) = \rho_0 \in \mathcal{D}_r(H), \tag{12.3}$$

where $\hat{P}_\rho : \mathcal{T}_1^{sa}(H) \to T_\rho \mathcal{D}_r(H)$ is the bounded projector of Proposition 11.4.1.

The equation (12.3) is the quantum specialisation of the Dirac–Frenkel equation (10.5) of Chapter 10, with $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ replaced by $\mathcal{D}_r(H)$, $\mathbf{V}_{D,\|\cdot\|_D}$ by $\mathcal{T}_1^{sa}(H)$, $\mathbf{F}$ by $\mathcal{L}$, and $\mathcal{P}_{\mathbf{v}_r}$ by $\hat{P}_\rho$.

Theorem 12.3.1 (Local well-posedness, [Acevedo and Falcó, 2026, Theorem 4.6]) *Under Assumption 12.1.1, for every $\rho_0 \in \mathcal{D}_r(H)$ there exist $T > 0$ and a unique $\mathcal{C}^1$-curve*

$$\rho : [0, T) \longrightarrow \mathcal{D}_r(H)$$

satisfying (12.3) *with $\rho(0) = \rho_0$. The solution depends continuously on ρ_0 in the trace norm.*

Proof. We verify the three conditions of Theorem A.2.1 (Picard–Lindelöf on Banach manifolds, Appendix A).

Step 1: $\mathcal{D}_r(H)$ is a $\mathcal{C}^\infty$ Banach manifold.

This is Theorem 11.4.1. The model space for a chart at ρ_0 is $\mathcal{L}(\mathbb{C}^r, S_0^\perp) \times \mathcal{T}_1^{sa}(\mathbb{C}^r)$, a Banach space (infinite-dimensional in the first factor, finite-dimensional in the second).

Step 2: The reduced vector field is locally Lipschitz.

In the chart Φ_{S_0} at ρ_0, the reduced vector field in coordinates is

$$F(A,\sigma) := d\Phi_{S_0}^{-1}\big(\Phi_{S_0}(A,\sigma)\big)\big[\hat{P}_{\rho(A,\sigma)}\big(\mathcal{L}(\rho(A,\sigma))\big)\big],$$

where $\rho(A,\sigma) = V_A \sigma V_A^*$.

We verify local Lipschitz continuity in three sub-steps:

(a) $\mathcal{L}(\rho)$ is Lipschitz in $\|\cdot\|_1$. By Assumption 12.1.1(R1), $\mathcal{L}$ is bounded on $\mathcal{T}_1(H)$: $\|\mathcal{L}(\rho) - \mathcal{L}(\rho')\|_1 \le \|\mathcal{L}\|_{\mathcal{L}(\mathcal{T}_1)} \|\rho - \rho'\|_1$.

(b) $\rho \mapsto \hat{P}_\rho$ is locally Lipschitz. Writing $\hat{P}_\rho(X) = P_{\mathrm{supp}(\rho)} X P_{\mathrm{supp}(\rho)} + \ldots$, the dependence on ρ enters only through the orthogonal projection $P = P_{\mathrm{supp}(\rho)}$. In the chart, $\mathrm{supp}(\rho(A,\sigma)) = \mathrm{Im}(V_A)$ and $P = V_A V_A^*$. The map $A \mapsto V_A V_A^*$ is $\mathcal{C}^\infty$ by Step 1 (Theorem 11.4.1(i)), hence Lipschitz on bounded sets.

(c) The chart map is Lipschitz. $d\Phi_{S_0}^{-1}$ is bounded on bounded sets since Φ_{S_0} is a $\mathcal{C}^\infty$ diffeomorphism with bounded differential.

Combining (a)–(c) by the chain rule: F is locally Lipschitz on a neighbourhood of $(0, V_0^* \rho_0 V_0)$ in $\mathcal{L}(\mathbb{C}^r, S_0^\perp) \times \mathcal{T}_1^{sa}(\mathbb{C}^r)$.

Step 3: Local existence and uniqueness by Picard–Lindelöf.

By Theorem B.3.1 (Picard–Lindelöf in Banach spaces), applied to the coordinate ODE $\dot{z}(t) = F(z(t))$ in the Banach space $\mathcal{L}(\mathbb{C}^r, S_0^\perp) \times \mathcal{T}_1^{sa}(\mathbb{C}^r)$, there exist $T > 0$ and a unique $\mathcal{C}^1$ solution $z : [0,T) \to \mathcal{L}(\mathbb{C}^r, S_0^\perp) \times \mathcal{T}_1^{sa}(\mathbb{C}^r)$ with $z(0) = (0, V_0^* \rho_0 V_0)$. Pushing forward by Φ_{S_0}, we obtain $\rho(t) = \Phi_{S_0}(z(t))$, a $\mathcal{C}^1$ curve in $\mathcal{D}_r(H)$ satisfying (12.3).

Step 4: Continuous dependence on ρ_0.

Continuous dependence of $z(t)$ on $z(0)$ follows from Theorem B.3.1. Transported through the $\mathcal{C}^\infty$ chart map Φ_{S_0}, this gives continuous dependence of $\rho(t)$ on ρ_0 in the trace norm. □

Remark 12.3.1 (Choice of projector) The projector $\hat{P}_\rho$ in Definition 12.3.1 is the explicit block projector of Proposition 11.4.1. Other choices of bounded projector $\pi_\rho : \mathcal{T}_1^{sa}(H) \to T_\rho \mathcal{D}_r(H)$ lead to equivalent well-posedness results; the choice affects the trajectory $\rho(t)$ but not the structural preservation properties. In particular, in the Hilbert–Schmidt setting one may use the orthogonal projection in $\mathcal{T}_2(H)$, which gives the TDVP (time-dependent variational principle) projection of Le Bris and Rouchon [2013].

12.4 Structure preservation

Proposition 12.4.1 (Trace, rank, and positivity) *Let $\rho : [0,T) \to \mathcal{D}_r(H)$ be the solution of* (12.3). *Then for all $t \in [0,T)$:*

(i) Trace preservation: $\mathrm{Tr}(\rho(t)) = 1$.
(ii) Rank preservation: $\mathrm{rank}(\rho(t)) = r$.
(iii) Positivity: $\rho(t) \geq 0$.

Proof. (i): Differentiating, $\frac{d}{dt}\mathrm{Tr}(\rho(t)) = \mathrm{Tr}(\dot\rho(t)) = \mathrm{Tr}(\hat P_{\rho(t)}(\mathcal{L}(\rho(t)))) = 0$, since $\hat P_\rho(X) \in T_\rho \mathcal{D}_r(H)$ satisfies $\mathrm{Tr}(\hat P_\rho(X)) = 0$ by (12.2).

(ii) and (iii): The solution $\rho(t)$ is a curve in $\mathcal{D}_r(H)$ by Theorem 12.3.1. Since $\mathcal{D}_r(H) \subset \mathcal{D}(H)$, every point satisfies $\rho(t) \geq 0$ and $\mathrm{rank}(\rho(t)) = r$. □

Remark 12.4.1 (Complete positivity is not guaranteed) Proposition 12.4.1 guarantees that $\rho(t) \in \mathcal{D}_r(H) \subset \mathcal{D}(H)$ (positive, trace one, rank r) along the trajectory. It does *not* guarantee that the reduced flow $\rho_0 \mapsto \rho(t)$ is a CPTP map (quantum channel). The projected dynamics is a nonlinear ODE on $\mathcal{D}_r(H)$; the linearity and complete positivity of the full Lindblad semigroup $(T_t)_{t\geq 0}$ are not inherited by the projected flow. Constructing a CPTP-preserving rank-r approximation requires additional structure, such as Kraus-form parametrisations; see Open Problem 13.5.3.

12.5 Kinematic equations in factorised coordinates

Stiefel-core parametrisation

Every $\rho \in \mathcal{D}_r(H)$ admits a factorisation

$$\rho = V\sigma V^*, \quad V \in \mathrm{St}(r,H),\ \sigma \in \mathcal{D}_+(\mathbb{C}^r), \tag{12.4}$$

where $\mathrm{St}(r,H) = \{V \in \mathcal{L}(\mathbb{C}^r, H) : V^*V = I_{\mathbb{C}^r}\}$ is the Stiefel manifold of isometries. The parametrisation is unique up to the unitary gauge action $(V,\sigma) \sim (VU, U^*\sigma U)$ for $U \in U(r)$.

Differentiating (12.4):

$$\dot\rho = \dot V\sigma V^* + V\dot\sigma V^* + V\sigma\dot V^*. \tag{12.5}$$

The Stiefel constraint $V^*V = I$ differentiates to give $V^*\dot V + \dot V^* V = 0$, so $V^*\dot V$ is skew-Hermitian. Imposing the *horizontal (parallel transport) gauge*

$$V^*\dot V = 0 \quad \Longleftrightarrow \quad \dot V = (I - VV^*)\dot V, \tag{12.6}$$

the variation $\dot V$ lies in the complement of $\mathrm{Im}(V) = S$, i.e., $\dot V(\mathbb{C}^r) \subset S^\perp$.

Derivation of the kinematic system

Under the gauge condition (12.6), projecting the Lindblad vector field $\mathcal{L}(\rho)$ onto $T_\rho\mathcal{D}_r(H)$ via $\hat{P}_\rho$ gives, using the block formula (11.9):

Equation for the core $\sigma(t)$**.** Multiplying $\dot{\rho} = \hat{P}_\rho(\mathcal{L}(\rho))$ on the left by V^* and on the right by V, and using $V^*\dot{V} = 0$:

$$\dot{\sigma}(t) = V(t)^*\, \hat{P}_{\rho(t)}\big(\mathcal{L}(\rho(t))\big)\, V(t). \tag{12.7}$$

Since $\hat{P}_\rho(X) = PXP + \text{off-diagonal} - \frac{\mathrm{Tr}(PXP)}{r}P$ and $P = VV^*$, this simplifies to

$$\dot{\sigma}(t) = V^*\mathcal{L}(\rho)V - \frac{\mathrm{Tr}(V^*\mathcal{L}(\rho)V)}{r}\, I_{\mathbb{C}^r} = V^*\mathcal{L}(\rho)V, \tag{12.8}$$

where the last equality uses $\mathrm{Tr}(\mathcal{L}(\rho)) = 0$ (trace preservation of the full Lindblad semigroup), which gives $\mathrm{Tr}(V^*\mathcal{L}(\rho)V) = \mathrm{Tr}(VV^*\mathcal{L}(\rho)) = \mathrm{Tr}(P\mathcal{L}(\rho)) = 0$ when the $(S^\perp, S^\perp)$ block of $\mathcal{L}(\rho)$ vanishes on the tangent space.

Equation for the frame $V(t)$**.** Applying $(I - VV^*)$ on the left and $V\sigma^{-1}$ on the right to $\dot{\rho} = \hat{P}_\rho(\mathcal{L}(\rho))$:

$$\dot{V}(t) = (I - V(t)V(t)^*)\, \hat{P}_{\rho(t)}\big(\mathcal{L}(\rho(t))\big)\, V(t)\, \sigma(t)^{-1}. \tag{12.9}$$

This is well-defined since $\sigma(t) \in \mathcal{D}_+(\mathbb{C}^r)$ is invertible (positive definite) along the trajectory.

Remark 12.5.1 (Equations (12.8)–(12.9) *as a special case of the MCTDH system)* Equations (12.8)–(12.9) are the quantum analogue of the MCTDH kinematic system (10.7a)–(10.7b) of Chapter 10. Precisely:

- $\sigma(t) \in \mathcal{D}_+(\mathbb{C}^r)$ plays the role of the core tensor $C^{(D)}(t)$: the finite-dimensional degrees of freedom within the fixed support.
- $V(t) \in \mathrm{St}(r, H)$ plays the role of the basis $\{u^{(\alpha)}_{i_\alpha}(t)\}$ of $U^{\min}_\alpha(\mathbf{v}_r(t))$: the isometric frame of the support subspace.
- The gauge condition $V^*\dot{V} = 0$ is the quantum analogue of the tangential constraint (10.11).
- The mean-field force $F^{(\alpha)}$ of (10.12) becomes here $V^*\mathcal{L}(\rho)V$ in the core equation and $(I - VV^*)\mathcal{L}(\rho)V\sigma^{-1}$ in the frame equation.

The key difference is the additional constraint $\mathrm{Tr}(\dot{\sigma}) = 0$ (trace zero on the core), which has no analogue in the pure-state MCTDH case.

The closed kinematic system

Combining (12.8) and (12.9), the Dirac–Frenkel projected Lindblad dynamics in Stiefel-core coordinates reads:

$$\dot{\sigma}(t) = V(t)^* \, \mathcal{L}\big(V(t)\sigma(t)V(t)^*\big) \, V(t), \tag{12.10a}$$

$$\dot{V}(t) = \big(I - V(t)V(t)^*\big) \, \mathcal{L}\big(V(t)\sigma(t)V(t)^*\big) \, V(t) \, \sigma(t)^{-1}, \tag{12.10b}$$

with initial conditions $V(0) = V_0 \in \mathrm{St}(r, H)$ and $\sigma(0) = V_0^* \rho_0 V_0 \in \mathcal{D}_+(\mathbb{C}^r)$.

Proposition 12.5.1 (Well-posedness of the kinematic system) *Under Assumption 12.1.1, the system* (12.10) *has a unique local solution* $(V(t), \sigma(t))$ *with* $V(t) \in \mathrm{St}(r, H)$ *and* $\sigma(t) \in \mathcal{D}_+(\mathbb{C}^r)$ *for* $t \in [0, T)$. *The reconstructed density operator* $\rho(t) = V(t)\sigma(t)V(t)^*$ *satisfies* (12.3).

Proof. In the chart Φ_{S_0} centred at ρ_0, the system (12.10) is a locally Lipschitz ODE on $\mathcal{L}(\mathbb{C}^r, S_0^\perp) \times \mathcal{T}_1^{sa}(\mathbb{C}^r)$ (a Banach space, finite-dimensional in the core direction and infinite- dimensional in the frame direction). Local existence and uniqueness follow from Theorem B.3.1. Positivity and trace of $\sigma(t)$ are preserved as in Proposition 12.4.1. □

Remark 12.5.2 (Complexity and numerical implementation) In a finite-dimensional truncation $H \cong \mathbb{C}^d$ (dimension d, rank $r \ll d$), the kinematic system (12.10) requires storing only $V(t) \in \mathbb{C}^{d\times r}$ and $\sigma(t) \in \mathbb{C}^{r\times r}$ at cost $O(dr + r^2)$, compared to $O(d^2)$ for the full state $\rho(t)$. Each time step of (12.10) costs $O(|J|dr^2 + dr^2 + r^3)$, where $|J|$ is the number of Lindblad operators. The dominant cost is the evaluation of $\mathcal{L}(\rho)$ projected into the frame coordinates, which requires computing $L_j V(t) \in \mathbb{C}^{d\times r}$ for each j. This is a factor d/r reduction compared to the full simulation.

12.6 Historical notes

The Lindblad equation.

The GKSL master equation was derived independently by Gorini, Kossakowski, and Sudarshan (1976) and by Lindblad (1976) as the most general Markovian, completely positive, trace-preserving quantum master equation. It is the quantum analogue of the Fokker–Planck equation for classical Markov processes. The bounded-generator case (Assumption 12.1.1) is completely understood; the unbounded case remains an active research area.

Dirac–Frenkel for open systems.

The application of the Dirac–Frenkel variational principle to the Lindblad equation — projecting onto a finite-rank manifold — appears in the numerical literature as the time-dependent variational principle (TDVP) for density operators, used in the simulation of open quantum spin chains via matrix product operators (MPO). The mathematical foundations, including the well-posedness theorem

(Theorem 12.3.1) and the explicit bounded projector (Proposition 11.4.1), are established in the companion paper Acevedo and Falcó [2026].

Low-rank methods for Lindblad equations.

Numerical low-rank methods for the Lindblad equation were developed by Le Bris, Rouchon, and Roussel Le Bris and Rouchon [2013], Le Bris et al. [2015] in the Hilbert–Schmidt setting, and by Gravina and Savona Gravina and Savona [2024] using adaptive rank strategies. The CPTP-preserving approach via Kraus parametrisations was developed by Appelö and Cheng Appelö and Cheng [2024]. The present chapter provides a unified geometric framework for all these methods.

Connection to tensor network methods.

For multipartite open systems $H = \bigotimes_{j=1}^{d} H_j$, the density operator $\rho(t)$ is a positive trace-class operator in $\mathcal{T}_1^{sa}(\bigotimes_j H_j)$. Approximating $\rho(t)$ by a matrix product operator (MPO) of bond dimension r is a special case of the tree-based tensor format of Chapters 7–9, applied to the operator space $\mathcal{T}_1^{sa}(\bigotimes_j H_j) \cong \bigotimes_j \mathcal{L}(H_j)$. The Dirac–Frenkel principle on this space gives a rigorous foundation for TDVP-MPO algorithms for open systems. This connection is explored in Open Problem 13.5.4.

Chapter 13
Open Problems and Perspectives

The results established in this monograph open several directions for further research. This chapter collects the most natural open problems, organised by theme, together with a discussion of connections to other active areas of mathematics and its applications. For each problem we describe the state of the art, identify the main technical obstacle, and indicate how it connects to the results of the monograph.

13.1 Global topology of tensor manifolds

Connectedness of tree-based manifolds

Proposition 9.8.1 establishes that each Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is connected for any Banach spaces V_α and any admissible rank $\mathfrak{r}$. The proof uses the explicit chart maps of Section 8.2 to construct a path between any two points of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.

For tree-based manifolds with $\mathrm{depth}(T_D) \geq 2$, connectedness in finite dimensions has been established using algebraic geometry: $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is the smooth part of a determinantal variety (Uschmajew Uschmajew and Vandereycken [2013a] for HT, Holtz–Rohwedder–Schneider Holtz et al. [2012] for TT). These proofs rely on the irreducibility of certain algebraic sets and do not extend directly to infinite-dimensional Banach spaces, where algebraic geometry methods are unavailable.

Open Problem 13.1.1 Let T_D be a dimension partition tree with $\mathrm{depth}(T_D) \geq 2$, V_α infinite-dimensional Banach spaces, and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$ with $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ a proper non-empty set. Is $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ connected?

Approach. The intersection representation (Theorem 5.4.1) writes $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ as the intersection of Tucker manifolds at each level of the tree. Each Tucker manifold is connected; the difficulty is showing that the intersection retains this property. A sufficient condition would be that the fibres of the map $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \to \mathfrak{M}_{\mathfrak{r}_1}(\mathbf{V}_D, \mathcal{P}_1)$ are connected.

Higher homotopy groups

The Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a fibre bundle over $\prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ with fibre $\mathbb{R}_*^{\times r_\alpha}$ (Theorem 8.3.1, Section 8.6). The long exact sequence of homotopy groups for this bundle gives

$$\cdots \to \pi_n(\mathbb{R}_*^{\times r_\alpha}) \to \pi_n(\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)) \to \pi_n\Big(\prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)\Big) \to \pi_{n-1}(\mathbb{R}_*^{\times r_\alpha}) \to \cdots$$

For $V_\alpha = \mathbb{R}^{n_\alpha}$ finite-dimensional, the homotopy groups of the real Grassmannian $\mathbb{G}_r(\mathbb{R}^n)$ are known (they stabilise as $n \to \infty$ to the homotopy groups of $BO(r)$), but $\pi_1(\mathbb{R}_*^{\times r_\alpha})$ is non-trivial whenever some $r_\alpha \geq 2$, making the exact sequence non-degenerate.

Open Problem 13.1.2 Compute $\pi_1(\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D))$ and $\pi_1(\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D))$ in the finite-dimensional case $V_\alpha = \mathbb{R}^{n_\alpha}$. Is $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ simply connected for $r_\alpha \geq 2$?

Relevance. The fundamental group controls the global behaviour of parallel transport around closed loops in the fibre bundle. In the quantum setting (Part **??**), the holonomy of the Tucker bundle is the structure group $G_{\mathfrak{r}} = \prod_\alpha \mathrm{GL}(r_\alpha)\mathbb{R}$ (Remark 8.6.1), and non-trivial π_1 would imply non-trivial holonomy — physically, a Berry phase for the Dirac–Frenkel dynamics.

Rank stratification and Whitney regularity

The decomposition $\mathbf{V}_D = \bigsqcup_{\mathfrak{r}} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a stratification of the tensor space. The boundary $\overline{\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)} \setminus \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) = \mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D) \setminus \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ consists of tensors with strictly smaller rank in at least one direction. Theorem 4.5.1 identifies $\mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D)$ as weakly closed in $\mathbf{V}_{D,\|\cdot\|_D}$ under condition (Inj), but the local geometry at boundary points is not yet understood.

Open Problem 13.1.3 Describe the tangent cone to $\mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D)$ at a boundary point $\mathbf{v}_0 \in \mathfrak{M}_{\mathfrak{r}_0}(\mathbf{V}_D)$ with $\mathfrak{r}_0 < \mathfrak{r}$ (componentwise). Is the stratification $\{\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)\}_{\mathfrak{r}}$ Whitney regular?

Known results. In finite dimensions, the stratification of matrices by rank is Whitney regular (classical), but for $d \geq 3$ the analogous result is open even in finite dimensions. Whitney regularity would imply that the metric projection $\mathit{\Pi}_{\mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D)}$ is continuous at boundary points, which is the key property needed for adaptive rank algorithms.

13.2 Riemannian and Finsler geometry on tensor manifolds

In the Hilbert case, $\mathbf{V}_{D,\|\cdot\|_D} = \bigotimes_j L^2(I_j)$ is a Hilbert space, and $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ inherits a Riemannian metric from the ambient inner product. Riemannian gradient methods on Tucker manifolds (ALS with retraction, Riemannian gradient descent, trust-region methods) have been developed and analysed by Uschmajew and Vandereycken Uschmajew and Vandereycken [2020], who establish local and global convergence under appropriate non-degeneracy conditions on the rank.

In the Banach setting, there is no canonical inner product on $\mathbf{V}_{D,\|\cdot\|_D}$, and the natural replacement is a *Finsler metric*: a continuously varying norm on each tangent space $T_{\mathbf{v}}\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.

Open Problem 13.2.1 Equip $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ with the Finsler metric $F(\mathbf{v}, \dot{\mathbf{v}}) := \|\dot{\mathbf{v}}\|_D$ (the restriction of the ambient norm to $T_{\mathbf{v}}\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \cong Z^{(D)}(\mathbf{v})$). Determine:

(a) *Geodesics and exponential map.* Are Finsler geodesics on $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ related to the retraction (2.5) of Section 2.6?
(b) *Curvature.* Does the Dirac–Frenkel flow preserve the Finsler structure (i.e., is the flow by isometries)?
(c) *Convexity.* Are the sublevel sets $\{\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) : \|\mathbf{v} - \mathbf{v}_0\|_D \leq R\}$ geodesically convex?

13.3 Algorithms and convergence analysis

Projector-splitting integrators on tree-based manifolds

For the TT format, Lubich, Oseledets, and Vandereycken Lubich et al. [2015] introduced the *projector-splitting integrator*, which approximates the Dirac–Frenkel ODE by splitting the projection onto the tangent space $Z^{(D)}(\mathbf{v}_r(t))$ into a sequence of rank-one updates. For each step of the algorithm, the update is exact on the ambient space and involves only one factor at a time. This makes the algorithm highly parallelisable and avoids the ill-conditioning of the full rank map.

The convergence analysis of this integrator in the Hilbert setting gives second-order accuracy in time for smooth solutions Lubich et al. [2015]. In the Banach setting, the analysis requires understanding the stability of the splitting under the metric projection $\mathcal{P}_{\mathbf{v}_r}$ of Chapter 10, which in general is neither orthogonal nor idempotent.

Open Problem 13.3.1 Extend the projector-splitting integrator to Banach tensor spaces. Specifically:

(a) Define the splitting of $\mathcal{P}_{\mathbf{v}_r}$ into factor-wise projections using the duality mapping J of Chapter 2, and prove that the split projections are well-defined and bounded.
(b) Establish error bounds for the resulting integrator in terms of the curvature of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (Open Problem 13.2.1) and the condition number of the rank map $\varrho_{\mathfrak{r}}$ at the current iterate.
(c) Analyse the special case $V_\alpha = L^p(I_\alpha)$, $p \neq 2$, where the duality mapping is non-linear.

Alternating least squares in Banach spaces

The ALS algorithm minimises a functional $J : \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D) \to \mathbb{R}$ by fixing all factor matrices except one and solving a linear subproblem. In the Hilbert setting, each subproblem is a linear least squares problem with a unique solution (when the rank map is non-degenerate), and convergence to a critical point follows from the monotone decrease of J.

In the Banach setting, each subproblem requires minimising J over a closed subspace of V_α, which in a non-Hilbert space may require solving a nonlinear dual problem (via the duality mapping). The condition for convergence involves the geometry of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$, specifically whether the tangent spaces $Z^{(D)}(\mathbf{v})$ vary Lipschitz continuously.

Open Problem 13.3.2 For a convex, coercive, and Fréchet-differentiable functional $J : \mathbf{V}_{D,\|\cdot\|_D} \to \mathbb{R}$ and a reflexive strictly convex $\mathbf{V}_{D,\|\cdot\|_D}$, prove convergence of ALS for minimisation over $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$. Determine the convergence rate in terms of the condition number of the Gram matrices $M_{ij}^{(\alpha)} = \langle \mathbf{U}_i^{(\alpha)}, \mathbf{U}_j^{(\alpha)} \rangle_{[\alpha]}$ at the current iterate (defined in Section 10.5).

13.4 Beyond Banach spaces

Quasi-Banach and Fréchet spaces

The L^p spaces for $0 < p < 1$ are quasi-Banach spaces: they satisfy a weaker triangle inequality $\|x + y\|^p \leq \|x\|^p + \|y\|^p$. These spaces arise in sparse approximation and compressed sensing, where $p < 1$ norms encourage sparsity. The natural component spaces for multi-dimensional sparse approximation are $\ell^p(\mathbb{N})$ or $L^p(\Omega)$ with $0 < p < 1$.

The theory of minimal subspaces (Chapter 4) requires reflexivity of $\mathbf{V}_{D,\|\cdot\|_D}$ for the existence of best Tucker approximations (Theorem 4.5.2). Quasi-Banach spaces with $p < 1$ are not locally convex and are not reflexive, so the proof breaks down. However, the algebraic definition of minimal subspaces (Chapter 3) and the manifold structure (Chapter 8) might still be recoverable under weaker hypotheses.

Open Problem 13.4.1 Develop a theory of minimal subspaces and Tucker manifolds for quasi-Banach component spaces $V_\alpha = L^p(I_\alpha)$ with $0 < p < 1$. In particular:

(a) Does the Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ retain its $\mathcal{C}^\infty$-Banach manifold structure when the V_α are quasi-Banach spaces? The key difficulty is whether finite-dimensional subspaces of L^p $(p < 1)$ are complemented; this fails for $p < 1$ by a result of Kalton (1984).
(b) Does a best Tucker approximation exist when $p < 1$? Or does the failure of reflexivity lead to non-existence as in the L^1 case?

Infinite-dimensional index sets

All results in this monograph assume $D = \{1, \ldots, d\}$ is finite. For quantum field theory, statistical mechanics on infinite lattices, and PDEs in infinitely many variables (e.g., parametric PDEs with countably many parameters as in Nouy [2017]), one requires D to be countably infinite.

The algebraic tensor product over $D = \mathbb{N}$ is well-defined as the inductive limit $\varinjlim_{d\in\mathbb{N}} \bigotimes_{j=1}^{d} V_j$, but the completion and the norm theory are more subtle: different orderings of the factors may give inequivalent completions.

Open Problem 13.4.2 Extend the tree-based tensor framework to infinite index sets $D = \mathbb{N}$. Concretely:

(a) What is the natural analogue of a dimension partition tree for $D = \mathbb{N}$? (A natural candidate: an infinite rooted binary tree with $\mathcal{L}(T_D) = \{\{j\} : j \in \mathbb{N}\}$.)
(b) Under what conditions on the norms $\|\cdot\|_j$ does the inductive limit $\varinjlim_d \mathfrak{M}_{\mathfrak{r}_d}(\bigotimes_{j=1}^{d} V_j)$ converge to a well-defined Banach manifold?
(c) Does the best approximation theorem (Theorem 4.5.2) extend to the infinite case, and under what reflexivity conditions?

13.5 Connections with quantum information and open systems

Tensor networks and entanglement geometry

In quantum information, the *area law* for ground states of gapped local Hamiltonians Vidal [2003] states that the entanglement entropy $S(\alpha) = -\mathrm{Tr}(\rho_\alpha \log \rho_\alpha)$ of a bipartition $D = \alpha \cup \alpha^c$ grows at most as $|\partial\alpha|$ (boundary area), not $|\alpha|$

(volume). This is the physical motivation for tree-based tensor formats: the rank constraints in $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ directly control $S(\alpha) \le \log r_\alpha$ for all $\alpha \in T_D$ (Remark 11.3.1 in Chapter 11).

The tensor network formalism extends beyond trees to general graphs (PEPS, MERA), where the entanglement structure is governed by the graph's topology. The manifold theory of this monograph covers only tree-based networks.

Open Problem 13.5.1 Formulate the Dirac–Frenkel variational principle for general tensor network states (not necessarily tree-based) in the language of C^*-algebras. What is the correct Banach manifold structure for PEPS (projected entangled pair states), where the graph has cycles? Does the immersion theorem (Theorem 8.5.1) extend to non-tree tensor networks, and under what conditions on the graph?

Lindblad dynamics: unbounded generators

Theorem 12.3.1 establishes well-posedness of the projected Lindblad dynamics under Assumption 12.1.1, which requires the generator $\mathcal{L}$ to be bounded on $\mathcal{T}_1^{sa}(H)$. The physically important cases — position decoherence ($L = \sqrt{2D}\, x$), momentum diffusion ($L = \sqrt{\gamma}\, p$), and scattering-decoherence models — all involve unbounded Lindblad operators. In these cases, $\mathcal{L}$ is well-defined only on a dense domain $\mathrm{Dom}(\mathcal{L}) \subset \mathcal{T}_1^{sa}(H)$, and the rank-$r$ stratum $\mathcal{D}_r(H)$ must be replaced by a domain-adapted version $\mathcal{D}_r(\mathcal{H})(H) := \{\rho \in \mathcal{D}_r(H) : \mathrm{supp}(\rho) \subset \mathrm{Dom}(\mathcal{H})\}$ (Section 6.4 of Acevedo and Falcó [2026]).

Open Problem 13.5.2 Extend Theorem 12.3.1 to unbounded Lindblad generators. Specifically:

(a) Construct a smooth Banach manifold structure on $\mathcal{D}_r(\mathcal{H})(H)$ using graph-norm Grassmannians, where support subspaces are required to lie in $\mathrm{Dom}(\mathcal{H})$ equipped with the graph norm $\|\psi\|_{\mathcal{H}} = \|\psi\| + \|\mathcal{H}\psi\|$.
(b) Construct a bounded tangent projector $\hat{P}_\rho^{\mathcal{H}} : \mathcal{T}_1^{sa}(H) \to T_\rho\mathcal{D}_r(\mathcal{H})(H)$ that enforces both the rank and domain constraints.
(c) Prove local well-posedness of the projected dynamics $\dot{\rho} = \hat{P}_\rho^{\mathcal{H}}(\mathcal{L}(\rho))$ for an appropriate class of initial data in $\mathcal{D}_r(\mathcal{H})(H)$.

Test case. The Joos–Zeh position-decoherence model $\dot{\rho} = -D[x, [x, \rho]]$ with $L = \sqrt{2D}\, x$ on $L^2(\mathbb{R})$, where $\mathrm{Dom}(\mathcal{H}) = W^{2,2}(\mathbb{R})$ and $\mathcal{D}_r(\mathcal{H})(L^2(\mathbb{R})) = \{\rho \in \mathcal{D}_r(L^2(\mathbb{R})) : \mathrm{supp}(\rho) \subset W^{2,2}(\mathbb{R})\}$.

CPTP-preserving rank reduction

The projected Lindblad dynamics of Chapter 12 preserves positivity, trace, and rank (Proposition 12.4.1), but the projected flow is a nonlinear ODE on $\mathcal{D}_r(H)$

and is not itself a completely positive trace-preserving (CPTP) map. This means the reduced dynamics does not correspond to a valid quantum channel, which is a serious limitation for quantum information applications.

Recent work by Appelö and Cheng Appelö and Cheng [2024] proposes Kraus-form low-rank integrators that preserve CPTP at the discrete level, but the relationship between their schemes and the continuous Dirac–Frenkel dynamics remains unclear.

Open Problem 13.5.3 Characterise the conditions under which the projected Lindblad flow $\rho_0 \mapsto \rho(t)$ is completely positive:

(a) Under what conditions on $\mathcal{L}$ and $\hat{P}_\rho$ does the flow $t \mapsto \rho(t)$ define a CPTP semigroup on $\mathcal{D}_r(H)$?
(b) Is there a choice of bounded projector $\hat{P}_\rho : \mathcal{T}_1^{sa}(H) \to T_\rho \mathcal{D}_r(H)$ that guarantees CPTP preservation, beyond the block projector of Proposition 11.4.1?
(c) Relate the continuous Dirac–Frenkel dynamics to the Kraus-form integrators of Appelö and Cheng [2024] in the limit of small time steps.

Multipartite open systems and tensor networks

For multipartite open quantum systems $H = \bigotimes_{j=1}^d H_j$, the density operator $\rho(t)$ is a positive trace-class operator on $\bigotimes_j H_j$. Approximating $\rho(t)$ by a *matrix product operator* (MPO) of bond dimension r is standard in numerical quantum physics, but the geometric foundations — what manifold structure does the set of rank-r MPOs carry, and is the Dirac–Frenkel principle well-posed on it? — have not been rigorously established.

The key difficulty is that $\rho \in \mathcal{T}_1^{sa}(\bigotimes_j H_j)$ is simultaneously an element of a tensor product (the space $\bigotimes_j \mathcal{L}(H_j)$ via the isomorphism $\mathcal{T}_1(\bigotimes_j H_j) \cong \bigotimes_j \mathcal{T}_1(H_j)$ under appropriate conditions) and a positive operator, and these two structures interact in a non-trivial way.

Open Problem 13.5.4 Develop the Banach manifold theory for matrix product operators of bond dimension r on $\mathcal{T}_1^{sa}(\bigotimes_{j=1}^d H_j)$:

(a) Identify the correct tree structure T_D for the MPO format and show that MPOs of bond dimension r correspond to elements of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ in the operator space $\mathcal{T}_1^{sa}(\bigotimes_j H_j)$.
(b) Prove that the set of fixed-bond-dimension MPOs is a smooth Banach manifold, extending the results of Chapter 9 to operator spaces.
(c) Formulate the Dirac–Frenkel principle for Lindblad dynamics on this manifold and prove local well-posedness, extending Theorem 12.3.1.

13.6 Learning on tensor manifolds

Statistical learning theory

Tensor network states arise naturally in machine learning as function approximation schemes for high-dimensional data: each tensor $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ defines a function $f_{\mathbf{v}} : \prod_j I_j \to \mathbb{R}$ via $f_{\mathbf{v}}(x_1, \ldots, x_d) = \mathbf{v}(x_1, \ldots, x_d)$, and the hypothesis class $\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$ is a subset of a Banach space.

The Rademacher complexity of a hypothesis class $\mathcal{H} \subset \{f : \mathcal{X} \to \mathbb{R}\}$ controls the generalisation error via $R_n(\mathcal{H}) = \mathbb{E}_\sigma \sup_{f \in \mathcal{H}} \frac{1}{n} \sum_i \sigma_i f(x_i)$, where σ_i are independent Rademacher random variables. For the Tucker format in finite dimensions, Goldberg and Marks (2007) and Grasedyck et al. Grasedyck et al. [2013] have obtained entropy bounds using the parametrisation by factor matrices, but these bounds depend on the choice of parametrisation rather than the intrinsic geometry of $\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$.

Open Problem 13.6.1 Compute the metric entropy (Kolmogorov ε-entropy) of $\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$ in the norm of $\mathbf{V}_{D,\|\cdot\|_D}$ in terms of the tree-based rank $\mathfrak{r}$, the depth $\mathrm{depth}(T_D)$, and the geometry of the Tucker manifold (specifically, the diameter and curvature of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ in the Finsler metric of Problem 13.2.1). Derive generalisation bounds for empirical risk minimisation over $\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$.

Optimal tree selection

Given a target function $f \in \mathbf{V}_{D,\|\cdot\|_D}$, the approximation error $\inf_{\mathbf{v} \in \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)} \|f - \mathbf{v}\|_D$ depends on both the rank bound $\mathfrak{r}$ and the tree structure T_D. Different trees give different approximation classes: the Tucker tree T_D^{Tucker} is optimal for functions with low multilinear rank, the TT tree T_D^{TT} for functions with sequential correlation structure, and balanced HT trees for functions with hierarchical correlation structure.

The inclusion chain (5.5) (Chapter 5) shows that deeper trees give smaller sets for fixed rank $\mathfrak{r}$, so the question is not which tree gives the *smallest* set but which gives the best *trade-off* between approximation error and computational cost.

Open Problem 13.6.2 Given $f \in \mathbf{V}_{D,\|\cdot\|_D}$ and computational budget B (e.g., $B = O(dr^k n)$ operations for some fixed k), find the dimension partition tree T_D^* and rank $\mathfrak{r}^*$ that minimise $\|f - \mathbf{v}_{T_D^*, \mathfrak{r}^*}\|_D$ subject to the budget constraint. Is this combinatorial-geometric optimisation problem tractable (e.g., via a greedy algorithm on the tree structure)? Can it be solved in the case $\mathbf{V}_{D,\|\cdot\|_D} = L^2(\prod_j I_j)$ using the singular value structure of the partial trace operators $\mathrm{Tr}_{\alpha^c}(\rho_f)$ (Chapter 11)?

Appendix A
Banach Manifolds: Reference Summary

This appendix is a reference guide to the differential-geometric notions used in the monograph. The full exposition — with complete proofs and examples — is in Chapter 2. This appendix records the precise statements for quick lookup, and adds the material on connections and curvature that is not needed in the main proofs but is relevant to the open problems of Chapter 13.

Standard references: Lang [1999] and Upmeier [1985].

A.1 Quick-reference index to Chapter 2

Concept	**Location**
Banach manifold, atlas, AT1–AT3	Def. 2.4.1, §2.4
Tangent space and tangent map	Def. 2.4.2, §2.4
Immersion, split image condition	Def. 2.4.3, §2.4
Split (complemented) subspaces	Prop. 2.2.1, §2.2
Finite-dimensional subspaces are split	Prop. 2.2.2, §2.2
Nilpotent algebra $\mathcal{L}_{(U,W)}(X)$, $\exp(L) = \mathrm{id}_+ L$	Prop. 2.2.3, §2.2
Grassmann–Banach manifold, chart map $\Psi_{U\oplus W}$	Thm. 2.3.1, §2.3
Tangent space $T_U\mathbb{G}_r(X) \cong \mathcal{L}(U,W)$	Prop. 2.3.2, §2.3
Continuous multilinear maps are $\mathcal{C}^\infty$	Prop. 2.5.1, §2.5
$\mathcal{C}^1 \Leftrightarrow$ analytic (complex)	Prop. 2.5.2, §2.5
Tucker fibre bundle, trivialisations, structure group $G_{\mathfrak{r}}$	§8.6
Quotient construction fails in ∞-dim.	§8.6, §8.8

A.2 ODEs on Banach manifolds

The following theorem is used in Chapter 10 (proof of Theorem 10.4.1) but does not appear explicitly in Chapter 2.

A *smooth vector field* on a $\mathcal{C}^\infty$-Banach manifold $\mathbb{M}$ is a $\mathcal{C}^\infty$-map $F : \mathbb{M} \to T\mathbb{M}$ with $F(m) \in T_m\mathbb{M}$. An *integral curve* of F through m_0 is a $\mathcal{C}^1$-curve $\gamma : (-\varepsilon, \varepsilon) \to \mathbb{M}$ with $\gamma(0) = m_0$ and $\dot{\gamma}(t) = F(\gamma(t))$.

Theorem A.2.1 (Existence and uniqueness; [Lang, 1999, Theorem 4.1.5]) *Let $\mathbb{M}$ be a $\mathcal{C}^\infty$-Banach manifold and F a $\mathcal{C}^\infty$ vector field. For every $m_0 \in \mathbb{M}$ there exist $\varepsilon > 0$ and a unique integral curve $\gamma : (-\varepsilon, \varepsilon) \to \mathbb{M}$ through m_0. The flow map $(t, m_0) \mapsto \gamma(t)$ is $\mathcal{C}^\infty$.*

Remark A.2.1 In local coordinates the theorem reduces to Picard–Lindelöf in Banach spaces (Theorem B.3.1). The $\mathcal{C}^\infty$ flow follows from smooth dependence on initial data; see [Lang, 1999, Ch. 4].

A.3 The tangent bundle

The *tangent bundle* of a $\mathcal{C}^p$-manifold $\mathbb{M}$ $(p \geq 1)$ is

$$T\mathbb{M} := \bigsqcup_{m \in \mathbb{M}} T_m\mathbb{M}, \tag{A.1}$$

with projection $\pi_{T\mathbb{M}} : (m, v) \mapsto m$ and zero section $\sigma_0 : m \mapsto (m, 0)$. Each chart (M_i, u_i) on $\mathbb{M}$ induces a chart $Tu_i : \pi^{-1}(M_i) \to u_i(M_i) \times X_i$, making $T\mathbb{M}$ a $\mathcal{C}^{p-1}$-Banach manifold modelled on $X_i \times X_i$.

Proposition A.3.1 ([Lang, 1999, Chapter 3]) *$T\mathbb{M}$ is a $\mathcal{C}^{p-1}$-Banach manifold; $\pi_{T\mathbb{M}}$ is a $\mathcal{C}^{p-1}$-submersion; σ_0 is a $\mathcal{C}^{p-1}$-embedding. For $f : \mathbb{M} \to \mathbb{N}$ of class $\mathcal{C}^p$, the tangent map $Tf : T\mathbb{M} \to T\mathbb{N}$ satisfies $T(g \circ f) = Tg \circ Tf$.*

Remark A.3.1 (Tucker manifold) The fibre $T_{\mathbf{v}}\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \cong \mathbf{E}_D$ (Theorem 8.3.1(i)); the tangent map of $\mathfrak{i} : \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \hookrightarrow \mathbf{V}_{D,\|\cdot\|_D}$ has image $Z^{(D)}(\mathbf{v})$ (Proposition 8.4.1); and the immersion theorem (Theorem 8.5.1) says this map has closed split image.

A.4 Connections, parallel transport, and curvature

This section is included for completeness and for Chapter 13; none of this material is used in the proofs of the main text.

Connections on vector bundles

A *(linear) connection* on $\pi : E \to \mathbb{M}$ is a bilinear map $\nabla : \mathfrak{X}(\mathbb{M}) \times \Gamma(E) \to \Gamma(E)$ satisfying $\nabla_{fX} s = f \nabla_X s$ and $\nabla_X(fs) = (Xf)s + f\nabla_X s$ for all $f \in \mathcal{C}^\infty(\mathbb{M})$. In a local trivialisation, $\nabla_X s = X(s) + \omega(X)(s)$ where ω is the $\mathcal{L}(F)$-valued connection 1-form.

Parallel transport along $\gamma : [0,1] \to \mathbb{M}$ gives isomorphisms $P_t : E_{\gamma(0)} \to E_{\gamma(t)}$ by solving $\nabla_{\dot\gamma} s = 0$, $s(0) = e_0$.

Levi-Civita connection and curvature

If $\mathbb{M}$ is Riemannian (each $T_m\mathbb{M}$ has a smoothly varying inner product), the unique torsion-free metric connection is the *Levi-Civita connection*; its curvature is $R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$.

Remark A.4.1 (Relevance to tensor manifolds) When $\mathbf{V}_{D,\|\cdot\|_D} = \bigotimes_j L^2(I_j)$, the Tucker manifold inherits a Levi-Civita connection from $\mathbf{V}_{D,\|\cdot\|_D}$; its curvature controls the accuracy of geometric integration schemes (Open Problem 13.3.1). In the non-Hilbert Banach setting no natural Riemannian structure is available. The *holonomy group* at m (parallel transport around closed loops) equals $U(r_\alpha)$ for $\mathbb{G}_{r_\alpha}(H_\alpha)$ in the Hilbert case; see Open Problem 13.1.2.

Appendix B
Functional Analysis: Reference Summary

This appendix is a reference guide to the Banach space results used in the monograph. The full exposition — with complete proofs, examples, and motivation — is in Chapters 2 and 4. This appendix records the precise statements for quick lookup, and contains the Picard–Lindelöf theorem (used in Chapter 10) and the Hahn–Banach theorem (used implicitly throughout).

Standard references: Hackbusch [2019], Megginson [1998], Ryan [2002].

B.1 Quick-reference index

Concept	Location
Strict/uniform convexity, smoothness	Defs. 2.1.1–2.1.3, §2.1
Duality mapping $J : X \to 2^{X^*}$	§10.3
Metric projection P_C, generalised projection Π_C	§10.3
Characterisation of metric projection via J	Thm. 10.3.1, §10.3
Reflexivity: definition, James's theorem	Thm. 4.1.1, §4.1
Reflexivity stability, reflexivity of tensor products	Props. 4.1.1–4.1.1, §4.1
Injective norm $\\|\cdot\\|_\varepsilon$, condition (Inj)	Def. 4.5.1, §4.5
Dual bound $\\|\varphi_1 \otimes \cdots \otimes \varphi_d\\|_* \leq \prod \\|\varphi_\alpha\\|$	Prop. 4.1.2, §4.1
Eberlein–Šmulian theorem	Used in proof of Thm. 4.5.2, §4.5
Mazur's theorem (convex closed = weakly closed)	Used in Thm. 4.5.1, §4.5
Weak lower semicontinuity of the norm	Used in Thm. 4.5.2, §4.5
Crossnorms, (Cross), (Inj) examples	Rem. 4.5.1, §4.5
Grothendieck inequality, equivalence in Hilbert case	Rem. 3.4.1, §**??**
Split operators, Fredholm theory	§2.2

B.2 The Hahn–Banach theorem

The Hahn–Banach theorem is used implicitly throughout the monograph (in the definition of the injective norm and in duality arguments). We state it here for reference.

Theorem B.2.1 (Hahn–Banach extension theorem) *Let X be a real vector space, $p : X \to \mathbb{R}$ sublinear, $Y \subset X$ a subspace, and $f : Y \to \mathbb{R}$ linear with $f \leq p|_Y$. Then there exists $F : X \to \mathbb{R}$ linear with $F|_Y = f$ and $F \leq p$.*

In particular: for any normed space X, subspace $Y \subset X$, and $f \in Y^*$, there exists $F \in X^*$ with $F|_Y = f$ and $\|F\|_{X^*} = \|f\|_{Y^*}$.

Corollary B.2.1 (Separation of points) *For any normed space X:* (i) *for every $x \neq 0$ there exists $f \in X^*$ with $\|f\| = 1$ and $f(x) = \|x\|$;* (ii) *X^* separates points of X.*

Theorem B.2.2 (Hahn–Banach separation) *In a locally convex space X: if $A, B \subset X$ are convex, A open, and $A \cap B = \emptyset$, then there exists $f \in X^*$ and*

$c \in \mathbb{R}$ with $f(a) < c \leq f(b)$ for all $a \in A$, $b \in B$. A closed convex set is the intersection of all closed half-spaces containing it.

B.3 The Picard–Lindelöf theorem in Banach spaces

This theorem underlies the existence result for ODEs on Banach manifolds (Theorem A.2.1) and is used in Chapter 10.

Theorem B.3.1 (Picard–Lindelöf in Banach spaces) *Let X be a Banach space, $U \subset X$ open, and $F : [0, T] \times U \to X$ continuous with $F(t, \cdot)$ locally Lipschitz uniformly in t. For every $x_0 \in U$ there exist $\varepsilon > 0$ and a unique $\mathcal{C}^1$-solution $x : [0, \varepsilon) \to U$ to $\dot{x}(t) = F(t, x(t))$, $x(0) = x_0$.*

Applied to the projected vector field $F(t, \mathbf{v}_r) = \mathcal{P}_{\mathbf{v}_r}(\mathbf{F}(t, \mathbf{v}_r))$ on $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ in Chapter 10. Local Lipschitz continuity follows from the $\mathcal{C}^\infty$-regularity of the manifold (Theorem 8.3.1).

Appendix C
Operators on Hilbert Spaces: Reference Summary

This appendix is a reference guide to the operator-theoretic vocabulary used in Part **??**. The full mathematical development — with complete proofs and motivation — is in Chapters 11 and 12. This appendix is intended for two purposes: as a *quick reference* for readers who are familiar with operator theory and want to locate a specific result, and as a *self-contained introduction* for readers without prior background in quantum mechanics who wish to read Part **??**.

Throughout this appendix, H denotes a separable complex Hilbert space of *arbitrary* (possibly infinite) dimension. Standard references: Reed and Simon [1972], Bratteli and Robinson [1987], and Ryan [2002] for the trace-class tensor-product perspective.

C.1 Quick-reference index to Part ??

Concept	**Location**
Inner product, Riesz isomorphism $J : H \to H^*$	§C.2
Dual H^* vs. conjugate space $\overline{H}$	Remark C.2.1
Dirac notation (not used; comparison only)	Remark C.2.2
Adjoint A^*, self-adjoint, positive, unitary operators	§C.3
Orthogonal projections and closed subspaces	Prop. 2.2.1
Rank-one operators $u \otimes \varphi \in \mathcal{L}(H)$	§C.3, Remark C.3.1
Compact operators, spectral theorem	§C.4, Thm. C.4.1
Singular values, SVD	§C.4, Thm. C.4.2
Hilbert–Schmidt operators $\mathcal{T}_2(H)$, $\langle A, B\rangle_{\mathrm{HS}} = \mathrm{Tr}(A^*B)$	§C.5, Def. C.5.1
$\mathcal{T}_2(H) \cong H \otimes_H H^*$ isometrically	Thm. C.5.1, Remark C.5.1
Trace-class operators $\mathcal{T}_1(H)$, trace norm $\lVert \cdot \rVert_1$	§C.6, Def. C.6.1
$\mathcal{T}_1(H) \cong H \otimes_\pi H^*$, (Inj) and $\lVert \cdot \rVert_1$	Remark C.6.1
Partial trace $\mathrm{Tr}_{\alpha^c}(\rho)$	§C.6
Density operators $\mathcal{D}(H)$, pure/mixed states	§C.7, Def. C.7.1
$\mathrm{supp}(\rho_\alpha) = U_\alpha^{\min}(v)$ — the fundamental connection	Thm. 11.1.1 in Ch. 11
Rank stratification $\mathcal{D}(H) = \bigsqcup_r \mathcal{D}_r(H)$	Prop. 11.2.1 in Ch. 11
Manifold structure of $\mathcal{D}_r(H)$, tangent space	Thm. 11.4.1 in Ch. 11
Bounded projector $\hat{P}_\rho : \mathcal{T}_1^{sa}(H) \to T_\rho \mathcal{D}_r(H)$	Prop. 11.4.1 in Ch. 11
Lindblad (GKSL) equation	§12.1 in Ch. 12
Well-posedness of projected Lindblad dynamics	Thm. 12.3.1 in Ch. 12
Kinematic equations in Stiefel-core coordinates	§12.5 in Ch. 12

C.2 Complex Hilbert spaces and the dual

A *complex Hilbert space* H carries an inner product $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{C}$, conjugate-linear in the first argument and linear in the second, positive definite, and complete. The Riesz representation theorem gives the isometric conjugate-linear isomorphism

$$J : H \longrightarrow H^*, \qquad J(v) = \langle v, \cdot \rangle, \tag{C.1}$$

where $H^* = \mathcal{L}(H, \mathbb{C})$ is the topological dual. This appendix works throughout with H^* (defined by the norm alone), not the conjugate space $\overline{H}$ (defined by the linear structure). The two are isomorphic as Banach spaces via J, but not canonically as complex Banach spaces.

Remark C.2.1 (Dual vs. conjugate space) In finite dimensions $H^* \cong H \cong \overline{H}$ as complex vector spaces. In infinite dimensions, the Riesz map $J : H \to H^*$ is conjugate-linear (not linear), making H and H^* non-isomorphic as complex Banach spaces. The identification $\mathcal{T}_1(H) \cong H \otimes_\pi H^*$ (Remark C.6.1) requires H^* as the second factor to preserve the $\mathbb{C}$-bilinearity of the tensor product; using $\overline{H}$ instead introduces a conjugate-linear identification that is inconsistent with the Banach-space tensor framework of Chapter 3.

Remark C.2.2 (Dirac notation) In the physics literature, $v \in H$ is written $|v\rangle$ (a *ket*) and $J(v) \in H^*$ is written $\langle v|$ (a *bra*), so $\langle u, v\rangle_H = \langle u|v\rangle$. The rank-one operator $u \otimes J(v)$ is written $|u\rangle\langle v|$. This monograph uses exclusively the standard mathematical notation: $\langle u, v\rangle$ for the inner product, $J(v) \in H^*$ for the Riesz dual, and $u \otimes \varphi \in \mathcal{L}(H)$ for rank-one operators.

C.3 Bounded operators, adjoint, and special classes

$\mathcal{L}(H)$ with the operator norm is a C*-algebra. The adjoint $A^* \in \mathcal{L}(H)$ of A is characterised by $\langle A^*u, v\rangle = \langle u, Av\rangle$ for all $u, v \in H$, with $\|A^*A\| = \|A\|^2$. Key classes: self-adjoint ($A = A^*$), positive ($A \geq 0$: $\langle u, Au\rangle \geq 0$), strictly positive ($A > 0$), unitary ($A^*A = AA^* = I$), orthogonal projection ($A = A^* = A^2$). Every closed subspace $S \subset H$ has an orthogonal projection P_S and $H = S \oplus S^\perp$ (cf. Proposition 2.2.2, which holds for *all* closed subspaces in Hilbert spaces, not just finite-dimensional ones).

For $u \in H$ and $\varphi \in H^*$, the rank-one operator $u \otimes \varphi \in \mathcal{L}(H)$ is defined by $(u \otimes \varphi)(w) \coloneqq \varphi(w)\, u$.

Remark C.3.1 (Rank-one operators and the tensor product $H \otimes H^*$) The finite-rank operators form the algebraic tensor product $H \otimes_a H^*$ inside $\mathcal{L}(H)$. The map $(u, \varphi) \mapsto u \otimes \varphi$ is $\mathbb{C}$-bilinear, making this consistent with the tensor product framework of Chapter 3. The Hilbert–Schmidt completion gives $\mathcal{T}_2(H) \cong H \otimes_H H^*$ and the trace-class completion gives $\mathcal{T}_1(H) \cong H \otimes_\pi H^*$; see Sections C.5–C.6.

C.4 Compact operators, spectral theorem, and SVD

$A \in \mathcal{L}(H)$ is *compact* if it maps the unit ball to a relatively compact set. The set $\mathcal{K}(H)$ of compact operators is a closed two-sided ideal, equal to the norm-closure of finite-rank operators (in separable H).

Theorem C.4.1 (Spectral theorem for compact self-adjoint operators, [Reed and Simon, 1972, Thm. VI.16]) *Let $A \in \mathcal{K}(H)$ with $A = A^*$. Then there exist an orthonormal sequence $\{u_k\}$ in H and real $\lambda_k \to 0$ with*

$$A = \sum_k \lambda_k \, u_k \otimes J(u_k), \tag{C.2}$$

convergent in operator norm. The λ_k are the non-zero eigenvalues of A.

The *singular values* of $A \in \mathcal{K}(H)$ are the eigenvalues $s_k(A) \geq 0$ of $|A| = (A^*A)^{1/2}$.

Theorem C.4.2 (Singular value decomposition) *For $A \in \mathcal{K}(H)$ with non-zero singular values $(s_k(A))$, there exist orthonormal sequences $\{u_k\}$, $\{v_k\}$ in H with*

$$A = \sum_k s_k(A) \, v_k \otimes J(u_k), \tag{C.3}$$

convergent in operator norm. For $d = 2$ and $v \in H_1 \otimes_H H_2$, this is the Schmidt decomposition; the Tucker rank $r_1(v) = r_2(v)$ equals the number of non-zero singular values (Remark 11.1.1 in Chapter 11).

C.5 Hilbert–Schmidt operators

Definition C.5.1 $A \in \mathcal{L}(H)$ is *Hilbert–Schmidt* if $\|A\|_2^2 := \sum_k \|Ae_k\|^2 = \mathrm{Tr}(A^*A) < \infty$ for some (equivalently every) orthonormal basis $\{e_k\}$. The space $\mathcal{T}_2(H)$ with $\langle A, B\rangle_{\mathrm{HS}} := \mathrm{Tr}(A^*B)$ is a Hilbert space, a two-sided ideal in $\mathcal{L}(H)$, with $\|A\| \leq \|A\|_2 = (\sum_k s_k(A)^2)^{1/2}$.

Theorem C.5.1 ($\mathcal{T}_2(H) \cong H \otimes_H H^*$) *There is a canonical isometric isomorphism of Hilbert spaces*

$$\mathcal{T}_2(H) \cong H \otimes_H H^*, \tag{C.4}$$

under which the elementary tensor $u \otimes \varphi \in H \otimes_a H^*$ corresponds to the rank-one operator $w \mapsto \varphi(w)\, u$.

Remark C.5.1 (Why H^* and not $\overline{H}$) The identification (C.4) uses H^* to preserve $\mathbb{C}$-bilinearity. In finite dimensions (or when $H = \overline{H}$ is identified via J) one also sees $\mathcal{T}_2(H) \cong H \otimes_H \overline{H}$ in the literature, but this conflates a linear and a conjugate-linear isomorphism. The present convention is consistent with the projective tensor norm identification in the next section.

C.6 Trace-class operators

Definition C.6.1 $A \in \mathcal{L}(H)$ is *trace class* if $\|A\|_1 := \mathrm{Tr}|A| = \sum_k s_k(A) < \infty$. The space $\mathcal{T}_1(H)$ with $\|\cdot\|_1$ is a Banach space (not Hilbert), a two-sided ideal in $\mathcal{L}(H)$, with $\mathcal{T}_1(H) \subset \mathcal{T}_2(H) \subset \mathcal{K}(H)$ and $\|A\| \leq \|A\|_2 \leq \|A\|_1$. The *trace* $\mathrm{Tr}(A) := \sum_k \langle e_k, Ae_k \rangle$ is absolutely convergent and basis-independent. The duality $\mathcal{T}_1(H)^* \cong \mathcal{L}(H)$ holds via $\langle A, X \rangle = \mathrm{Tr}(A^* X)$.

Remark C.6.1 ($\mathcal{T}_1(H) \cong H \otimes_\pi H^*$ and (Inj)*)* The identification

$$\mathcal{T}_1(H) \cong H \otimes_\pi H^* \tag{C.5}$$

holds isometrically Grothendieck [1953], Ryan [2002]. On $H \otimes_a H^*$:

$$\|\cdot\|_\varepsilon \leq \|\cdot\|_H \leq \|\cdot\|_1 = \|\cdot\|_\pi,$$

where $\|\cdot\|_H$ is the Hilbert–Schmidt norm and $\|\cdot\|_\varepsilon$ the injective norm. Both $\|\cdot\|_H$ and $\|\cdot\|_\pi$ satisfy condition (Inj) of Definition 4.5.1. The trace norm $\frac{1}{2}\|\rho - \sigma\|_1$ equals the total variation distance between quantum states ρ and σ — the maximum probability of distinguishing them by any measurement — which is why it is the natural norm for state estimation and compression.

The *partial trace* $\mathrm{Tr}_{\alpha^c}(A) \in \mathcal{T}_1(H_\alpha)$, for $A \in \mathcal{T}_1(H_1 \otimes_H H_2)$, is characterised by $\mathrm{Tr}_{H_1}(B\,\mathrm{Tr}_2(A)) = \mathrm{Tr}_H((B \otimes I_2)\,A)$ for all $B \in \mathcal{L}(H_1)$.

C.7 Density operators and quantum states

Definition C.7.1 A *density operator* is $\rho \in \mathcal{T}_1^{sa}(H) := \{A \in \mathcal{T}_1(H) : A = A^*\}$ with $\rho \geq 0$ and $\mathrm{Tr}(\rho) = 1$. The *state space* is $\mathcal{D}(H) := \{\rho \in \mathcal{T}_1^{sa}(H) : \rho \geq 0,\, \mathrm{Tr}(\rho) = 1\}$.

$\mathcal{D}(H)$ is convex and trace-norm closed, but not globally a manifold (rank singularities, boundary phenomena). A state is *pure* if $\mathrm{rank}(\rho) = 1$ (equivalently $\rho = u \otimes J(u)$ for a unit vector $u \in H$) and *mixed* otherwise.

The *rank-r stratum* $\mathcal{D}_r(H) := \{\rho \in \mathcal{D}(H) : \mathrm{rank}(\rho) = r\}$ is a smooth Banach manifold (Theorem 11.4.1 in Chapter 11). The stratification $\mathcal{D}(H) = \bigsqcup_r \mathcal{D}_r(H) \cup \mathcal{D}_\infty(H)$ mirrors the Tucker rank decomposition $\mathbf{V}_D = \bigsqcup_{\mathfrak{r}} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ of Theorem 4.4.1.

The fundamental connection with minimal subspaces (Theorem 11.1.1) is: for $v \in \bigotimes_{j=1}^d H_j$ with $\|v\| = 1$,

$$\mathrm{supp}\big(\mathrm{Tr}_{\alpha^c}(v \otimes J(v))\big) = U_\alpha^{\min}(v), \qquad \mathrm{rank}\big(\mathrm{Tr}_{\alpha^c}(v \otimes J(v))\big) = r_\alpha(v). \tag{C.6}$$

This is the cornerstone of Chapter 11.

C.8 Historical notes

Operator-theoretic foundations of quantum mechanics.

Von Neumann von Neumann [1932] established the functional-analytic framework using density operators on Hilbert spaces. Schatten Schatten [1950] introduced the Schatten p-classes $\mathcal{T}_p(H)$ with norm $\|A\|_p = (\sum_k s_k(A)^p)^{1/p}$; the cases $p = 1$ (trace class) and $p = 2$ (Hilbert–Schmidt) are central to Part **??**.

Tensor norms and the trace.

The identification $\mathcal{T}_1(H) \cong H \otimes_\pi H^*$ (Remark C.6.1) is due to Grothendieck Grothendieck [1953], connecting the trace norm to the projective tensor norm of Chapter 3. The Hilbert–Schmidt identification $\mathcal{T}_2(H) \cong H \otimes_H H^*$ (Theorem C.5.1) is the non-commutative analogue of the Fubini isomorphism $L^2(\Omega_1) \otimes_H L^2(\Omega_2) \cong L^2(\Omega_1 \times \Omega_2)$ (cf. [Reed and Simon, 1972, Chapter II]).

Minimal subspaces and partial traces.

Equation (C.6) makes precise a connection implicit in the quantum information literature since Schmidt (1907). Its systematic treatment in the tensor-Banach-space framework is the contribution of Falcó–Hackbusch Falcó and Hackbusch [2012] and Acevedo–Falcó Acevedo and Falcó [2026].

Appendix D
Computational Aspects and Algorithms

The theory developed in this monograph — minimal subspaces, Tucker and tree-based manifolds, the Dirac–Frenkel variational principle — has direct computational consequences. This appendix describes the main algorithms used in practice, explains their connection to the mathematical structures of the main text, and identifies precisely where the Banach-space generalisations of Chapters 4–10 modify or constrain the algorithmic picture.

Each algorithm is stated with pseudocode, its cost is analysed, and its relationship to the abstract theory is made explicit. The principal references are Hackbusch [2019] (systematic treatment of tensor numerics), Bachmayr et al. [2016] (survey of tensor network methods), and Uschmajew and Vandereycken [2020] (Riemannian optimisation on tensor manifolds).

D.1 Storage complexity of tensor formats

The curse of dimensionality, quantified

Let $V_j = \mathbb{R}^n$ for $j = 1, \ldots, d$. The full tensor space $\mathbf{V}_D = \bigotimes_{j=1}^d \mathbb{R}^n$ has dimension n^d: intractable for moderate n and d. Tensor formats replace this by structured low-dimensional parametrisations. The table below summarises storage cost in terms of maximal rank r and mode size n:

Format	Parameters	Storage	Tree
Full	n^d entries	$O(n^d)$	—
Tucker	core r^d + d factor matrices $n\times r$	$O(r^d + dnr)$	T_D^{Tucker}
HT (balanced binary)	transfer tensors at $O(d)$ nodes, each $r\times r^2$	$O(dr^3 + dnr)$	T_D^{HT}
TT (Tensor Train)	d core matrices, each $r\times n\times r$	$O(dr^2 n)$	T_D^{TT}

Remark D.1.1 (Exponential vs. polynomial scaling) The Tucker format reduces factor storage to $O(dnr)$ (linear in d) but retains the core at $O(r^d)$. The HT and TT formats remove this remaining exponential cost by imposing rank constraints at each level of the tree, reducing core storage to $O(dr^3)$ and $O(dr^2n)$ respectively. This is precisely the content of the intersection representation (Theorem 5.4.1): $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ imposes rank constraints at *every* level of T_D, not only at the leaves.

Parameter count for tree-based formats

For a general dimension partition tree T_D and rank tuple $\mathfrak{r}$, the total parameter count is:

$$P(\mathfrak{r}, T_D) = \sum_{j\in D} r_{\{j\}} n_j + \sum_{\alpha\in T_D\setminus(\mathcal{L}(T_D)\cup\{D\})} r_\alpha \prod_{\beta\in S(\alpha)} r_\beta + \prod_{\beta\in S(D)} r_\beta, \tag{D.1}$$

where the three terms count, respectively, the leaf factor matrices, the internal transfer tensors, and the root tensor.

D.2 The Higher-Order SVD

Setup and goal

The *Higher-Order SVD* (HOSVD) of De Lathauwer, De Moor, and Vandewalle De Lathauwer et al. [2000] computes a Tucker approximation. Given $\mathbf{v} \in \mathbf{V}_D = \bigotimes_{j=1}^d \mathbb{R}^{n_j}$ and target ranks $\mathfrak{r} = (r_j)_{j\in D}$, it produces:

(i) Factor matrices $U_j \in \mathbb{R}^{n_j\times r_j}$ with orthonormal columns, approximating $U_j^{\min}(\mathbf{v})$.
(ii) A core tensor $C^{(D)} \in \mathbb{R}^{r_1\times\cdots\times r_d}$ such that $\mathbf{v} \approx \sum_{(i_j)} C^{(D)}_{(i_j)} \bigotimes_j u^{(j)}_{i_j}$.

Algorithm

Algorithm D.1 (HOSVD).
Input: tensor $\mathbf{v} \in \mathbb{R}^{n_1 \times \cdots \times n_d}$, target ranks $(r_1, \dots, r_d)$.
Output: core tensor $C^{(D)}$, factor matrices $U_1, \dots, U_d$.

1. **Mode unfoldings.** For each $j = 1, \dots, d$:
   a. Compute the j-mode unfolding $M_j = \mathcal{M}_{\{j\}}(\mathbf{v}) \in \mathbb{R}^{n_j \times \prod_{k \neq j} n_k}$.
   b. Compute the truncated SVD: $M_j \approx U_j \Sigma_j V_j^\top$, retaining the r_j largest singular values. Set $U_j \in \mathbb{R}^{n_j \times r_j}$ (left singular vectors).
2. **Core compression.** Compute
$$C^{(D)}_{i_1,\dots,i_d} = \sum_{k_1,\dots,k_d} (U_1)_{k_1 i_1} \cdots (U_d)_{k_d i_d}\, \mathbf{v}_{k_1,\dots,k_d},$$
   i.e., project $\mathbf{v}$ onto the subspaces $\operatorname{span}(U_j)$ in each mode.
3. **Output.** Return $(C^{(D)}, U_1, \dots, U_d)$.

Cost and approximation quality

The dominant cost is d SVDs of unfolding matrices of size $n_j \times \prod_{k \neq j} n_k$, giving arithmetic cost $O(dn^{d+1})$, which still scales exponentially in d. For tensors already in a low-rank format, recompression costs $O(dr^{d+1})$ or $O(dr^3)$ depending on the tree structure.

Theorem D.2.1 (HOSVD quasi-optimality; [De Lathauwer et al., 2000, Theorem 4.2]) *Let* $\mathbf{v} \in \mathbf{V}_D$ *and let* $\mathbf{w}$ *be the output of* Algorithm D.1 *with ranks* $(r_j)_j$. *Then*

$$\|\mathbf{v} - \mathbf{w}\|_F \leq \sqrt{d} \min_{\mathbf{u} \in \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)} \|\mathbf{v} - \mathbf{u}\|_F. \tag{D.2}$$

The HOSVD is not a best-approximation algorithm, but provides a quasi-optimal approximation with factor $\sqrt{d}$.

Remark D.2.1 (Connection to minimal subspaces) The factor matrix U_j is the best rank-r_j approximation to the column space of the j-mode unfolding M_j, whose column space is exactly $U_j^{\min}(\mathbf{v})$ when $r_j = r_j(\mathbf{v})$ (the exact rank). Algorithm D.1 is therefore a computational procedure for approximating the minimal subspaces of Chapter 4.

In the Banach setting (non-Hilbert V_j), the SVD is replaced by a best rank-r_j approximation in the operator norm of $\mathcal{L}(V_j^*, \mathbf{V}_\alpha^{[j]})$. Its existence is guaranteed by Theorem 4.5.2, but the quasi-optimality factor $\sqrt{d}$ may worsen depending on the Banach–Mazur distance of $\mathbf{V}_{D,\|\cdot\|_D}$ from a Hilbert space.

D.3 Alternating Least Squares

The ALS problem and microiterations

Given $\mathbf{u} \in \mathbf{V}_{D,\|\cdot\|_D}$ and rank tuple $\mathfrak{r}$, the best Tucker approximation problem is

$$\min_{\mathbf{w} \in \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)} \|\mathbf{u} - \mathbf{w}\|_D^2. \tag{D.3}$$

Alternating Least Squares (ALS) solves this by fixing all factor matrices but one and optimising over the remaining one, cycling over all modes.

Algorithm D.2 (ALS for Tucker format).
Input: $\mathbf{u} \in \mathbf{V}_{D,\|\cdot\|_D}$, ranks $\mathfrak{r}$, initial factor matrices $\{U_j^{(0)}\}_{j=1}^d$.
Repeat until convergence:

1. For $j = 1, \dots, d$:
   a. **Fix** all U_k for $k \neq j$ at their current values.
   b. **Form the reduced right-hand side:** compute the array
   $$\left(R_j\right)_{i_j,(i_k)_{k \neq j}} := \sum_{(k_l)} \mathbf{u}_{k_1,\dots,k_d} \prod_{l \neq j} (U_l)_{k_l,i_l},$$
   which is the tensor $\mathbf{u}$ projected onto all modes except j. This gives $R_j \in \mathbb{R}^{n_j \times \prod_{k \neq j} r_k}$.
   c. **Solve the mode-j subproblem:**
   $$U_j^{\text{new}} = \operatorname*{arg\,min}_{U \in \mathbb{R}^{n_j \times r_j}} \left\| R_j - U\, H_j \right\|_F^2,$$
   where $H_j \in \mathbb{R}^{r_j \times \prod_{k \neq j} r_k}$ is the Gram factor assembled from the other modes.
   d. **Orthonormalise:** $U_j \leftarrow U_j^{\text{new}}$ after a QR step.
2. **Update the core:** $C^{(D)}_{i_1,\dots,i_d} \leftarrow \sum_{k_1,\dots,k_d} \mathbf{u}_{k_1,\dots,k_d} \prod_j (U_j)_{k_j,i_j}$.

Convergence

Theorem D.3.1 (ALS convergence in Hilbert spaces; [Uschmajew and Vandereycken, 2020, Theorem 4.1]) *Let $\mathbf{V}_{D,\|\cdot\|_D}$ be a Hilbert tensor space and let $(\mathbf{w}^{(k)})$ be the sequence generated by* Algorithm D.2. *Then:*

(i) *The residuals $\|\mathbf{u} - \mathbf{w}^{(k)}\|$ are monotonically non-increasing.*
(ii) *Every accumulation point $\mathbf{w}^*$ satisfies the first-order optimality conditions in each mode direction: $\langle \mathbf{u} - \mathbf{w}^*, v \otimes \mathbf{w}^*_{[j]} \rangle = 0$ for all $v \in V_j$, $j = 1, \dots, d$.*

(iii) *Global convergence to a best approximation is not guaranteed; ALS may converge to a local minimum or saddle point.*

Remark D.3.1 (ALS in Banach spaces) In a reflexive Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$, the mode-$j$ subproblem in step 1(c) becomes: find the best rank-r_j operator approximation in the norm $\|\cdot\|_D$ projected onto mode j. This is a convex problem whose unique solution exists by Theorem 4.5.2. Monotone decrease of part (i) holds in full generality. The characterisation of accumulation points in part (ii) must be rephrased using the duality mapping J (Appendix B) in place of the inner product:

$$\langle v \otimes \mathbf{w}^*_{[j]},\, J(\mathbf{u} - \mathbf{w}^*)\rangle = 0 \quad \forall\, v \in V_j,\ j = 1, \dots, d.$$

The convergence analysis in this Banach setting is Open Problem 13.3.2.

D.4 The projector-splitting integrator

The Dirac–Frenkel ODE

Consider the Dirac–Frenkel reduced model on $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (Chapter 10):

$$\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}\big(\mathbf{F}(t, \mathbf{v}_r(t))\big), \quad \mathbf{v}_r(0) = \mathbf{v}_0. \tag{D.4}$$

Direct integration requires evaluating $\mathcal{P}_{\mathbf{v}_r(t)}$ at each step. The *projector-splitting integrator* of Lubich et al. [2015] avoids this by decomposing the projection into a sequence of simpler updates.

The TT case

For the TT format ($T_D = T_D^{\mathrm{TT}}$, bond dimensions $r_1, \dots, r_{d-1}$), the tangent space projection splits as

$$\mathcal{P}_{\mathbf{v}_r} = \sum_{k=1}^{d} P_k - \sum_{k=1}^{d-1} P_{k,k+1}, \tag{D.5}$$

where P_k projects onto the subspace with the k-th component free and $P_{k,k+1}$ is the correction at the $(k, k{+}1)$ interface.

Algorithm D.3 (Projector-splitting, one step h).
Input: $\mathbf{v}_r^n$ in TT format, vector field $\mathbf{F}$, step h.
Output: $\mathbf{v}_r^{n+1}$ in TT format.

1. **Forward sweep** ($k = 1, \dots, d$): integrate the k-th component ODE

$$\dot{C}^{(k)}(t) = \big\langle \mathbf{F}(t, \mathbf{v}_r(t)),\, P_k(\,\cdot\,)\big\rangle$$

from t^n to $t^n + h$; left-orthogonalise via QR.

2. **Correction sweep** ($k = d-1, \ldots, 1$): apply the correction steps $-P_{k,k+1}$ symmetrically.
3. **Return** the updated TT tensor $\mathbf{v}_r^{n+1}$.

Theorem D.4.1 (Error bound in Hilbert spaces; [Lubich et al., 2015, Theorem 4.1]) *Let $\mathbf{V}_{D,\|\cdot\|_D}$ be a Hilbert tensor space and $\mathbf{F}$ Lipschitz with constant L. Then there exists $C > 0$ (depending on L, the curvature of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$, and the initial data) such that*

$$\|\mathbf{u}(t^n) - \mathbf{v}_r^n\| \le Ch + O\big(\delta_{\mathfrak{r}}(\mathbf{u})\big), \tag{D.6}$$

where $\delta_{\mathfrak{r}}(\mathbf{u}) = \inf_{\mathbf{w}\in\mathfrak{M}_{\le\mathfrak{r}}(\mathbf{V}_D)} \|\mathbf{u} - \mathbf{w}\|$ is the best-approximation error.

The curvature term in C is the quantity whose Banach-space analogue appears in Open Problem 13.3.1. For general tree-based formats, the splitting integrator of Ceruti et al. [2021] applies a level-by-level sweep along T_D at cost $O(\mathrm{depth}(T_D) \cdot dr^3)$ per time step.

D.5 DMRG and the Tucker manifold

DMRG as ALS on the TT manifold

The *Density Matrix Renormalisation Group* (DMRG), introduced by White White [1992], is the most widely used algorithm for ground-state computation in quantum many-body systems. In its modern formulation Holtz et al. [2012], DMRG is equivalent to ALS on the TT manifold applied to the eigenvalue problem $H\mathbf{u} = \lambda\mathbf{u}$: each microiteration updates one TT component while fixing all others, producing an effective eigenvalue problem of size $r^2 n$.

Remark D.5.1 (Gauge conditions and the Tucker bundle) The *left-orthogonal gauge* in DMRG requires each component tensor $C^{(k)} \in \mathbb{R}^{r_{k-1}\times n_k \times r_k}$ to satisfy

$$\sum_{i_k=1}^{n_k} \big(C^{(k)}\big)^{\top}_{:,i_k,:} \big(C^{(k)}\big)_{:,i_k,:} = I_{r_k}.$$

This is the computational analogue of fixing the complement W_k at each tree node to be the orthogonal complement of $\mathrm{span}(C^{(k)})$, giving a canonical trivialisation of the Tucker bundle (Section 8.6). The gauge group $\prod_k \mathrm{GL}(r_k)\mathbb{R}$ of Remark 6.4.1 is quotiented out by the QR orthogonalisation step.

Two-site DMRG and rank adaptation

One-site DMRG cannot change the bond dimensions r_k. *Two-site DMRG* optimises two adjacent components simultaneously, forming an effective tensor of rank $r_k \cdot r_{k+1}$ which is then truncated by SVD to obtain updated bond dimensions. Geometrically, this corresponds to moving between strata of the stratification $\mathbf{V}_D = \bigsqcup_{\mathfrak{r}} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (Theorem 4.4.1).

D.6 Retraction-based optimisation on tensor manifolds

Riemannian gradient methods

In the Hilbert case, the Tucker manifold is a Riemannian submanifold of $\mathbf{V}_{D,\|\cdot\|_D}$. Riemannian gradient descent Uschmajew and Vandereycken [2020] for minimising $f : \mathbf{V}_{D,\|\cdot\|_D} \to \mathbb{R}$ over $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ reads:

$$\mathbf{v}_{k+1} = \mathrm{Ret}_{\mathbf{v}_k}\big(-\alpha_k\, \mathcal{P}_{\mathbf{v}_k}(\nabla f(\mathbf{v}_k))\big), \tag{D.7}$$

where $\mathcal{P}_{\mathbf{v}}(\nabla f(\mathbf{v}))$ is the Riemannian gradient and $\mathrm{Ret}_{\mathbf{v}} : Z^{(D)}(\mathbf{v}) \to \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a retraction.

Definition D.6.1 (Retraction) A *retraction* on a Banach manifold $\mathbb{M}$ is a $\mathcal{C}^1$-map $\mathrm{Ret} : T\mathbb{M} \to \mathbb{M}$ such that, for each $m \in \mathbb{M}$:

(i) $\mathrm{Ret}_m(0_m) = m$,
(ii) $D(\mathrm{Ret}_m)(0_m) = \mathrm{id}_{T_m\mathbb{M}}$.

Example D.6.1 (Tucker retraction via HOSVD truncation) A practical retraction on $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is:

$$\mathrm{Ret}_{\mathbf{v}}(\dot{\mathbf{v}}) = \mathrm{HOSVD}_{\mathfrak{r}}(\mathbf{v} + \dot{\mathbf{v}}),$$

i.e., apply Algorithm D.1 to $\mathbf{v} + \dot{\mathbf{v}}$ and truncate to rank $\mathfrak{r}$. This satisfies (i) and (ii) with quasi-optimality constant $\sqrt{d}$ (Theorem D.2.1), and remains valid in Banach spaces as long as the best-approximation step is solvable — guaranteed by Theorem 4.5.2.

The Banach case: duality-mapping gradient

In a non-Hilbert Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$, the Riemannian gradient is replaced by the *duality-mapping gradient*: the unique element $\nabla_B f(\mathbf{v}) \in Z^{(D)}(\mathbf{v})$ satisfying

$$\big\langle \nabla_B f(\mathbf{v}),\, J\big(\mathbf{v} - \mathbf{u}\big)\big\rangle = Df(\mathbf{v})[\mathbf{v} - \mathbf{u}] \qquad \forall\, \mathbf{u} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D), \tag{D.8}$$

where $J : \mathbf{V}_{D,\|\cdot\|_D} \to (\mathbf{V}_{D,\|\cdot\|_D})^*$ is the normalised duality mapping (Appendix B, §B.1). The iteration (D.7) becomes a generalised gradient step using the generalised projection $\Pi_{\mathbf{v}}$ of Appendix B in place of $\mathcal{P}_{\mathbf{v}}$. Convergence analysis in this setting is related to Open Problem 13.2.1.

D.7 Historical notes

Tucker decomposition and HOSVD.

The Tucker decomposition Tucker [1966] was introduced as a psychometric tool and has been used in data analysis since the 1970s. The HOSVD as an algorithm for best Tucker approximations was introduced by De Lathauwer, De Moor, and Vandewalle De Lathauwer et al. [2000]; the quasi-optimality bound (D.2) is from the same paper. A comprehensive survey of tensor decompositions is given in Kolda and Bader Kolda and Bader [2009].

Tensor Train and DMRG.

The TT decomposition was introduced independently by Vidal Vidal [2003] (matrix product states, MPS) and by Oseledets and Tyrtyshnikov Oseledets [2011]. The DMRG algorithm of White White [1992] predates both and was recognised in Holtz et al. [2012] as ALS on the TT manifold. The TT-SVD of Oseledets [2011] provides quasi-optimal TT approximations analogous to the HOSVD for Tucker.

Projector-splitting integrators.

The projector-splitting integrator for TT formats was introduced by Lubich, Oseledets, and Vandereycken Lubich et al. [2015], building on the matrix case of Koch and Lubich Koch and Lubich [2007] (MCTDH). The extension to general tree-based formats is in Ceruti et al. [2021]. Convergence in the Banach setting is Open Problem 13.3.1.

Riemannian optimisation.

Riemannian gradient methods on Tucker and TT manifolds were systematically developed by Uschmajew and Vandereycken Uschmajew and Vandereycken [2020], using the Riemannian structure in a way that parallels Chapters 8–9 but restricted to the Hilbert case. The Banach generalisation is related to Open Problem 13.2.1.

Appendix E
Index of Notation

All macros are defined in `notation.tex`. Entries are grouped thematically and listed in the order they first appear in the text. For each entry, the rightmost column indicates the chapter or appendix where the symbol is first introduced; a blank entry means the symbol is standard notation used throughout without introduction.

Sets, fields, and standard spaces

$\mathbb{K}$	field ($\mathbb{R}$ or $\mathbb{C}$)	
$\mathbb{R}$, $\mathbb{C}$	real and complex numbers	
$\mathbb{N}$, $\mathbb{N}_0$, $\mathbb{Z}$	positive integers, non-negative integers, integers	
$D = \{1, \ldots, d\}$	dimension index set	Ch. 3
$d = \#(D)$	number of dimensions	Ch. 3
$\alpha^c = D \setminus \alpha$	complement of α in D	Ch. 3
2^D	power set of D	Ch. 5
$\#(S)$	cardinality of set S	
$\overline{S}$	closure of S	
$B_X = \{x \in X : \|x\| \leq 1\}$	closed unit ball of X	App. B
$L^p(\Omega)$, $W^{k,p}(\Omega)$	Lebesgue and Sobolev spaces	Ch. 2
$\mathcal{HS}(H_1, H_2)$	Hilbert–Schmidt operators $H_1 \to H_2$	App. B
K_G	Grothendieck constant	App. B

Norm conditions and crossnorms

The following conditions on a norm $\|\cdot\|_D$ on $\mathbf{V}_D$ are used throughout; they are formally introduced in Chapter 4 and App. B.

(Cross)	$\lVert \bigotimes_j v_j \rVert_D = \prod_j \lVert v_j \rVert_j$ (crossnorm condition)	Ch. 3
(Inj)	$\lVert\cdot\rVert_D \geq \lVert\cdot\rVert_\varepsilon$ (reasonable crossnorm / injective condition)	Ch. 4
$\lVert \mathbf{v} \rVert_\varepsilon$	injective (spatial / Grothendieck) norm	Ch. 4
$\lVert \mathbf{v} \rVert_\pi$	projective (nuclear) norm	Ch. 3
$\lVert \cdot \rVert_H$	Hilbert tensor norm on $\bigotimes_j H_j$	App. B
(Inj)[T_D]	T_D-local version of (Inj), one condition per level of tree T_D	Ch. 7

Banach spaces and linear maps

$(X, \lVert\cdot\rVert)$	normed (Banach) space	Ch. 2
$X^* = \mathcal{L}(X, \mathbb{K})$	topological dual of X	Ch. 2
X'	algebraic dual of X	Ch. 2
$\langle x, f \rangle$	duality pairing, $f \in X^*$, $x \in X$	
$\mathcal{L}(X, Y)$	bounded linear maps $X \to Y$	Ch. 2
$\mathcal{L}(X)$	$\mathcal{L}(X, X)$	
$\lVert A \rVert_{Y \leftarrow X}$	operator norm of $A \in \mathcal{L}(X, Y)$	Ch. 2
$\mathrm{GL}(X)$	bounded linear automorphisms of X (general linear group)	Ch. 2
$\mathrm{GL}(r)\mathbb{R}$	invertible $r \times r$ real matrices	App. A
id_X	identity map on X	
$P_{U \oplus W}$	projection onto U along W	Ch. 2
$J : X \to 2^{X^*}$	normalised duality mapping	Ch. 10
$J_X : X \to X^{**}$	canonical embedding into bidual	App. B
$\phi(x, y)$	Lyapunov functional $\lVert x \rVert^2 - 2\langle x, J(y) \rangle + \lVert y \rVert^2$	Ch. 10
$\mathcal{L}_{(U,W)}(X)$	nilpotent algebra associated to splitting $X = U \oplus W$	Ch. 2

Grassmannians and subspaces

$\mathbb{G}(X)$	Grassmannian of all split subspaces of X	Ch. 2
$\mathbb{G}_r(X)$	r-dimensional split subspaces of X	Ch. 2
$\mathbb{G}(W, X)$	split subspaces of X with complement W (chart domain of $\mathbb{G}(X)$)	Ch. 2
$\Psi_{U \oplus W}$	Grassmann chart map at U with complement W	Ch. 2
$T_U \mathbb{G}_r(X) \cong \mathcal{L}(U, W)$	tangent space to Grassmannian	Ch. 2
$\mathrm{Fr}(\mathbb{M})$	frame bundle of manifold $\mathbb{M}$	App. A

Component spaces and tensor products

V_j, V_α	component spaces ($j \in D$, $\alpha \subset D$)	Ch. 3
$\|\cdot\|_j$, $\|\cdot\|_\alpha$	norms on V_j, V_α	
$\mathbf{V}_D = \bigotimes_{j \in D} V_j$	algebraic tensor product	Ch. 3
$\mathbf{V}_\alpha = \bigotimes_{j \in \alpha} V_j$	partial algebraic tensor product	Ch. 3
$v \otimes_a w$	algebraic tensor product of vectors	Ch. 3
$\mathbf{v}, \mathbf{u}, \mathbf{w}$	elements of $\mathbf{V}_D$ (tensors)	
$\mathbf{V}_{D,\|\cdot\|_D}$	completion of $\mathbf{V}_D$ under $\|\cdot\|_D$	Ch. 3
$\|\cdot\|_D$	norm on $\mathbf{V}_D$ (or $\mathbf{V}_{D,\|\cdot\|_D}$)	
$H_1 \otimes_H H_2$	Hilbert tensor product (completion under $\|\cdot\|_H$)	App. B
$\mathcal{M}_\alpha(\mathbf{v})$	α-matricisation (unfolding) of $\mathbf{v}$	Ch. 4

Minimal subspaces and Tucker format

$U_\alpha^{\min}(\mathbf{v})$	minimal subspace of $\mathbf{v}$ w.r.t. direction α	Ch. 4
$W_\alpha^{\min}(\mathbf{v})$	complement of $U_\alpha^{\min}(\mathbf{v})$ used in local charts	Ch. 8
$r_\alpha(\mathbf{v}) = \dim U_\alpha^{\min}(\mathbf{v})$	α-rank of $\mathbf{v}$	Ch. 4
$\mathfrak{r} = (r_\alpha)_{\alpha \in D}$	Tucker rank tuple	Ch. 4
$\mathcal{AD}(\mathbf{V}_D)$	admissible rank tuples for $\mathbf{V}_D$	Ch. 4
$\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$	Tucker tensors with fixed rank $\mathfrak{r}$	Ch. 4
$\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$	Tucker tensors with bounded rank $\leq \mathfrak{r}$	Ch. 4
$\varrho_\mathfrak{r}$	rank map $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D) \to \prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$	Ch. 8
$Z^{(D)}(\mathbf{v})$	tangent subspace of $\mathbf{V}_{D,\|\cdot\|_D}$ at $\mathbf{v} \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$	Ch. 8
$\mathbf{E}_D$	model Banach space of Tucker manifold, $\prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \times \mathbb{R}^{\times r_\alpha}$	Ch. 8
$\xi_\mathbf{v}$	local chart map at $\mathbf{v} \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$	Ch. 8
$\mathcal{U}(\mathbf{v})$	chart domain at $\mathbf{v}$	Ch. 8
$\chi_\mathbf{v}$	fibre-bundle trivialisation map at $\mathbf{v}$	App. A
$F_\mathfrak{r} = \mathbb{R}_*^{\times_\alpha r_\alpha}$	typical fibre of the Tucker bundle (full-rank core tensors)	App. A

Trees and tree-based formats

T_D	dimension partition tree on D	Ch. 5
$S(\alpha)$	sons of internal vertex α in T_D	Ch. 5
$\mathcal{L}(T_D)$	leaves of T_D (singletons $\{j\}$, $j \in D$)	Ch. 5
$\mathrm{level}(\alpha)$	level of vertex α	Ch. 5
$\mathrm{depth}(T_D)$	depth of the tree T_D	Ch. 5
$\mathcal{P}_k(T_D)$	level-k partition of D induced by T_D	Ch. 5
T_D^{Tucker}	Tucker tree (star, depth 1)	Ch. 5
T_D^{TT}	Tensor Train tree (linear chain, depth $d-1$)	Ch. 5
T_D^{HT}	Hierarchical Tucker tree (complete binary, depth $\lceil \log_2 d \rceil$)	Ch. 5
$\mathfrak{r} = (\mathfrak{r}_\alpha)_{\alpha \in T_D \setminus \mathcal{L}(T_D)}$	tree-based rank tuple	Ch. 6
$\mathcal{AD}(\mathbf{V}_D, T_D)$	admissible tree-based rank tuples	Ch. 6
$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$	= tree-based tensors with fixed rank $\mathfrak{r}$	Ch. 6
$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$		
$\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$	tree-based tensors with bounded rank $\leq \mathfrak{r}$	Ch. 7
$C^{(\alpha)}$	transfer tensor at vertex $\alpha \in T_D$	Ch. 6
Δ_k	Laplacian-like map at level k of T_D, used in construction of tree-based atlas	Ch. 9

Differential geometry and tangent bundle

$\mathbb{M}$, $\mathbb{N}$	Banach manifolds	Ch. 2
$T_m\mathbb{M}$	tangent space at $m \in \mathbb{M}$	Ch. 2
$T\mathbb{M}$	tangent bundle of $\mathbb{M}$, $\bigsqcup_m T_m\mathbb{M}$	App. A
$\pi_{T\mathbb{M}} : T\mathbb{M} \to \mathbb{M}$	bundle projection	App. A
$\sigma_0 : \mathbb{M} \to T\mathbb{M}$	zero section, $m \mapsto (m, 0)$	App. A
$Tf(m)$	tangent map (derivative) of f at m	Ch. 2
$Df(m)$	Fréchet derivative of f at m	Ch. 2
$\mathcal{C}^p$	p times continuously Fréchet differentiable	
$\mathcal{C}^\omega$	analytic (real or complex)	
$\mathfrak{X}(\mathbb{M})$	smooth vector fields on $\mathbb{M}$	App. A
$\Gamma(E)$	smooth sections of vector bundle E	App. A
∇	linear connection on a vector bundle	App. A
$R(X,Y)Z$	curvature tensor of connection ∇	App. A
$\mathrm{Hol}_m(\nabla)$	holonomy group of ∇ at m	App. A

Dirac–Frenkel variational principle

$\mathbf{F}(t,\mathbf{v})$	vector field (right-hand side of ODE)	Ch. 10
$\mathbf{v}_r(t)$	solution curve on $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$	Ch. 10
$\mathcal{P}_{\mathbf{v}}$	metric projection onto $Z^{(D)}(\mathbf{v})$	Ch. 10
$\Pi_{\mathbf{v}}$	generalised projection onto $Z^{(D)}(\mathbf{v})$ (Banach analogue)	Ch. 10
$P_C(x)$, $\Pi_C(x)$	metric and generalised projection onto convex set C	App. B
$\arg\min$	argument of the minimum	Ch. 10

Assumptions and conditions (summary)

The following labelled conditions appear repeatedly across chapters. A condition tagged with $[T_D]$ indicates the tree-based version, which specialises to the Tucker version when $T_D = T_D^{\mathrm{Tucker}}$.

(Cross)	tensor product map is continuous ($\|\cdot\|_D$ is a crossnorm)	Ch. 3
(Inj)	$\|\cdot\|_D \geq \|\cdot\|_\varepsilon$ (reasonable crossnorm; implies (Cross))	Ch. 4
(Inj)$[T_D]$	(Inj) holds at every level of T_D: $\|\cdot\|_\alpha \geq \|\cdot\|_{\varepsilon,\alpha}$ for $\alpha \in T_D$	Ch. 7

Common abbreviations

TT	Tensor Train
HT	Hierarchical Tucker
TBT	Tree-Based Tensor
HOSVD	Higher-Order Singular Value Decomposition
MCTDH	Multiconfiguration Time-Dependent Hartree
MPS	Matrix Product State
DMRG	Density Matrix Renormalisation Group
ALS	Alternating Least Squares
ODE	Ordinary Differential Equation
PDE	Partial Differential Equation
SVD	Singular Value Decomposition

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